Manual

Service Interface

SCS-SIF 8.1

Version 1.1.23372

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Contents

1 HYDRA Service Interface

Purpose

You use the HYDRA Service Interface to connect external systems and applications via web services. Using the services, you can request, receive and process data and therefore use all functionalities that are available in the server. The Service Interface provides the following functionalities:

- Receiving service requests via http/https. Parameters are transferred in format JavaScript Object Notation (JSON).
- Providing results and information from HYDRA in format JavaScript Object Notation (JSON).
- Transmitting data to the HYDRA database and processing this data in the installed and licensed HYDRA applications.
- The interface is based on service requests via http/https and REST protocol (level 0 and 1).
- The PDM dialogs of the former HYDRA Production Data Manager (SCS-PDM) in default format are included in the HYDRA Service Interface.
- The basic license includes the technical interface, the documentation of the released services and a service tester to easily test service requests during test operation.

Requirements

- Software status HYDRA MW 3.x, service pack 12. You require service pack 13 for the file download and upload.
- Two licenses are required: the basic license SCS-SIF and the client license SCS-SIC "Service Interface" for each client using services.
- User administration: to communicate via interface, a user login with a HYDRA user is required. You can use existing users or separate communication users. The user selected for login must have function authorizations:
  - From MW 4.0$_{pe}$: Each service called up must be enabled for the user by function authorizations. The service tester uses the "MDUser.list" service when logging in, so the user must have at least the function authorization "Svc:MDUser.list". For more information on the function authorizations for services, refer to the chapter on Interface Technology (in the same document or in the Service Interface manual if you use the Service Tester documentation as a separate document).
  - User must have the function authorization „svcitf.login“ for MW 3.x.
- The external systems meet the technical requirements to process http/https and JSON.
- The customer and MPDV discuss and coordinate the planned data exchange in detail (chargeable service).
- You require the respective licenses of the HYDRA applications that you use via the interface.

Implementation notes

You use this component, if the following is true:

- You want to connect external systems or clients to the HYDRA system.
- You use web services via http(s) to exchange data.

MPDV offers individual workshops to integrate and use the HYDRA Service Interface efficiently. It has proven its worth to implement the Service Interface as part of a "Development Consulting" because the customer benefits of a customized support. After a short training in theory, you can directly start to implement your requirements with the support of our experienced trainers.

Integration

The data interface extends HYDRA 8 / MES Weaver 3.0 (MW 3.0). Using the interface, you can request HYDRA services via RESTful architecture (Representational State Transfer). The interface supports the released services and also PDM dialogs.

As of version MES Weaver 3.x, service pack 12, the data interface is included in the Web Service Provider WSP.

2 Interface Technology

2.1 Overview

The Service Interface enhances the Web Service Provider WSP and provides the possibility to request web services using REST via http or https. This documentation provides a technical description of how to use the REST interface using client software.

For the examples in the following, it is assumed that the interface is implemented for system number 1. In case of system number 1, you can locally access the interface via the address

https://localhost:8080

If you want to get access from other hosts, you must replace localhost with the precise host name. If required, you must replace port 8080 with the right port for the system number, e.g. 801 for system number 2.

To keep the http session alive, you must specify the "Fully Qualified Host Name" (FQHN). This is either the host name (sometimes with subnet and domain) or the IP address. Examples of addresses with "Fully Qualified Host Names":

- "https://myserver.example.com:8080" (if no subnet exists)
- "https://myserver.mysubnet.example.com:8080" (with subnet)
- "https://192.0.2.123:8080"

Representational State Transfer

Wikipedia

REpresentational State Transfer (REST) is an architectural style that defines a set of constraints and properties based on HTTP. Web Services that conform to the REST architectural style, or RESTful web services, provide interoperability between computer systems on the Internet. REST-compliant web services allow the requesting systems to access and manipulate textual representations of web resources by using a uniform and predefined set of stateless operations. Other kinds of web services, such as SOAP web services, expose their own arbitrary sets of operations.

[…]

By using a stateless protocol and standard operations, REST systems aim for fast performance, reliability, and the ability to grow, by re-using components that can be managed and updated without affecting the system as a whole, even while it is running.

Note on https:

The Service Interface supports https. The communication is then encoded. You can configure the communication via https during installation/implementation or later on the server. This communication then affects not only the Service Interface, but also all other clients that communicate via http with the server, e.g. the MOC.

If you create own clients, you must decide if your client knows and checks the certificate or not. The technical aspects only depend on the technology and development environment you use on your client..

2.2 General

2.2.1 Formats and standards used

The following formats and standards are used and are required when using the interface:

- HTTP authentication with "Basic Authentication" according to RFC 2617
- Base64 encoding of binary data
- Date format ISO 8601 e.g. '2015-04-28T14:47:55+02:00'

Note: the date format according to ISO 8601 can have different formats. As of service pack 15, the time stamps also include milliseconds if you enable this accuracy in the system, e.g. '2020-04-28T14:47:55.123+02:00'. This display complies with ISO 8601.

Only very few services of the process data processing currently use time stamps including milliseconds (2020).

2.2.2http Session / Authorization concept

2.2.2.1 User login and session management

The user login is performed via Basic Authentication with a user who is created in the user configuration and who has the required authorizations for responsibility areas and function authorizations.

The function authorizations required depend on the relevant system version. Details are described in the following section.

When you request data, the system checks the responsibility areas the user is authorized for. Condition: The requested data is assigned to a responsibility area and the check is implemented in the web service.

You can only log in and open a session, if the interface is licensed (e.g. license SCS-SIF) and a sufficient number of client licenses is available in the system.

When you first call a service with user login via Basic Authentication, the server returns a cookie with a session ID. From the second call on, you must pass the cookie with the session ID in the http request instead of a Basic Authenitcation.

If you perform several service requests one after the other, always use the session of the first call from the second call on. If you do not use this session, a new session is created with each service request that is only closed once a specific time has elapsed without activity.

A great number of open sessions uses resources and has a negative effect on the system performance.

A request may include **either** user and password for Basic Authentication **or** the session cookie.

This ensures, that you can correctly identify expired sessions on your client and do not open many new sessions unintentionally.

2.2.2.2 Function authorizations

As of MW 4.0$_{pe}$ and MIP:

A user must possess a function authorization for each service or PDM dialog that the user must be able to call.

- The function authorizations are built as follows: the prefix "Svc:" plus name of the service the authorization applies to.
- With PDM dialogs, the authorizations are built as follows: prefix "Svc:DLG_" plus dialog.

- If the PDM dialog includes special characters, e.g. a semicolon, you must replace these characters with underscores in the function authorization. A dot in the dialog remains a dot in the function authorization.

You can assign function authorizations for single services or for groups of services or PDM dialogs. For groups of services or PDM dialogs, you can use the placeholders "*" and "?" in the service name.

- * : the asterisk stands for no or any number of characters
- ? : the question mark stands for one character

Function authorizations for the service interface are edited in the system just like the other function authorizations. You can assign single function authorizations to users or create profiles and assign a profile to a user.

The Service Tester of MPDV uses a request of the service MDUser.list for the user login. If you want to work with the Service Tester, the user requires the function authorization "Svc:MDUser.list"!

Examples:

Function authorization	Effect
Svc:MDUnits.list	Users with this function authorization can call the service MDUnits.list.
Svc:MDUnits.*lock	Users with this function authorization can call the services MDUnits.lock and MDUnits.unlock.
Svc:MDUnits.*	Users with this function authorization can call all services of the object MDUnits.
Svc:DLG_A_UN	Users with this function authorization can call the PDM dialog "DLG=A_UN|…".
Svc:DLG_EXAMPLE.UPDATE	Users with this function authorization can call the PDM dialog "DLG=EXAMPLE.UPDATE|…".
Svc:DLG_LIST_11	Users with this function authorization can call the PDM dialog "DLG=LIST;11|…".
Svc:*	Users with this function authorization can call all released services and PDM dialogs (**not recommended!**). This function authorization is the equivalent of the function authorization "svcitf.login" used with MW 3.x.
Svc:DLG_*	Users with this function authorization can call any released PDM dialog (**not recommended!**).

As of MW 4.0pe, the user does not need the function authorization "svcitf.login" any more that is used with MW 3.x because the system now checks the function authorizations for each service.

Up to MW 3.x:

The user must have the function authorization "svcitf.login". The user cannot log in without this function authorization. Further function authorizations are not checked. The user can call all released services.

2.2.2.3 Identifying clients via AccessId

When the clients log in, the clients pass a so-called "AccessId" to the system.

The AccessId is used to identify the implemented clients. The AccessId helps to detect errors and provides some protection against unauthorized access.

The client must pass the AccessId either as URL parameter or as http header when the session is created via Basic Authentication. For details, refer to the sections that show the general structure of service calls.

Customers and partners use different forms of AccessId:

- As a customer with licensed service interface, you use the installation ID stored in the basic settings for your self-developed clients. To pass the installation ID to the server, this ID must have an 8-digit format with leading zeros.
- MPDV assigns an individual AccessId to the partners. To pass the individually assigned AccessId to the server, this ID must have a 6-digit format with leading zeros. The AccessId of the partner must be released in the customer's system via a license. The partner provides this release to the customer.

With MW 4.0pe, the AccessId is mandatory. It is checked by the system.

Systems with MW 3.x do not necessarily require the AccessId. You can optionally pass the AccessId either with the URL or as http header. It is then ignored by the system.

This means that you can develop clients independent of the MesWeaver version. If you always pass the AccessId to the system, the clients work with MW 3.x and also with MW 4.0pe.

2.2.2.4 Lifetime and logout

To keep the http session alive, you must specify the "Fully Qualified Host Name" (FQHN). This is either the host name (sometimes with subnet and domain) or the IP address. Examples of addresses with "Fully Qualified Host Names":

- "https://myserver.example.com:8080" (if no subnet exists)
- "https://myserver.mysubnet.example.com:8080" (with subnet)
- "https://192.0.2.123:8080"

To keep the http session alive, you must send the cookies that you have received with the logon with each new request.

Each http session takes up resources in the server. You can avoid to reserve resources unnecessarily, if you end the client session via logout as soon as the connection to the server is no longer needed.

Use the following request to log out:

https://<hostname>:<port>/logout

For example:

https://localhost:8080/logout

The session is automatically closed after 30 minutes without activity. When the session has expired or the session cookie is invalid, the following error message with http status code 401 is shown when a service is requested:


```
{
  "timestamp": 1562053321775,
  "status": 401,
  "error": "Unauthorized",
  "message": "Full authentication is required to access this resource",
  "path": "/data/MDUnits/list"
}
```


After error message with status code 401, the client must re-log in with Basic Authentication.

2.2.3 Recommended frameworks for the client

To implement clients, a great number of frameworks, libraries and tools is available on the Internet to manage the http connections and to manage the information in JSON format. To implement your client, you can use any tools that guarantee a correct operation of the http connection and provide valid JSON contents.

To map the JSON format transmitted via REST to objects of the used programming language, MPDV recommends the following frameworks, e.g.

- For Java 'jackson' https://github.com/FasterXML/jackson
- For C++ 'rapidjson' https://github.com/miloyip/rapidjson
- The frameworks .NET as of version 4.x, JavaScript and Python include the functions by default.

To implement the http client in Java, MPDV recommends a basic solution using the packages HttpClient and HttpCore of Apache (https://hc.apache.org/). These two packages offer a much easier support of cookies for the session management than the native Java classes for http. For the programming examples, we used the two packages HttpClient and HttpCore.

2.2.4 http operations

REST-compliant services provide different operations to deliver resources or information. The REST interface uses the two operations GET and POST.

- GET
  The operation GET retrieves information using the web service. The information is delivered in JSON format. You can also run services using GET, but you cannot specify any parameters.
- POST
  The operation POST executes the web service and provides the requested data or results in JSON format. Using the POST request, you can transfer data in the request body. For example, you can pass selection criteria for list services or data to create or change objects.

You can directly enter a service request (without passing additional parameters, e.g. filter parameters) in the command line (= URL) of any Internet browser (e.g. Firefox, Chrome) (http method "GET"). If parameters are passed for the service, e.g. filter criteria with "list" services or values with "insert"/"update" services, these parameters are passed in the body of a POST request. A browser requires additional applications here, e.g. Postman in Chrome.

In later sections, you will find examples to request web services in an Internet browser without additional program code or external programs. The document also gives examples how to request web services and PDM dialogs from external programs.

2.3 Communication process

The following sections include general communication examples. In our examples, we use the Internet browser to request services.

We recommend to install a plug-in or an extension in the browser to be able to read the results issued by the services in JSON format more easily. For Chrome, this would be e.g. "JSON Formatter".

2.3.1 Passing the AccessId

You can pass the AccessId either as URL parameter or in a header. Note: the ID is a character string in the server. For this reason, you must format the AccessId with leading zeros.

- As a customer, format the installation ID stored in the basic settings as AccessId with **8 digits** and leading zeros.
- The partners format the individually assigned AccessId with **6 digits** and leading zeros.

As a parameter of the URL

```
https://<hostname>:<port>/meta?X-Access-Id=<AccessId>
```

Example (using the installation ID of the basic settings as AccessId):

```
https://<hostname>:<port>/meta?X-Access-Id=00012345
```

Below, you will find many examples of transferring the AccessId as URL parameter.

As http header

```
X-Access-Id=<AccessId>
```

Example (using the installation ID of the basic settings as AccessId):

```
X-Access-Id=00012345
```

For further information, refer to the programming examples below.

2.3.2 Reading service information

2.3.2.1 Overview of all existing services

Use the function "meta" to show a list of all existing services:

URL

https://<hostname>:<port>/meta?X-Access-Id=<AccessId>

Example

https://localhost:8080/meta?X-Access-Id=01234567

Result

```
[
...
"InspectionOrderCharacteristic.deleteDocument",
"Kanban.stock",
```

```
"CostCenters.lock",
"InterpolationPoint.list",
"SYSScheduler.list",
"PlanningKeyfigure.lock",
"QMDepartments.list",
"QMCharacteristic.lockDocument",
"BOResource.workplaceOverview",
"WageTypeStatisticsConfiguration.copy",
"InspectionRequirement.cancel",
"Clockings.lock",
"PersonalAccountsBooking.new",
"MDReasonText.delete",
"MDPerformanceAccount.lock",
"WorkflowManagementRole.insert",
...
]
```

Note: Not all existing services are released for use by the Service Interface.

- SCS-SIF: The released services are listed in a separate section in the product documentation.
- MIP: For information on the released services, refer to the object documentation.

2.3.2.2 Service definition

Use the function "meta/<domain>/<service>" to show the definition of a precise service.

URL

https://<hostname>:<port>/meta/<domain>/<service>?X-Access-Id=<AccessId>

Example

Example: Description of the service 'MDUnits.list':

https://localhost:8080/meta/MDUnits/list?X-Access-Id=01234567

Result (shortened)

```
{
  "name": "MDUnits.list",
  "function": "list",
  "parameters": [
    {
      "acronym": "units.unit",
      "webServiceType": "STRING",
      "isMandatory": false,
      "supportedOperators": [
        "BETWEEN",
        "BETWEEN_OR_NULL",
        "EQUAL",
        "EQUAL_OR_NULL",
        "GT",
        "GTE",
        "GTE_OR_NULL",
        "GT_OR_NULL",
        "IN",
        "IN_OR_NULL",
        "LIKE",
```

```
        "LIKE_OR_NULL",
        "LT",
        "LTE",
        "LTE_OR_NULL",
        "LT_OR_NULL",
        "NOT_EQUAL",
        "NOT_EQUAL_OR_NULL"
      ]
    },
    {
      "acronym": "units.classification",
      "webServiceType": "STRING",
      "isMandatory": false,
      "supportedOperators": [
        "BETWEEN",
        "BETWEEN_OR_NULL",
        "EQUAL",
        "EQUAL_OR_NULL",
        "GT",
        "GTE",
        "GTE_OR_NULL",
        "GT_OR_NULL",
        "IN",
        "IN_OR_NULL",
        "LIKE",
        "LIKE_OR_NULL",
        "LT",
        "LTE",
        "LTE_OR_NULL",
        "LT_OR_NULL",
        "NOT_EQUAL",
        "NOT_EQUAL_OR_NULL"
      ]
    },
    {
      "acronym": "units.designation",
      "webServiceType": "STRING",
      "isMandatory": false,
      "supportedOperators": [
        "BETWEEN",
        "BETWEEN_OR_NULL",
        "EQUAL",
        "EQUAL_OR_NULL",
        "GT",
        "GTE",
        "GTE_OR_NULL",
        "GT_OR_NULL",
        "IN",
        "IN_OR_NULL",
        "LIKE",
        "LIKE_OR_NULL",
        "LT",
        "LTE",
        "LTE_OR_NULL",
        "LT_OR_NULL",
        "NOT_EQUAL",
        "NOT_EQUAL_OR_NULL"
      ]
    }
  ],
  "requestableColumns": [
    {
      "acronym": "units.unit",
      "webServiceType": "STRING"
    },
    {
      "acronym": "units.classification",
      "webServiceType": "STRING"
    },
    {
      "acronym": "units.designation",
      "webServiceType": "STRING"
    }
  ]
}
```

2.3.3 Service request

A service request includes the server address with port, the key expression "data", the function name and a request "body". To enter the "body" and to be able to transmit additional parameters (HTTP method "POST"), you need an appropriate editor as plug-in application in the browser (e.g. the Postman application in Chrome) or you must carry out the request via an interface program, e.g. JavaScript, .Net. You will find further examples for different programming environments in the sections that follow.

Potential parameters for the service request in the "body":

- **params** (optional) ➔ a list of filter parameters (filter values or input data)
- **columns** (optional) ➔ the list of the return columns of the request
- **requestId** (optional) ➔ unique consecutive number
  You need this number that is unique for each session to precisely identify a request. Using this number, you can assign the server returns and log outputs on the server to the original request. Using this unique number, you can also cancel a request, e.g. if processing of the request is too long.
  If the client does not send a requestId, the server automatically assigns internal consecutive numbers.

The requestId is optional. But for each logon, the client may either send service requests always **without** or always **with** own requestId to avoid conflicts between the requestIds generated on the client and on the server.

If the client wants to use requestIds, the client must also administer the Ids for each service request.

- After logon, the client must start with the value 1.
- With each service request, the client must increment the value by one.
- The requestId can have a maximum value of 2147483647 (32 bit signed integer). Before the maximum value is exceeded, the client must restart with a requestId with value 1.

- **returnAsObject** (optional) ➔ true/false. The default value is "false". If you set this option to "true", the result is returned as complete object in JSON format. If the result data is returned as objects, parsing of result data in the client is easier because each data row is a fully qualified JSON object. Otherwise you always require the META information of the result to interpret a data row. The option "returnAsObject" increases the result size by the factor 2 and therefore slightly reduces the performance of the data transfer.

2.3.3.1 Service request without parameters

Use the key expression "data" to request a service:

https://<hostname>:<port>/data/<domain>/<service>?X-Access-Id=<AccessId>

Request of the service 'MDUnits.list' via

https://localhost:8080/data/MDUnits/list?X-Access-Id=01234567

with the request body:

Request body

```
{
  "params": [],
  "columns": [
            "units.unit",
    "units.modified_by",
    "units.si_unit",
    "units.unitiso",
    "units.designation",
    "units.classification",
    "units.modified_ts"
  ],
          "requestId": 11,
  "returnAsObject": false
}
```

To show the data, the system checks the responsibility areas the logged on user is authorized for. Precondition: The requested data is assigned to a responsibility area and the check is implemented in the service.

Result (returnAsObject=false)


```
[ {
  "__rowType" : "META",
  "__type" : "units",
  "data" : [ {
    "name" : "units.unit",
    "index" : 0,
    "type" : "STRING"
  }, {
    "name" : "units.modified_by",
    "index" : 0,
    "type" : "STRING"
  }, {
    "name" : "units.unitiso",
    "index" : 0,
    "type" : "STRING"
  }, {
    "name" : "units.designation",
    "index" : 0,
    "type" : "STRING"
  }, {
    "name" : "units.classification",
    "index" : 0,
    "type" : "STRING"
  }, {
    "name" : "units.modified_ts",
```

```
    "index" : 0,
    "type" : "DATETIME"
  }, {
    "name" : "units.si_unit",
    "index" : 0,
    "type" : "BOOLEAN"
  } ]
}, {
  "__rowType" : "DATA",
  "__type" : "units",
  "data" : [ "C", "12345", "C", "Temperature", "Temperature", "2018-02-21T13:03:16+01:00", null ]
}, {
  "__rowType" : "DATA",
  "__type" : "units",
  "data" : [ "CM", "12345", "CMT", "Centimeter", "Length", "2004-07-16T00:00:00+02:00", false ]
}, {
  "__rowType" : "DATA",
  "__type" : "units",
  "data" : [ "F", "12345", null, "Fahrenheit", "Temperature", "2004-07-16T00:00:00+02:00", false ]
}, {
  "__rowType" : "DATA",
  "__type" : "units",
  "data" : [ "FT", "12345", "FOT", "Foot", "Length", "2004-07-16T00:00:00+02:00", false ]
}
]
```


Result (returnAsObject=true)


```
[ {
  "__rowType" : "META",
  "__type" : "units",
  "data" : [ {
    "name" : "units.unit",
    "index" : 0,
    "type" : "STRING"
  }, {
    "name" : "units.modified_by",
    "index" : 0,
    "type" : "STRING"
  }, {
    "name" : "units.unitiso",
    "index" : 0,
    "type" : "STRING"
  }, {
    "name" : "units.designation",
    "index" : 0,
    "type" : "STRING"
  }, {
    "name" : "units.classification",
    "index" : 0,
    "type" : "STRING"
  }, {
    "name" : "units.modified_ts",
    "index" : 0,
    "type" : "DATETIME"
  }, {
    "name" : "units.si_unit",
    "index" : 0,
    "type" : "BOOLEAN"
  } ]
}, {
  "__rowType" : "OBJECT",
  "__type" : "units",
  "data" : {
    "units.unit" : "C",
    "units.modified_by" : "Schreiber",
    "units.unitiso" : "C",
    "units.designation" : "Temperatur",
    "units.classification" : "Temperatur",
    "units.modified_ts" : "2018-02-21T13:03:16+01:00"
  }
}, {
  "__rowType" : "OBJECT",
  "__type" : "units",
  "data" : {
    "units.unit" : "CM",
    "units.modified_by" : "12345",
    "units.si_unit" : false,
    "units.unitiso" : "CMT",
    "units.designation" : "Zentimeter",
    "units.classification" : "Länge",
    "units.modified_ts" : "2004-07-16T00:00:00+02:00"
  }
}, {
  "__rowType" : "OBJECT",
  "__type" : "units",
  "data" : {
    "units.unit" : "F",
    "units.modified_by" : "12345",
    "units.si_unit" : false,
    "units.designation" : "Fahrenheit",
    "units.classification" : "Temperatur",
    "units.modified_ts" : "2004-07-16T00:00:00+02:00"
  }
```

```
}, {
  "__rowType" : "OBJECT",
  "__type" : "units",
  "data" : {
    "units.unit" : "FT",
    "units.modified_by" : "12345",
    "units.si_unit" : false,
    "units.unitiso" : "FOT",
    "units.designation" : "Fuß",
    "units.classification" : "Länge",
    "units.modified_ts" : "2004-07-16T00:00:00+02:00"
  }
} ]
```


2.3.3.2 JSON data structure of result

The format with returnAsObject=true is selfexplanatory. For returnAsObject=false, the following applies:

The result is an array of 'rows'. Each row has a type *__rowType*.

Requests with errors

If a "row" of *rowType* "ERROR" is included, then an error occurred in the service request. The data of the "row" includes further information on the error. See also the below description of error handling.

Requests without errors

In a request without errors, the first "row" has *rowType* "META". The row includes an array of objects that show column name and data type for each column of the result set.

All other *rows* have *rowType* "DATA". The content depends on the option "returnAsObject":

- returnAsObject=false: the *row* includes an array that contains for each column the value for the META data.
- returnAsObject=false: the *row* includes a complete JSON object.

A service request can return several *result sets*, e.g. a list of persons including data from the HR master data and another list including planning data for the persons. In case of services with several *result sets*, you can identify connected data via the *__type.* In case of services with only one result set, the system uses a generated value for the *__type.*

2.3.3.3 Error handling

If you use the interface in an incorrect way and the server cannot perform a proper processing, you get an http status code unequal to 200 (OK) as result.

If the server can interpret the request, you get the status code 200 (OK) even if errors were detected in the further processing, e.g. missing mandatory parameters. Also with http status code 200, you must therefore always check in the body of the http request if errors occurred.

2.3.3.3.1 http status code

http status code	Meaning
200	OK or error in the content when the service is executed. You'll find further information in the body of the http result. The body is usually a JSON object. See the description in the following.
401	"Unauthorized" Possible reasons - Wrong user/password - Session has expired or wrong session cookie - Wrong or not licensed AccessId You'll find further information in the result body. See the description in the following.

500	Another error occurred. You'll find further information on the error in the body of the http result.
	
	Examples of errors that can occur during data download:
	
	`The file 'F:\hydra3\3\spool\xxx_persons_list.dat' does not exist!`
	
	`User '12345' has no 'READ' permission for file '3\spool\persons_list.dat'!`

2.3.3.3.2 Data in the body of the http result

Explanation using the example of an invalid combination of user and password

In this example, the http status code is 401.

```
[
  {
  "__rowType" : "ERROR",
  "__type" : "ERROR",
  "data" : {
    "level" : "ERROR",
    "type" : "SYSTEM",
      "messages": [
        {
          "languageKey": "Could not authenticate",
          "parameters": null
        }
      ],
    "cause" : null,
      "stackTrace": null
    },
    "message": "Could not authenticate"
  }
]
```

Result[0].__rowType = "ERROR"

The rowType ERROR indicates that an error occurred.

Result[0].data.message

The first or the main error. The output is in English.

Result[0].data.messages

Array with all errors identified. If available, the attribute does not include a long text but only a language key. Depending on the language of the user, the client then shows a readable text for this language key. You can use the services TranslationService.listLanguages and TranslationService.list to identify the languages and texts stored in the system.

Example: invalid AccessId

In this example, the http status code is 401.


```
[
  {
  "__rowType" : "ERROR",
  "__type" : "ERROR",
  "data" : {
    "level" : "ERROR",
    "type" : "SYSTEM",
      "messages": [
        {
          "languageKey": "REST interface usage is not allowed due to access ID: [0012345] not authorized",
          "parameters": null
        }
      ],
    "cause" : null,
      "stackTrace": null
    },
    "message": "REST interface usage is not allowed due to access ID: [0012345] not authorized"
  }
]
```


Example: expired or wrong session cookie

In this example, the http status code is 401.


```
[
  {
  "__rowType" : "ERROR",
  "__type" : "ERROR",
  "data" : {
    "level" : "ERROR",
    "type" : "SYSTEM",
      "messages": [
        {
          "languageKey": "Full authentication is required to access this resource",
          "parameters": null
        }
      ],
    "cause" : null,
      "stackTrace": null
    },
    "message": "Full authentication is required to access this resource"
  }
]
```


Example: missing function authorization for a service

In this example, the http status code is 200.


```
[
  {
  "__rowType" : "ERROR",
  "__type" : "ERROR",
  "data" : {
    "level" : "ERROR",
    "type" : "SYSTEM",
      "messages": [
        {
          "languageKey": "lkServiceMissingFctAuth",
```

```
          "parameters": [
            "MDUser.insert"
          ]
        }
      ],
      "cause": {
    "level" : "ERROR",
    "type" : "SYSTEM",
        "messages": [
          {
            "languageKey": "lkServiceMissingFctAuth",
            "parameters": [
              "MDUser.insert"
            ]
          },
          {
            "languageKey": "de.mpdv.jtp.restauthorization.RestAuthorizationException: lkServiceMissingFctAuth ...",
            "parameters": []
          }
        ],
    "cause" : null,
        "stackTrace": null
      },
      "stackTrace": "de.mpdv.hydra.common.exceptions.businessService.UnCheckedBusinessException: REST interface usage is not
allowed due to missing authorization for user: [12345] for service: [MDUser.insert]\r\n\tat
de.mpdv.jtp.restauthorization.RestAuthorizationManager. ...... 999 more\r\n"
    },
    "message": "REST interface usage is not allowed due to missing authorization for user: [12345] for service:
[MDUser.insert]"
  }
]
```


In addition to the known error information (Result[0].data.message and Result[0].data.messages), further information is output in this case, e.g. a stack trace. This information then allows further analyses, if required.

In this example, you can identify the error text of the language key "lkServiceMissingFctAuth" using the service TranslationService.list, e.g. for the language "en-US". You then get the text "Missing function authorization for service '{0}'". The texts can include placeholders (e.g. "{0}"). The client can replace the placeholders with the values of the array "parameters" to obtain an individual error message.

Placeholders	Value
{0}	First element of the array "parameters"
{1}	Second element of the array "parameters"
…	…

Example: missing mandatory parameter (1)

(ServiceType InterpretedWrapper, InterpretedJavaService2, InterpretedJavaService, …)

In this example, the http status code is 200.


```
[
  {
  "__rowType" : "ERROR",
  "__type" : "ERROR",
  "data" : {
    "level" : "ERROR",
      "type": "LOGIC",
      "messages": [
        {
          "languageKey": "lkParamMissing",
          "parameters": [
            "units.classification"
          ]
        }
      ],
```

```
    "cause" : null,
      "stackTrace": "de.mpdv.commoncontracts.serviceexecution.ServiceExecutionException: Mandatory parameter pair
units.classification.op and units.classification.param is missing\r\n\tat de.mpdv.commoncontracts. ... 999 more\r\n"
    },
    "message": "Mandatory parameter pair units.classification.op and units.classification.param is missing"
  }
]
```


Example: missing mandatory parameter (2)

(ServiceType InterpretedBapiService)


```
[ {
  "__rowType" : "ERROR",
  "__type" : "ERROR",
  "data" : {
    "level" : "ERROR",
    "type" : "LOGIC",
    "messages" : [ {
      "languageKey" : "lkParamMissing",
      "parameters" : [ "category.id" ]
    }, {
      "languageKey" : "lkParamMissing",
      "parameters" : [ "category.application" ]
    }, {
      "languageKey" : "lkParamMissing",
      "parameters" : [ "category.designation" ]
    } ],
    "cause" : null,
    "stackTrace" : "de.mpdv.commoncontracts.serviceexecution.ServiceExecutionException: ... "
  },
  "message" : "Validation error occurred"
} ]
```


In this example, the content of Result[0].data.message is quite general and Result[0].data.messages then includes all validation errors found.

Example: service not licensed


```
[ {
  "__rowType" : "ERROR",
  "__type" : "ERROR",
  "data" : {
    "level" : "ERROR",
    "type" : "SYSTEM",
    "messages" : [ {
      "languageKey" : "lkServiceUnavailable",
      "parameters" : [ "category.insert" ]
    } ],
    "cause" : {
      "level" : "ERROR",
      "type" : "SYSTEM",
      "messages" : [ {
        "languageKey" : "lkServiceUnavailable",
        "parameters" : [ "category.insert" ]
      }, {
        "languageKey" : "de.mpdv.jtp.restauthorization.RestAuthorizationException: lkServiceUnavailable\u0000REST interface
usage is not allowed due to missing license for service: [category.insert]\u0000category.insert\r\n\tat
de.mpdv.jtp.restauthorization.RestAuthorizationChecker.validateServiceIsLicensed(RestAuthorizationChecker.java:140)\r\n\t ...
",
      "parameters" : [ ]
      } ],
    "cause" : null,
      "stackTrace" : null
    },
    "stackTrace" : "de.mpdv.hydra.common.exceptions.businessService.UnCheckedBusinessException: REST interface usage is not
allowed due to missing license for service: [category.insert]\r\n\tat
de.mpdv.jtp.restauthorization.RestAuthorizationManager.surroundWithErrorHandling(RestAuthorizationManager.java:102 ... 999
more\r\n"
  },
  "message" : "REST interface usage is not allowed due to missing license for service: [category.insert]"
} ]
```

Example: error message from InterpretedWrapper


```
[ {
  "__rowType" : "ERROR",
  "__type" : "ERROR",
  "data" : {
    "level" : "ERROR",
    "type" : "LOGIC",
    "messages" : [ {
      "languageKey" : "lkRetCodeHydra",
      "parameters" : [ "1661" ]
    }, {
      "languageKey" : "lkShortTextHydra",
      "parameters" : [ "Missing parameter" ]
    }, {
      "languageKey" : "lkLongTextHydra",
      "parameters" : [ "Missing parameter FZ.DATB              " ]
    }, {
      "languageKey" : "MESSAGE_DATA",
      "parameters" : [ "id\\,1001661,P_KEY_FZ.DATB\\,FZ.DATB" ]
    }, {
      "languageKey" : "HYDRA_RESPONSE_DATA",
      "parameters" : [ "RET\\,1661,KT\\,Missing parameter,LT\\,Missing parameter FZ.DATB              " ]
    }, {
      "languageKey" : "HYDRA_REQUEST_DATA",
      "parameters" : [ "DLG\\,FZ.INSERT,MODUL\\,FZ.INSERT,CLIENT.SID\\,61701ccc-f6fe-4554-b0a7-
62227e049645,CLIENT.DEVID\\,100003@10.10.10.117,DEVICE_ID\\,100003@10.10.10.117,BEARB\\,12345,DAT\\,09/03/2019,ZEI\\,53891" ]
    } ],
    "cause" : null,
    "stackTrace" : null
  },
  "message" : "HYDRA signaled error by return code != 0. Return code is: 1661, short message is: Missing parameter, long
message is: Missing parameter FZ.DATB              "
} ]
```


You can identify this variant because in Result[0].data.messages an object with the attribute "languageKey" : "lkRetCodeHydra" is included. The error number included is unique. You can find a text for this error in Result[0].data.messages in the object with attribute "languageKey" : "lkLongTextHydra".

2.3.3.4 Service request including a filter

Example: request of service 'MDUnits.list' with a filter:

If you want to know which filter parameters are available for a service and which operators a filter supports, you can request the meta information on a service as described above:

https://<hostname>:<port>/meta/<domain>/<service>?X-Access-Id=<AccessId>

Example:

https://localhost:8080/meta/MDUnits/list?X-Access-Id=01234567

Filters are defined in the request body. And you can specify in the body which columns are shown.

URL

The URL is the same as the URL of a request without filter.

https://localhost:8080/data/MDUnits/list?X-Access-Id=01234567

Request body

```
{
  "params": [
    {
      "acronym": "units.unit",
    "value" : "CM",
    "operator" : "EQUAL"
    }
  ],
  "columns": [
              "units.unit",
    "units.designation",
  ],
          "requestId": 5,
  "returnAsObject": false
}
```

or

```
{
  "params": [
    {
      "acronym": "units.unit",
      "value": [
        "CM",
        "C",
        "F",
        "FT"
      ],
      "operator": "IN"
    }
  ],
  "columns": [
              "units.unit",
    "units.modified_by",
    "units.si_unit",
    "units.unitiso",
    "units.designation",
    "units.classification",
    "units.modified_ts"
  ],
  "requestId": 6,
  "returnAsObject": false
}
```

Result

The result corresponds to the request without filter.

2.3.3.5 Service request to create data

Technically, the request of a service to create or change data is equal to the request of a service to show data using a filter. In this case, the parameters of the service do not include filters, but new data.

Example: Create a new unit "abc" with the classification "ABC" using the following URL:

https://localhost:8080/data/MDUnits/insert?X-Access-Id=01234567

Request body


```
{
          "requestId": 12,
  "params": [
    {
      "acronym": "units.classification",
      "value": "ABC",
    "operator" : "EQUAL"
    },
    {
      "acronym": "units.unit",
      "value": "abc",
    "operator" : "EQUAL"
    }
  ],
  "columns": [
    "units.unit"
  ]
}
```


Setting parameters to empty (database "null"):

If you want to set a parameter of a service to empty, which creates or changes data (e.g. "Insert" or "Update"), you must pass the operator "EQUAL" and leave out the element "value". This is valid for all data types.


```
{
  "acronym": "units.classification",
     "operator" : "EQUAL"
}
```


Result


```
[
  {
    "__rowType": "META",
    "__type": "units",
    "data": [
      {
        "name": "units.unit",
        "type": "STRING"
      }
    ]
  },
  {
    "__rowType": "DATA",
    "__type": "units",
```

```
    "data": [
      "abc"
    ]
  }
]
```

You can only use the functions, e.g. insert, update, delete, etc., that are defined for the respective service.

2.3.3.6 Multilingual database contents

Multilingual database contents must be enabled on the server. This document does not deal with the activation of multilingual database contents.

The installation manual of your MIP or your MES Weaver system (MW) provides further information for system administrators. Contact MPDV if you need help with the activation of multilingual database contents on the server.

If multilingual database contents are activated in your system, then you can use the request to specify for the objects with multilingual database contents which of the activated and edited languages is displayed.

Attribute in the request body:

- **language** (optional) ➔ ISO language key (en, de, fr, …).

To identify the ISO language keys of the available and activated languages, use the command hyd_lang with option -l (lower case L) on the server.

- Windows: hyd_lang.exe –l
- Linux: hyd_lang.out –l

If you enter an invalid language key, the native language is used.

Request body

Example for the service MDPerformanceAccount.list:

```
{
  "params" : [ ],
  "columns" : [ "rpa.abbreviation", "rpa.color", "rpa.number", "rpa.text"
],
          "requestId": 223,
  "language" : "en",
  "returnAsObject": false
```

2.3.3.7 Cancelling a request

To cancel a previous request (in our example, the requestId 12), request the service 'System.requestCancel' using the same requestId. Proceed as follows:

https://localhost:8080/data/System/requestCancel?X-Access-Id=01234567

Request body

```
{
           "requestId": 12,
  "params": [],
  "columns": []
}
```

2.3.3.8 Requesting PDM dialogs

PDM dialogs are only available with MES Weaver systems (MW 3.x, MW 4.0 pe, ..).

The PDM dialogs are not available for MIP systems without MES Weaver functions.

Using the service interface, you can also request the older PDM dialogs. To request PDM dialogs, a POST request with body is possible.

https://server.domain.local:8080/dlg/command?X-Access-Id=01234567

The request body then includes the PDM dialog as string (no JSON).

The result body includes the return of the PDM dialog as string (no JSON).

If you call PDM dialogs, which create list files on the server, you can download the list file in a further step from the server after having requested the PDM dialog. See section "2.3.4 Download and upload of files".

Examples are provided with the examples of command line tools and with the Java programming example.

2.3.4 Download and upload of files

2.3.4.1 Overview and requirements

In some use cases, you must download files from the server to the client or upload files from the client to the server. The http requests are provided for this purpose.

To protect data on the server from unauthorized access, the access rights for users on the server are stored in a configuration file "filePermissions.json". In your company, the system administrator must edit this configuration file before you can download or upload files.

The download and upload of files is available using the Web Service Provider (WSP) as of Service Pack 13 (October 2018).

2.3.4.2 Configuration of the access rights "filePermissions.json"

You use a configuration file on the server to specify which user has read and/or write access to which files. The configuration file has the name "filePermissions.json" and is stored on the server:

x:\mip\jdir\MOC\<systemNbr>\filePermissions.json

Example:

\\myserver\mip1\jdir\MOC\1\filePermissions.json

The system administrator manually edits the file using a text editor. The file includes data in JSON format.

If you change authorizations in the file, you must restart the WSP service to activate the changes.

Structure of the JSON object

The JSON object includes an array with any number of different authorization rules. Each authorization rule consists of the following elements:

users

Array of one or several users that are configured in the system in the user administration. If an asterisk "*" is used as user, the rule applies to all users.

permission

String "READ" or "WRITE". The attribute specifies if the rule grants only read access or read and write access. Write access automatically includes read access.

files

Array of file names with paths to which the rule applies. You can specify the paths relative to the installation directory of the system. You can use the classic placeholders for file names:

* The asterisk stands for any number of any character

? The question mark stands for exactly one random character.

comment

Comment to explain the rule

Example:

```
[ { "users"     :[ "userId", "userId2", "userId3" ],
    "permission":"READ",
    "files"     :[ "./?/spool/persons_list.dat", "./?/spool/machines_list.dat", "./?/spool/*_interface.dat" ],
    "comment"   :"Access rights for RestDemoJava"
  },
  { "users"     :[ "userId" ],
    "permission":"WRITE",
    "files"     :[ "./?/spool/persons_list_uploaded.dat" ],
    "comment"   :"Access rights for RestDemoJava"
  }
]
```

2.3.4.3 Download of files from server to client

You use the key expression "dlg/fileDownload" in the URI to download files.

The file name is passed as parameter of the URL. The slash "/" is always expected as the directory separator, regardless of the operating system. File names are specified relative to the installation directory of the system.

https://<hostname>:<port>/dlg/fileDownload?file=<myFileName>&X-Access-Id=<AccessId>

Download example of file spool/persons_list.dat:

https://myserver:8080/dlg/fileDownload?file=spool%2Fpersons_list.dat&X-Access-Id=01234567

Please note the masking of special characters in the parameters in the http request with "URL encoding". For example, a slash is a "%2F". Development environments usually provide appropriate classes or APIs for coding, e.g. URIBuilder().setParameter() in Java.

Result

The content of the downloaded file is the content of the http response. See programming example in Java.

If you enter the URI in the browser, the browser saves the file in the download directory or asks for the storage location.

Result

http status code	Meaning

200	OK

500	Error has occurred. You'll find further information on the error in the body of the http result. Examples:
	
	`The file 'F:\hydra3\3\spool\xxx_persons_list.dat' does not exist!`
	
	`User '12345' has no 'READ' permission for file '3\spool\persons_list.dat'!`

2.3.4.4 Upload of files from client to server

You use the key expression "dlg/fileUpload" in the URI to upload files.

The file name on the server is passed as parameter. The slash "/" is always expected as the directory separator, regardless of the operating system. File names are specified relative to the installation directory of the system.

https://<hostname>:<port>/dlg/fileUpload?fileName=<myFileName>&X-Access-Id=<AccessId>

Upload example of file spool/persons_list_uploaded.dat:

https://myserver:8080/dlg/fileUpload?fileName=spool%2Fpersons_list_uploaded.dat&X-Access-Id=01234567

Please note the masking of special characters in the parameters in the http request with "URL encoding". For example, a slash is a "%2F". Development environments usually provide appropriate classes or APIs for coding, e.g. URIBuilder().setParameter() in Java.

The upload is an "http multipart request". The file content of the uploaded file is passed as multipart entity. Sie also "multipart/form-data" at https://www.w3.org/TR/html401/interact/forms.html#h-17.13.4.2.

For implementation examples, refer to the programming examples for Java and C#.

If the file already exists on the server, the file is overwritten without confirmation.

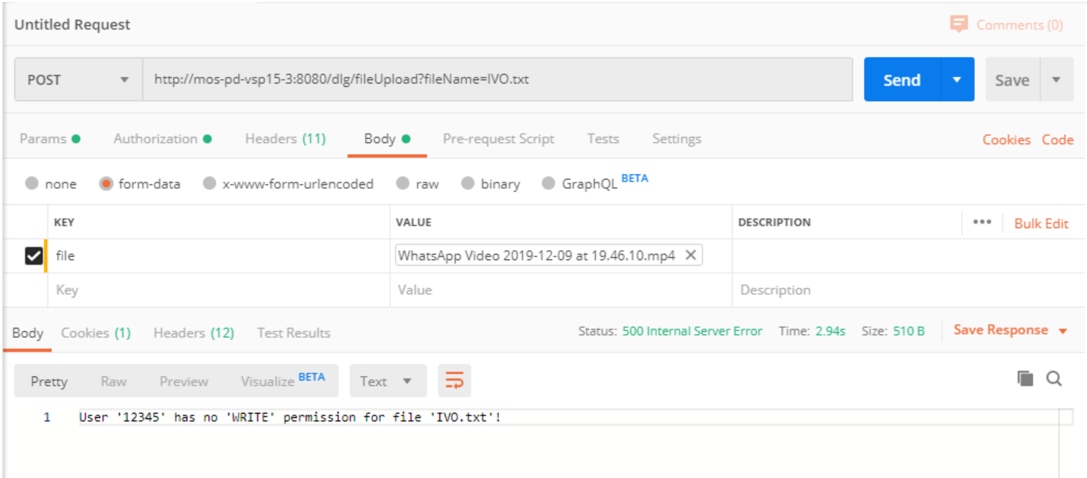
If the target directory does not exist, it is created.

Result

http status code	Meaning
200	OK

500	Error has occurred. You'll find further information on the error in the body of the http result. Examples: `User '12345' has no 'WRITE' permission for file '3\spool\persons_list.dat'!`

To test uploads, you can optionally use the free tool "Postman".



2.4 Programming examples

2.4.1 General

The examples in the following sections partly show simplified programs to demonstrate the basic interface communication. Especially the further processing of the data in JSON format is not completely implemented.

Security aspects of user and password management need to be better protected in professional applications. Important: Do not store user name and password in a fixed and unencrypted form in the program code. For the examples in the following sections, we stored user name and password in the program code for better understanding only.

The presented source codes are completely contained here and can be copied from the PDF version of this document. MPDV also provides the source code files in file form as part of the trainings and services of the Service Interface.

2.4.2 Command line

2.4.2.1 Overview

The sections in the following provide information on the tools used.

The examples below show three requests:

- The first request performs a user login with Basic Authentication, saves the session cookie in a file and gets information on the logged in user via the service interface.
- The second request uses the cookie of the first request and needs not perform a Basic Authentication. In the example, a service is used to request the "units".
- The third request is a PDM dialog and uses the session cookie of the first request.

2.4.2.2 "curl" and "wget"

https://en.wikipedia.org/wiki/Wget (07/2019):
GNU **Wget** (or just **Wget**, formerly Geturl, also written as its package name, **wget**) is a computer program that retrieves content from web servers. It is part of the GNU Project. Its name derives from World Wide Web and get. It supports downloading via HTTP, HTTPS, and FTP.

"**wget**" is available on most current Linux distributions. Refer to the Internet for downloads with Windows and further information.

The examples below have been created using "**wget**" version 1.11.4 of the year 2008.

You can use the command line programs "curl" and "wget" to call URLs. They provide all functions that are required for the service interface:

- Communication via HTTPS or HTTP
- User login with Basic Authentication
- Cookie management for the session cookie
- POST requests with JSON data in the body
- The results can be saved in a file.

There are only few differences between the programs "curl" and "wget" that can be important when used with the service interface:

- "wget" is licensed via GPL v3. "curl" is licensed via MIT
- "wget" requires GnuTLS or OpenSSL for the support of SSL/TLS.
- "curl" is available for more (exotic) server platforms.

- "wget" can be entered with the left hand only on most keyboards with Latin alphabet.

"curl" and "wget" are not products of MPDV Mikrolab GmbH.

- MPDV does not provide any warranty, support or maintenance for the two programs.
- MPDV does not provide a complete documentation for the two programs.

2.4.2.3 HTTPS with "curl" and "wget"

If the protocol of the URL is "https:", then both programs automatically use HTTPS.

Without further configuration, the communication via https is encrypted. Depending on the versions of "curl" and "wget", the programs check the validity of the certificate by default or not.

You can also configure or control via command line parameters that the certificate validity is checked. Use the usual search engines on the Internet to get further information.

2.4.2.4 Requesting a service with user login and start of session

In the first example, we want to log in a user using the service MDUser.list that provides information on the user directly with the login.

The result returned includes the user information provided by the service and a cookie with session ID.

In the example, we use the user 12345. When you try the examples on your system, you must everywhere replace the user 12345 with a user of your system or at least change the password, if required. And you must use the URL of your system.

If you perform several service requests one after the other, always use the session of the first call from the second call on. If you do not use this session, a new session is created with each service request that is only closed once a specific time has elapsed without activity.

A great number of open sessions uses resources and has a negative effect on the system performance.

Request of "curl" (all in one row):

```
curl --user "12345:secret" -j -c curl_cookies.txt -X POST -H "Content-Type: application/json" -H "X-Access-Id: 01234567" -d
@request1.json -o curl_result1.json -k https://server.domain.local:8080/data/MDUser/list
```

Description of the command line parameters:

--user "12345:secret"

User login with Basic Authentication. In the example: User "12345" with password "secret". If user and password do not contain any special characters, you can omit the quotation marks.

-j

Re-initialize cookie memory.

-c curl_cookies.txt

Saves the cookies received from the server in the specified file.

-X POST

Using this parameter, "curl" performs a POST request. Standard is a GET request.

-H "Content-Type: application/json"

HTTP header that specifies the content type of the body as JSON.

-H " X-Access-Id: 01234567"

Transfer of the AccessId in the http header. The example uses an installation ID with 8 digits.

-d @request1.json

Data for the body of the POST request in the specified file. The data includes the input parameters for the service requested.

-o curl_result1.json

The program saves the output in the file specified.

-k

Disables the certificate check with https. Depending on the "curl" version, the program automatically checks the certificate with https connections. You can then provide the certificate (see documentation of "curl") or use the option -k to disable the certificate check for the https connection.

https://server.domain.local:8080/data/MDUser/list

Server URL with specification of the service that is requested, in the example MDUser.list. If the protocol is "https:", HTTPS is automatically used.

Request of "wget" (all in one row):

```
wget --user="12345" --password="secret" --save-cookies wget_cookies.txt --keep-session-cookies --header="X-Access-Id:
01234567" --header="Content-Type: application/json" --post-file=request1.json --output-document=wget_result1.json --no-check-
certificate https://server.domain.local:8080/data/MDUser/list
```

Description of the command line parameters:

--user="12345"

--password="secret"

User login with Basic Authentication. In the example: User "12345" with password "secret". If user and password do not contain any special characters, you can omit the quotation marks.

--save-cookies cookies.txt

Saves the cookies received from the server in the specified file.

--keep-session-cookies

Also saves session cookies. Without this parameter, "wget" does not save the session cookies in the cookie file because they are only valid for a limited period of time.

--header="Content-Type: application/json"

HTTP header that specifies the content type of the body as JSON.

--header=" X-Access-Id: 01234567"

Transfer of the AccessId in the http header. The example uses an installation ID with 8 digits.

--post-file=request1.json

"wget" uses the method POST for the request. The data for the body of the POST request is read in the file specified. The data includes the input parameters for the service requested.

--output-document=wget_result1.json

The data read is saved in the file specified.

https://server.example.com.com:8080/data/MDUser/list

Server URL with specification of the service that is requested, in the example MDUser.list. If the protocol is "https:", HTTPS is automatically used.

Content of the request file request1.json:


```
{
  "params": [
    {
      "acronym": "user.id",
      "value": "12345",
      "operator": "EQUAL"
    }
  ],
  "columns": [
    "user.person.id",
    "user.id",
    "user.language",
    "user.designation"
  ],
  "returnAsObject": false
}
```


Content of the result file *_result1.json (formatted):


```
[
  {
    "__rowType": "META",
    "__type": "user",
    "data": [
      {
        "name": "user.designation",
```

```
        "type": "STRING"
      },
      {
        "name": "user.id",
        "type": "STRING"
      },
      {
        "name": "user.language",
        "type": "STRING"
      },
      {
        "name": "user.person.id",
        "type": "INTEGER"
      }
    ]
  },
  {
    "__rowType": "DATA",
    "__type": "user",
    "data": [
      "Administrator",
      "12345",
      "1",
      50014
    ]
  }
]
```


Content of the cookie file *_cookies.txt:

The cookie file includes the session ID after successful request. The file format is the format originally defined by Netscape. The example has been generated using "curl". The cookie file of "wget" is similar.

```
# Netscape HTTP Cookie File
# https://curl.haxx.se/docs/http-cookies.html
# This file was generated by libcurl! Edit at your own risk.

#HttpOnly_ server.domain.local FALSE    /          FALSE      0          JSESSIONID          DF3FF42E899B281B8B69453302CB57CC
```

2.4.2.5 Requesting a service in a running session

In the second example, we request a random service in a running session, in the example MDUnits.list.

Request of "curl" (all in one row):

```
curl -b curl_cookies.txt -X POST -H "Content-Type: application/json" -d @request2.json -o curl_result2.json -k
https://server.domain.local:8080/data/MDUnits/list
```

The command line parameters are similar to the ones of the first request, but with the following changes:

No user login with "--user 12345:secret"

No transfer of the AccessId

A user login with Basic Authentication and the transfer of the AccessId are not required because you use the session of the first request.

-b curl_cookies.txt

Cookies are read in the file specified and passed to the server with the request. The file includes the cookie with the session ID of the first request.

Request of "wget" (all in one row):

```
wget --load-cookies wget_cookies.txt --header="Content-Type: application/json" --post-file=request2.json --output-
document=wget_result2.json --no-check-certificate https://server.domain.local:8080/data/MDUnits/list
```

The command line parameters are similar to the ones of the first request, but with the following changes:

No user login with -user und --password

No transfer of the AccessId

A user login with Basic Authentication and the transfer of the AccessId are not required because you use the session of the first request.

--load-cookies wget_cookies.txt

Cookies are read in the file specified and passed to the server with the request. The file includes the cookie with the session ID of the first request.

Content of the request file request2.json:

```
{
  "params": [
    {
      "acronym": "units.classification",
      "value": "area",
      "operator": "EQUAL"
    }
  ],
  "columns": [
    "units.unit",
    "units.si_unit",
    "units.unitiso",
    "units.designation",
    "units.classification"
  ],
  "returnAsObject": true
}
```

Content of the result file *_result2.json (formatted):

```
[
  {
    "__rowType": "META",
    "__type": "units",
    "data": [
      {
        "name": "units.classification",
        "type": "STRING"
      },
      {
        "name": "units.designation",
        "type": "STRING"
      },
      {
        "name": "units.unit",
        "type": "STRING"
      },
      {
        "name": "units.unitiso",
        "type": "STRING"
      },
      {
        "name": "units.si_unit",
        "type": "BOOLEAN"
      }
    ]
  },
  {
    "__rowType": "OBJECT",
    "__type": "units",
    "obj": {
      "units.unit": "CM2",
      "units.designation": "square centimeter",
      "units.classification": "area",
```

```
      "units.si_unit": false
    }
  },
  {
    "__rowType": "OBJECT",
    "__type": "units",
    "obj": {
      "units.unit": "FT2",
      "units.designation": "square feet",
      "units.classification": "area",
      "units.si_unit": false
    }
  },
  {
    "__rowType": "OBJECT",
    "__type": "units",
    "obj": {
      "units.unit": "M2",
      "units.unitiso": "MTK",
      "units.designation": "square meter",
      "units.classification": "area",
      "units.si_unit": true
    }
  }
]
```


Behavior with expired or non-existing session

When the session has expired or the session cookie is invalid, the following error message with http status code 401 is shown when a service is requested:


```
{
  "timestamp": 1562053321775,
  "status": 401,
  "error": "Unauthorized",
  "message": "Full authentication is required to access this resource",
  "path": "/data/MDUnits/list"
}
```


A (new) login is required, as described in the section above.

2.4.2.6 Requesting a PDM dialog in a running session

In the third example, we request a PDM dialog in a running session, in the example DLG=HYDRA.SELECT.

Request of "curl" (all in one row):

```
curl -b curl_cookies.txt -X POST -H "Content-Type: text/plain; charset=UTF-8" -d @request3.txt -o curl_result3.txt -k
https://server.domain.local:8080/dlg/command
```

The command line parameters are similar to the ones of the previous request, but with the following changes:

-H "Content-Type: text/plain; charset=UTF-8"

If you request PDM dialogs, you use "text/plain" with character set UTF8 instead of JSON.

https://server.domain.local:8080/dlg/command

The URL for PDM dialogs is always "…/dlg/command".

-d @request3.txt

The body of the PUT request includes the dialog string.

Request of "wget" (all in one row):

```
wget --load-cookies cookies.txt --header="Content-Type: text/plain; charset=UTF-8" --post-file=request3.txt --output-
document=result3.txt --no-check-certificate https://server.domain.local:8080/dlg/command
```

--header="Content-Type: text/plain; charset=UTF-8"

If you request PDM dialogs, you use "text/plain" with character set UTF8 instead of JSON.

https://server.domain.local:8080/dlg/command

The URL for PDM dialogs is always ".../dlg/command".

--post-file=request3.txt

The body of the PUT request includes the dialog string.

Content of the request file request3.txt:

```
DLG=HYDRA.SELECT|
```

Content of the result file *_result3.txt (example):

(all in one row)


```
RET=0|KT=|LT=|DATA=HYDRA|OPT:ABBRNR=J|OPT:MNRFTYP=A|OPT:ADEKNRPLAUS=J|OPT:ERFAMENGE=J|OPT:AUSMANU=J|OPT:AUSNR=J|OPT:PZEAUTO=N
|OPT:KDVLIST=J|OPT:WILLESYS= |OPT:UCHGEN=N|OPT:TRMGEN= |OPT:KORRRMNGEN= |OPT:EDITBRMSPERR= |OPT:AUSPAUPDAUER= |BDEOPT:5=
|BDEOPT:9=N|BDEOPT:10= |BDEOPT:11= |BDEOPT:12=N|BDEOPT:13=N|BDEOPT:16=N|BDEOPT:17=N|BDEOPT:19=J|BDEOPT:20=
|BDEOPT:21=N|BDEOPT:22=N|BDEOPT:23=
|BDEOPT:30=N|OPT:RMGEN=|OPT:PDAUERBMKVERB=JJJJJJJJJJJN|ANR:GKP=|ANR:GKM=|OPT:EVENTEDIT=N|FLSOPT:6= |FLSOPT:7=
|OPT:ADATF=N|OPT:AART=K|OPT:RESMGT=
|OPT:RESUWG=M|OPT:RESOWG=M|OPT:GUTMANU=J|OPT:PZPLAUS=J|OPT:MDEGEN=N|OPT:MDEKNRPLAUS=J|OPT:EMSAKTIV=N|OPT:MDETLG=N|OPT:ANTVERB
PM=J|OPT:ERFPMENGE=J|PNRFUELL=0|OPT:PZEPLAUS=N|OPT:PDAUERSKMOD=N|OPT:RMIRUEZ=N|OPT:SAG=N|OPT:TGLKTOBEGR=N|OPT:VLIST=J|OPT:VGZ
=N|OPT:VLISTMOD=M|OPT:WSTEMP=N|OPT:WSTEMPFGR=J|OPT:ZST=N|SIGN:STMP=S|OPT:PLNAPZMGRP=N|OPT:PZEBERECHT= |OPT:PZEINFOFORM=
|OPT:ZKSBERECHT= |OPT:CNRAUTOGEN=J|ADEKAR:BMKNR=0|ADEKAR:AKTIV=N|ADEKAR:DAUER=0|ADEKAR:DAUERMAX=-
327|MDEKAR:BMKPLAN=0|MDEKAR:BMKNPLAN=0|MDEKAR:OG=0|MDEKAR:UG=0|MDEKAR:AKTIV=N|LEN:AUNR=8|LEN:AGNR=4|LEN:KNR=4|LEN:PNR=8|LEN:C
NR=10|LEN:AFOLG=0|LEN:UAGNR=0|LEN:SPLNR=0|LEN:RMNR=8|KK1:AKTIV=N|KK1:VER=0|KK2:AKTIV=N|KK2:VER=0|SYSNR=9999|SEARCHSTRING=|KER
NELVER=1|MAXDAUERSTEMP=50400|KDNR=99999|PRJ=HYDRA|VER=8.3|SAP:AUNRVON=1|SAP:AUNRBIS=12|SAP:KNRVON=1|SAP:KNRBIS=8|SAP:KAPARTVO
N=0|SAP:KAPARTBIS=0|SAP:KSTVON=0|SAP:KSTBIS=0|SAP:PNRVON=1|SAP:PNRBIS=8|SAP:SPLITNRBIS=0|SAP:SPLITNRVON=0|SAP:UVGNRVON=0|SAP:
UVGNRBIS=0|SAP:VGFOLGVON=5|SAP:VGFOLGBIS=6|SAP:VGNRVON=1|SAP:VGNRBIS=4|SAP:WERK=|SAP:ZBER=|SAP:ZFIR=|SAPOPT:2=N|SAPOPT:3=N|SA
POPT:4=N|SAPOPT:5=N|MWEVENT=Y|DISTRIBUTORTIME=1200|CFGMTIME=60|TABONLC=10|TRANSSIZE=10000|TRANSTIME=600|TIMELEAK=10|REVERSETI
ME=3600|VISON=N|TRANSFKT=F|SPCININT=N|MASDATTARG=D|EXPPATH=PDVARC|TRANSPATH=PDVTRANS|CNRFIX:WE=WE|CNRFIX:PR=PR|MPL:PARAM1=|MP
L:PARAM2=|CNRAUTOGEN=J|HOSTNAME:SMTP=webmail.mpdv.local|PORT:SMTP=|TMOUT:SMTP=1000|SENDER:SMTP=hydadm@mpdv.com|SMTPPATH=|MOBI
LTELPATH=|PAGERPATH=|LOGINMETHOD:SMTP=0|USER:SMTP=|PWD:SMTP=LTlrgmex46zz2QtM6ieR6Q==|MODUL=|SMENGEFAKT=-32|MSPERREWARTZ=-
327|WAEHRUNG=EUR|PLANVL=0|PLANHORIZ=5|SIMHORIZ=30|FIXHORIZ=1|SKYLINEHORIZ=0|OPT:LV=J|STREUBREITE=1|FIXPKTTERM=S|DEFTERMR=R|PR
IONBER=0|HLSKENNZ:1= |HLSKENNZ:2= |OFSLST=0|OPT:SPLITTEN= |BEARB=12345|BEARBDAT=08/01/2019|BEARBZEI=38676|
```


If you call PDM dialogs, which create list files on the server, you can download the list file in a further step from the server after having requested the PDM dialog. See section "0 Examples are provided with the examples of command line tools and with the Java programming example.

Download and upload of files".

2.4.3 Java

2.4.3.1 Requirements for executing the example

2.4.3.1.1 Configuration file for access rights

In the file "filePermissions.json", you must replace deviating users before the test.

```
[
  { "users": [ "12345" ],
    "permission": "READ",
    "files": [ "?/spool/persons_list.dat" ],
    "comment": "Test Download"
  },
  { "users": [ "12345" ],
    "permission": "WRITE",
    "files": [ "?/spool/persons_list_uploaded.dat" ],
    "comment": "Test Upload"
  }
]
```

Create the configuration file on the server and restart the WSP

Example: \\myserver\mip1\jdir\MOC\1\filePermissions.json

For further information on the configuration file "filePermissions.json", refer to section "2.3.4 Download and upload of files".

2.4.3.1.2 Function authorizations for services

For the example, the user 12345 requires the following function authorizations to call services:

- Svc:MDUnits.list
- Svc:DLG_A_AN
- Svc:DLG_HYDRA.SELECT
- Svc:DLG_PNR.LOCK
- Svc:DLG_PNR.LIST

2.4.3.2 Source code class "Main"

```
package com.company;

import java.io.File;

public class Main
{
    public static void main(final String[] args)
    {
        // TODO: Replace by your protocol here (hhtp or https)
        final String exampleProtocol = "https";
        // TODO: Replace by your server here
        final String exampleServer = "MOS-PD-vMW4-01.mpdv.local";
        // TODO: Replace by your port here
        final Integer examplePort  = Integer.valueOf(8080); // Instance 3 --> Port 8082
        // TODO: Replace by your AccessId here (formatted to 8 digits with leading zeros)
        final String exampleAccessId   = "00099994";
        // TODO: Replace by your user id here
        final String exampleUser   = "12345";
        // TODO: Replace by your user's password here
        final String examplePwd    = "mpdv";

        final RestConnectionDemo restConnectionDemo = new RestConnectionDemo();
```

```
        // Login. Session is stored in object restConnectionDemo in HTTP connection info.
        restConnectionDemo.login(exampleProtocol, exampleServer, examplePort, exampleAccessId, exampleUser, examplePwd);

        // Example of web service call.
        // Preparing parameters. JSON set up may be solved smarter ...
        final String paramsJson =
            "{" +
                " \"params\" : [ { " +
                "   \"acronym\" : \"units.designation\", " +
                "   \"value\" : \"%e%\", " +
                "  \"operator\" : \"LIKE\" " +
                " }, { " +
                "   \"acronym\" : \"units.modified_by\", " +
                "   \"value\" : [ \"12345\", \"Schreiber\" ], " +
                "   \"operator\" : \"IN\" " +
                " } ], " +
                " \"columns\" : [ \"units.unit\", \"units.si_unit\", \"units.designation\", \"units.unitiso\", " +
                    "\"units.classification\", \"units.modified_by\", \"units.modified_ts\" ] " +
                "};";
        restConnectionDemo.webservicCallExample("/data/MDUnits/list", paramsJson);

        // Example of PDM dialog calls
        restConnectionDemo.dlgCallExample("DLG=A_AN|ANR=47114711|MNR=0815|PNR=999999|");
        // ==> RET=90 = invalid machine id

        restConnectionDemo.dlgCallExample("DLG=HYDRA.SELECT|DATEI=spool/hydra.dat|");
        // ==> RET=0 = OK. , return sting contains data

        restConnectionDemo.dlgCallExample("DLG=PNR.LOCK|DATEI=spool/person_lock.dat|");
        // ==> RET=1700 = missing person id

        restConnectionDemo.dlgCallExample("DLG=PNR.LIST|BEARB=12345|");
        // ==> Missing DATEI, list is stored on server in default file name, for example ...mip1\1\spool\hyb9999.ls

        restConnectionDemo.dlgCallExample("DLG=PNR.LIST|BEARB=12345|DATEI=spool/persons_list.dat|");
        // ==> RET=0 = OK. File on server contains data.

        final File targetFile = new File(System.getProperty("user.home") + File.separator + "persons_list.dat");
        restConnectionDemo.fileDownloadExample("spool/persons_list.dat", targetFile.toPath().toString());

        restConnectionDemo.fileUploadExample(targetFile.toPath().toString(), "spool/persons_list_uploaded.dat");

        restConnectionDemo.logout();
    }
}
```

2.4.3.3 Source code class "RestConnectionDemo"

```
package com.company;

// Preconditions:
// Load packages HttpClient and HttpCore from Apache and add as library to project

import org.apache.http.HttpEntity;
import org.apache.http.HttpHeaders;
import org.apache.http.HttpResponse;
import org.apache.http.auth.AuthScope;
import org.apache.http.auth.BasicUserPrincipal;
import org.apache.http.auth.Credentials;
import org.apache.http.client.HttpRequestRetryHandler;
import org.apache.http.client.methods.*;
import org.apache.http.client.utils.URIBuilder;
import org.apache.http.entity.ContentType;
import org.apache.http.entity.StringEntity;
import org.apache.http.entity.mime.HttpMultipartMode;
import org.apache.http.entity.mime.MultipartEntityBuilder;
import org.apache.http.impl.client.BasicCookieStore;
import org.apache.http.impl.client.BasicCredentialsProvider;
import org.apache.http.impl.client.CloseableHttpClient;
import org.apache.http.impl.client.HttpClients;

import javax.net.ssl.SSLContext;
import javax.net.ssl.SSLException;
import javax.net.ssl.TrustManager;
import javax.net.ssl.X509TrustManager;
import java.io.*;
import java.net.URI;
import java.net.URISyntaxException;
import java.net.UnknownHostException;
import java.nio.file.Files;
import java.nio.file.StandardCopyOption;
import java.security.KeyManagementException;
import java.security.NoSuchAlgorithmException;
import java.security.Principal;
import java.security.SecureRandom;
import java.security.cert.X509Certificate;

public class RestConnectionDemo {
    private String protocol;
    private String serverName;
    private Integer port;
    private String accessId;
    private CloseableHttpClient httpClient;
```

```
    // convert InputStream to String
    private static String getStringFromInputStream(InputStream is) {

        BufferedReader br = null;
        StringBuilder sb = new StringBuilder();

        String line;
        try {
            br = new BufferedReader(new InputStreamReader(is, "UTF-8"));
            while ((line = br.readLine()) != null) {
                sb.append(line);
            }

        } catch (IOException e) {
            e.printStackTrace();
        } finally {
            if (br != null) {
                try {
                    br.close();
                } catch (IOException e) {
                    e.printStackTrace();
                }
            }
        }

        return sb.toString();
    }

    public void login(final String parProtocol,
                      final String parServerName,
                      final Integer parPort,
                      final String parAccessId,
                      final String userName,
                      final String passwordStr) {
        protocol = parProtocol;
        serverName = parServerName;
        port = parPort;
        accessId = parAccessId;

        System.out.println("-- login ----------------------------------------------------------------------");
        System.out.println("  Server   " + parServerName);
        System.out.println("  Port     " + parPort);
        System.out.println("  AccessId " + parAccessId);
        System.out.println("  User     " + userName);
        System.out.println("  PWD      " + passwordStr);

        if (httpClient != null) {
            System.out.println("logout first");
            return;
        }

        final BasicCredentialsProvider credentialsProvider = new BasicCredentialsProvider();
        credentialsProvider.setCredentials(AuthScope.ANY, new Credentials() {
             @Override
            public Principal getUserPrincipal() {
                return new BasicUserPrincipal(userName);
            }

             @Override
            public String getPassword() {
                return passwordStr;
            }
        });
        final HttpRequestRetryHandler retryHandler = (exception,
                                                      executionCount,
                                                      context) ->
        {
            if (executionCount >= 5) // Do not retry if over max retry count
            {
                return false;
            }
            if (exception instanceof InterruptedIOException) // Timeout
            {
                return false;
            }
            if (exception instanceof UnknownHostException) // Unknown host
            {
                return false;
            }
            return !(exception instanceof SSLException);
        };
        SSLContext sslContext = null;
        try {
            sslContext = SSLContext.getInstance("SSL");
            sslContext.init(null, new TrustManager[]{new X509TrustManager() {
                public X509Certificate[] getAcceptedIssuers() {
                    return null;
                }

                public void checkClientTrusted(X509Certificate[] certs,
                                               String authType) {
                }

                public void checkServerTrusted(X509Certificate[] certs,
                                               String authType) {
                }
            }}, new SecureRandom());
        } catch (KeyManagementException | NoSuchAlgorithmException e) {
            e.printStackTrace();
        }
```

```
    httpClient = HttpClients.custom()
            .setRetryHandler(retryHandler)
            .setDefaultCookieStore(new BasicCookieStore())
            .setDefaultCredentialsProvider(credentialsProvider)
            .setSslcontext(sslContext)
            .setSSLHostnameVerifier((s, sslSession) -> true).build();

    System.out.println();
}

public String webservicCallExample(final String webService,
                                   final String paramsJson) {
    if (httpClient == null) {
        System.out.println("not logged in");
        return null;
    }

    final String uriString = protocol+"://" + serverName + ":" + port + webService;

    System.out.println("-- webServiceCall -------------------------------------------------------------");

    final URI uri;
    try {
        uri = new URI(uriString);
    } catch (final URISyntaxException e) {
        System.out.println("Error: Invalid URI syntax: " + uriString + "  " + e);
        throw new RuntimeException();
    }

    String retVal = "";
    try {
        final RequestBuilder builder = RequestBuilder.post()
                .setUri(uri)
                .setHeader(HttpHeaders.CONTENT_TYPE, "application/json; charset=UTF-8").
                setHeader("X-Access-Id", accessId );

        builder.setEntity(new StringEntity(paramsJson, ContentType.APPLICATION_JSON));
        final HttpUriRequest post = builder.build();

        System.out.println("execute " + post.getURI() + " ...");
        try (CloseableHttpResponse response = httpClient.execute(post)) {
            System.out.println(response.getStatusLine());
            retVal = getStringFromInputStream(response.getEntity().getContent());
            System.out.println(retVal);
        }
    } catch (final RuntimeException ex) {
        throw ex;
    } catch (final Exception ex) {
        System.out.println("error.communicationexeption" + "\r\n" + ex);
    }

    System.out.println();
    return retVal;
}

public String dlgCallExample(final String dlgDataStr) {
    System.out.println("-- dlgDataCall -----------------------------------------------------------------");

    if (httpClient == null) {
        System.out.println("not logged in");
        return null;
    }
    String retVal = "";
    try {
        final URI uri;
        final String uriString = protocol+"://" + serverName + ":" + port + "/dlg/command";
        try {
            uri = new URI(uriString);
        } catch (final URISyntaxException e) {
            System.out.println("Error: Invalid URI syntax: " + uriString + "  " + e);
            throw new RuntimeException();
        }

        final RequestBuilder builder = RequestBuilder.post()
                .setUri(uri)
                .setHeader(HttpHeaders.CONTENT_TYPE, "text/plain; charset=UTF-8");
        builder.setEntity(new StringEntity(dlgDataStr, ContentType.TEXT_PLAIN));

        final HttpUriRequest post = builder.build();

        System.out.println("execute " + post.getURI() + " (" + dlgDataStr + ") ...");
        try (CloseableHttpResponse response = httpClient.execute(post)) {
            System.out.println(response.getStatusLine());
            retVal = getStringFromInputStream(response.getEntity().getContent());
            System.out.println(retVal);
        }
    } catch (final Exception ex) {
        System.out.println("error.communicationexeption" + "\r\n" + ex);
    }

    System.out.println();
    return retVal;
}

public void fileDownloadExample(final String serverFileName, final String localFileName) {
    if (httpClient == null) {
        System.out.println("not logged in");
        return;
    }
```

```
        final String uriString = protocol+"://" + serverName + ":" + port + "/dlg/fileDownload";
        final URIBuilder builder;

        System.out.println("-- fileDownloadExample -------------------------------------------------------");

        try {
            builder = new URIBuilder(uriString);
            builder.setParameter("file", serverFileName);
            final HttpGet get = new HttpGet(builder.build());

            final HttpResponse response = httpClient.execute(get);
            try (final InputStream initialStream = response.getEntity().getContent()) {
                if (response.getStatusLine().getStatusCode() != 200) {
                    System.out.println("Error Downloading " + serverFileName + ": " + response.getStatusLine());
                    System.out.println(getStringFromInputStream(response.getEntity().getContent()));
                } else {
                    final File targetFile = new File(localFileName);
                    Files.copy(initialStream, targetFile.toPath(), StandardCopyOption.REPLACE_EXISTING);
                    System.out.println("Success: Downloaded " + serverFileName + " to " + localFileName);
                }
            }
        } catch (final IOException e) {
            e.printStackTrace();
        } catch (final Exception ex) {
            System.out.println("error.communicationexeption" + "\r\n" + ex);
        }

        System.out.println();
    }

    public void fileUploadExample(final String localFileName, final String serverFileName) {
        if (httpClient == null) {
            System.out.println("not logged in");
            return;
        }

        final String uriString = protocol+"://" + serverName + ":" + port + "/dlg/fileUpload";
        final URIBuilder builder;

        System.out.println("-- fileUploadExample ---------------------------------------------------------");
        System.out.println(uriString);

        try {
            builder = new URIBuilder(uriString);
            builder.setParameter("fileName", serverFileName);

            final File file = new File(localFileName);
            if (!file.exists()) {
                System.out.println("File '" + file.getAbsolutePath() + "' does not exist.");
        return;
            }
            final HttpPost post = new HttpPost(builder.build());

            HttpEntity entity = MultipartEntityBuilder
                    .create()
                    .setMode(HttpMultipartMode.BROWSER_COMPATIBLE)
                    .addBinaryBody("file", file, ContentType.MULTIPART_FORM_DATA, serverFileName)
                    .build();
            post.setEntity(entity);
            final HttpResponse response = httpClient.execute(post);
            if (response.getStatusLine().getStatusCode() != 200) {
                System.out.println("Error Uploading to " + serverFileName + ": " + response.getStatusLine());
                System.out.println(getStringFromInputStream(response.getEntity().getContent()));
        } else {
                System.out.println("Success: Uploaded " + localFileName + " to " + serverFileName);
            }
        } catch (final Exception ex) {
            System.out.println("error.communicationexeption" + "\r\n" + ex);
        }
        System.out.println();
    }

    public void logout() {
        System.out.println("-- logout --------------------------------------------------------------------");

        if (httpClient == null) {
            System.out.println("not logged in");
            return;
        }

        try {
            final URI uri;
            final String uriString = protocol+"://" + serverName + ":" + port + "/logout";
            try {
                uri = new URI(uriString);
            } catch (final URISyntaxException e) {
                System.out.println("Error: Invalid URI syntax: " + uriString + "  " + e);
                throw new RuntimeException();
            }

            final HttpUriRequest post = RequestBuilder.post()
                    .setUri(uri).build();
            try (CloseableHttpResponse response = httpClient.execute(post)) {
                System.out.println(response.getStatusLine());
                // Result code 302 is OK and normal behaviour
                if ((response.getStatusLine().getStatusCode() != 200) &&
                        (response.getStatusLine().getStatusCode() != 302)) {
                    System.out.println("Error logout: " + response.getStatusLine());
                    System.out.println(getStringFromInputStream(response.getEntity().getContent()));
                } else {
```

```
                    System.out.println("Logout OK.");
                }
            }
            httpClient.close();
            httpClient = null;
        } catch (final Exception e) {
            System.out.println("error.communicationexeption" + "\r\n" + e);
        }
    }
}
```

2.4.3.4 Source code explanations

Requirements

The example uses the Apache packages HttpClient and HttpCore to ensure an easy operation of the http connection in Java. Use the search function to find both packages on the Internet and to download them from Apache. Please respect the current license terms.

Class Main

The class "Main" contains sample requests of functions of class "RestConnectionDemo".

- Login
- Example of web service requests
- Example of requests of historical PDM dialogs
- Examples for download and upload of files
- Logout

Class RestConnectionDemo

Class BasicCookieStore

This class is used to save the session cookie.

Function "getStringFromInputStream"

You use this function to read the content of the http result from a stream and to return it as string. The function is used several times in the following functions:

Function "login"

This function shows how to log in the user to start the http session. The session ID of the login is saved in a cookieStore (class BasicCookieStore).

Function "webservicCallExample"

This function demonstrates the web service request. The result in variable retVal is in JSON format. The structure has been described previously. For further interpretation, you can e.g. use the framework "jackson". You can download this framework from the Internet.

Function "dlgCallExample"

This function shows the request of PDM dialogs. The variable retVal contains the PDM return string.

Function "fileDownloadExample"

This function shows the download of files from the server. To ensure that the download is successful, the correct access rights in the configuration file "filePermissions.json" must have been assigned.

Function "fileUploadExample"

This function shows the upload of files to the server. To ensure that the upload is successful, the correct access rights in the configuration file "filePermissions.json" must have been assigned.

Function "logout"

Ends the http session.

2.4.4 .NET / C#

2.4.4.1 Requirements for executing the example

2.4.4.1.1 Configuration file for access rights

In the file "filePermissions.json", you must replace deviating users before executing the example.

```
[
  { "users": [ "12345" ],
    "permission": "READ",
    "files": [ "?/spool/persons_list.dat" ],
    "comment": "Test Download"
  },
  { "users": [ "12345" ],
    "permission": "WRITE",
    "files": [ "?/spool/persons_list_uploaded.dat" ],
    "comment": "Test Upload"
  }
]
```

Create the configuration file on the server and restart the WSP

Example: \\myserver\mip1\jdir\MOC\1\filePermissions.json

For further information on the configuration file "filePermissions.json", refer to section "2.3.4 Download and upload of files".

2.4.4.1.2 Function authorizations for services

For the example, the user 12345 requires the following function authorizations to call services:

- Svc:MDUnits.list
- Svc:DLG_HYDRA.SELECT
- Svc:DLG_PNR.LIST

2.4.4.2 Source code

```
using System;
using System.IO;
using System.Net;
using System.Text;

// Reflection and Collections in this example are only needed for debug output of cookie container
using System.Collections;
using System.Reflection;

namespace RestExample
{
  internal static class Program
  {
    // TODO: Replace by your server here
    static string Url = "MOS-PD-vMW4-HR.mpdv.local";
    // TODO: Replace by your AccessId here
    static string AcccessId = "00099963";
    // Instance 3 --> Port 8080 // TODO: Replace by your port here
    static string Port = "8080";
    // User login information
    // TODO: Replace by your user id here
    static string User = "12345";
    // TODO: Replace by your user's password here
    static string Password = "mpdv";

    // beautifying
    static string Separator =
      "--------------------------------------------------------------------------------";

    // Cookies to keep session.
    private static CookieContainer _cookies;
    private static string _sesCookieName = "";
    private static string _sesCookieValue = "";

    private static void MyRequestMDUnitsList(string unitSelection)
    {
      var json = "{" +
                 "\"params\":[" +
                 "{" +
                 "\"acronym\":\"units.unit\"," +
                 "\"value\":\"" + unitSelection + "\"," +
                 "\"operator\":\"EQUAL\"" +
                 "}" +
                 "]," +
                 "\"columns\":[\"units.unit\", \"units.designation\", \"" +
                 "units.modified_by\", \"units.modified_ts\"] " +
                 "}";

      MyServiceRequest("MDUnits.list", json);
    }

    // The service File.uploadByPath is not released to the public.
    /*
    private static void MyRequestFileUploadByPath(string pathToFile, string hydraPath)
        {
            var json = "{" +
                       "\"params\":[" +
                       "{" +
                       "\"acronym\":\"file.data\"," +
                       "\"value\":\"" + Convert.ToBase64String(File.ReadAllBytes(pathToFile)) + "\"," +
                       "\"operator\":\"EQUAL\"" +
                       "}," +
                       "{" +
                       "\"acronym\":\"file.name\"," +
                       "\"value\":\"" + Path.GetFileName(pathToFile) + "\"," +
                       "\"operator\":\"EQUAL\"" +
                       "}," +
                       "{" +
                       "\"acronym\":\"file.pathname\"," +
                       "\"value\":\"" + hydraPath + "\"," +
                       "\"operator\":\"EQUAL\"" +
                       "}" +
                       "]," +
                       "\"columns\":[]" +
                       "}";

            MyServiceRequest("File.uploadByPath", json);
        }
*/

    private static void MyHandleRequestSessionAuth(HttpWebRequest request)
    {
      // Set AccessId header
      request.Headers["X-Access-Id"] = AcccessId;

      if (_cookies == null) {
        // New session. Basis Auth only on fist request
        request.Headers["Authorization"] = "Basic " + Convert.ToBase64String(Encoding.UTF8.GetBytes(User + ":" + Password));
        // set cookieContainer to store cookie with session ID of HTTP result
        request.CookieContainer = new CookieContainer();
        _cookies = request.CookieContainer;
        Console.WriteLine("Basic authentication for user {0}.", User);
      } else {
        // Existing session, session ID by cookie instead of basic auth.
        request.CookieContainer = _cookies;
```

```
        // Display cookies in container
        var table = (Hashtable)request.CookieContainer.GetType().InvokeMember("m_domainTable",
                      BindingFlags.NonPublic | BindingFlags.GetField | BindingFlags.Instance,
                      null,
                      request.CookieContainer,
                      new object[] { });
        foreach (var key in table.Keys) {
          foreach (Cookie cookie in request.CookieContainer.GetCookies(new Uri("http://"+key+"/"))) {
            Console.WriteLine("Send cookie: Name = {0} ; Value = {1} ; Domain = {2}", cookie.Name, cookie.Value,
              cookie.Domain);
          }
        }
      }
    }

    private static void MyHandleResultCookies(WebResponse response)
    {
      foreach (Cookie cook in ((HttpWebResponse)response).Cookies) {
        Console.WriteLine("Received cookie: " + cook.Name + " = " + cook.Value);
        if (cook.Name.Equals("JSESSIONID")) {
          _sesCookieName = cook.Name;  // JSESSIONID
          _sesCookieValue = cook.Value; // C607A26DF635B2BB57A24F7C9F23122F
          Console.WriteLine("Stored session cookie.");
        }
      }
    }

    private static void MyPrintWebException(WebException e)
    {
      Console.WriteLine();
      Console.WriteLine("Error message: " + e.Message);
      if (e.Response != null) {
        var statusNumber = (int)((HttpWebResponse)e.Response).StatusCode;
        Console.WriteLine("Status Code {0}: {1}", statusNumber, ((HttpWebResponse)e.Response).StatusCode);
        var reader = new StreamReader(e.Response.GetResponseStream());
        var responseBody = reader.ReadToEnd();
        reader.Close();
        Console.WriteLine("Response body: \n" + responseBody);
      }
    }

    private static void MyServiceRequest(string serviceName, string json)
    {
      // URL scheme to call services:
      //    "http://<server.fully.qualified>:<port>/data/<domain>/<service>",
      //     e. g. "http://myserver.example.com:8080/data/MDUnits/list"
      var split = serviceName.Split('.');
      var request = (HttpWebRequest)WebRequest.Create("http://" + Url + ":" + Port + "/data/" + split[0] + "/" + split[1]);
      request.Method = "POST";
      request.ContentType = "application/json; charset=UTF-8";

      Console.WriteLine(Separator);
      MyHandleRequestSessionAuth(request);

      using (var streamWriter = new StreamWriter(request.GetRequestStream())) {
        streamWriter.Write(json);
        streamWriter.Flush();

        // Display the request
        Console.WriteLine("Request:");
        Console.WriteLine(request.RequestUri);

        Console.WriteLine(json);

        try {
          // Get the response.
          var response = request.GetResponse();

          // Display the status.
          Console.WriteLine("Status: " + ((HttpWebResponse)response).StatusCode + " " +
((HttpWebResponse)response).StatusDescription);
          Console.WriteLine();

          // Get the stream containing content returned by the server.
          var dataStream = response.GetResponseStream();

          var reader = new StreamReader(dataStream);
          var responseFromServer = reader.ReadToEnd();
          Console.WriteLine("Response:");
          MyHandleResultCookies(response);
          Console.WriteLine(responseFromServer);
          Console.WriteLine();

          // Cleanup
          reader.Close();
          dataStream.Close();
        } catch (WebException  e) {
          MyPrintWebException(e);
        }
        // Cleanup
        streamWriter.Close();
      }
    }

    private static void MyDlgCall(string dlgDataStr)
    {
      var request = (HttpWebRequest)WebRequest.Create("http://" + Url + ":" + Port + "/dlg/command");
      request.Method = "POST";
      request.ContentType = "text/plain; charset=UTF-8";
```

```
      Console.WriteLine(Separator);
      MyHandleRequestSessionAuth(request);

      using (var streamWriter = new StreamWriter(request.GetRequestStream())) {
        streamWriter.Write(dlgDataStr);
        streamWriter.Flush();

        // Display the request
        Console.WriteLine("Request:");
        Console.WriteLine(request.RequestUri);
        Console.WriteLine(dlgDataStr);

        // Get the response.
        try {
          var response = request.GetResponse();

          // Display the status.
          Console.WriteLine();
          Console.WriteLine("Status: " + ((HttpWebResponse)response).StatusCode + " " +
((HttpWebResponse)response).StatusDescription);
          Console.WriteLine();

          var dataStream = response.GetResponseStream();
          var reader = new StreamReader(dataStream);
          var responseFromServer = reader.ReadToEnd();

          Console.WriteLine("Response:");
          MyHandleResultCookies(response);
          Console.WriteLine(responseFromServer);
          Console.WriteLine();

          // Cleanup
          response.Close();
          dataStream.Close();
          reader.Close();
        } catch (WebException  e) {
          MyPrintWebException(e);
        }
        // Cleanup
        streamWriter.Close();
      }
    }

    private static void MyFileDownload(string serverFileName, string localFileName)
    {
      var reqUrl = "http://" + Url + ":" + Port + "/dlg/fileDownload" + "?file=" + serverFileName;

      var request = (HttpWebRequest)WebRequest.Create(reqUrl);

      Console.WriteLine(Separator);
      MyHandleRequestSessionAuth(request);

      // Display the request
      Console.WriteLine("Request:");
      Console.WriteLine(request.RequestUri);
      Console.WriteLine("Download " + serverFileName + " to " + localFileName);

      try {
        // Get the response.
        var response = request.GetResponse();

        // Display the status.
        Console.WriteLine();
        Console.WriteLine("Status: " + ((HttpWebResponse)response).StatusCode + " " +
((HttpWebResponse)response).StatusDescription);
        Console.WriteLine();

        // Get the stream containing content returned by the server.
        // and copy content to file
        var dataStream = response.GetResponseStream();
        var fileStream = File.Create(localFileName);
        dataStream.CopyTo(fileStream);
        fileStream.Dispose();

        // Cleanup
        response.Close();
        dataStream.Close();
      } catch (WebException  e) {
        MyPrintWebException(e);
      }
    }

    private static void MyFileUpload(string localFileName, string serverFileName, string fileContentType)
    {
      var reqUrl = "http://" + Url + ":" + Port + "/dlg/fileUpload" + "?fileName=" + serverFileName;
      string contentType = "multipart/form-data";

      var webClient = new WebClient();
      string boundary = "---F-i-l-e-U-p-l-o-a-d-B-o-u-n-d-a-r-y--------------------" + DateTime.Now.Ticks.ToString("x");
      webClient.Headers.Add("Content-Type", contentType + "; boundary=" + boundary);
      webClient.Headers.Add(HttpRequestHeader.Cookie, _sesCookieName + "=" + _sesCookieValue);
      webClient.Headers.Add("X-Access-Id", AcccessId);

      Console.WriteLine(Separator);
      Console.WriteLine("Send cookie " + _sesCookieName + " = " + _sesCookieValue);
      Console.WriteLine("Request:");
      Console.WriteLine(reqUrl);
      Console.WriteLine("Upload " + localFileName + " to " + serverFileName);
```

```
      // for "multipart/form-data" see https://www.w3.org/TR/html401/interact/forms.html#h-17.13.4.2
      var requestArray = File.ReadAllBytes(localFileName);
      var fileData = webClient.Encoding.GetString(requestArray);
      var package = string.Format(
                      "--{0}\r\n" +
                      "Content-Disposition: form-data; name=\"file\"; filename=\"{1}\"\r\n" +
                      "Content-Type: {2}; charset=ISO-8859-1\r\n" +
                      "\r\n" +
                      "{3}\r\n" +
                      "--{0}--\r\n",
                      boundary, serverFileName, fileContentType, fileData);

      // Show full content of the http request body: Console.WriteLine( System.Text.Encoding.ASCII.GetString(nfile));
      try {
        var responseArray = webClient.UploadData(reqUrl, "POST", webClient.Encoding.GetBytes(package));
        // Decode and display the response.
        Console.WriteLine();
        Console.WriteLine("Status: OK");
        Console.WriteLine();
        var responseBody = System.Text.Encoding.ASCII.GetString(responseArray);
        if (!responseBody.Equals("")) {
          Console.WriteLine("Response body:");
          Console.WriteLine(responseBody);
          Console.WriteLine();
        }
      } catch (WebException e) {
        MyPrintWebException(e);
      }
    }

    private static void MyLogout()
    {
      var request = (HttpWebRequest)WebRequest.Create("http://" + Url + ":" + Port + "/logout");
      request.Method = "POST";
      request.ContentType = "text/plain; charset=UTF-8";

      Console.WriteLine(Separator);
      MyHandleRequestSessionAuth(request);

      try {
        using (var streamWriter = new StreamWriter(request.GetRequestStream())) {
          streamWriter.Write("");
          streamWriter.Flush();

          // Display the request
          Console.WriteLine("Request:");
          Console.WriteLine(request.RequestUri);

          // Get the response.
          var response = request.GetResponse();

          // Display the status.
          Console.WriteLine();
          Console.WriteLine("Status: " + ((HttpWebResponse)response).StatusCode + " " +
((HttpWebResponse)response).StatusDescription);
          Console.WriteLine();

          // Cleanup
          response.Close();
          streamWriter.Close();
        }
      } catch (WebException e) {
        var statusNumber = (int)((HttpWebResponse)e.Response).StatusCode;
        if ((statusNumber == 200 ) || // Some status codes are normal result
            (statusNumber == 302 ) ||
            (statusNumber == 401)) {
          Console.WriteLine();
          Console.WriteLine("Status: OK, logged out successfully ({0}).", statusNumber);
          Console.WriteLine();
          _cookies = null;
        } else {
          MyPrintWebException(e);
        }
      }
    }



    public static void Main()
    {
      var usersFolder = Environment.GetFolderPath(Environment.SpecialFolder.Personal);

      Console.WriteLine("Service Interface Example:");

      MyRequestMDUnitsList("CM");
      MyRequestMDUnitsList("M");
      // The service File.uploadByPath is not released to the public.
//  MyRequestFileUploadByPath(usersFolder+System.IO.Path.DirectorySeparatorChar+"persons_list.dat", "SPOOL"); //TODO Choose
an appropriate path; the path has to be configured in the system!
      MyDlgCall("DLG=HYDRA.SELECT|DATEI=spool/hydra.dat|");
      MyDlgCall("DLG=PNR.LIST|BEARB=12345|DATEI=spool/persons_list.dat|");
      MyFileDownload("spool/persons_list.dat", usersFolder + Path.DirectorySeparatorChar + "persons_list.dat");
      MyFileUpload(usersFolder + Path.DirectorySeparatorChar + "persons_list.dat", "spool/persons_list_uploaded.dat",
"text/plain");
      MyLogout();

      Console.WriteLine(Separator);
      Console.Write("Press any key to continue ... ");
      Console.ReadKey(true);
    }
```

```
    }
}
```

2.4.4.3 Explanatory notes

Please note the handling of the session via cookies using the methods MyHandleRequestSessionAuth() and MyHandleResultCookies().

The authorization header passes the authorization with the first service request in the form "user name:password" Base64 encoded to the server.

Example:

```
Authorization: Basic dXNlcjpwd2Q=
```

"dXNlcjpwd2Q=" is the Base64 encoding of "user:pwd" and therefore stands for user name "user", password "pwd".

A disadvantage of this procedure is that user name and password are only encoded for technical reasons, but they are not encrypted. From a security point of view, this procedure is just as insecure as if the password were transmitted in plain text. With SSL/TLS encryption for HTTPS, an encrypted connection is established before the password is transmitted, so that the password cannot be intercepted even with Basic Authentication.

In this example, the AccessId is passed as header:

```
            // Set AccessId header
            request.Headers["X-Access-Id"] = acccessId;
```

2.4.5 MS Excel Macro (Visual Basic)

2.4.5.1 Overview

The file RestDemo_VB.xlsm provides the example.

The example does not support the correct processing of the session cookie for permanent connections. It is only suitable for occasional manual calls and not for permanent use in a productive environment with a large data throughput.

2.4.5.2 Source code

```
' Demo code for Web Service calls by REST interface
Option Explicit

Sub Test()

    'CallRestService ActiveSheet, "http://wsp_server:8080/data/MDUnits/list", "12345", "xyz"
    CallRestService ActiveSheet, "http://wsp_server:8080/data/PersonalAttendanceOverview/list", "12345", "xyz"

' Possible error messages
    ' "Die angegebene Ressource wurde nicht gefunden"
    ' "The specified resource was not found"
    '   => REST Service is not available
    ' "Objekt unterstützt Eigenschaft oder Methode nicht"
    ' "Object does not support property or method'
    '   => Invalid user or password OR invalid reasult value (called /meta insted of /data ?)


End Sub


Sub CallRestService(ws As Worksheet, strURL As String, strUser As String, strPassword As String)
    ' Calls the given URL as (REST) Service and fills Excel given worksheet with returned values

    Dim XMLHttp As Object
    Dim strMethod As String, bolAsync As Boolean, varMessage

    ' Create Microsoft XML HTTP object
    Set XMLHttp = CreateObject("MSXML2.XMLHTTP")

    ' ---------------- GET-Request, returning JSON ----------------
    strMethod = "GET"
    bolAsync = False
    varMessage = ""

    ' Send request
    Call XMLHttp.Open(strMethod, strURL, bolAsync, strUser, strPassword)
    Call XMLHttp.send(varMessage)

    ' Transfer result into JSON Object
    Dim sc As Object, jsonObj As Object
    Set sc = CreateObject("ScriptControl")
    sc.Language = "JScript"
    Set jsonObj = sc.Eval("(" + XMLHttp.responseText + ")")

    ' Output of result
    Dim aCount As Long, bCount As Long, item As Object, data As Object, content As Variant, target As Range, header As
String, headerCount As Long
    Set item = VBA.CallByName(jsonObj, 0, VbGet)
    Set data = VBA.CallByName(item, "data", VbGet)

    ' Clear excel worksheet
    ws.Cells.Clear

    ' Output header line
    headerCount = VBA.CallByName(data, "length", VbGet) - 1
    For aCount = 0 To headerCount
       Set item = VBA.CallByName(data, aCount, VbGet)
       header = VBA.CallByName(item, "name", VbGet)
       ws.Cells(1, aCount + 1) = header
    Next

    ' Transfer data row by row into Excel worksheet
    For aCount = 1 To VBA.CallByName(jsonObj, "length", VbGet) - 1
        Set item = VBA.CallByName(jsonObj, aCount, VbGet)
        Set data = VBA.CallByName(item, "data", VbGet)
        content = Split(data, ",")

  If (UBound(content) = headerCount) Then
            ' Row can be set at once (fast)
            Set target = Range(ws.Cells(aCount + 1, 1), ws.Cells(aCount + 1, UBound(content)))
            target.Value = content
      Else
            ' Row data containss columns separators.
            ' Row data must be set column by column (slower).
            For bCount = 0 To headerCount
                ws.Cells(aCount + 1, bCount + 1) = VBA.CallByName(item.data, bCount, VbGet)
            Next
        End If
    Next

End Sub
```

2.4.5.3 Explanatory notes

The example uses an object "MSXML2.XMLHTTP" to communicate via REST interface.

Workflow:

- The method "open" opens a session including user logon and URL specification.
- Optionally, you can use the method "setRequestHeader" to set up a header.
- Call the method "send" to perform the request.
- You can interpret the result using response attributes, in particular the attribute "reponseText".

In the example, the JSON document included in the "responseText" is interpreted. The returned data is used to transfer the header line and all data rows into the Excel table.

For further processing, the JSON arrays with data in the *DATA* row are inserted into a string as comma-separated value and converted into a VB array using "Split()". This procedure ensures a very quick transfer of the individual data rows into the Excel table. As the data itself can contain commas, it is checked in the following line, if the number of elements in the VB array is identical to the number of elements in the JSON array.

```
  If (UBound(content) = headerCount) Then
```

If the number of elements is not identical, the data contained commas. In this case, the Excel row must be filled column by column. It is much slower to proceed column by column than row by row.

2.4.6 JavaScript

2.4.6.1 Source code (HTML file)

HTML file RestDemo_js.html:

```
<!DOCTYPE html>
<html>

<head>
  <title>REST interface example JavaScript</title>

  <script>
    // processes result of service call
    function processResult( result )
    {
      var resultData = JSON.parse( result );
      var rowCount = resultData.length;
      var columnCount = 0;
      var resultHtml = '';

      // Create HTML table by processing header and data rows of result JSON
      resultHtml = resultHtml + '<table>  <tbody> ';
      for (var rowIdx = 0; rowIdx < rowCount; rowIdx++) {
        resultHtml = resultHtml + '<tr>';
        if( rowIdx == 0 ) {
          // process header row
          columnCount = resultData[rowIdx].data.length;
          for (var columnIdx = 0; columnIdx < columnCount; columnIdx++) {
            resultHtml = resultHtml + '<td>' + resultData[rowIdx].data[columnIdx].name + '</td>';
          }
        }
        else {
          // process data row
          for (var columnIdx = 0; columnIdx < columnCount; columnIdx++) {
            resultHtml = resultHtml + '<td>' + resultData[rowIdx].data[columnIdx] + '</td>';
          }
        }
        resultHtml = resultHtml + '</tr>';
      }
      resultHtml = resultHtml + '</tbody> </table>  ';

      document.getElementById("result_area").innerHTML = resultHtml;
    }
```

```
    // Click of button "Execute service call"
    function doServiceCall () {
      var serverAndPort = document.getElementById("serverAndPort").value;
      var username = document.getElementById("login").value;
      var password = document.getElementById("pass").value;
      var serviceName = 'MDUnits.list';

      var parameters = {
        "params" : [ {
          "acronym" : "units.designation",
          "value" : "%e%",
          "operator" : "LIKE"
        }, {
          "acronym" : "units.modified_by",
          "value" : [ "12345", "Schreiber" ],
              "operator": "IN"
        } ],
        "columns" : [ "units.unit", "units.si_unit", "units.designation", "units.unitiso", "units.classification",
"units.modified_by", "units.modified_ts" ]
      };

      document.getElementById("result_area").innerText = '<running>';

      callService(serverAndPort, username, password, serviceName, parameters, processResult );
    }

    // Perform http service call
    function callService(sysAndPort, username, password, serviceName, dataObj, callback){
      var url = "http://" + sysAndPort + "/data/" + serviceName.replace(".", "/");


      xhttp = new XMLHttpRequest()
      xhttp.open("POST", url, true, username, password );
      xhttp.withCredentials = true;
      xhttp.setRequestHeader("Content-type", "application/json");

      xhttp.onreadystatechange = function () {
        if (this.readyState == 4 ){
          if( (this.status == 200 || this.status == 400)) {
            callback(this.responseText);
          }
          else {
            var tmpResult = [
                          {
                            "__rowType": "META",
                            "__type": "ws result",
                            "data": [
                              {
                                "name": "http.status",
                                "type": "STRING"
                              },
                              {
                                "name": "http.statustext",
                                "type": "STRING"
                              }
                            ]
                          },
                          {
                            "__rowType": "DATA",
                            "__type": "ws result",
                            "data": [
                              this.status,
                              this.statusText
                            ]
                          }
                        ];
            callback( JSON.stringify(tmpResult) );
          }
        }
      };

      xhttp.send(JSON.stringify(dataObj));
    }

    window.onload = function () {
      document.getElementById("execute_call").onclick = doServiceCall;
    };
  </script>
</head>

<body>
  <main>
    <h1>REST interface example JavaScript</h1>

    <h2>Preconditions</h2>
    The REST interface does not allow &quot;cross-origin resource sharing&quot;.
    Therefore, this example HTML file must be stored on the server and run with
    a browser on the server. In case of error in the result section
    &quot;http status: 0&quot; is shown.

    <h2>Parameter</h2>
<table>
      <tbody>
        <tr>
          <td align=left> <label>Server:port</label> </td>
          <td> <input id="serverAndPort" name="serverAndPort" value="servername:8080"> </td>
        </tr>
        <tr>
          <td> <label for="login">User</label> </td>
          <td> <input id="login" name="login" value="12345"> </td>
```

```
        </tr>
        <tr>
          <td> <label for="pass">Password</label> </td>
          <td> <input id="pass" name="pass" type="password"> </td>
        </tr>
        <tr>
          <td> </td>
          <td> <button type="button" id="execute_call">Execute service call</button>  </td>
        </tr>
      </tbody>
</table>

    <h2>Result</h2>
    <div id="result_area">
      Result of service call will be showed here
    </div>
  </main>
</body>
</html>
```

2.4.6.2 Explanatory notes

The example shown here only shows rudimentary basics.

In the standard, the service interface does not support a Cross-Origin Resource Sharing (CORS). This means that the HTML page must be on the server and must also be displayed there.

In the example, the session management with cookies is missing and a new session is generated with each request.

For the use with Node.js, an own tutorial "Call services with Node.js" is available describing all relevant aspects.

The function "doServiceCall()" prepares the request. The function "callService()" then sends the request with an object "XMLHttpRequest".

Data output is performed similarly to the callback function. The callback function is implemented via the "processResult()" function and provides the result set as HTML table in the section (<div>) "result_area".

2.5 Troubleshooting

To analyze unexpected behavior, you can use the log file of the Web Service Provider WSP on the server.

The log file is stored in the WSP log directory, e.g.

- \\server\mip1\jdir\MOC\1\err\mip-java-1.log
- \\server\hydra1\JHydraDir\MOC\1\err\hydra-1.log

If the debug level is set to "info", all requests via service interface and the result are written in the log file.

You can change the log level in the configuration file of the WSP. The configuration file is also stored in the WSP directory, e.g.:

- \\server\mip1\jdir\MOC\1\config.properties
- \\server\hydra1\JHydraDir\MOC\1\config.properties

If you require more detailed information, you can set the log level to "trace" or "debug" instead of "info".

In normal operation, set the log level to "error". Because the more detailed the log level, the greater the effect on the performance.

3 Tutorial: Using Node.js to Call Services

3.1 Objective of the tutorial

This tutorial illustrates how to call system services using JavaScript in Node.js.

3.2 Requirements

- To proceed with the tutorial, you need access to an installed system. Therefore, you need the credentials.
- The JavaScript runtime environment Node.js (tested with version 8.9.4) must be installed on the local computer.

3.3 Installation

The source code is available support portal to download. It is located in the ZIP file in the subdirectory "tutorial_service_call_javascript".

The source code is also included in the MIP-SDK as a separate ZIP file.

Unpack the ZIP file or the sub directory " „tutorial_service_call_javascript“ aus der ZIP-Datei in ein Verzeichnis Ihrer Wahl (z. B. C:\Users\<username>\Tutorial).

3.4 Operation and result

As part of the tutorial, the sample program carries out the following steps:

1. Add a unit with typo (parameter *units.designation* is "pice“ instead of "piece“) using the service MDUnits.insert
    - Display the created unit using the service MDUnits.list

2. Correct the spelling mistake using the service MDUnits.update
   - Display the created unit using the service MDUnits.list
3. Delete the unit using the service MDUnits.delete
   - Display the created unit using the service MDUnits.list
4. Add a unit where a mandatory parameter is missing (*units.classification*) using the service MDUnits.insert

Change the connection data to the server to include your system. The file "settings.json" of the directory where you copied the contents of the ZIP archive includes the connection data. For example: "C:\Users\<UserName>\Tutorial\settings.json". The contents of the file *settings.json* are described here.

Open a command line on the local computer to run the sample program. In the command line, go to the directory where you unpacked the source code of the tutorial. Execute the sample program:

Command	$TUTORIALINSTALLATIONDIR$/npm start

The output generated while executing the sample program shows the single steps and their result.

```
PS C:\Projects\Tutorials\tutorial_service_call_javascript> npm start

> tutorial-service-call-javascript@1.0.0 start C:\Projects\Tutorials\tutorial_service_call_javascript
> node index.js

Step 1: insert an unit with a typo
    insertMDUnits succeeded
List the unit
    listMDUnits succeeded
    Print units:
    units.unit            | units.modified_by     | units.unitiso         | units.designation     | units.classification  | units.modified_ts         | units.si_unit         |
    PC                    | 12345                 | PC                    | pice                  | Pieces                | 2018-04-24T13:57:49+02:00 | null                  |
Step 2: update the unit to correct the typo
    updateMDUnits succeeded
List the unit
    listMDUnits succeeded
    Print units:
    units.unit            | units.modified_by     | units.unitiso         | units.designation     | units.classification  | units.modified_ts         | units.si_unit         |
    PC                    | 12345                 | PC                    | piece                 | Pieces                | 2018-04-24T13:57:53+02:00 | null                  |
Step 3: delete the unit
    deleteMDUnits succeeded
List the unit
    listMDUnits succeeded
    No record available
Step 4: insert an unit with a missing mandatory parameter
    insertMDUnits: Not null value is null for parameter units.classification
PS C:\Projects\Tutorials\tutorial_service_call_javascript>
```

3.5 Https

You can use the secure transfer protocol https to operate the sample program. To run the example program with https you have to put the file "ca-root.pem" into the directory "$TUTORIALINSTALLATIONDIR$" of the tutorial. The "ca-root.pem" file contains the root certificate.

If you want to run the example program with https, open a command line as described in 1.4. Execute the sample program at this point:

Command	$TUTORIALINSTALLATIONDIR$/npm run start_https

3.6 Source code description of the sample program

The sample program mainly consists of two modules:

- util
- editUnits

A new unit is added, changed, displayed and deleted in the installed sample program (section 1.3 describes how to install the program). Additionally, an attempt is made to add an erroneous unit.

3.6.1 Util

The *util* module includes the following functions:

- Defining the access ID (constant: access_ID)
- Loading http settings from a JSON file (*loadHttpSettings*)
- Getting the used http settings (*getHttpSettings*)
- Processing the callback of requests (*requestCallback*)
- Displaying data after calling a list service (*printData*)
- Sending a request (*sendRequest*)

3.6.1.1 Defining access ID

The access ID is defined as constant in the util module and transferred as header information when a request is sent to the server (sendRequest). If this access ID is missing, an error is issued.

Define the access ID in section "constants" in the upper part of the module.

```
…
//Constants
const tab = "    "; //Indent of the lines
const pad_end = 25; //Width of the columns
const access_ID = "00012345"; //Access-ID
…
```

3.6.1.2 loadHttpSettings

The function *loadHttpSettings* imports the JSON file *settings.json*. The function uses the content of the JSON file to generate a JavaScript object. The file *settings.json* must be stored in the root directory of the program.

The contents of the file *settings.json* are described here.

3.6.1.3 getHttpSettings

The function *getHttpSettings* is a getter for the settings object.

3.6.1.4 requestCallback

The sample program calls the function *requestCallback* after an http request. The function includes, among others, the response and a callback function as parameters. The job of this function is to parse the response. The method generates the command line output according to the result.

```
try {
    var ob = JSON.parse(data);
    var status = null;
    if(!Array.isArray(ob)) {
        console.log(ob);
        status = {"message": ob};
    }
    else if(ob.length === 0 || (typeof ob[1] !== "undefined" && ob[1]["__rowType"] === "DATA")) {
        console.log(tab + functionName + " succeeded");
    }
    else if(ob[0]["__rowType"] === "ERROR") {
        console.error(tab + functionName + ": " + ob[0].message);
        status = {"message":ob[0].data.messages[0].parameters[0]};
    }
    else {
        console.log(functionName + ": Something unexpected happened");
        status = {"message": "Something unexpected happened"};
    }

    if(callCallback) {
        //Return the error code
        callback(status);
    }
}
catch(error) {
    console.error(functionName + ": " + error);
}
```

As indicated in the listing above, a check is made to identify whether the response is an array or an object. It is an http error if the response is an object.

If successful and depending on the request, the service returns either an empty array or an array with at least two elements. The attribute *__rowType* distinguishes the elements. The first element includes meta information (*__rowType* is "META“). The other elements include the requested data (*__rowType* is "DATA“).

In case of error, an array including only one single element is returned (*__rowType* is "ERROR“). The object's attribute *message* includes the error message. You can find the error code in the attribute *data.messages[0].parameters[0]*.

The function parameter *callback* is a function. After evaluating the response, the function *requestCallback* calls the *callback* function. If successful, the argument is *null.* In case of an error, the transferred argument is an object with the attribute *message*.

3.6.1.5 printData

The function *printData* displays the response of a *list request* in the command line. The function expects as parameter the response array as JSON string. Data is displayed in a table. In the table view, the parameters of the *META* element provide the header and the *DATA* elements provide the rows.

3.6.1.6 sendRequest

The function *sendRequest* creates a *Post Request* and sends it to a server of the system. The parameter *parameters* includes the body of the request. The parameter has the type *String*. The parameter *path* is the path of the request (e.g. /data/BOOrder/insert).

The sample program sends an empty *cookie* header field with the first request. The service sends a cookie in the header of the response. The function *sendRequest* of the sample program reads this cookie from the header field *set-cookie* and stores it. The stored cookie is transferred along with the headers of the next requests. This prevents new sessions from being created on the server with every request.

The function sendRequest calls the callback, once a response has arrived.

```
var req = http.request(options, (res) => {
    //Save the cookie sent from server
    if(typeof res.headers["set-cookie"] !== "undefined") {
        cookie = res.headers["set-cookie"][0];
    }

    //Read the data
    let data = "";
    res.on("data", (result) => {
        data += result;
    });

    //Data is complete, call the callback
    res.on("end", () => {
        if(typeof callback === "function") {
            callback(null, data);
        }
    });
});
```

3.6.2 editUnits

The *editUnits* module includes functions to execute requests. To do so, the function of the *workplaceStatusEvaluation* module calls the sendRequest method of the *util* module. The functions prepare the request bodies. The functions transfer the request bodies along with the corresponding path and a callback.

The module includes the following functions:

- Create a new unit (*insertMDUnits*)
- Change a unit (*updateMDUnits*)
- Display units (*listMDUnits*)
- Delete a unit (*deleteMDUnits*)

All these functions have the same structure. See the following example of *deleteMDUnits*:

```
function deleteMDUnits(unit = {}, callback = ()=>{}) {
    //Check if fields exists
    if( typeof unit.classification === "undefined" ||
        typeof unit.unit === "undefined" ||
        typeof unit.unitiso === "undefined" ||
        typeof unit.designation === "undefined"
    ) {
```

```
        callback({"message": "insertMDUnits: field undefined"});
        return;
    }
    //The parameters are sent as a JSON string
    var parameters = JSON.stringify({
        "params": [
            {
                "acronym": "units.unit",
                "value" : unit.unit,
                "operator" : "EQUAL"
            }
        ]
    });

    util.sendRequest(parameters, "/data/MDUnits/delete", callback);
}
```

The above listing shows that the function expects two parameters:

- *Unit*: an object
- *Callback:* a callback function

The function uses the parameter *unit* to generate a JSON string. The function transfers this string with the path "/data/MDUnits/delete“ and the callback to the method *sendRequest* of the *util* module.

The functions *insertMDUnits, updateMDUnits, listMDUnits and deleteMDUnits* are encapsulated with wrappers (e. g. *insertUnit*) in the index file of the *editUnits* module (*/app/index.js*). For a sequential process, the previously processed wrapper receives an (anonymous) function as callback. This anonymous function calls the next wrapper. The following listing shows an example of how to create and display a unit.

```
insertUnit(unit, () => {
    listUnit(unit, () => {});
});
```

4 Service Tester

4.1 Overview

The Service Tester is a tool to test the services available in the system.

- You can easily select from the available services.
- You can set the service parameters.
- The results of a service request are clearly presented in a table.

An SQL client is available to check the contents of the database and to change them, if required.

You can also run PDM dialogs with the respective installations.

The Service Tester is a tool for experts. Each user is responsible for the data integrity in case of services that modify data.

4.2 Requirements

- Web Service Provider WSP in version Service Pack 13
- You must have licensed the service interface to connect the service test with the service. You can process the licensing with the licenses SCS-SIF and SCS-SIC.
- The user selected for login must have function authorizations:
  - From MW 4.0_{pe}: Each service called up must be enabled for the user by function authorizations. The service tester uses the "MDUser.list" service when logging in, so the user must have at least the function authorization "Svc:MDUser.list". For more information on the function authorizations for services, refer to the chapter on Interface Technology (in the same document or in the Service Interface manual if you use the Service Tester documentation as a separate document).
  - User must have the function authorization „svcitf.login“ for MW 3.x.
- You can obtain the Service Tester if you attend a training at MPDV's or via the download area of the support portal in form of a packed ZIP file.

4.3 Installation

Copy the content of the ZIP file including the Service Tester into a target directory of your choice. A special installation routine is not required.

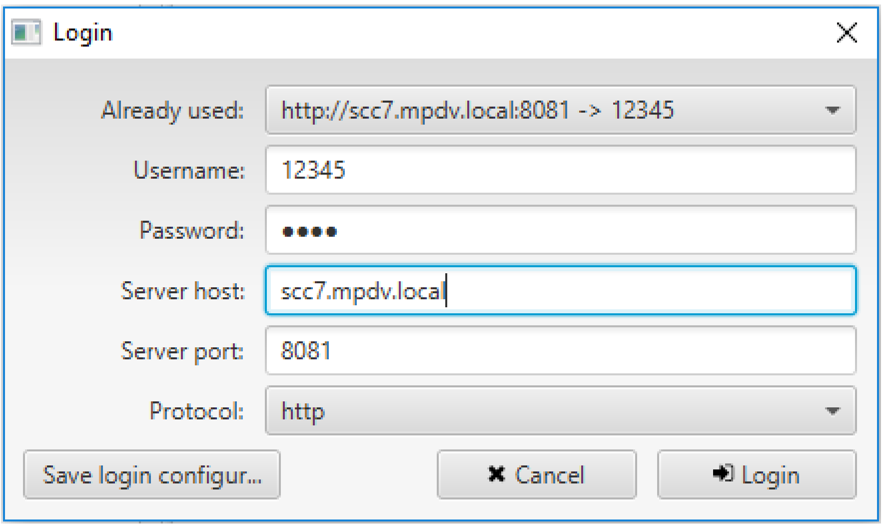
4.4 Program start and login

Program start

Double-click the program file to start the Service Tester.

Login window

After the start, the login window opens. Enter the user name, the respective password, the server address, the server port and the protocol in the login window.



Server host:

To keep the http session alive, you must specify the "Fully Qualified Host Name" (FQHN). This is either the host name (sometimes with subnet and domain) or the IP address. Examples of addresses with "Fully Qualified Host Names":

- "http://myserver.example.com:8080" (if no subnet exists)
- "http://myserver.mysubnet.example.com:8080" (with subnet)
- "http://192.0.2.123:8080"

If you only entered the server name during login, the server tester tries to extend the address by a domain and a subnet.

Protocol http/https:

The service test supports https. The communication is then encoded. However, the service tester does not check the validity of the certificate used on the server, since the service tester is designed as a test and development tool for use in the internal secure area only.

You can configure the communication via https during installation/implementation or later on the server. This communication then affects not only the Service Interface, but also all other clients that communicate via http with the server, e.g. the MOC.

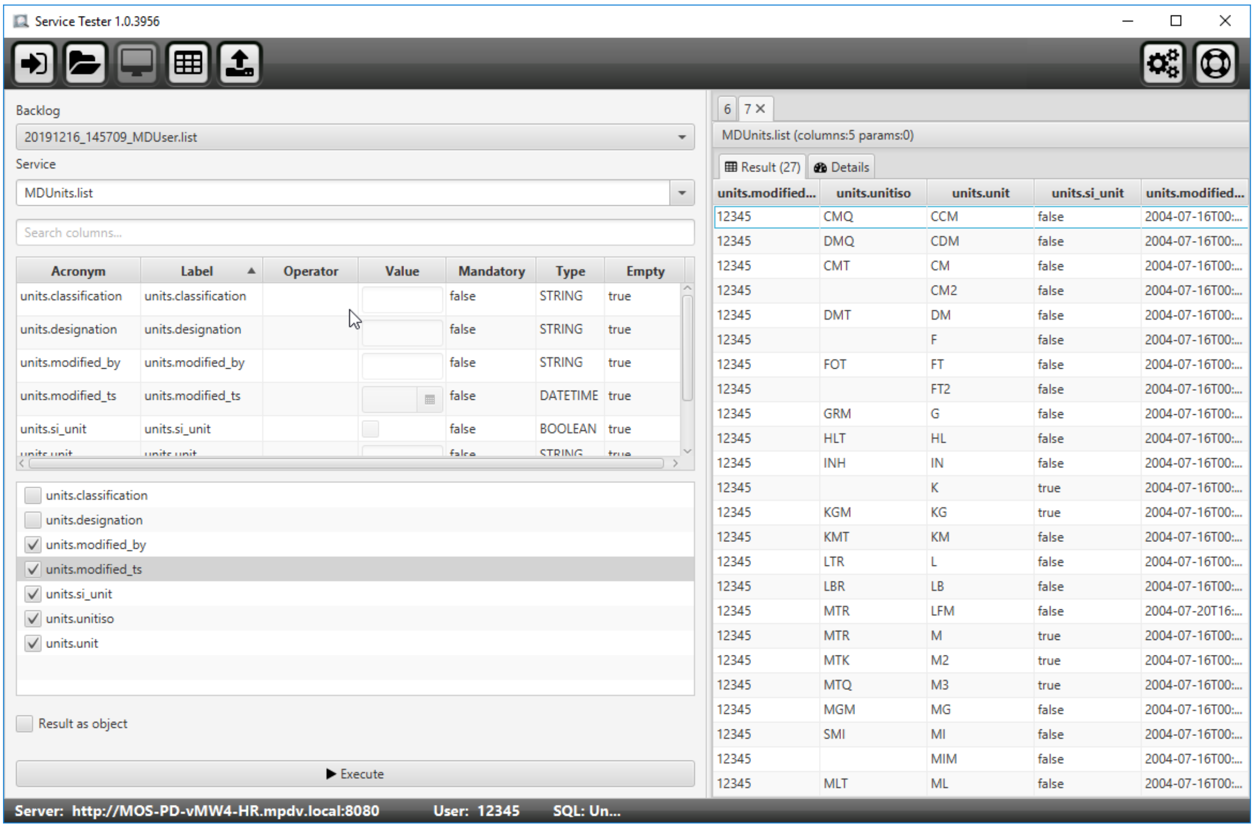
If you create own clients, you must decide if your client knows and checks the certificate or not. The technical aspects only depend on the technology and development environment you use on your client..

Button "Save login configuration"

You can use the button "Save login configuration" to create a JSON file that includes the currently displayed login information. The password is saved in encrypted form. The Service Tester does not require this file with the login information. But the file can be used by other development tools. See documentation of these development tools.

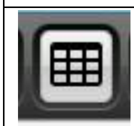
4.5 Main view

If you confirm the login, the main view of the Service Tester opens. On top of the main view, there is a button bar. On the left-hand side, several input fields provide the possibility to select a service and to define parameters. The right-hand side displays the output of the service that is run.



4.5.1 Buttons

The button bar on top provides further features of the Service Tester.

	Use this button to logout. The login window opens. See "Program start and login".
	Use this button to execute a request saved in JFX format.
	Click this button to return to the main view. See main view.
	Click this button to open the view for executing SQL statements. See "SQL request".

	Click this button to open the view for requesting PDM dialogs. See "DLG request".

4.5.2 Request and test services

4.5.2.1 Service and service parameters

On the left-hand side of the main view, select the service and specify the service parameters.

Backlog: Rerun service request from the history

From the "backlog" combo box on top, you can call the services that you have last executed. If you select a service request in the list, the service parameters and result columns, that are described in the following, are automatically populated.

Service: Selecting a service

The second combo box "Service" lists all services available in the server. If you select a service, the available service parameters and result columns are displayed in the lists below.

Search function for acronyms

Use the search field "Search..." to search for acronyms in the service parameter table and in the list of result columns below.

Table of service parameters

The parameter table shows all columns of the result. You can edit the parameters and include conditions in the service. This narrows down the result. A red row is a mandatory parameter, a green row is a parameter that is already set. Right-click to open a context menu where you can edit or delete the conditions.

Use drag and drop to load selected requests from the WSP log or an MOC client into the Service Tester.

Table of the result columns

You can select individual columns from the list of result columns that are then shown as result. You can use the mouse to mark several rows. Select the rows marked using the context menu (right-click) or disable the selection of the rows marked.

Option "Result as Object"

If you check the option "Result as Object", the result is returned as object in JSON format.

If the result data is returned as objects, parsing of result data in the client is easier because each data row is a fully qualified JSON object. Otherwise, you always require the META information of the result to interpret a data row.

The option "Result as Object" increases the result size by a factor of about 2 and therefore slightly diminishes the performance of the service execution.

Execute: Request service

You use the button "Execute" to perform the service request.

4.5.2.2 The result

On the right hand side, the results of the last service call are displayed. Each result is structured in several tabs.

Tab *Result*

Tab *Result* shows the results in a table. Click the column title to sort by the respective column. Press the shift key to sort by several columns. Right-click to call the context menu. You can use the context menu to copy the selected data as CSV (*comma-separated values*) to the clipboard or to open the data in Excel.

Details

Tab *Details* includes two buttons:

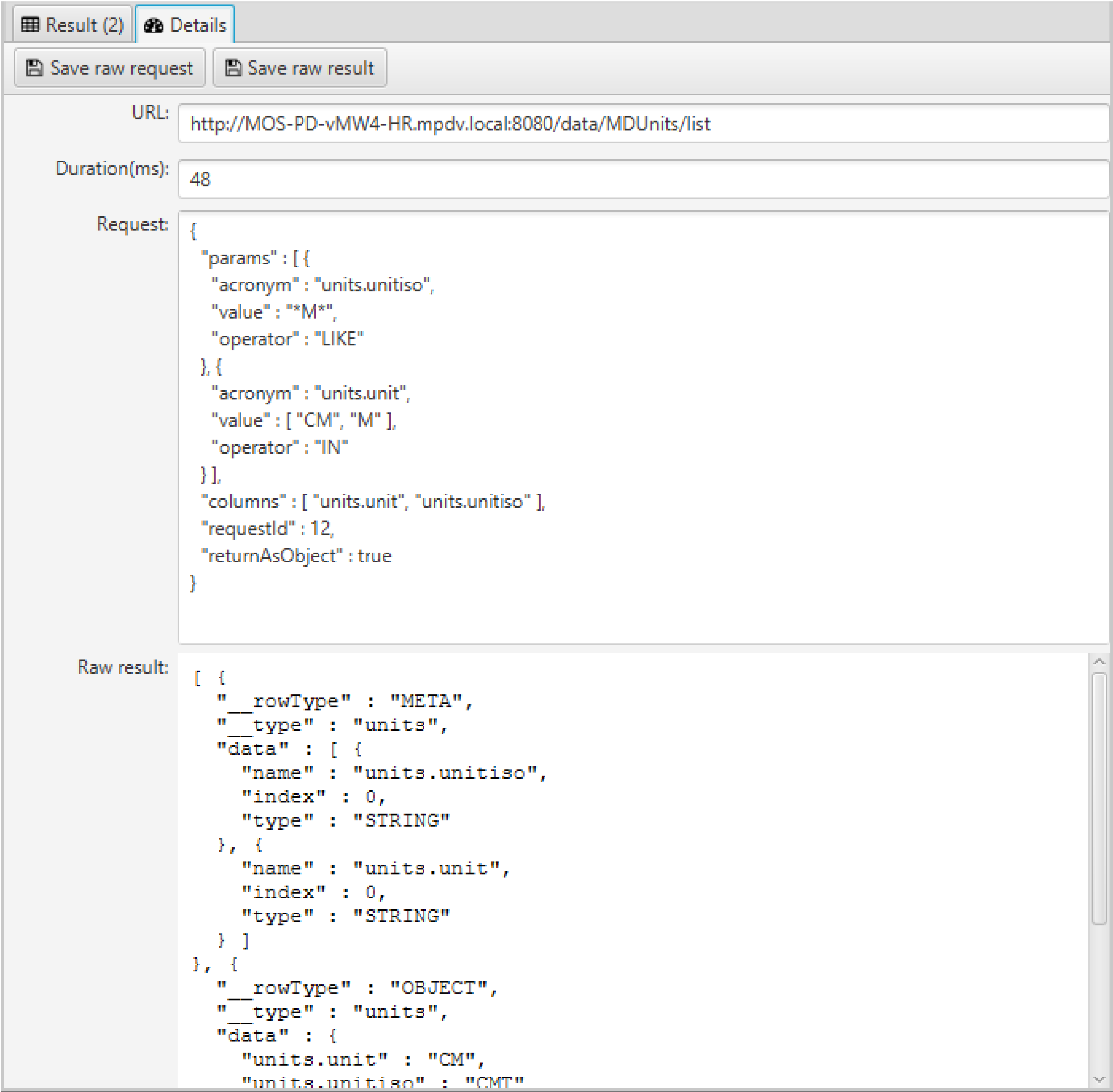
Save raw request	The service request is saved in JSON format (file extension *.jfx).
Save raw result	The service request and the result are saved in JSON format (file extension *.jfx).

Tab *Details* also includes four text fields.

URL	URL to call the service. You can copy this URL into your own client implementation. From MW 4.0$_{pe,}$ you must attach AccessId to the request in your own client. Format the AccessId in your own client with leading zeros: • As a customer, format the installation ID stored in the basic settings as AccessId with **8 digits** and leading zeros.

	• The partners format the individually assigned AccessId with **6 digits** and leading zeros.
	
	`https://<host>:<port>/data/MDUser/list?X-Access-Id=<AccessId>`
	Example (using the installation ID of the basic settings as AccessId):
	
	`https://<host>:<port>/data/MDUser/list?X-Access-Id=00012345`
	You can find further information on the AccessId in the chapter on Interface Technology (in the same document or in the Service Interface Manual if you use the Service Tester documentation as a separate document).

Duration (ms)	Specifies the service duration.
Request	Represents the body of the http request (JSON). You can copy the content of the text field in the body of your own client implementation anf change it, if required.
Raw Result	Presents the result as JSON.



4.6 SQL request

Click the following button to open the SQL window:

If you want to use SQL requests, you must first specify the connection information to the database in the window "Settings" described below.

In the SQL window, you can send SQL requests to the respective database. You can only execute SQL statements that display data. Statements that change data or modify the database structure are not permitted.

4.7 DLG request

Note: PDM dialogs are not available in all systems.

Click the following button to open the window for the execution of PDM dialogs.

The DLG request window sends PDM dialogs to the server and displays the result as text.

Example:

The following PDM dialog shows the basic system settings:

```
DLG=HYDRA.SELECT|BEARB=12345|
```

```
RET=0|KT=|LT=|DATA=HYDRA|OPT:ABBRNR=J|OPT:MNRFTYP=A|OPT:ADEKNRPLAUS=J|OPT:ERFAMENGE=J|OPT:AUSMANU=J|OPT:AUSNR=J|OPT:PZEAUTO=N
|OPT:KDVLIST= |OPT:WILLESYS= |OPT:UCHGEN=N|OPT:TRMGEN=J|OPT:KORRRMNGEN= |OPT:EDITBRMSPERR= |OPT:AUSPAUPDAUER= |BDEOPT:5=
|BDEOPT:9=N|BDEOPT:10= |BDEOPT:11= |BDEOPT:12=N|BDEOPT:13=N|BDEOPT:16=N|BDEOPT:17=N|BDEOPT:19=N|BDEOPT:20=
|BDEOPT:21=J|BDEOPT:22=J|BDEOPT:23=N|BDEOPT:30=N|OPT:RMGEN=N|OPT:PDAUERBMKVERB=NNNNNNNNNNNNN|ANR:GKP=GKP100020100|ANR:GKM=|OP
T:EVENTEDIT=J|FLSOPT:6= |FLSOPT:7= |OPT:ADATF=J|OPT:AART=K|OPT:RESMGT=
|OPT:RESUWG=M|OPT:RESOWG=M|OPT:GUTMANU=J|OPT:PZPLAUS=N|OPT:MDEGEN=J|OPT:MDEKNRPLAUS=J|OPT:EMSAKTIV=N|OPT:MDETLG=N|OPT:ANTVERB
PM=N|OPT:ERFPMENGE=J|PNRFUELL=0|OPT:PZEPLAUS=N|OPT:PDAUERSKMOD=N|OPT:RMIRUEZ=N|OPT:SAG=A|OPT:TGLKTOBEGR=N|OPT:VLIST=J|OPT:VGZ
=|OPT:VLISTMOD=M|OPT:WSTEMP=N|OPT:WSTEMPFGR=J|OPT:ZST=N|SIGN:STMP=S|OPT:PLNAPZMGRP= |OPT:PZEBERECHT= |OPT:PZEINFOFORM=
|OPT:ZKSBERECHT=
|OPT:CNRAUTOGEN=J|ADEKAR:BMKNR=9|ADEKAR:AKTIV=J|ADEKAR:DAUER=5|ADEKAR:DAUERMAX=240|MDEKAR:BMKPLAN=0|MDEKAR:BMKNPLAN=0|MDEKAR:
OG=0|MDEKAR:UG=0|MDEKAR:AKTIV=N|LEN:AUNR=8|LEN:AGNR=4|LEN:KNR=4|LEN:PNR=8|LEN:CNR=20|LEN:AFOLG=0|LEN:UAGNR=0|LEN:SPLNR=1|LEN:
RMNR=|KK1:AKTIV=N|KK1:VER=0|KK2:AKTIV=
|KK2:VER=0|SYSNR=9999|SEARCHSTRING=|KERNELVER=1|MAXDAUERSTEMP=50400|KDNR=99999|PRJ=ALCAN_MW20|VER=8.1|SAP:AUNRVON=|SAP:AUNRBI
S=|SAP:KNRVON=1|SAP:KNRBIS=8|SAP:KAPARTVON=0|SAP:KAPARTBIS=0|SAP:KSTVON=0|SAP:KSTBIS=0|SAP:PNRVON=1|SAP:PNRBIS=8|SAP:SPLITNRB
IS=0|SAP:SPLITNRVON=0|SAP:UVGNRVON=|SAP:UVGNRBIS=|SAP:VGFOLGVON=|SAP:VGFOLGBIS=|SAP:VGNRVON=|SAP:VGNRBIS=|SAP:WERK=|SAP:ZBER=
|SAP:ZFIR=|SAPOPT:2=N|SAPOPT:3=N|SAPOPT:4=N|SAPOPT:5=N|MWEVENT=Y|DISTRIBUTORTIME=1200|CFGMTIME=60|TABONLC=10|TRANSSIZE=10000|
TRANSTIME=600|TIMELEAK=10|REVERSETIME=3600|VISON=N|TRANSFKT=F|SPCININT=N|MASDATTARG=D|EXPPATH=PDVARC|TRANSPATH=PDVTRANS|CNRFI
X:WE=WE|CNRFIX:PR=PR|MPL:PARAM1=|MPL:PARAM2=|CNRAUTOGEN=N|HOSTNAME:SMTP=webmail.mpdv.local|PORT:SMTP=0|TMOUT:SMTP=0|SENDER:SM
TP=automail@bsp.de|SMTPPATH=|MOBILTELPATH=SMS|PAGERPATH=PAGER|MODUL=|SMENGEFAKT=0|MSPERREWARTZ=0|WAEHRUNG=EUR|PLANVL=10800|PL
ANHORIZ=30|SIMHORIZ=999|FIXHORIZ=1|SKYLINEHORIZ=0|OPT:LV=J|STREUBREITE=1|FIXPKTTERM=S|DEFTERMR=V|PRIONBER=0|HLSKENNZ:1=
|HLSKENNZ:2= |OFSLST=0|OPT:SPLITTEN=J|BEARB=12345|BEARBDAT=04/10/2018|BEARBZEI=45805|
```

4.8 Settings

Click the following button to open the "Settings" window:

You can save different settings in the "Settings" window.

4.8.1 Tab *Database*

Tab *Database* includes different settings on the database connection. You can use the button "Try Connection" to test if the connection information entered is correct. If the connection information is correct, the button is displayed in green. If an error occurs, the button is displayed in red and an error message is issued.

Database connection to MS SQL server

Driver	net.sourceforge.jtds.jdbc.Driver
URL	Schema: jdbc:jtds:<server_type>://<server>[:<port>][/<database>][;<property>=<value>[;...]] Examples: jdbc:jtds:sqlserver://scc7/hydra2;instance=hydms2 jdbc:jtds:sqlserver://win2008-7/hydra3;instance=hydms3
User name	DB user. Default: hydadm
Password	Password of DB user.

Database connection to Oracle

Driver	oracle.jdbc.driver.OracleDriver
URL	Schema: jdbc:oracle:thin:@[HOST][:PORT]:SID or jdbc:oracle:thin:@//[HOST][:PORT]/SERVICE Example: jdbc:oracle:thin:@linux9:1521:HYD1
User name	DB user. Default: hydadm
Password	Password of DB user.

4.9 Service Tester in Batch Mode

4.9.1 Overview

You can also start the Service Tester without user interface using the command line in order to execute services. To this end, the services to be requested are listed in so-called request files. The information on the server connection is included in a file with connection information. Both files are passed to the Service Tester as command line parameters. The request file includes a list of service requests to be executed and optionally also older PDM dialogs.

The Service Tester writes the results of the executed services in a file that can be checked manually or automatically.

This function has a double purpose:

- You can run test scenarios for self-developed applications.
- You can import bigger amounts of data into the system. To this end, another system automatically creates the files using services that are executed.

4.9.2 Program start

To start the Service Tester in Batch Mode, you request a Java JAR file. The Service Tester is then available for all platforms. You require a Java Runtime environment (JRE) with at least Java version 10. ObenJDK is used preferably. You can use the JRE for Windows, which is included in the Service Tester, by calling the java.exe with absolute path. A batch file is also available in Windows simplifying the call.

Via command line parameters, the Java program then obtains further information on the connection to the server, on the executing request file and the result file.

Overview of the command line parameters

Parameters	Description
-c <path>	Login Configuration: Path to the JSON file that includes the login information
-r <path>	Request file in JSON format
-l <path>	Log file or folder
-a	Processing is stopped at first error
-v	Show program version

Description of the command line parameters

-c <path>

Login Configuration: The parameter must be specified. Path to the file that includes the connection information (e.g. serverconfig.json). This file includes the login information for a Web Service Provider WSP. The login information includes e.g. server name, system, user login and database connection. The structure is described in detail below.

-r <path>

Path to the request file

-l <path>

Path to a log file or a folder. The results of the requests are written in a log file or a folder. If you specify a file, the results are saved as list in JSON format in the file. If you specify a folder, the results are written in separate files. The file names are numbered consecutively with each request.

-a

This parameter is optional. If this parameter is entered, the processing of a file is stopped as soon as the service request produces an error. If the parameter is not entered, the system continues with the next service if an error occurs.

-v

This parameter is optional. If this parameter is entered, the Java program shows the version number on the screen.

Examples

The examples are made using an installation in the directory "e:\DevTools\ServiceTester".

Request including all parameters for Batch Mode (all in one row)

java -jar e:\DevTools\ServiceTester\Service_Tester.jar -c
e:\ServiceTesterData\examples\serverconfig.json -r
e:\ServiceTesterData\examples\requestfile.json -l e:\ServiceTesterData\examples\logfile.json

Call with windows batch file

The call must be issued from the installation directory of the Service Tester.

```
e:
cd DevTools\ServiceTester
Service_Tester.bat -c e:\ServiceTesterData\examples\serverconfig.json -r
e:\ServiceTesterData\examples\requestfile.json -l
e:\ServiceTesterData\examples\logfile.json
```

Only display of version

java -jar e:\DevTools\ServiceTester\Service_Tester.jar -v

Request with absolute path to the JRE

e:\DevTools\ServiceTester\jre\bin\java.exe -jar e:\DevTools\ServiceTester\Service_Tester.jar –v

Return value of the program

You can use the return value of a program to identify if the processing was error-free or if errors occurred.

Value	Meaning
0	Entire program flow and all requests error-free
7	With at least one request, the server returned an error.
Other	Other errors, see log file or screen output

```
P:\>javaw -jar e:\DevTools\ServiceTester\Service_Tester.jar -c e:\ServiceTesterData\examples\serverconfig.json -r
e:\ServiceTesterData\examples\requestfile.json -l e:\ServiceTesterData\examples\logfile.json

P:\>echo Exit Code is %errorlevel%
Exit Code is 0
```

4.9.3 Processing with Batch Mode

The Service Tester uses the information included in the specified file with connection information to connect to a Web Service Provider (WSP) on a server.

The service requests or PDM dialogs listed in the request file passed are processed in the specified order one after the other. The results are written in a result file.

Depending on the parameter -a, the processing is stopped in case of an error or the processing continues using the next request (default).

4.9.4 Creating file with connection information

The Service Tester requires connection information to be able to connect to a Web Service Provider (WSP). The connection information is saved in a JSON file. You can save several files with connection information. Depending on the purpose, you then pass the required file to the Service Tester via command line.

You can use the Service Tester to create the files that include connection information:

1. Start the Service Tester in "normal mode" from the user interface (double-click ServiceTester.exe).
2. Select from the field "Already used" a connection that has already been used successfully.
3. Use the button "Save login configuration" to create the file for the selected connection and save the file at an appropriate location.

For example, the file can have the following structure:

```
{
    "protocol": "http",
  "serverHost" : "myserver.company.local",
    "serverPort": 8080,
    "user": "12345",
    "passwordCrypted": "leNvHRFTgbeNBnOsASMyBA==",
    "sqlPassword": "PuxPGSRf2YUNN+/RJO2Qow==",
    "sqlDriver": "net.sourceforge.jtds.jdbc.Driver",
  "sqlUsername" : "mydbuser",
  "sqlConnectionString" : "jdbc:jtds:sqlserver://myserver/hydra1;instance=HYDMS1"
}
```

4.9.5 Request files

Request files include service requests. Also requests of PDM dialogs are supported. The contents are in JSON format.

You manually create request files using a text editor or automatically using your own software. If you manually create a request file, you can copy the JSON notation of a single JTP request object from the GUI of the Service Tester from tab "Details" of the service request result.

The Service Tester ignores unknown keys in the request file. You can therefore insert further fields in the objects that can be used as comment, for example. Though, unknown keys in the request file are not transferred into the log file output.

The request file includes an array of objects. Each array element includes either an object "jtpRequest" or "dlgRequest":

<table>
<tr><th>Object</th><th>Content</th></tr>
<tr><td>jtpRequest</td><td>JTP Request Object, e.g.:<pre>
{
        "params" : [{
      "acronym": "units.unit",
    "value" : "CM",
    "operator" : "EQUAL"
  }],
  "columns":["units.si_unit", "units.unitiso", "units.designation"],
          "requestId": 1,
          "returnAsObject": false
}
</pre></td></tr>
<tr><td>dlgRequest</td><td>String with the PDM dialog data, e.g.:<pre>
DLG=PNR.MODIFY|PNR.PNR=667|PNR.KST=5187|
</pre></td></tr>
</table>

Example: requestfile.json

```
[
  {
      "jtpRequest": {
        "domain": "MDUnits",
        "service": "list",
        "request": {
        "params" : [ {
      "acronym": "units.unit",
          "value" : [ "CM", "KG", "G", "M" ],
              "operator": "IN"
        } ],
```

```
        "columns" : [ "units.unit", "units.modified_by", "units.si_unit", "units.unitiso", "units.modified_ts" ],
          "requestId": 1,
          "returnAsObject": false
        }
      }
    },
    {
      "dlgRequest" : "DLG=HYDRA.SELECT|BEARB=12345|ID=2|"
    }
]
```

Recommendation for the requestId

Number the different requests in field **requestId** in a sequence. The Service Tester also shows the **requestId** in the log file. You can therefore easily and uniquely match the request in the request file to the result in the log file.

With PDM dialogs, you can use the identification "**|ID=<n>|".**

4.9.6 Log files

In case of errors with the program start or syntactic errors in the request file, the log file includes further information on the error cause, for example:

```
e:\ServiceTesterData\examples\invalid_requestfile.json (Das System kann die angegebene Datei nicht finden)
```

or

```
Unrecognized token 'SomeInvalidJson': was expecting ('true', 'false' or 'null')
 at [Source: e:\ServiceTesterData\examples\requestfile.json; line: 2, column: 19]
 at [Source: e:\ServiceTesterData\examples\requestfile.json; line: 2, column: 3] (through reference chain:
java.lang.Object[][0])
```

The log file of an error-free program flow is structured in JSON. It includes an object with three further objects

```
{
  "serverConfig": {<...>},
  "requests": [<...>],
  "resultTotal": {<...>}
}
```

serverConfig

This object includes the connection information from the file passed with connection information. This way, you can see which server has executed the requests recorded in the following.

requests

This object includes an array with objects for each request. The following further objects are included:

```
  "requests": [
    {
      "jtpRequest": {<...>},
      "result": [<...>],
      "startTime": "2018-09-07T08:54:36.968Z"
    },
    {
      "dlgRequest": "DLG=HYDRA.SELECT|BEARB=12345|ID=2|",
```

```
      "result": "RET=0|KT=|LT=|DATA=HYDRA|OPT:ABBRNR=J|OPT:MNRFTYP=A|...",
      "startTime": "2018-09-07T08:54:37.215Z"
    }
  ],
```

- **jtpRequest** or **dlgRequest**: Repetition of the request from the request file
- **result**: request result. This object includes the entire result information of the service request. If the service request produces errors, further information on the error are included here.
- **startTime**: Time stamp of the service request.

resultTotal

This object includes a summary of all executed requests in the elements "countTotal", "countSuccessFull", "countFailed" and "endTime".

```
  "resultTotal": {
    "countTotal": 2,
    "countSuccessFull": 2,
    "countFailed": 0,
    "endTime": "2018-09-07T08:54:37.237Z"
  }
```

Example of a log file:


```
{
  "serverConfig": {
    "protocol": "http",
    "serverHost": "mos-swe-test-01.mpdv.local",
    "serverPort": 8081,
    "user": "12345",
    "passwordCrypted": "leNvHRFTgbeNBnOsASMyBA==",
    "sqlPassword": "PuxPGSRf2YUNN+/RJO2Qow==",
    "sqlDriver": "net.sourceforge.jtds.jdbc.Driver",
    "sqlUsername": "hydadm",
    "sqlConnectionString": "jdbc:jtds:sqlserver://mos-swe-test-01/hydra2;instance=HYDMS2"
  },
  "requests": [
    {
      "jtpRequest": {
        "domain": "MDUnits",
        "service": "list",
        "request": {
  "params": [
            {
      "acronym": "units.unit",
              "value": [
            "CM",
            "KG",
            "G",
                "M"
              ],
              "operator": "IN"
            }
          ],
          "columns": [
            "units.unit",
            "units.modified_by",
            "units.si_unit",
            "units.unitiso",
            "units.modified_ts"
          ],
          "requestId": 1,
          "returnAsObject": false
        }
      },
      "result": [
        {
                          "__rowType": "META",
          "__type": "units",
          "data": [
            {
              "name": "units.unit",
              "index": 0,
        "type": "STRING"
            },
            {
              "name": "units.modified_by",
              "index": 0,
        "type": "STRING"
            },
            {
              "name": "units.unitiso",
              "index": 0,
        "type": "STRING"
            },
            {
              "name": "units.modified_ts",
              "index": 0,
              "type": "DATETIME"
            },
            {
              "name": "units.si_unit",
              "index": 0,
              "type": "BOOLEAN"
            }
          ]
        },
        {
    "__rowType": "DATA",
          "__type": "units",
          "data": [
            "CM",
            "12345",
            "CMT",
            "2004-07-16T00:00:00+02:00",
false
          ]
        },
        {
    "__rowType": "DATA",
          "__type": "units",
          "data": [
            "G",
            "12345",
            "GRM",
            "2004-07-16T00:00:00+02:00",
false
          ]
        },
        {
```

```
    "__rowType": "DATA",
          "__type": "units",
          "data": [
            "KG",
            "12345",
            "KGM",
            "2004-07-16T00:00:00+02:00",
true
          ]
        },
        {
    "__rowType": "DATA",
          "__type": "units",
          "data": [
            "M",
            "12345",
            "MTR",
            "2004-07-16T00:00:00+02:00",
true
          ]
        }
      ],
      "startTime": "2018-09-07T08:54:36.968Z"
    },
    {
      "dlgRequest": "DLG=HYDRA.SELECT|BEARB=12345|ID=2|",
      "result": "RET=0|KT=|LT=|DATA=HYDRA|OPT:ABBRNR=J|OPT:MNRFTYP=A|...",
      "startTime": "2018-09-07T08:54:37.215Z"
    }
  ],
  "resultTotal": {
    "countTotal": 2,
    "countSuccessFull": 2,
    "countFailed": 0,
    "endTime": "2018-09-07T08:54:37.237Z"
  }
}
```

5 Licensing

5.1 License model

5.1.1 Overview

The license model of the HYDRA Service Interface is based on two components:

SCS-SIF: Basic License Service Interface

The basic license releases the interface. The basic license also includes development tools and the option to request released services for test purposes. The released services and PDM dialogs are listed in section "5.2 Released services and PDM dialogs".

SCS-SIC: Client License Service Interface

For each connected client, you need an SCS-SIC license ("named device" licensing). Clients are all devices, programs and software components that request services or implement a user interface, e.g.

- BDE terminal
- Operating unit
- Data collection stations
- Machine control of a machine
- Devices and software to visualize machines/systems
- Client software for users
- Client of a third-party software with connection to the Service Interface
- Office PCs
- IPCs
- …

5.1.2 Examples of how to identify client licenses

5.1.2.1 Example 1

System description

The IT department of a customer has developed a GUI (e.g. with C# / .NET) that a user can use to log on and off operations to or from a workplace in the MES. This application is installed in small industrial PCs that are placed near the workplaces.

Identifying the number of client licenses

You require one client license SCS-SIC for each industrial PC.

5.1.2.2 Example 2

System description

At 5 machining centers in a production, the manufacturer has implemented an extension in the machine control. The malfunctions of the machining centers identified via the Service Interface and the return to the production status are transferred as machine status change to the MES system.

Identifying the number of client licenses

You require a client license for each of the 5 machines.

5.1.2.3 Example 3

System description

There are monitors at 15 places in production where a web browser shows graphic reports and evaluations on the current production status of the site with 200 machines via a web server that the customer has implemented themselves. The web server regularly requests the current production data from the MES via the service interface and further processes this data.

Identifying the number of client licenses

You require a client license SCS-SIC for each of the 15 monitors showing graphic evaluations.

5.1.2.4 Example 4

System description

The following process is implemented as customization in an MES system:

- Specific operator actions on a shop floor terminal directly trigger a processing in the ERP system via interface to the ERP.
- As a processing result, the ERP system calls a service via the service interface to change an attribute for the machine in the MES.

Identifying the number of client licenses

The operator action on the shop floor terminal triggers the communication via service interface. For this reason, you require a client license SCS-SIC for each shop floor terminal triggering the process.

5.1.2.5 Example 5

System description

The company wants to transfer the orders that must be produced from an ERP system to the MES.

Identifying the number of client licenses

To transfer production orders from higher-level systems such as ERP systems, there are specific products to connect these higher-level systems. These products must be used here. (As of 2020, these are e.g. EIS-ERP, EIS-LUG, …). You require one EIS license for each connected higher-level system.

If the data is transferred via the Service Interface, you require the EIS licenses and additionally a basic license SCS-SIF and a single client license SCS-SIC to release the communication via the Service Interface.

5.2 Released services and PDM dialogs

5.2.1 Categorization of services and dialogs

You can classify the services and dialogs in three categories. The category specifies to what extent the system's business logic processes the service results.

Category	Description
DC = Data Collection	Data collection: Basic services of data collection, e.g. log an operation on, change machine status, clock-in, create or change master data
SM = Status Management	Status Management: Services and dialogs that provide information on the status information managed in HYDRA, e.g. list of the currently running operations or of the persons logged on.
DP = Data Processing	Data Processing: Services and dialogs that process and provide data or store it in the system, e.g. an OEE report or the HR labor time calculation.

5.2.2 Customer-specific services and dialogs

Customer-specific services and dialogs in the customer's namespace are generally released and can be requested via the interface. You can easily identify customer-specific services and dialogs. The name of these services and dialogs are prefixed by "U_".

It does not make a difference whether the customer-specific service or dialog has been created by MPDV of by the customer using the MES Development Suite.

Customer-specific services and dialogs are classified in the category "DP".

5.2.3 Global services and dialogs

Description	Name	Type	Cat.
User login	BEARB.LOGIN	Dialog	DC
User logout	BEARB.LOGOUT	Dialog	DC
List user field definitions	MDUserfieldConfiguration.list (*SP16)	Service	DC

Description	Name	Type	Cat.
List user field definitions for resource family	MDUserfieldConfiguration.listByResource (*SP16)	Service	DC
List user field definitions (outdated! Use service!)	USRFLDELEM.LIST	Dialog	DC
Manage user	MDUser.copy	Service	DC
Manage user	MDUser.delete	Service	DC
Manage user	MDUser.insert	Service	DC
Manage user	MDUser.list	Service	DC
Manage user	MDUser.lock	Service	DC
Manage user	MDUser.unlock	Service	DC
Manage user	MDUser.update	Service	DC
Manage user (outdated): Edit user	BEARB.COPY	Dialog	DC
Manage user (outdated): Edit user	BEARB.DELETE	Dialog	DC
Manage user (outdated): Edit user	BEARB.INSERT	Dialog	DC
Manage user (outdated): Edit user	BEARB.LOCK	Dialog	DC
Manage user (outdated): Edit user	BEARB.NEW	Dialog	DC
Manage user (outdated): Edit user	BEARB.SELECT	Dialog	DC
Manage user (outdated): Edit user	BEARB.UNLOCK	Dialog	DC
Manage user (outdated): Edit user	BEARB.UPDATE	Dialog	DC
Manage user (outdated): List of users	BEARB.LIST	Dialog	DC
Client login	CLIENT.LOGIN	Dialog	DC
Client logout	CLIENT.LOGOUT	Dialog	DC
Create dialog error log	SCMD;52	Dialog	DC
Manage units	MDUnits.delete	Service	DC
Manage units	MDUnits.insert	Service	DC
Manage units	MDUnits.list	Service	DC
Manage units	MDUnits.lock	Service	DC
Manage units	MDUnits.unlock	Service	DC
Manage units	MDUnits.update	Service	DC
eReport Manager: Request status	PrintService.state (*)	Service	DC
eReport Manager: List available printers	PrintService.listPrinters (*)	Service	DC
eReport Manager: List available reports	PrintService.listReports (*)	Service	DC
eReport Manager: Request meta data for report	PrintService.reportMetaById (*)	Service	DC
eReport Manager: Request meta data for report	PrintService.reportMetaByName (*)	Service	DC
eReport Manager: Create report	PrintService.report (*)	Service	DP
Edit escalations	EscalationMessages.close	Service	DC
Edit escalations	EscalationMessages.forward	Service	DC
Edit escalations	EscalationMessages.list	Service	DC

Description	Name	Type	Cat.
Edit escalations	EscalationMessages.read	Service	DC
Trigger escalation	EscalationMessages.insertHms	Service	DC
Trigger escalation (outdated) (available as of SP 16)	ESKMSG.INSERT	Dialog	DC
Read basic settings	Setup.list (SP16)	Service	DC
Read basic settings (outdated! Use service!)	HYDRA.SELECT	Dialog	DC
Check function authorization	SYSAuthorization.checkAuthorization (*SP16)	Service	SM
List function authorizations	SYSAuthorization.list (*SP16)	Service	DC
List function authorizations (outdated! Use service!)	BEARBFKT.LIST	Dialog	DC
Edit function authorizations	BEARBFKT.COPY	Dialog	DC
Edit function authorizations	BEARBFKT.DELETE	Dialog	DC
Edit function authorizations	BEARBFKT.INSERT	Dialog	DC
Edit function authorizations	BEARBFKT.LOCK	Dialog	DC
Edit function authorizations	BEARBFKT.NEW	Dialog	DC
Edit function authorizations	BEARBFKT.SELECT	Dialog	DC
Edit function authorizations	BEARBFKT.UNLOCK	Dialog	DC
Edit function authorizations	BEARBFKT.UPDATE	Dialog	DC
List function profiles	FKTPROF.LIST	Dialog	DC
Edit function profiles	FKTPROF.COPY	Dialog	DC
Edit function profiles	FKTPROF.DELETE	Dialog	DC
Edit function profiles	FKTPROF.INSERT	Dialog	DC
Edit function profiles	FKTPROF.LOCK	Dialog	DC
Edit function profiles	FKTPROF.NEW	Dialog	DC
Edit function profiles	FKTPROF.SELECT	Dialog	DC
Edit function profiles	FKTPROF.UNLOCK	Dialog	DC
Edit function profiles	FKTPROF.UPDATE	Dialog	DC
Delete locked data records	LOCK.DELETE	Dialog	DC
Edit INI configuration	INI.COPY	Dialog	DC
Edit INI configuration	INI.DELETE	Dialog	DC
Edit INI configuration	INI.EXPORT	Dialog	DC
Edit INI configuration	INI.IMPORT	Dialog	DC
Edit INI configuration	INI.INSERT	Dialog	DC
Edit INI configuration	INI.LOCK	Dialog	DC
Edit INI configuration	INI.NEW	Dialog	DC
Edit INI configuration	INI.SELECT	Dialog	DC
Edit INI configuration	INI.UNLOCK	Dialog	DC
Edit INI configuration	INI.UPDATE	Dialog	DC

Description	Name	Type	Cat.
List INI configurations	INI.LIST	Dialog	DC
List INI data configurations (outdated! Use service!)	INIDATA.LIST	Dialog	DC
List INI data configurations	IniDataConfiguration.list (*SP16)	Service	DC
Edit INI data configurations	INIDATA.COPY	Dialog	DC
Edit INI data configurations	INIDATA.DELETE	Dialog	DC
Edit INI data configurations	INIDATA.INSERT	Dialog	DC
Edit INI data configurations	INIDATA.LOCK	Dialog	DC
Edit INI data configurations	INIDATA.NEW	Dialog	DC
Edit INI data configurations	INIDATA.SELECT	Dialog	DC
Edit INI data configurations	INIDATA.UNLOCK	Dialog	DC
Edit INI data configurations	INIDATA.UPDATE	Dialog	DC
List categories	Category.list (*SP16)	Service	DC
List cost centers	CostCenters.list (*SP16)	Service	DC
List licenses / authorization keys	SYSLicensing.managedIds (*SP16)	Service	DC
List licenses (outdated! Use service!)	LIC.LIST	Dialog	DC
Edit licenses (please use user interface!)	LIC.DELETE	Dialog	DC
Edit licenses (please use user interface!)	LIC.INSERT	Dialog	DC
Create logging entry	SCMD;51	Dialog	DC
List machines/workplaces	MNR.LIST	Dialog	DC
Edit machine configuration	MNR.COPY	Dialog	DC
Edit machine configuration	MNR.DELETE	Dialog	DC
Edit machine configuration	MNR.INSERT	Dialog	DC
Edit machine configuration	MNR.LOCK	Dialog	DC
Edit machine configuration	MNR.NEW	Dialog	DC
Edit machine configuration	MNR.SELECT	Dialog	DC
Edit machine configuration	MNR.UNLOCK	Dialog	DC
Edit machine configuration	MNR.UPDATE	Dialog	DC
Create MLE outbound segment	HSODATA.INSERT	Dialog	DC
Number range: Manage definition	MDNumberRanges.delete	Service	DC
Number range: Manage definition	MDNumberRanges.delete	Service	DC
Number range: Manage definition	MDNumberRanges.insert	Service	DC
Number range: Manage definition	MDNumberRanges.list	Service	DC
Number range: Manage definition	MDNumberRanges.lock	Service	DC
Number range: Manage definition	MDNumberRanges.unlock	Service	DC
Number range: Manage definition	MDNumberRanges.update	Service	DC
Number range: Create new number	MDNumberRanges.createnr	Service	DC

Description	Name	Type	Cat.
Number range: Edit definition (outdated! Use service!)	NRKREIS.DELETE	Dialog	DC
Number range: Edit definition (outdated! Use service!)	NRKREIS.INSERT	Dialog	DC
Number range: Edit definition (outdated! Use service!)	NRKREIS.LOCK	Dialog	DC
Number range: Edit definition (outdated! Use service!)	NRKREIS.UNLOCK	Dialog	DC
Number range: Edit definition (outdated! Use service!)	NRKREIS.UPDATE	Dialog	DC
Number range: Create new number (use service!)	NRKREIS.CREATENR	Dialog	DC
Manage assignment of personnel function profile	PersonalFunction.delete (*SP16)	Service	DC
Manage assignment of personnel function profile	PersonalFunction.insert (*SP16)	Service	DC
Manage assignment of personnel function profile	PersonalFunction.list (*SP16)	Service	DC
Manage assignment of personnel function profile	PersonalFunction.lock (*SP16)	Service	DC
Manage assignment of personnel function profile	PersonalFunction.unlock (*SP16)	Service	DC
Manage assignment of personnel function profile	PersonalFunction.update (*SP16)	Service	DC
Manage personnel function profile	PersonalFunctionProfile.delete (*SP16)	Service	DC
Manage personnel function profile	PersonalFunctionProfile.insert (*SP16)	Service	DC
Manage personnel function profile	PersonalFunctionProfile.list (*SP16)	Service	DC
Manage personnel function profile	PersonalFunctionProfile.lock (*SP16)	Service	DC
Manage personnel function profile	PersonalFunctionProfile.unlock (*SP16)	Service	DC
Manage personnel function profile	PersonalFunctionProfile.update (*SP16)	Service	DC
Edit HR master data (outdated! Use service!)	PNR.DELETE	Dialog	DC
Edit HR master data (outdated! Use service!)	PNR.INSERT	Dialog	DC
Edit HR master data (outdated! Use service!)	PNR.LIST	Dialog	DC
Edit HR master data (outdated! Use service!)	PNR.LOCK	Dialog	DC
Edit HR master data (outdated! Use service!)	PNR.MODIFY	Dialog	DC
Edit HR master data (outdated! Use service!)	PNR.UNLOCK	Dialog	DC
Edit HR master data (outdated! Use service!)	PNR.UPDATE	Dialog	DC
Manage HR master data	BOPerson.delete	Service	DC
Manage HR master data	BOPerson.insert	Service	DC
Manage HR master data	BOPerson.list	Service	DC
Manage HR master data	BOPerson.lock	Service	DC
Manage HR master data	BOPerson.login	Service	DC
Manage HR master data	BOPerson.modify	Service	DC
Manage HR master data	BOPerson.unlock	Service	DC
Manage HR master data	BOPerson.update	Service	DC
Manage HR master data	BOPerson.updatePinCode	Service	DC
List paths	PATH.LIST	Dialog	DC
Edit paths	PATH.COPY	Dialog	DC

Description	Name	Type	Cat.
Edit paths	PATH.DELETE	Dialog	DC
Edit paths	PATH.INSERT	Dialog	DC
Edit paths	PATH.LOCK	Dialog	DC
Edit paths	PATH.NEW	Dialog	DC
Edit paths	PATH.SELECT	Dialog	DC
Edit paths	PATH.UNLOCK	Dialog	DC
Edit paths	PATH.UPDATE	Dialog	DC
List ReferenceData for selection lists/radio groups	SYSReferenceData.list (*SP16)	Service	DC
Send terminal status	SCMD;41	Dialog	DC
Shift day models for machines	MDMFDayModel.list (*SP17)	Service	DC
Shift year models für machines	MDMFYearModel.list (*SP17)	Service	DC
Synchronize terminal	SCMD;53	Dialog	DC
Terminal administration	TNR.ADMIN	Dialog	DC
Edit terminal configuration	TNR.COPY	Dialog	DC
Edit terminal configuration	TNR.DELETE	Dialog	DC
Edit terminal configuration	TNR.INSERT	Dialog	DC
Edit terminal configuration	TNR.LOCK	Dialog	DC
Edit terminal configuration	TNR.NEW	Dialog	DC
Edit terminal configuration	TNR.SELECT	Dialog	DC
Edit terminal configuration	TNR.UNLOCK	Dialog	DC
Edit terminal configuration	TNR.UPDATE	Dialog	DC
List terminal configurations	TNR.LIST	Dialog	DC
Terminal list	LIST;45	Dialog	DC
Restart terminal	TNR.NEUSTART	Dialog	DC
Manage terminals	Terminal.list	Service	DC
Update terminal	TNR.PROGLADEN	Dialog	DC
Read server time	SCMD;44	Dialog	DC
List assignment of responsibility areas	MDResponsibilityArea.list (*SP16)	Service	DC
List assignment of responsibility areas (outdated! Use service!)	BEARBVABPROF.LIST	Dialog	DC
Edit assignment of responsibility areas	BEARBVABPROF.COPY	Dialog	DC
Edit assignment of responsibility areas	BEARBVABPROF.DELETE	Dialog	DC
Edit assignment of responsibility areas	BEARBVABPROF.INSERT	Dialog	DC
Edit assignment of responsibility areas	BEARBVABPROF.LOCK	Dialog	DC
Edit assignment of responsibility areas	BEARBVABPROF.NEW	Dialog	DC
Edit assignment of responsibility areas	BEARBVABPROF.SELECT	Dialog	DC
Edit assignment of responsibility areas	BEARBVABPROF.UNLOCK	Dialog	DC

Description	Name	Type	Cat.
Edit assignment of responsibility areas	BEARBVABPROF.UPDATE	Dialog	DC
List responsibility profiles	VABPROF.LIST	Dialog	DC
Edit responsibility profiles	VABPROF.COPY	Dialog	DC
Edit responsibility profiles	VABPROF.DELETE	Dialog	DC
Edit responsibility profiles	VABPROF.INSERT	Dialog	DC
Edit responsibility profiles	VABPROF.LOCK	Dialog	DC
Edit responsibility profiles	VABPROF.NEW	Dialog	DC
Edit responsibility profiles	VABPROF.SELECT	Dialog	DC
Edit responsibility profiles	VABPROF.UNLOCK	Dialog	DC
Edit responsibility profiles	VABPROF.UPDATE	Dialog	DC

(*) For technical reasons, the service might not be available in all systems.

(*SP16) The service is available as of service pack 16.

(*SP17) The service is available as of service pack 17.

5.2.4 BDE services and dialogs

Description	Name	Type	Cat.
List deviation reasons	LIST;23	Dialog	DC
Log operation off	A_AB	Dialog	DC
Log operation on	A_AN	Dialog	DC
Deallocate operation	ANR.AUSPLANEN	Dialog	DC
Finish operation	A_BE	Dialog	DC
Plan operation	ANR.EINPLANEN	Dialog	DC
Unlock operation	ANR.ENTSPERREN	Dialog	DC
Release operation	ANR.FREIGEBEN	Dialog	DC
Lock operation	ANR.SPERREN	Dialog	DC
Split operation	ANR.SPLITCREATE	Dialog	DC
Log operation and person on	A_P_AN	Dialog	DC
Interrupt operation	A_UN	Dialog	DC
Operation: Currently logged on operations	BOOperation.listCurrentlyLoggedOn	Service	SM
Manage operations	BOOperation.delete	Service	DC
Manage operations	BOOperation.insert	Service	DC
Manage operations	BOOperation.list	Service	DC
Manage operations	BOOperation.lock	Service	DC

Description	Name	Type	Cat.
Manage operations	BOOperation.unlock	Service	DC
Manage operations	BOOperation.update	Service	DC
Manage operations: long texts	BOOperation.deleteNotes (*SP16)	Service	DC
Manage operations: long texts	BOOperation.insertNotes (*SP16)	Service	DC
Manage operations: long texts	BOOperation.listNotes (*SP16)	Service	DC
Manage operations: long texts	BOOperation.lockNotes (*SP16)	Service	DC
Manage operations: long texts	BOOperation.unlockNotes (*SP16)	Service	DC
Manage operations: long texts	BOOperation.updateNotes (*SP16)	Service	DC
Operations: Currently logged on operations	BOOperation.overviewParallelLogon (*SP16)	Service	SM
Cancel operation split	ANR.SPLITDELETE	Dialog	DC
Extended operation split	ANR.ADVSPLITCREATE	Dialog	DC
Operation overview	BOOperation.overview (*SP16)	Service	SM
List work plans	MDWorkplanOrder.list (*SP16)	Service	DC
Work plans: Generate order from work plan	MDWorkplanOrder.generateOrder (*SP16)	Service	DP
Update order	ANR.AKTUALISIEREN	Dialog	DC
Unlock order/operation for editing purposes	ANR.UNLOCK	Dialog	DC
Lock order/operation for editing purposes	ANR.LOCK	Dialog	DC
Edit order/operation	ANR.COPY	Dialog	DC
Edit order/operation	ANR.DELETE	Dialog	DC
Edit order/operation	ANR.INSERT	Dialog	DC
Edit order/operation	ANR.SELECT	Dialog	DC
Edit order/operation	ANR.UPDATE	Dialog	DC
Change order/operation status	ANR.SETSTATUS	Dialog	DC
Manage orders	BOOrder.delete	Service	DC
Manage orders	BOOrder.insert	Service	DC
Manage orders	BOOrder.list	Service	DC
Manage orders	BOOrder.lock	Service	DC
Manage orders	BOOrder.unlock	Service	DC
Manage orders	BOOrder.update	Service	DC
Manage orders: long texts	BOOrder.deleteNotes (*SP16)	Service	DC
Manage orders: long texts	BOOrder.insertNotes (*SP16)	Service	DC
Manage orders: long texts	BOOrder.listNotes (*SP16)	Service	DC
Manage orders: long texts	BOOrder.lockNotes (*SP16)	Service	DC
Manage orders: long texts	BOOrder.unlockNotes (*SP16)	Service	DC
Manage orders: long texts	BOOrder.updateNotes (*SP16)	Service	DC
List orders/operations	ANR.LIST	Dialog	DC

Description	Name	Type	Cat.
List order types	LIST;87	Dialog	DC
Manage order-related postings	DCAdeLogRecord.authorize	Service	DC
Manage order-related postings	DCAdeLogRecord.copy	Service	DC
Manage order-related postings	DCAdeLogRecord.delete	Service	DC
Manage order-related postings	DCAdeLogRecord.insert	Service	DC
Manage order-related postings	DCAdeLogRecord.list	Service	DC
Manage order-related postings	DCAdeLogRecord.lock	Service	DC
Manage order-related postings	DCAdeLogRecord.unlock	Service	DC
Manage order-related postings	DCAdeLogRecord.update	Service	DC
List order components (material list)	LIST;74	Dialog	DC
Order list	LIST;11	Dialog	DC
Update order network	ANETZ.AKTUALISIEREN	Dialog	DC
Create order network	ANETZ.INSERT	Dialog	DC
Unlock order network for editing purposes	ANETZ.UNLOCK	Dialog	DC
Lock order network for editing purposes	ANETZ.LOCK	Dialog	DC
Delete order network	ANETZ.DELETE	Dialog	DC
Edit order network	ANETZ.UPDATE	Dialog	DC
Order shift log	RPOperation.shiftlog (*SP16)	Service	DP
Order overview	BOOrder.overview (*SP16)	Service	SM
List scrap reasons	LIST;84	Dialog	DC
Scrap profile	RPOperation.quantityControl (*SP16)	Service	DP
Scrap statistics	RPOperation.quantityStatistics (*SP16)	Service	DP
Approve BDE posting	ADEPRO.SIGN	Dialog	DC
Edit BDE posting	ADEPRO.COPY	Dialog	DC
Edit BDE posting	ADEPRO.DELETE	Dialog	DC
Edit BDE posting	ADEPRO.INSERT	Dialog	DC
Edit BDE posting	ADEPRO.LOCK	Dialog	DC
Edit BDE posting	ADEPRO.SELECT	Dialog	DC
Edit BDE posting	ADEPRO.UNLOCK	Dialog	DC
Edit BDE posting	ADEPRO.UPDATE	Dialog	DC
List operator positions	MDOperatorPosition.list (*SP16)	Service	DC
List operator positions of machines	LIST;14	Dialog	DC
List resource performance accounts	MDPerformanceAccount.list (*SP16)	Service	DC
Manage production resources and tools (tools, documents, etc.)	BOPProductionResources.delete	Service	DC
Manage production resources and tools (tools, documents, etc.)	BOPProductionResources.insert	Service	DC
Manage production resources and tools (tools, documents, etc.)	BOPProductionResources.list	Service	DC

Description	Name	Type	Cat.
Manage production resources and tools (tools, documents, etc.)	BOPProductionResources.lock	Service	DC
Manage production resources and tools (tools, documents, etc.)	BOPProductionResources.unlock	Service	DC
Manage production resources and tools (tools, documents, etc.)	BOPProductionResources.update	Service	DC
List production variants	LIST;50	Dialog	DC
List reasons	MDReason.list (*SP16)	Dialog	DC
List reasons (outdated! Use service!)	GR.LIST	Dialog	DC
Edit reasons	GR.COPY	Dialog	DC
Edit reasons	GR.DELETE	Dialog	DC
Edit reasons	GR.INSERT	Dialog	DC
Edit reasons	GR.LOCK	Dialog	DC
Edit reasons	GR.NEW	Dialog	DC
Edit reasons	GR.SELECT	Dialog	DC
Edit reasons	GR.UNLOCK	Dialog	DC
Edit reasons	GR.UPDATE	Dialog	DC
List reason texts	MDReasonText.list (*SP16)	Service	DC
List reason texts (outdated! Use service!)	GRTXT.LIST	Dialog	DC
Edit reason texts	GRTXT.COPY	Dialog	DC
Edit reason texts	GRTXT.DELETE	Dialog	DC
Edit reason texts	GRTXT.INSERT	Dialog	DC
Edit reason texts	GRTXT.LOCK	Dialog	DC
Edit reason texts	GRTXT.NEW	Dialog	DC
Edit reason texts	GRTXT.SELECT	Dialog	DC
Edit reason texts	GRTXT.UNLOCK	Dialog	DC
Edit reason texts	GRTXT.UPDATE	Dialog	DC
List comments on operations	LIST;61	Dialog	DC
Manage bill of materials (BOM)	BOPComponents.delete	Service	DC
Manage bill of materials (BOM)	BOPComponents.insert	Service	DC
Manage bill of materials (BOM)	BOPComponents.list	Service	DC
Manage bill of materials (BOM)	BOPComponents.lock	Service	DC
Manage bill of materials (BOM)	BOPComponents.unlock	Service	DC
Manage bill of materials (BOM)	BOPComponents.update	Service	DC
Cost control	RPOperation.costControl (*SP16)	Service	DC
List machine information	LIST;10	Dialog	DC
Change machine status	M_MST	Dialog	DC
Machine status list	LIST;16	Dialog	DC
List of Material Requirements	BOOperation.materialRequirementList	Service	DP

Description	Name	Type	Cat.
Create material list	MATLIST.INSERT	Dialog	DC
Unlock material list for editing purposes	MATLIST.UNLOCK	Dialog	DC
Lock material list for editing purposes	MATLIST.LOCK	Dialog	DC
Delete material list	MATLIST.DELETE	Dialog	DC
Edit material list	MATLIST.UPDATE	Dialog	DC
Quantity upload	A_MR	Dialog	DC
Log person off	P_AB	Dialog	DC
Log person on	P_AN	Dialog	DC
Personnel list	LIST;12	Dialog	DC
Personnel shift log	RPPerson.shiftlog (*SP16)	Service	DP
Persons: Currently logged on persons	BOPerson.listCurrentlyLoggedOn (*SP16)	Service	SM
Persons: Currently logged on persons	RPPerson.personnelReport (*SP16)	Service	DP
Persons: Log off all persons from machine	P_AAB	Dialog	DC
List premium indicators	LIST;24	Dialog	DC
Logging of production lock	M_PSPERRE	Dialog	DC
Setup change list	BOOperation.setupChangeList	Service	DP
Create merged operation	ANR.SAGINSERT	Dialog	DC
Delete merged operation	ANR.SAGDELETE	Dialog	DC
Beginning of shift	A_AAN	Dialog	DC
End of shift	A_AUN	Dialog	DC
Read shift information of the machine	MNR.SKINFO	Dialog	DC
Read shift calendar	LIST;38	Dialog	DC
Shift change	A_ASW	Dialog	DC
Change target quantity	A_SMG	Dialog	DC
Change target cycle	M_SZY	Dialog	DC
Automatic status update	M_AST	Dialog	DC
Change partitioning	M_TLG	Dialog	DC
Posting of part quantity (partial confirmation)	A_TR	Dialog	DC
Counter: List configuration	LIST;131	Dialog	DC
List providing the assignment of workplaces to MDE shop floor clients	LIST;140	Dialog	DC

(*SP16) The service is available as of service pack 16.

5.2.5 MDE services and dialogs

Description	Name	Type	Cat.
Workplaces/Machines	BOResource.workplaceOverview	Service	SM
List groups	MDResourceGroup.list (*SP16)	Service	DC
List groups (outdated! Use service!)	GRP.LIST	Dialog	DC
Edit groups	GRP.COPY	Dialog	DC
Edit groups	GRP.DELETE	Dialog	DC
Edit groups	GRP.INSERT	Dialog	DC
Edit groups	GRP.LOCK	Dialog	DC
Edit groups	GRP.NEW	Dialog	DC
Edit groups	GRP.SELECT	Dialog	DC
Edit groups	GRP.UNLOCK	Dialog	DC
Edit groups	GRP.UPDATE	Dialog	DC
List groups	GRP.LIST	Dialog	DC
List group assignments	MDResourceGroupAssignment.list (*SP16)	Service	DC
List group assignments (outdated! Use service!)	GRPRES.LIST	Dialog	DC
Edit group assignment	GRPRES.COPY	Dialog	DC
Edit group assignment	GRPRES.DELETE	Dialog	DC
Edit group assignment	GRPRES.INSERT	Dialog	DC
Edit group assignment	GRPRES.LOCK	Dialog	DC
Edit group assignment	GRPRES.NEW	Dialog	DC
Edit group assignment	GRPRES.SELECT	Dialog	DC
Edit group assignment	GRPRES.UNLOCK	Dialog	DC
Edit group assignment	GRPRES.UPDATE	Dialog	DC
Efficiency report, OEE report	RPProductionReporting.efficiencyReport	Service	DP
Manage machine-related postings	Workplacebooking.delete	Service	DC
Manage machine-related postings	Workplacebooking.insert	Service	DC
Manage machine-related postings	Workplacebooking.list	Service	DC
Manage machine-related postings	Workplacebooking.lock	Service	DC
Manage machine-related postings	Workplacebooking.unlock	Service	DC
Manage machine-related postings	Workplacebooking.update	Service	DC
Machine history	RPResource.workplaceHistory	Service	DC
Edit machine statuses	MST.COPY	Dialog	DC
Edit machine statuses	MST.DELETE	Dialog	DC
Edit machine statuses	MST.INSERT	Dialog	DC
Edit machine statuses	MST.LOCK	Dialog	DC
Edit machine statuses	MST.NEW	Dialog	DC

Description	Name	Type	Cat.
Edit machine statuses	MST.SELECT	Dialog	DC
Edit machine statuses	MST.UNLOCK	Dialog	DC
Edit machine statuses	MST.UPDATE	Dialog	DC
Machine statuses: List	MST.LIST	Dialog	DC
Machine time profile	RPMachine.timeProfile	Service	DP
Material: Batch data overview	BOBatch.list (* SP17)	Service	DC
Material: Currently used material	BOBatch.listCurrentlyLoggedOnInput	Service	SM
Material: Currently produced material	BOBatch.listCurrentlyLoggedOnOutput	Service	SM
Create MDE posting	MDEPRO.COPY	Dialog	DC
Edit MDE posting	MDEPRO.COPY	Dialog	DC
Edit MDE posting	MDEPRO.DELETE	Dialog	DC
Edit MDE posting	MDEPRO.INSERT	Dialog	DC
Edit MDE posting	MDEPRO.LOCK	Dialog	DC
Edit MDE posting	MDEPRO.UNLOCK	Dialog	DC
Edit MDE posting	MDEPRO.UPDATE	Dialog	DC
Resources: Currently logged on	BOResource.listCurrentlyLoggedOn	Service	SM
Status: Current parallel statuses	BOResource.listCurrentlyStatus	Service	SM
Status analysis	RPStatusAnalysis.list	Service	DP
Edit status classes	STKL.DELETE	Dialog	DC
Edit status classes	STKL.INSERT	Dialog	DC
Edit status classes	STKL.LOCK	Dialog	DC
Edit status classes	STKL.SELECT	Dialog	DC
Edit status classes	STKL.UNLOCK	Dialog	DC
Edit status classes	STKL.UPDATE	Dialog	DC
Status classes: List	STKL.LIST	Dialog	DC
Status report	RPProductionReporting.statusReport	Service	DP
Status report (machine-related)	RPProductionReporting.standstillReport	Service	DP
List status texts	MDStatusText.list (*SP16)	Service	DC
List status texts (outdated! Use service!)	MSTTXT.LIST	Dialog	DC
Edit status texts	MSTTXT.DELETE	Dialog	DC
Edit status texts	MSTTXT.INSERT	Dialog	DC
Edit status texts	MSTTXT.LOCK	Dialog	DC
Edit status texts	MSTTXT.SELECT	Dialog	DC
Edit status texts	MSTTXT.UNLOCK	Dialog	DC
Edit status texts	MSTTXT.UPDATE	Dialog	DC
Maintenances: Current maintenances	BOPMaintenance.list	Service	SM

Description	Name	Type	Cat.
Edit counter configuration	MNRCTR.COPY	Dialog	DC
Edit counter configuration	MNRCTR.DELETE	Dialog	DC
Edit counter configuration	MNRCTR.INSERT	Dialog	DC
Edit counter configuration	MNRCTR.LOCK	Dialog	DC
Edit counter configuration	MNRCTR.MODIFY	Dialog	DC
Edit counter configuration	MNRCTR.NEW	Dialog	DC
Edit counter configuration	MNRCTR.SELECT	Dialog	DC
Edit counter configuration	MNRCTR.UNLOCK	Dialog	DC
Edit counter configuration	MNRCTR.UPDATE	Dialog	DC
Counter configuration: List	MNRCTR.LIST	Dialog	DC
List assignment of machine/WP to terminal	MNRTNR.LIST	Dialog	DC
Edit assignment of machine/WP to terminal	MNRTNR.DELETE	Dialog	DC
Edit assignment of machine/WP to terminal	MNRTNR.INSERT	Dialog	DC
Edit assignment of machine/WP to terminal	MNRTNR.SELECT	Dialog	DC
Cycle progression	RPResource.cycleProgression	Service	DP

(*SP16) The service is available as of service pack 16.

(*SP17) The service is available as of service pack 17.

5.2.6 CAQ services and dialogs

Description	Name	Type	Cat.
Inspection characteristics	InspectionCharacteristic.list	Service	DC
Failure mode analysis	FailureModeAnalysis.list	Service	DP
Measure tracking	TracingMeasures.list	Service	DC
Inspection requirements	InspectionRequirement.list	Service	DC
Inspection requirements	QMDefectAssignedInspectionRequirement.list	Service	DC
Inspection requirements	QMMeasureAssignedInspectionRequirement.list	Service	DC
Inspection plan	InspectionPlan.list	Service	SM
Inspection plan characteristics	InspectionPlanCharacteristic.list	Service	SM
Inspection points	InspectionPoint.list	Service	SM
Inspection steps	InspectionStep.list	Service	SM
Samples	QMSample.list	Service	SM
Sample values	QMSingleValue.list	Service	SM

5.2.7 HLS/PEP services and dialogs

Description	Name	Type	Cat.
Workplace schedule	WorkplaceSchedule.list	Service	DP
Assignments of persons to workplaces	PersonnelWorkplaceAssignment.delete	Service	DC
Assignments of persons to workplaces	PersonnelWorkplaceAssignment.insert	Service	DC
Assignments of persons to workplaces	PersonnelWorkplaceAssignment.list	Service	DC
Assignments of persons to workplaces	PersonnelWorkplaceAssignment.lock	Service	DC
Assignments of persons to workplaces	PersonnelWorkplaceAssignment.unlock	Service	DC
Assignments of persons to workplaces	PersonnelWorkplaceAssignment.update	Service	DC
Assignments of persons to workplaces	PersonnelWorkplaceAssignment.lock	Service	DC
planned inventory levels	InventoryLevelProjection.list	Service	DP
Production variants	ProductionVariants.list	Service	SM
Workforce requirements of workplaces	PersonnelRequirementOfWorkplaces.list	Service	SM
Operations that can be planned	MDPlanableOperations.list (*SP16)	Service	SM
Workplaces that can be planned	MDPlanableMachines.list (*SP16)	Service	SM
Planning variants	PlanningVariants.list	Service	SM
Save planning	MDShopFloorPlanning.save (*SP16)	Service	DP
Planning profiles	PlanningProfile.list (*SP16)	Service	DC
Qualifications	Qualifications.list	Service	DC
Setup change matrix	PlanningSetupChangeMatrix.list	Service	SM
Assignment of qualifications to staff	PersonnelQualificationsAssignment.list	Service	SM

5.2.8 HR services and dialogs

Description	Name	Type	Cat.
Current account balances	PersonalAccountBalances.list	Service	DP
Manage badges	Badges.delete	Service	DC
Manage badges	Badges.insert	Service	DC
Manage badges	Badges.list	Service	DC
Manage badges	Badges.lock	Service	DC
Manage badges	Badges.unlock	Service	DC
Manage badges	Badges.update	Service	DC
Autom. status clocking	P_AST	Dialog	DC
Absence reason clocking (advance or subsequent clocking)	P_FGR	Dialog	DC
Manage absence planning	PersonalAbsencePlanning.delete	Service	DC

Description	Name	Type	Cat.
Manage absence planning	PersonalAbsencePlanning.deleteByPerson (*Res)	Service	DC
List work equipment	WorkEquipment.list (*SP16)	Service	DC
Edit work equipment (also handout, return)	WorkEquipment.update (*SP16)	Service	DC
Manage absence planning	PersonalAbsencePlanning.insert	Service	DC
Manage absence planning	PersonalAbsencePlanning.list	Service	DC
Manage absence planning	PersonalAbsencePlanning.listPreviousIllnesses	Service	SM
Manage absence planning	PersonalAbsencePlanning.lock	Service	DC
Manage absence planning	PersonalAbsencePlanning.unlock	Service	DC
Manage absence planning	PersonalAbsencePlanning.update	Service	DC
Create absence planning	FZ.INSERT	Dialog	DC
Delete absence planning	FZ.DELETE	Dialog	DC
Delete absence planning	FZ.DELETE_ALL	Dialog	DC
Public holidays for access control	LIST;30	Dialog	DC
Clocking-out	P_GEH	Dialog	DC
Information on staff that can be planned	PersonalPlanningData.list	Service	DP
Clocking-in	P_KOM	Dialog	DC
Create account limit	PZEKTOG.INSERT	Dialog	DC
Wage type statistics	WageTypesStatistics.list	Service	DP
Monthly results, wage types	PersonalTimeMonthlyWagetypes.list	Service	DP
Monthly results, times and accounts	PersonalTimeMonthlyResults.list	Service	DP
Subsequent clocking: Clocking-in including reason.	P_NB	Dialog	DC
Show messages for the person	BOPerson.messages	Service	SM
opening hours	LIST;32	Dialog	DC
Online request of a person's account balances	SCMD;31	Dialog	DC
Online check of clocking authorization	SCMD;46	Dialog	DC
Online check of access authorizations	SCMD;49	Dialog	DC
Clocking out for break	P_PAU	Dialog	DC
Delete personal day type	PersonalDayTypes.delete	Service	DC
Create personal day type	PersonalDayTypes.insert	Service	DC
Show personal day type	PersonalDayTypes.list	Service	DC
Lock personal day type	PersonalDayTypes.lock	Service	DC
Change personal day type	PersonalDayTypes.update	Service	DC
Unlock personal day type	PersonalDayTypes.unlock	Service	DC
Labor time schedule	PersonalTimePlan.list	Service	DP
Labor time statistics	PersonalTimeStatistic.list	Service	DP

Description	Name	Type	Cat.
Access status logs	Z_PRO	Dialog	DC
Logging of an access	Z_ZUT	Dialog	DC
Logging of an access attempt	Z_ZV	Dialog	DC
Clocking archive	ClockingArchive.list	Service	SM
Edit clocking authorizations	ZBERECHT.DELETE	Dialog	DC
Edit clocking authorizations	ZBERECHT.INSERT	Dialog	DC
Clocking authorizations and infos of the Time and Attendance	ClockingAuthorizations.listTerminal (*SP16)	Service	DC
Clocking authorizations of the Time and Attendance (outdated! Use service ClockingAuthorizations.listTerminal!)	LIST;27	Dialog	DC
Manage clockings	Clockings.delete	Service	DC
Manage clockings	Clockings.execute	Service	DC
Manage clockings	Clockings.insert	Service	DC
Manage clockings	Clockings.list	Service	DC
Manage clockings	Clockings.lock	Service	DC
Manage clockings	Clockings.unlock	Service	DC
Manage clockings	Clockings.update	Service	DC
Advance clocking: Clocking-out including reason.	P_VB	Dialog	DC
Time management: Delete all clockings of a day	STMP.DELETE	Dialog	DC
Time management: Import working time models	GLZJMOD.MODIFY	Dialog	DC
Time management: Import planned working time and working time day types	GLZTMOD.INSERT	Dialog	DC
Time management: Import planned working time and working time day types	GLZTMOD.MODIFY	Dialog	DC
Time management: Import planned working time and working time day types	GLZTMOD.UPDATE	Dialog	DC
Time management: Import shift rhythm model	SCHZARTJMOD.MODIFY	Dialog	DC
Time management: Import clocking	STMP.LOAD	Dialog	DC
Time management: Import daily result	TGERG.MODIFY	Dialog	DC
List access points	LIST;46	Dialog	DC
Status of access point	Z_STA	Dialog	DC
Access authorizations	LIST;28	Dialog	DC
Edit access authorizations	ZPROFZGRP.DELETE	Dialog	DC
Edit access authorizations	ZPROFZGRP.INSERT	Dialog	DC
List access time models	LIST;29	Dialog	DC
Manage access profile	AccessProfiles.delete	Service	DC
Manage access profile	AccessProfiles.insert	Service	DC

Description	Name	Type	Cat.
Manage access profile	AccessProfiles.list	Service	DC
Manage access profile	AccessProfiles.lock	Service	DC
Manage access profile	AccessProfiles.unlock	Service	DC
Manage access profile	AccessProfiles.update	Service	DC
Access profile assignment to badge	AccessProfileAssignment.delete	Service	DC
Access profile assignment to badge	AccessProfileAssignment.insert	Service	DC
Access profile assignment to badge	AccessProfileAssignment.list	Service	DC
Access profile assignment to badge	AccessProfileAssignment.lock	Service	DC
Access profile assignment to badge	AccessProfileAssignment.unlock	Service	DC
Access profile assignment to badge	AccessProfileAssignment.update	Service	DC

(*Res) The service name is reserved. It is not yet available in the current release.

(*SP16) The service is available as of service pack 16.

5.2.9 MPL services and dialogs

Description	Name	Type	Cat.
Log off output batch	CA_AB	Dialog	DC
Log on output batch	CA_AN	Dialog	DC
Create cutting plan	BAHNVERT.CREATE	Dialog	DC
Batch change	CA_WL	Dialog	DC
Log input batch off	CE_AB	Dialog	DC
Log input batch on	CE_AN	Dialog	DC
Input batch change	CE_WL	Dialog	DC
Bill of materials (BOM)	LIST;74	Dialog	DC
Restore batch from archive	CNR.RESTORE	Dialog	DC
Split batch	CNR.SPLITCREATE	Dialog	DC
Merge batch	CNR.SUMMARIZE	Dialog	DC
Repost MES batch/ERP batch	C_UMB	Dialog	DC
Batch attributes of a material type	LIST;67	Dialog	DC
Edit batch inventory	CNR.COPY	Dialog	DC
Edit batch inventory	CNR.DELETE	Dialog	DC
Edit batch inventory	CNR.INSERT	Dialog	DC
Edit batch inventory	CNR.LOCK	Dialog	DC
Edit batch inventory	CNR.MODIFY	Dialog	DC

Description	Name	Type	Cat.
Edit batch inventory	CNR.NEW	Dialog	DC
Edit batch inventory	CNR.SELECT	Dialog	DC
Edit batch inventory	CNR.UNLOCK	Dialog	DC
Edit batch inventory	CNR.UPDATE	Dialog	DC
Create/change batches	CNR.MODIFY	Dialog	DC
Batch Logs	CNRPROT.LIST	Dialog	DC
Change batch status	C_STA	Dialog	DC
List batches with assignment to transport unit	LIST;u_l_mpl_hu_elose	Dialog	DC
Create batch assignment	CNRBAUM.INSERT	Dialog	DC
Material/batch list	LIST;13	Dialog	DC
Create material movements	MBEW.INSERT	Dialog	DC
List material movements	MaterialBooking.list (*SP16)	Service	DC
List material buffers	LIST;49	Dialog	DC
List material buffers	MaterialBuffer.list (*SP16)	Service	DC
Edit material buffers	MATPUF.COPY	Dialog	DC
Edit material buffers	MATPUF.DELETE	Dialog	DC
Edit material buffers	MATPUF.INSERT	Dialog	DC
Edit material buffers	MATPUF.LOCK	Dialog	DC
Edit material buffers	MATPUF.NEW	Dialog	DC
Edit material buffers	MATPUF.SELECT	Dialog	DC
Edit material buffers	MATPUF.UNLOCK	Dialog	DC
Edit material buffers	MATPUF.UPDATE	Dialog	DC
List material types	MaterialType.list (*SP16)	Service	DC
List material types	LIST;21	Dialog	DC
Edit material types	MATTYP.COPY	Dialog	DC
Edit material types	MATTYP.DELETE	Dialog	DC
Edit material types	MATTYP.INSERT	Dialog	DC
Edit material types	MATTYP.LOCK	Dialog	DC
Edit material types	MATTYP.NEW	Dialog	DC
Edit material types	MATTYP.SELECT	Dialog	DC
Edit material types	MATTYP.UNLOCK	Dialog	DC
Edit material types	MATTYP.UPDATE	Dialog	DC
Quantity change, not product-related	ACCMNT.CHANGE	Dialog	DC
Request new batch number	CNR.NEWCNR	Dialog	DC
Create new batch	CNRGEN.CREATENR	Dialog	DC
Packing/palleting: assign batches	CE_AN_PA	Dialog	DC

Description	Name	Type	Cat.
Packing/palleting: Delete batch assignment	CE_DEL_PA	Dialog	DC
Packing/palleting: complete TPU	CA_WL_PA	Dialog	DC
Create transport order	TRANR.CREATE	Dialog	DC
Log on transport order	TRANR.START	Dialog	DC
Finish transport order	TRANR.END	Dialog	DC
Reserve transport order	TRANR.RESERVE	Dialog	DC
List transport units	LIST;52	Dialog	DC
Edit transport units	TPE.COPY	Dialog	DC
Edit transport units	TPE.DELETE	Dialog	DC
Edit transport units	TPE.INSERT	Dialog	DC
Edit transport units	TPE.LOCK	Dialog	DC
Edit transport units	TPE.NEW	Dialog	DC
Edit transport units	TPE.SELECT	Dialog	DC
Edit transport units	TPE.UNLOCK	Dialog	DC
Edit transport units	TPE.UPDATE	Dialog	DC
Consumption posting	A_VERB	Dialog	DC
Goods movement	C_MBEW	Dialog	DC
Goods receipt batch	C_GEN	Dialog	DC

(*SP16) The service is available as of service pack 16.

5.2.10 PDV services and dialogs

Description	Name	Type	Cat.
Operation logon periods: identify statistical basic parameters	ProcessDataStatistics.aggregateByOperation	Service	DC
Operation logon periods: transfer from MF data	AnalysisOperationLogonPeriods.importFromManufacturin g	Service	DC
List logical channels	LOGCHAN.LIST	Dialog	DC
Edit logical channels	LOGCHAN.COPY	Dialog	DC
Edit logical channels	LOGCHAN.DELETE	Dialog	DC
Edit logical channels	LOGCHAN.INSERT	Dialog	DC
Edit logical channels	LOGCHAN.LOCK	Dialog	DC
Edit logical channels	LOGCHAN.SELECT	Dialog	DC
Edit logical channels	LOGCHAN.UNLOCK	Dialog	DC
Edit logical channels	LOGCHAN.UPDATE	Dialog	DC
Edit characteristic attribute	PAUMMAUSP.COPY	Dialog	DC

Description	Name	Type	Cat.
Edit characteristic attribute	PAUMMAUSP.DELETE	Dialog	DC
Edit characteristic attribute	PAUMMAUSP.INSERT	Dialog	DC
Edit characteristic attribute	PAUMMAUSP.LOCK	Dialog	DC
Edit characteristic attribute	PAUMMAUSP.NEW	Dialog	DC
Edit characteristic attribute	PAUMMAUSP.SELECT	Dialog	DC
Edit characteristic attribute	PAUMMAUSP.UNLOCK	Dialog	DC
Edit characteristic attribute	PAUMMAUSP.UPDATE	Dialog	DC
List characteristic attributes	PAUMMAUSP.LIST	Dialog	DC
Edit PDV event	PDVEVENTCFG.COPY	Dialog	DC
Edit PDV event	PDVEVENTCFG.DELETE	Dialog	DC
Edit PDV event	PDVEVENTCFG.INSERT	Dialog	DC
Edit PDV event	PDVEVENTCFG.LOCK	Dialog	DC
Edit PDV event	PDVEVENTCFG.SELECT	Dialog	DC
Edit PDV event	PDVEVENTCFG.UNLOCK	Dialog	DC
Edit PDV event	PDVEVENTCFG.UPDATE	Dialog	DC
List PDV events	PDVEVENTCFG.LIST	Dialog	DC
Process data specification: record change	ProcessDataSpecifications.insert	Service	DC
Process data specification: transfer change to target structures	ProcessDataSpecifications.distributeStandard	Service	DC
Process data specification: aggregate change to process data specifications	ProcessDataSpecifications.aggregatePeriodicStandard	Service	DC
Record process data value	ProcessValues.insert	Service	DC
Collect process data value (batch mode)	ProcessValues.batchInsert (*SP16)	Service	DC
Record process data value from PCC files	ProcessValues.importFromPccFiles	Service	DC
Transfer process data values in target structures	ProcessValues.distributeStandard	Service	DC
Process data values: aggregate curve progressions	ProcessValues.aggregateCurveProgression	Service	DC

(*SP16) The service is available as of service pack 16.

5.2.11 WRM services and dialogs

Description	Name	Type	Cat.
List resources logged on	LIST;129	Dialog	DC
Edit code group	HSPMGRP.COPY	Dialog	DC
Edit code group	HSPMGRP.DELETE	Dialog	DC
Edit code group	HSPMGRP.INSERT	Dialog	DC

Description	Name	Type	Cat.
Edit code group	HSPMGRP.LOCK	Dialog	DC
Edit code group	HSPMGRP.NEW	Dialog	DC
Edit code group	HSPMGRP.SELECT	Dialog	DC
Edit code group	HSPMGRP.UNLOCK	Dialog	DC
Edit code group	HSPMGRP.UPDATE	Dialog	DC
Download DNC resource to machine	RES_DOWNL	Dialog	DC
Upload DNC resource from machine	RES_UPLOAD	Dialog	DC
DNC versions	DNCVersions.list	Service	SM
Edit failure codes	HSPMCODE.COPY	Dialog	DC
Edit failure codes	HSPMCODE.DELETE	Dialog	DC
Edit failure codes	HSPMCODE.INSERT	Dialog	DC
Edit failure codes	HSPMCODE.LOCK	Dialog	DC
Edit failure codes	HSPMCODE.NEW	Dialog	DC
Edit failure codes	HSPMCODE.SELECT	Dialog	DC
Edit failure codes	HSPMCODE.UNLOCK	Dialog	DC
Edit failure codes	HSPMCODE.UPDATE	Dialog	DC
List free resource attributes	RESATTR.LIST	Dialog	DC
Edit free resource attributes	RESATTR.COPY	Dialog	DC
Edit free resource attributes	RESATTR.DELETE	Dialog	DC
Edit free resource attributes	RESATTR.INSERT	Dialog	DC
Edit free resource attributes	RESATTR.LOCK	Dialog	DC
Edit free resource attributes	RESATTR.NEW	Dialog	DC
Edit free resource attributes	RESATTR.SELECT	Dialog	DC
Edit free resource attributes	RESATTR.UNLOCK	Dialog	DC
Edit free resource attributes	RESATTR.UPDATE	Dialog	DC
Maintenance requirement	M_IH_ANF	Dialog	DC
Complete maintenance posting	M_IH_AB	Dialog	DC
List maintenance postings	LIST;53	Dialog	DC
Edit catalogs	HSPMCAT.COPY	Dialog	DC
Edit catalogs	HSPMCAT.DELETE	Dialog	DC
Edit catalogs	HSPMCAT.INSERT	Dialog	DC
Edit catalogs	HSPMCAT.LOCK	Dialog	DC
Edit catalogs	HSPMCAT.NEW	Dialog	DC
Edit catalogs	HSPMCAT.SELECT	Dialog	DC
Edit catalogs	HSPMCAT.UNLOCK	Dialog	DC
Edit catalogs	HSPMCAT.UPDATE	Dialog	DC

Description	Name	Type	Cat.
List comments on a resource	LIST;133	Dialog	DC
List downloadable DNC programs	LIST;83	Dialog	DC
Machines – List DNC family	LIST;82	Dialog	DC
Activate measure	RES_MASS	Dialog	DC
List measures for resources	LIST;117	Dialog	DC
Material posted to finished maintenance posting	M_IH_M_AAB	Dialog	DC
Material posted to maintenance order	M_IH_M_ANR	Dialog	DC
Material posted to maintenance posting	M_IH_M	Dialog	DC
List materials/batches and resources in combined form	LIST;132	Dialog	DC
Object list for machine and catalogs	LIST;97	Dialog	DC
Log resource off	RES_AB	Dialog	DC
Log resource on	RES_AN	Dialog	DC
Remove resource	RES_AUS	Dialog	DC
Integrate resource	RES_EIN	Dialog	DC
Release resource	RES_FREI	Dialog	DC
Repost resource	RES_UMB	Dialog	DC
List resource families	MDResourceFamily.list (*SP16)	Dialog	DC
List resource families (outdated! Use service!)	LIST;119	Dialog	DC
List resources (machines, tools, etc.)	BOResource.list	Service	DC
List resources	LIST;115	Dialog	DC
List resources	RES.LIST	Dialog	DC
Edit resources	RES.COPY	Dialog	DC
Edit resources	RES.DELETE	Dialog	DC
Edit resources	RES.INSERT	Dialog	DC
Edit resources	RES.LOCK	Dialog	DC
Edit resources	RES.NEW	Dialog	DC
Edit resources	RES.SELECT	Dialog	DC
Edit resources	RES.UNLOCK	Dialog	DC
Edit resources	RES.UPDATE	Dialog	DC
List resource types	MDResourceType.list (*SP16)	Service	DC
Edit assignment of required resources	RESBEDRES.COPY	Dialog	DC
Edit assignment of required resources	RESBEDRES.DELETE	Dialog	DC
Edit assignment of required resources	RESBEDRES.INSERT	Dialog	DC
List resource status	LIST;116	Dialog	DC
Set resource status	RES_STATUS	Dialog	DC
Set resource status after logging off OP	RES_ABSTA	Dialog	DC

Description	Name	Type	Cat.
Manage resource statuses	ResourceStatusAssignment.delete	Service	DC
Manage resource statuses	ResourceStatusAssignment.insert	Service	DC
Manage resource statuses	ResourceStatusAssignment.list	Service	DC
Manage resource statuses	ResourceStatusAssignment.lock	Service	DC
Manage resource statuses	ResourceStatusAssignment.unlock	Service	DC
Manage resource statuses	ResourceStatusAssignment.update	Service	DC
Resource list	RESLIST.LIST	Dialog	DC
Resource list including recursive tree view	ResourceList.list	Service	SM
Edit resource list	RESLIST.DELETE	Dialog	DC
Edit resource list	RESLIST.INSERT	Dialog	DC
Manage resource list	ResourceList.delete	Service	DC
Manage resource list	ResourceList.insert	Service	DC
Manage resource list	ResourceList.lock	Service	DC
Manage resource list	ResourceList.unlock	Service	DC
Manage resource list	ResourceList.update	Service	DC
List resource types	LIST;118	Dialog	DC
List resource maintenances	LIST;120	Dialog	DC
Hours posted to finished maintenance posting	M_IH_H_AAB	Dialog	DC
Hours posted to maintenance order	M_IH_H_ANR	Dialog	DC
Hours posted to maintenance posting	M_IH_H	Dialog	DC
Manage maintenances	BOPMaintenance.delete	Service	DC
Manage maintenances	BOPMaintenance.insert	Service	DC
Manage maintenances	BOPMaintenance.list	Service	DC
Manage maintenances	BOPMaintenance.lock	Service	DC
Manage maintenances	BOPMaintenance.reset	Service	DC
Manage maintenances	BOPMaintenance.unlock	Service	DC
Manage maintenances	BOPMaintenance.update	Service	DC
Manage maintenances (available as of SP 16)	RESWART.INSERT	Dialog	DC
Manage maintenances (available as of SP 16)	RESWART.UPDATE	Dialog	DC
Manage maintenances (available as of SP 16)	RESWART.DELETE	Dialog	DC
Manage maintenances (available as of SP 16)	RESWART.LIST	Dialog	DC
Manage maintenances (available as of SP 16)	RESWART.SELECT	Dialog	DC
Manage maintenances (available as of SP 16)	RESWART.LOCK	Dialog	DC
Manage maintenances (available as of SP 16)	RESWART.UNLOCK	Dialog	DC
List maintenance activities	LIST;91	Dialog	DC
Maintenance status and activations	RES_WART	Dialog	DC

(*SP16) The service is available as of service pack 16.

6 Repository Client

You use the MPDV Repository Client MRC to display and edit repository data. It provides a user-friendly access.

6.1 Quick start

This section provides a quick overview of how to work with the Repository Client. The individual steps are only briefly described. For further information on the individual steps, refer to the sections in the following, if required.

Installation

Requirements

To use the Repository Client, you must have installed the Microsoft DotNet framework (at least version 4.5.2).

Program installation

To install the program, just copy the folder including the binary files into your system. An installation program is not required.

Installation of developer license

If the developer license is not available, you can only read the data. You cannot save or export the data.

The developer license is handed out once you have attended a respective Customizing Training. The developer license is provided as *.lic file. This file is included in the data medium that you have received during the training: Folder "Repository Client", subfolder "`tools/licence`", e.g.
`x:\CUT-MOC_81_files\Tools\MPDVRepositoryClient\tools\licence\mpdvWrite.lic`.

Copy the folder "`licence`" with its content into the folder of the Repository Client in the roaming directory of the Windows user, e.g.:
`C:\Users\%User%\AppData\Roaming\MPDV\RepositoryClient\licence\mpdvWrite.lic`

This folder is automatically created on the first start of the Repository Client.

Before you start

Before you start working with the Repository Client, you must make sure that the Repository Client has been installed according to the installation instructions and that the required license files are stored as described there.

In general, the repository is empty when you start work. However, if you do not want to start with an empty repository, it is recommended to make sure that the data you want to work with is available.

First steps

Start the Repository Client.

➔ Start .\bin\mrc.exe

Now load or create a Workset.

A workset specifies the sources of the repository that you want to edit.

➔ Click the button *Load work set* in the file-based repository
-> select workingset.work

Click the button *Repository* → *Load Repository* to load the data from the sources specified in the workset.

How to proceed further depends on what you want to do via the Repository Client.

Tip: Working with perspectives

The Repository Client provides the possibility to use different perspectives for different tasks. Do not hesitate to close views you do not need or drag them to another open view in order to tab them and hence provide for more space and clarity. You can also use more than one view of the same type. Simply adapt your perspective to the requirements of your tasks.

Example:
If you create a service and you would like to check with an existing service how to populate the fields, you can simply open another service view. Thus, you do not need to destroy your current view and re-orient yourself later.

Default perspectives

The installation of the Repository Client provides default perspectives:

default

This is a good perspective to start with. It provides a combined view for server and client-related contents.

Select a domain on the top left. Via the included relations, the top right area shows the services, servicesGUI and properties of this domain. The bottom right area shows the ServiceParameters of the service selected above, the ServiceParameterGui of the ServiceGui selected above and the ControlDataSources, ReferenceData and Authorizations of the selected domain.

default client

Similar to the "default" perspective but limited to the contents concerning client development.

default server

Similar to the "default" perspective but limited to the contents concerning server development.

DB schema

Shows documentation of the database structure.

Validation

Use this perspective to validate contents.

Proceed as follows:

- Open perspective and load data from workset.
- Perform validation: tab *Repository* --> button *Validate*.
- A CSV file listing the identified irregularities is generated in the sub-directory "validation_logs" of the Repository Client's installation directory. The application that is linked to this file type in the operating system opens the CSV file.
- In the views for Domains, Properties, ServiceParameters, etc. the Repository Client only shows the entries with detected irregularities.

You should analyze and, if necessary, correct the detected irregularities. Not every irregularity leads to an error.

6.2 Start and exit Repository Client

Start the Repository Client via the Windows start menu, a link on the desktop or the command line. As soon as all required components have been loaded, the application window is shown.

You can start and run the Repository Client multiple times in parallel on a PC. You can access different repository data in each of the started instances of the Repository Client. For example, you can view several versions of the repository at the same time.

You can also start the client by opening one of the workset files. On start of the client, the system attempts to load the contents of the workset defined in this file. This option is available after the first start of the Repository Client.

Command line parameters

You may transfer parameters to the Repository Client upon the start. The following parameters are supported:

`-perspective/-p <perspectivename>` Use this parameter to start the Repository Client with a specific perspective. If you do not use this parameter, the last active perspective is started by default.

`-workset/-w <worksetfile>` Use this parameter to specify the workset to be loaded. If you do not specify the workset on start of the application, the last loaded workset is loaded.

`--autoload/--a` If you use this parameter, the Repository Client loads the repository defined in the last active workset or in the workset transferred via parameters. This repository is loaded directly on start of the application.

`--trim/--t` If you use this parameter, the Repository Client removes so-called leading and trailing "whitespaces" when loading the repository. Only select this option if needed, because the load time increases extremely.

Exit

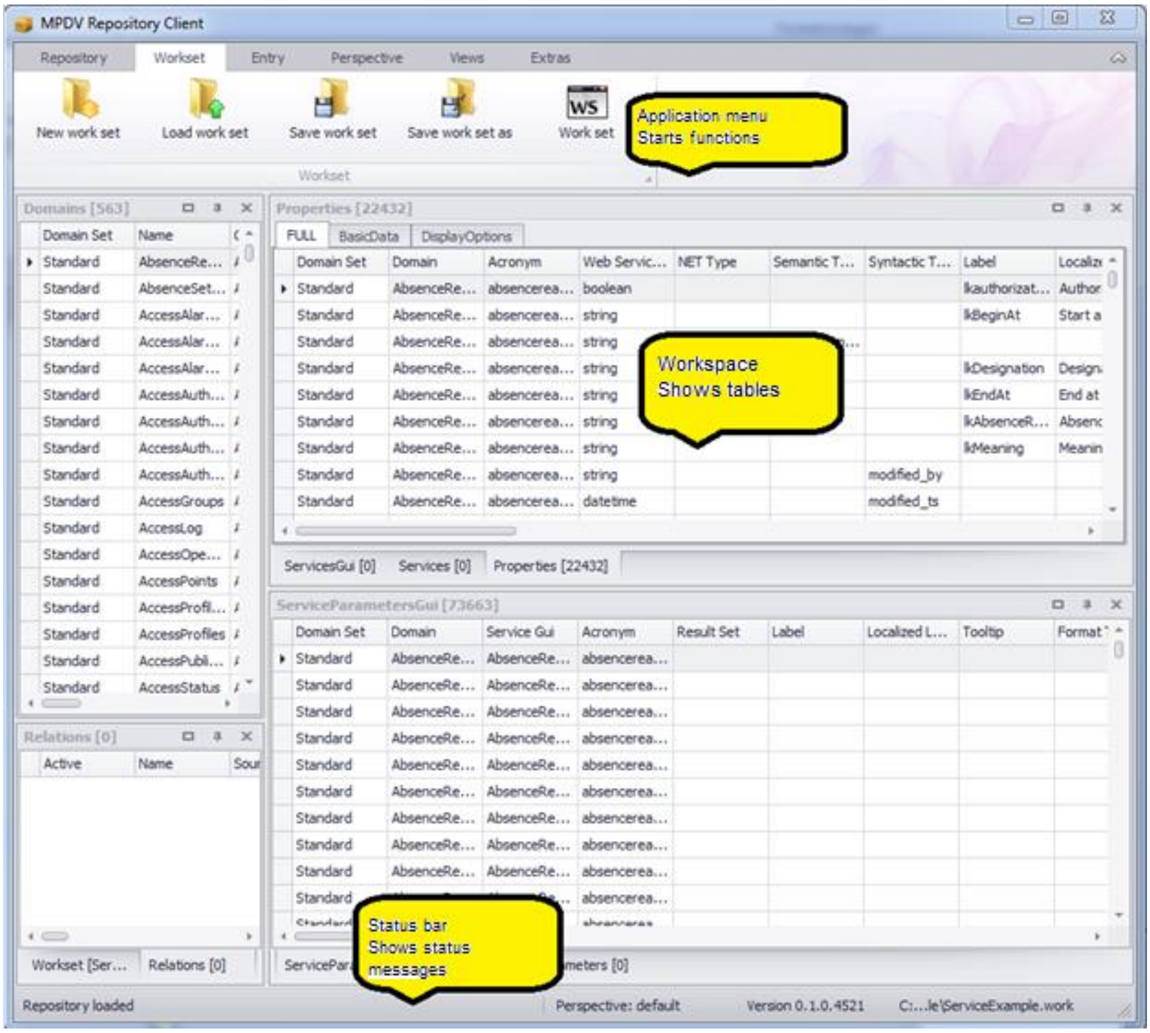
To exit the application, click in the title bar.

If the loaded repository includes active changes when you exit the application, a respective message is issued asking you to save the changes. If you do not want to save the changes, you can discard the changes or stop exiting the application.

Note: Changes to the workset and perspective are discarded when you exit the application, if you have not saved the changes.

6.3 The Application Window

The application window forms a framework for the display of different tables. It includes the application menu with control elements to call and control different functionalities. The menu is on top of the window. A status bar is at the bottom of the window. The status bar shows progress and event messages.



You can individually dock the grids/table views. To do so, click the title bar of a table view/grid and drag it out of the docking position. For orientation purposes, the system shows the docking positions where you can drop the table view. You can also drop a table view without docking it.

6.4 Grids/table views

Grids/table views are components to present data records in a table. You can change the tables in the Repository Client according to your requirements. For each grid/table view, the functions described below are available.

The settings, that you make in a table, are saved with the perspective. To undo changes, you can switch to the standard perspective (in the application menu: *Perspective* → *Change perspective*).

Sort table data

Click the table header to sort table data in descending order. If you click the table header once more, data is sorted in ascending order. The selected sorting option is shown.

You can sort data by several columns: Press the *Shift* key of your keyboard after sorting the first column. Then click the other column headings by which you want to sort.

You can also use the context menu of the table header to sort data.

Group data in the table

You can group table data if the *group by* area is shown. If the *group by* area is not shown, you can show the area via the context menu of the table header (*Show/Hide group by box*). To group by a column, click the column header and drag it to the grouping pane. Multiple grouping is also supported.

Optimum column width (best fit)

Select the option *"Best fit"* in the context menu of the table header to adjust the column width of the selected column to the optimum width. In this case, "optimum" means that the column is as wide as the largest entry in the selected column.

Optimum column width (all columns) / Best fit (all columns)

Click this function to adjust all columns to the optimum width.

Change column width

You can also change the column width using the mouse, i.e. move the space between two cells to the left or right.

Show and/or hide columns and entire categories

Use the context menu function *Select columns* to show and/or hide individual columns and entire categories. For this purpose, select the function in the context menu and then drag the required columns and/or categories from the table to the pool or from the pool to the table.

Change the sequence of columns and categories

Also use the mouse to change the display order of columns and categories. To do so, drag the column or category you want to move and drop it at the required location. The system will indicate the location to which the column and/or category will be allocated when you release the left mouse button.

Freezing columns to prevent horizontal scrolling

You can freeze columns at the left and right-hand side to keep the columns in view while scrolling. These column settings are included in the perspective and can be saved with the perspective. Right-click the column header and press one of the below-mentioned shortcuts to freeze columns:

- CTRL + right click: freeze at the left-hand side.
- ALT + right click: freeze at the right-hand side.
- SHIFT + right click: Unfreeze.

Filter table data

Click the filter icon in the column where you intend to set the filter and select the required filter option from the list; i.e. you select one of the values available in the table or compose a combination of values in the *user-defined* filter.

You can also set several filters in different columns. The table footer indicates that the table has been filtered and also shows the filter criteria. Select the function *Edit filter* on the right of the footer to open the filter editor. Use the filter editor to create complex filter criteria across all columns. You may also open the filter editor via the context menu of the table header.

Search box

Use the context menu of the table header to access the option *Show search box.* This option provides a search box within the table. Use this box to quickly search and/or filter the requested data. Simply start typing in this box and the system will only show those rows matching the data you typed. The more characters you enter, the more you narrow down the result.

Filter row

Use the context menu of the table header to open the option *Show filter row*. This option provides an additional row shown below the table header. You can enter a search term in any column, and the system will narrow down the displayed rows appropriately. The system supports wildcards. You can also combine search terms in various columns to restrict the search result.

Edit rows

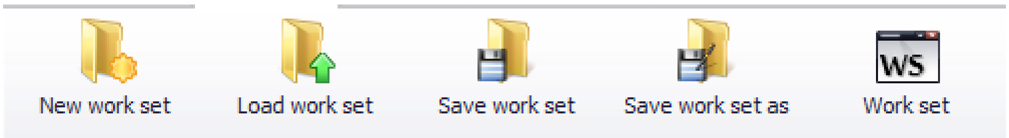
Double-click a row or press the "Enter" button to switch to the editing mode and edit the respective row. When you have finished editing and leave the row, the editing mode is terminated. Edited rows are highlighted in color.

6.5 The application menu

You can use the application ribbon menu of the Repository Client to control various functions of the tool. It includes several tabs that are described in the following.

Workset

Includes functions to administer worksets. A workset specifies the sources included in the repository that you want to edit. To display a workset, use the workset panel which consists of a grid/table view.



- **New workset**: This function creates a new workset. If a workset is loaded that has been modified, a dialog pops up asking you to save the changes.
  Click "Yes" to save the changes, "No" will discard them. In both cases, a new workset is created. Click "Cancel" to cancel the process of creating a new workset.
- **Load workset**: This functions loads a workset from an existing file. You can select the workset to be loaded via a file dialog. If the currently loaded workset has been modified, you can save the changes as described above.
- **Save workset**: This function saves the current workset in the file from which it was loaded. If the current workset is new, a file dialog pops up asking you to select the file in which you want to save the workset.
- **Save workset in**: Use this function to save the currently loaded workset in a file. You can select the files using a file dialog.
- **Workset**: Use this button to show and/or hide the grid/table view presenting the currently loaded workset. For details on the workset table, please refer to section Workset.

Repository

Includes functions to load and save data in the repository. In addition, you may display repository-specific data here.

- **Load repository**: Use this function to load the repository. The currently loaded workset specifies the data sources that are used to load the repository. If a repository is loaded that has been modified, a dialog pops up asking you to save the changes. Click "Yes" to save the changes, "No" will discard them. In both cases, the repository will be reloaded subsequently. Click "Cancel" to cancel the process of loading a repository.
- **Save repository: *only available if used in development mode**
  Use this function to save changes in the repository. If no changes have been made, an appropriate note will be displayed.
- **Export repository**: ***only available if used in development mode**
  In contrast to saving the repository, you can use this function to export parts of the repository. For details on this function, please refer to section **Error! Reference source not found.**.
- **Validate**: ***only available if used in development mode**
  You can use this function to validate your data records manually. (See section "Validation").
- **Value list**: Use this menu entry to show and/or hide the value list. The list includes permissible entries for specific fields of the repository.
- **References**: Use this entry to show and/or hide the table with repository references. For details on references, please refer to section References.

- **Changes**: ***only available if used in development mode**
  Use this button to show and/or hide the change view. This view shows the current modifications in the loaded repository.
- **Service documentation**
  Use this button to show the extended documentation of selected standard services. For further information on the service documentation, refer to section "6.9 Service documentation".

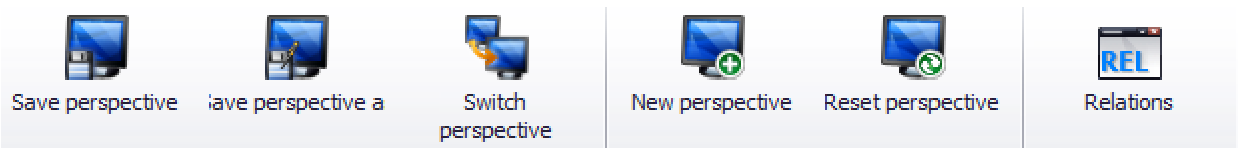
Data collection

The *Entry* tab summarizes the functions that you can use to edit the loaded repository. The entries refer to the currently focused table view/grid.

- **New entry**: Use this function to create a new entry. For details on this function, please refer to *Context menu* → *New*.
- **Copy entry**: Use this function to copy selected table entries. For details on this function, please refer to *Context menu* → *Copy*.
- **Cut entry**: Use this function to cut selected table entries. For details on this function, please refer to *Context menu* → *Cut*.
- **Paste entry**: Use this function to insert (paste) entries from the cache/clipboard. For details on this function, please refer to *Context menu* → *Insert*.
- **Advanced pasting**: Use this function to edit entries in the clipboard prior to inserting them. For details on this function, please refer to *Context menu* → *Advanced pasting*.
- **Delete entry**: Use this function to delete the selected entries. For details on this function, please refer to *Context menu* → *Delete*.
- **Show entry info**: Use this function to open a dialog showing information on the currently selected entry. For details on this function, please refer to *Context menu* → *Info*.
- **Get references**: Use this function to open a new grid/table view showing the currently selected data record including referenced values. For details on this function, please refer to *Context menu* → *Get references*.

Perspective

These entries of the menu refer to the administration of perspectives. A perspective is a layout of table views/grids and includes also the associated relations between table views/grids.



- **Save perspective**: Use this function to save the currently shown arrangement of tables.
- **Save perspective as**: Use this function to save the currently shown perspective under a different name. You can enter the new name in the displayed dialog.
- **Switch perspective**: Use this function to change the perspective. A dialog with all available perspectives is shown.
- **New perspective**: Use this function to create a new perspective. Similar to the "Save perspective as" function, you can select the name of the perspective in a dialog. After entering the name, you can immediately switch to the new perspective.
- **Reset perspective**: Use this function to reset the current perspective to the status saved last.
- **Relations**: Use this menu entry to show/hide the grid/table view showing the relations between the grids/table views. For details on relations, please refer to section Relations.

Note: Changes to the perspective are discarded when you exit the application, if you have not saved the changes explicitly.

Views

Use these entries to open table views/grids. The entries will open a new grid/table view each showing the relevant data records of the repository.

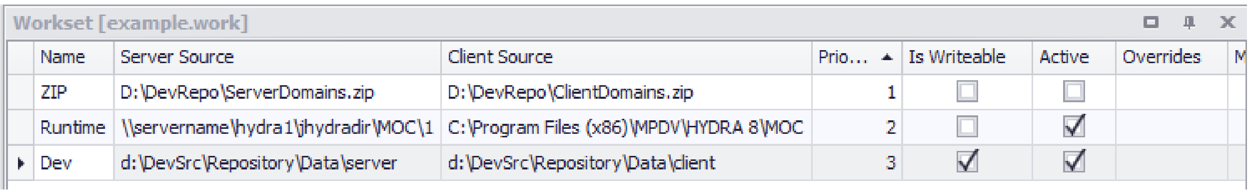
For clear identification of the data records shown in the tables, the *Parent* column is included in each of the tables. The *Parent* column includes the identifier for the father node in the repository tree. The other columns of these tables are defined by the repository documentation.

The *View* area additionally includes a group with entries for the remaining grids/table views of the application.

6.6 Workset

Worksets define a set of data sources that make up the repository that you want to edit. Use the workset management function to organize your work on different projects and create an appropriate workset for each of your projects. You can show/hide the workset table via the application menu (*Workset* → *Workset*).

Note: The workset loaded last will be loaded on start of the Repository Client.


Workset [example.work]

Name	Server Source	Client Source	Prio...	Is Writeable	Active	Overrides	M
ZIP	D:\DevRepo\ServerDomains.zip	D:\DevRepo\ClientDomains.zip	1	☐	☐		
Runtime	\\servername\hydra1\jhydradir\MOC\1	C:\Program Files (x86)\MPDV\HYDRA 8\MOC	2	☐	☑		
Dev	d:\DevSrc\Repository\Data\server	d:\DevSrc\Repository\Data\client	3	☑	☑		

The workset table includes the following columns:

Name

You can specify the domain set that you want to use to load the data source. The repository data include the information on the domain set that was used to load the data. You can copy data from a domain set into another domain set. Example: Copy existing applications from the domain set "Runtime" (read only) into your development directory, e.g. the domain set "Dev" (writable) where you can make changes.

Client Source

You can specify the data source that you want to use to load client data. The following options are available:

- Load data from the runtime structure of the client reference in the server
  In the server, the client configurations are stored as client reference with runtime structure. You can load the repository data from this structure.
  Example:
  HYDRA: x:\jhydradir\MaintenanceManager\rt\client\MOC
  MIP: x:\wsp_config\MaintenanceManager\rt\client\MOC
  The access is read-only.
- Load data from the runtime structure of a local MOC client
  If you enter a path to an MOC installation directory, the respective client data are directly loaded from the MOC runtime installation.
  Example:
  C:\Program Files (x86)\MPDV\HYDRA 8\MOC
  The access is read-only.
- Load data from a local directory with domain structure
  You use local directories with domain structure for the administration of your own developments. Using the Repository Client, you can read data in this directory and you can also save data into this directory.
  Example: d:\DevSrc\Repository\Data\client
- Load data from a ZIP archive
  You can enter a ZIP file (including path) that includes the data in domain structure. MPDV provides the ZIP archives as part of the trainings or on the support portal.
  The access is read-only.

Server Source

You can specify the server source that you want to use to load the server-specific data. The following options are available:

- Load data from the runtime structure in the server
  You can read the configurations for the server in a server installation. To this end, the configuration directory of the web service provider (WSP) up to the instance number is specified.

  Example:

  HYDRA: \\<servername>\<install_dir>\jhydradir\MOC\1

  MIP: \\<servername>\<install_dir>\jdir\MOC\1

  The configuration is loaded from the standard scope by default As an alternative, you can also load the configuration from the *custom* scope, if you enter the value "custom" in the field "name".

  The access is read-only.
- Load data from a local directory with domain structure
  You use local directories with domain structure for the administration of your own developments. Using the Repository Client, you can read data in this directory and you can also save data into this directory.
  Example: d:\DevSrc\Repository\Data\server
- Load data from a ZIP archive
  You can enter a ZIP file (including path) that includes the data in domain structure. MPDV provides the ZIP archives as part of the trainings or on the support portal.
  The access is read-only.

Priority

The priority of a data record specifies the loading sequence of sources that are allocated to the same data record. In the above example, data is first read from the local development directory "d:\DevSrc\Repository" and then from the runtime installations "Server" and "Client". Data records with low priority are overridden and consequently not loaded.

Is Writeable

In this column you can specify if a data source grants write access. You only require write access in workset entries where you want to make local developments.

Please note that ZIP archives do not grant write access. For this reasons, you must not enable "is Writeable" in case of a ZIP data source.

Please note that it is not supported to save data in the runtime structure (client directory or server runtime directory). You must therefore not enable "Is Writable" for data sources with runtime structure.

Overrides

You can specify in this column which domain set is overridden by the current one. This affects the resolving of references (for details please refer to section References).

An entry in the "Overrides" column does not have any effect on the loading of the data sources. See column "Priority".

Active

Use this option to enable or disable an entry.

6.7 Relations

Relations are a property of perspectives and define table filters. Tables that include relations are dynamically adapted to the selected values of another table by setting a filter. For example: Using relations, you can specify that only the service parameters of the service currently selected in another service view are displayed in a service parameter view. The *Relations* table lists the relations of the current perspective. You can call this table via the application menu (*Perspective* →*Relations*).

Relations [13]

Active	Name	Source	Target	Filter	Var0	Var1	Var2
☑	Authorizations for Domain	Domains	Authorizations	[Parent] = '$id->Authorizations'			
☑	ControlDataSources for Domain	Domains	ControlDataSources	[Parent] = '$id->ControlDataSources'			
☑	DataObject for Service	Services	Dataobjects	[DomainSet] = '$0' and [Domain] = '$1' AND [Name] = '$2'	DomainSet	Domain	Name
☑	Properties for Domain	Domains	Properties	[Parent] = '$id->Properties'			
☑	ReferenceData for Domain	Domains	ReferenceData	[Parent] = '$id->ReferenceData'			
☑	SchemaColumns for ServiceParameter	ServiceParameters	SchemaColumns	([TableName] = '$0' AND [ColumnName] = '$1' AND [TableNam…	DBTabelle	DBField	HydraAcronym
☐	ServiceParameter for ServiceParameterGui	ServiceParametersGui	ServiceParameters	[Domain] = '$0' AND [Service] = '$1' AND [Acronym] = '$2'	Domain	ServiceGui	Acronym
☑	ServiceParameters for Service	Services	ServiceParameters	[Parent] = '$id'			
☑	ServiceParametersGui for Service	Services	ServiceParametersGui	[DomainSet] = '$0' and [Domain] = '$1' AND [ServiceGui] = '$2'	DomainSet	Domain	Name
☐	ServiceParametersGui for ServiceGui	ServicesGui	ServiceParametersGui	[Parent] = '$id'			
☑	Services for Domain	Domains	Services	[Parent] = '$id->Services'			
☐	Service forServiceGui	ServicesGui	Services	[Name] = '$0'	Name		
☑	ServicesGui for Domain	Domains	ServicesGui	[Parent] = '$id->ServicesGui'			

The following columns are displayed:

Active

This checkbox specifies if the relation is used.

Name

The name of the relation – free choice.

Source

The table and its selection that are used to set the filters. If you edit an entry in this column, the currently possible assignments, i.e. all currently existing views are presented in a selection box.

Target

The table where the filter is applied. If you edit an entry in this column, the currently possible assignments, i.e. all currently existing views are presented in a selection box.

Filter

You can store the filter expression here that you want to apply to the target table. Variables ranging between $0 and $9 are supported. These variables are dynamically filled with the values of the columns Var[0-9].

Var[0-9]

In these columns, you can specify the columns of the source table that are used to adapt the filters dynamically.

As soon as a correct (and activated) relation is entered in this table, it is applied. If you close one of the referenced views of a relation, the relevant view is removed from the relation. If this results in a double entry in the *Relations* table, this entry is removed. You can therefore use this view to administer the relations between concrete table instances and to administer unbound relations that can serve as template for relations.

6.8 References

References show the inherent connections between data records of the repository. They are defined by the repository structure and cannot be edited in the Repository Client. For example: A value in the column "Syntactic Type" of the *Property* table references another data record in the *Property* table. The *References* table lists the defined references and may be activated via the application menu (*Repository* → *References*).

References [6]

Name	Source	Source Column	Dependency	Condition	Target	Filter	Priority
ControlDataSource1	Property	ControlDataSource	ControlDataSourceMode	Lookup	ControlDataSource	[Name] = $value and [Domainset] = $domset	-1
ControlDataSource2	Property	ControlDataSource	ControlDataSourceMode	Reference	ReferenceData	[type] = $value and [Domainset] = $domset	-1
SyntacticType	Property	SyntacticType			Property	[Acronym] = $value	0
SemanticType	Property	SemanticType			Property	[Acronym] = $value	1
Acronym1	ServiceParameter	Acronym			Property	[Acronym] = $value	3
Acronym2	GuiServiceParameter	Acronym			Property	[Acronym] = $value	3

The following columns are displayed:

- **Name:** Name of the reference
- **Source**: The repository object type that can include this reference.
- **SourceColumn**: The source type property that can include the reference.
- **Dependency**: Source type property specifying the reference target.

- **Condition:** Value that the property specified under *Dependency* must have. Only then, the reference is pursued. For example: The value of ControlDataSourceMode (lookup, reference) specifies the target of the reference (ControlDataSource or ReferenceData) which is specified in the ControlDataSource property.
- **Target**: The repository object type that is referenced.
- **Filter:** Filter that selects the referenced data from the overall quantity of this type of data. Such a filter can include the variable $value (value of column *Value* in the current row) and $parent (value of column *Parent* in the current row).
- **Priority**: Specifies the priority of the reference. You can find further details in section "Get references".

References provide two general functions:

Show reference: You can use this function to display the referenced data in a new table. You can call this function via the context menu (*Context menu* → *Show reference*).

Note: This function is only available in the context menu of cells which can include references.

Get references: Use this function to complete missing values of a data record with values of referenced data records. For example: You can use this function to show the inherited values of a property of the SemanticType or the SyntacticType.

You can call this function via the context menu (*Context menu* → *Get references*) of the table views/grids. A new data record is generated and shown in a new panel. The generated data record is a copy of the currently selected data record. The values that are not filled are filled by those in the referenced data records. The reference priority specifies the filling sequence.

6.9 Service documentation

The Repository Client contains an extended documentation for selected standard services. The documentation mainly includes services released in the system and to be used with the Service Interface (SCS-SIF).

The documentation is available as of MRC version 1.8.STD.65500 (beginning of 2019).

There are two options to access the service documentation:

- In the toolbar "Repository", you can use the button *Service Documentation* to open the table of contents of the services included. Via hyperlinks, you can navigate to the different services.
- In the table views "Services" and "ServicesGui", you can use the context menu to open the documentation of the selected service if it is available.

Printing a service documentation:

You can print the service documentation using the shortcut Ctrl-P.

7 The Repository

7.1 Overview

The data of the repository is used in multiple ways:

- The repository defines and describes the **interface between client and server**. The input parameters and the result sets of service requests are described.
- In case of interpreted service types, the **processing and the business logic of a service** is specified via configuration in the repository. Only in exceptional cases, an actual programming in the server is required.
- For the client, the repository defines **how the data is displayed on the client** and which GUI elements are used to enter data. The repository also defines how the client checks the user input. You can generate most of the applications on the client using the configurations of the repository. Here, programming on the client is not required.

The repository data is grouped and structured using **domains**. A domain summarizes all data that logically belongs to an application.

The domain contains hierarchically structured and typed data. A domain includes *services* and *service parameters*, the respective *GUI settings*, *properties*, *authorizations*, *ReferenceData* and *ControlDataSources*.

Find below a detailed description of the repository elements.

7.2 Domain

Domains have properties and provide services within the domain context.

A domain is the smallest software unit. You can update the domain using an update package. Create a separate domain for each application. This domain then includes the services implemented for this application. You can also use the services and client attributes of a domain in applications of other domains. For example, a client application in its own domain can use a service of a different domain.

You can assign global contents to a global domain: for example, client menu configurations or separate global syntactic types.

Name

Each domain has a unique name. For the name, you use the notation "UpperCamelCase".

7.3 Service

Services have transfer parameters and return values, which are often identical to the properties of the domains.

7.3.1 Name

Name of a service. The service name usually consists of the domain name that includes the service and the **function**, separated by a dot.

7.3.2 Function

This field describes the requested service function. Typical functions are list, update, insert, delete, new, ...

7.3.3 ServiceType

There are several service types.

InterpretedJavaService2: Services of this type are used to display lists and evaluations. The services are interpreted at runtime using repository data. Contrary to the *InterpretedJavaService,* the services of type *InterpretedJavaService2* are prepared to stream data and provide more elegant options for Java user exits.

InterpretedJavaService *(obsolete)*: Services of this type are interpreted at runtime using repository data. These services have been replaced with the service type *InterpretedJavaService2.*

InterpretedBAPIService: You use services of this type to edit data. The services are interpreted at runtime using repository data.

ExternalJavaService: Services of this type are completely implemented in Java. You can use these services to implement lists or editing functions. You use these services if the possibilities of the interpreting service types are not sufficient and the logic must be converted into Java programming.

InterpretedWrapper: Services of this type are interpreted at runtime using the repository data. The service is implemented as wrapper of an existing PDM dialog and is therefore subject to specific limitations, e.g. it does not support any dynamic Where.

Wrapper *(obsolete)*: Services of this type are programmed and wrap an existing BAPI function. They are therefore subject to specific limitations, e.g. no dynamic Where.

JavaService (obsolete): Services of this type are completely implemented in Java.

Recommendation:

- The type *InterpretedJavaService2* is recommended for services that you use to read data.
- The type *InterpretedBAPIService* is recommended for services that you use to write data.
- If the interpreted service types cannot meet the requirements (or only with great effort) even if they include Java user exits, you should use the services implemented in Java of type *ExternalJavaService*.
- The other service types are older technologies and should not be used for new developments.

7.3.4 ListMode

For services of type *Wrapper* or *InterpretedWrapper*: This column must be populated for each service. The column specifies whether the requested PDM dialog returns a file as result or whether it is only a return string. "Y" => The result is a file, otherwise only a string.

7.3.5 DLG

For all services of **ServiceType** *InterpretedWrapper* or *Wrapper,* you must fill in either **DLG** or **SystemCall.** You fill in DLG, if **ServiceType** is *Wrapper* or *InterpretedWrapper* and if the service requests a PDM dialog with the structure "DLG=<content in this column>|..."

7.3.6 SystemCall

For all services of **ServiceType** *InterpretedWrapper* or *Wrapper*, you must fill in either **DLG** or **SystemCall.** Fill in SystemCall, if the you want to run a program in the server. In the column, the name of the external program is specified. The result is a PDM dialog with the structure: "DLG=SYSTEM.CALL|PROG=<content of this column>|...".

7.4 ServiceGui

The ServiceGui data define the use and the presentation of the services on a client. You can clearly allocate the ServiceGui to a service via their name.

7.4.1 Name

The name of the service for which this data record provides presentation information.

7.4.2 Package

This field is obsolete and must be left empty.

7.4.3 Extended

This field is obsolete and must be left empty.

7.4.4 AdditionalDataLogics

This field is obsolete and must be left empty.

7.4.5 ApplicationID

Application ID used for generating applications in the client. In case of editing applications, the ApplicationID is edited with the main data source of the application that you want to generate.

7.4.6 ApplicationTitle

Language key for the title of the generated application. In case of editing applications, the ApplicationTitle is edited with the main data source of the application that you want to generate.

7.4.7 ApplicationHelpFile

File name of help file (including file extension) of the generated applications. In case of editing applications, the ApplicationHelpFile is edited with the main data source of the application that you want to generate.

The name of the help file should be independent of the technology of a used client. The client should therefore put a prefix in front of the file name. You can then design the help file displayed according to the client's technology.

Example for the client MOC: In ApplicationHelpFile, you enter "Article.pdf". The client MOC then loads the document "MOC_Article.pdf" as online help. The client automatically uses the prefix "MOC_".

7.4.8 ApplicationHelpIndex

Bookmark that is activated when Help is opened. In the main application, it is usually "Overview". You must only edit this bookmark for the main data source of the application that you want to generate.

7.4.9 Description

7.4.9.1 General

Language key for short description of service.

You can show this description on the client when the selection of services is displayed.

7.4.9.2 Processing in the MOC client

The MOC shows the description if you add a data source while configuring an application.

7.5 ServiceParameter

ServiceParameters specify the parameters of a service. They provide information on the data source and value ranges.

The service parameters include selection criteria and the columns of the result set. A service parameter can be a selection criterion or be included in the result set. The attributes described below specify if a service parameter is used as selection criterion and/or is included in the result set.

7.5.1 Acronym

Name of the parameter. The combination of **Acronym** and **ResultSet** must be unique for each service.

7.5.2 ResultSet

If the associated service returns more than one **ResultSet**, a name must be indicated here. This way, you can return results in parallel that have been calculated at the same time but have a different structure. The combination of **Acronym** and **ResultSet** must be unique for each service.

7.5.3 WebServiceType

Data type of the parameter *(decimal, integer, string, boolean, binary, datetime).* This value must be identical to the configured value of the property configuration. **IMPORTANT:** *binary* parameters are not supported by default. You can only use these parameters in user exits.

7.5.4 DefaultValue

Specifies a service default value for a parameter.

7.5.5 IsResult

Specifies whether this service parameter is part of the ResultSet (return value). If you want to use the **DefaultValue**, do not set this field (IsResult).

In case of services ot type InterpretedWrapper, you must only set the column IsResult to "Y" for UPDATE, LOCK, UNLOCK, DELETE, INSERT and COPY, if the BAPI actually returns a value, e.g. a new internal_id when you create new data records.

7.5.6 IsDynamicResult

Required for the generation of the Java function (for dynamic ResultSets, the column number must automatically be extended to the fixed number). Missing columns are added as empty columns (i.e. these columns are not computed).

7.5.7 InputAsArray

The client must transfer values in form of an array. **InputAsArray** is only reasonable in case of a quantity input parameter, i.e. if at least one of the two columns, **IsSpecialParameter** and **IsFilterParameter**, is set and a quantity operator such as BETWEEN or IN is possible.

Specify if a field is an array or not (with filters always *yes* except for Boolean type).
If true and no array or empty, then exception. Is currently only verified in case of mandatory special parameters.

7.5.8 IsSpecialParameter

Specifies whether or not the parameter is a special type controlling the service functionality (i.e. is not a filter parameter). For the **ServiceType** *Wrapper*, this is the only possible parameter type. In case of the **ServiceType** *JavaService*, it represents a special parameter not directly included in the WHERE condition but with different "controlling" effects. If you want to use the *Default Value* on the server side, do not set this field. In addition to the defined special parameters of standard processing, you can also use other special parameters in user exits.

7.5.9 IsFilterParameter

Specifies whether it is a filter parameter. If you want to use the **DefaultValue** on the server side, do not set this field.

7.5.10 IsMandatory

Specifies whether it is a mandatory parameter for the service. If true and parameter is missing, an exception is thrown. Is currently only checked for special parameters.

7.5.11 Can* (filter) operators

This option specifies whether the service supports the relevant filter operator for this parameter. Set the "Can*" fields for filter parameters.

Available operators:

- **CanEqual**
- **CanLike**
- **CanBetween**
- **CanIn**
- **CanNotEqual**
- **CanLt (Can Less Than)**

- **CanLte (Can Less Than or Equal To)**
- **CanGt (Can Greater Than)**
- **CanGte (Can Greater Than or Equal To)**

For technical reasons, each operator has a second operator that you should select is a data record must be selected, if the operator is applicable or if the comparative value is NULL. The operator **CanEqual** will only return a data record in case of equal values, **CanEqualOrNull** in case of equal values or if the data record value is NULL. Accordingly, there are the following operators:

- **CanEqualOrNull**
- **CanLikeOrNull**
- **CanBetweenOrNull**
- **CanInOrNull**
- **CanNotEqualOrNull**
- **CanLtOrNull**
- **CanLteOrNull**
- **CanGtOrNull**
- **CanGteOrNull**

Especially with List Services you should make sure that generally all parameters support all operators in order to achieve the highest possible selectivity. In general, the framework supports this for Java services.

- You may only set **CanIn**, **CanBetween**, **CanBetweenOrNull** and **CanInOrNull,** if **InputAsArray** is also set.
- **CanLike** is only useful if the **WebServiceType** is *string*.
- With WebServiceType boolean, only **CanEqual** is useful.
- With WebServiceType string, all operators are possible.
- With all other types, all operators except for **CanLike** and **CanLikeOrNull** are useful.

Before you set wrappers, you must check which operators are actually supported by the PDM dialog or the system command.

7.5.12 HydraAcronym

With service type *InterpretedWrapper,* the HYDRA acronym is specified.

7.5.13 HydraResultAcronym

If the acronym of the selection criterion is different to the acronym in the result file, you can enter an acronym that is different to the HydraAcronym for the service type InterpretedWrapper and ListMode=Y.

7.5.14 TransferEmptyValuesToHydra

Specifies whether blank values, too, are to be transferred to the server, or whether the ID is simply omitted. "Y" => blank values are transferred, otherwise => ID is completely omitted.

Note: You must set this field for Insert and Update (editing screens). Only then, you can enter blank values and/or overwrite existing values with blank values.

7.5.15 HydraShiftPart

The following components are combined with the **Reference** field: Start of shift date, start of shift time, end of shift, end of shift time stamp, start of shift time stamp. These components are marked as belonging together. The column "HydraShiftPart" can include the following values:

- beginDate
- beginTime
- beginDatetime
- endTime
- endDatetime

Important: The column can only be populated if the parameter is part of a group that includes the following five components: Start of shift date, start of shift time, end of shift, end of shift time stamp, start of shift time stamp. The column must not be populated if it is only a group of three components including date, time and date + time field. In this case, ONLY populate the **Reference** column.

7.5.16 Reference

Is used to generate a DateTime data type from one field each for the date and the time (in seconds after midnight) and to identify the shift parameters.

7.5.17 TransformationType

Use this field to specify transformations for input and result parameters for List Services/wrappers (e.g. convert Bool to J/N and vice-versa or correct filtering for DateTime fields that consist of two fields in the database). For further details on this field, refer to section 7.10.

7.5.18 PlugName

Specifies whether the result parameter for this service is directly derived from the specified DataObject or whether it is added to the DataObject via plug.

Example:

Service A.List uses a plug of service B.List in the service parameter b. Consequently, the following configuration applies to service A.List:

ServiceParameter	DataObjectName	PlugName
a	A.List	
b	A.List	B.List

If the field *PlugName* includes a value, the Interpreter replaces the values of the ServiceParameter with those values of the plugged service when creating the SQL statement.

In the special case where an interpreted List Service does not use an own table but only plugs, and subsequently adds fields via user exit, these fields should state USEREXIT!

If you create new services, it is recommended to avoid plugs and to provide data directly via the DataObject via Join. Dependencies between several services are thus avoided.

7.5.19 DBField

Database field that you use to make a selection. Write the database field in lower case. You can either enter simply the field name or (for complex expressions) the expression with placeholders for the alias (e.g. hydadm.get_datetime(%1$s.bearb_date,%1$s.bearb_time) or {fn substring(%1$s.field,2,1)}).

Proceed as follows for joins to other tables:

Entry: <ALIAS>.<DBfield>

Example:

DB field: STA1.status_bez

Acronym: gage.status.designation

Table: caq_status (STA1)

Conditions: status_typ = ‘PMSTATUS’, status_nr = status

7.5.20 DBAlias

The alias for the table that is used to select the value for the acronym.

7.5.21 DBTabelle

The table that is used to select the value for the acronym.

7.5.22 DBFieldAlternative

If you cannot use the **DBField** because the **ConditionalFieldKey** is not applicable, you use the **DBFieldAlternative**.

You can enter a number, ''null, 'string', {fn ...} or another field / subselect. If it is another field or subselect, you MUST enter %1$s for the alias of the table.

If **DBFieldAlternative** is empty, but you require an alternative field, NULL is selected.

7.5.23 DataObjectName

If a service uses several data sources to identify its data, you can store the data source (= DataObject = DO) that issues the result parameter in this field. For example: A service includes the parameters a, b and c:
- a is computed,
- b is identified using data object (DO) F and
- c is identified using data object (DO) G.
For a: the field is blank. For b: the field contains F. For c: the field contains G. Is used as reference for the ...do.xml configuration.

7.5.24 ConditionalFieldKey

This field specifies if a DB field is only conditionally available. The ConfigurationManager checks the condition for the existence of the field. Enter the feature key of the Configuration Manager **(feature set)** in this repository field to enable the check.

If a parameter is a conditional field and the condition is not fulfilled, the entries for the MOC acronym are removed from the ComplexSelectMap and the SpecialFilterMap.

As a result, the changes in the Special Filter Map via user exits and transformation type are also lost!

7.5.25 Constraints

Constraints are processing parameters that are used for **ServiceType** *InterpretedBAPIService*. Constraints are structured as keys with optional values. The separator between keys is the pipe character (|). You use a semicolon to separate various values. You use the equal sign (=) to separate key and value. The general structure is as follows:

Key1=Value;Value;Value|Key2|Key3=Value|

The following constraints are available:

Constraint Key	**Constraint value(s)**	**Description**
KEY	exactly one number between 1 and 5	Define field as key including key number for hyd_lock table
SERIAL	none	Field is a SERIAL (and/or auto-increment)
SEP_DATETIME	1st parameter refers to the date field 2nd parameter refers to the time field	Allows processing of separate date and time fields
BOOL	1st parameter is the value to be entered into the DB if true 2nd parameter is the value to be entered into the DB if false 3rd parameter is the value to be entered into the DB if null (null for null) 4th parameter is the type of DB field, e.g. BOOL=J;N;null;string| BOOL=1;0;null;integer|	Use this to write Boolean values into a string or Integer Field.
MODIFY_TS	None	
MODIFY_BY	None	
CREATE_TS	None	
CREATE_BY	None	

7.6 ServiceParameterGui

The ServiceParameterGui define how ServiceParameters are displayed on the client. Use **Acronym** and **ResultSet** to clearly allocate ServiceParameterGui to a service parameter.

A number of settings exist in the ServiceParameterGui and in the properties. You normally use the properties to define how data is displayed on the client. You only fill the respective field in the ServiceParameterGui if you want to display specific services on the client in a way that is different to the settings in the properties. The ServiceParameterGui fields overwrite the property fields of the same name.

7.6.1 Acronym

Name of the parameter for which this data record provides presentation information. There must be a corresponding property for each **acronym** of a parameter.

7.6.2 ResultSet

See **ResultSet** with ServiceParameter.

7.6.3 Label

7.6.3.1 General

The label includes a language key. Using this language key, the parameter on the user interface is identified (by default), e.g. as label text of a column header. Overwrites the value from the property configuration.

7.6.3.2 Processing in the MOC client

The label is displayed as label text of a field or a column title.

7.6.4 Tooltip

Specifies a specific tooltip for the parameter in the service context. Entry as language key.

7.6.5 FormatType

Use this field to overwrite specific values of a property in relation to the service (currently **Label, Length**, **ControlType**, **ControlTypeMode**, **ControlDataSource**, **ControlDataSourceMode**, **ControlDataSourceResult**).

For example: If you enter *workplace.id* as **FormatType** for the parameter *resource.id*, you can define for the parameter to be a *resource.id* in this service, however its length, label and control properties are taken from *workplace.id*.

In this case (other than in case of semantic and syntactic types), the value from **FormatType** takes priority. For this reason, we have a new hierarchy:

- Value from **FormatType**
- Value from ServiceParameterGUI
- Value from Property
- Value from **SemanticType**
- Value from **SyntacticType**

7.6.6 ClientDefaultValue

Input fields have a **ClientDefaultValue** property. The value entered here is displayed as default value when the control is initialized. "From" and "to" values are separated by semicolons.

Set checkbox: If the value of this field is set to *true* during a CheckEdit, the checkbox is set after initializing.

Preallocation of text fields with "from" and "to" values (InputAsArray): set value1;value2 to prepopulate the 'from' and 'to' fields during a text edit.

Date fields: In case of date fields, the field can be preallocated with an offset. If you set default values for date fields, you must absolutely specify the type of offset. The following offsets are possible:

- h (hours)
- d (days)
- w (weeks)
- m (months)
- y (years)

The 'to' value is always **relative to the 'from' value.** The default value is always a DateTime object. The presentation depends on the output format of the relevant field.

You can put "[" and "]" in front and at the end of the relevant value to specify the start and end of a period of time. Consequently, e.g. "[0d;0d]" means that 12:00:00 AM is entered in the 'from' field today and 11:59:59 PM is entered in the 'to' field today. "[-1w;0w]" means from Monday last week up to Sunday last week.

Examples

Current date:

```
0d
```

From today to the day after tomorrow:

```
0d;2d
```

From today to one week from today:

```
0d;1w
```

From yesterday to tomorrow:

```
-1d;2d
```

From one year ago today to one year from today:

```
-1y;2y
```

Year shortlists: You can configure a year shortlist by **ControlDataSource** = YearList and **ControlDataSourceMode** = Script, or even by standard "Service-ControlDataSource". In this case, you can use the following default values:

- Current year: 0y and/or currentyear
- Last year: -1y
- Following year: 1y
- 4 years ago: -4y
- Year that was current 10 months ago: y-10m → this is mostly the case when the relevant year field is used in combination with a month shortlist.

If you want to preallocate two fields **(ShowSecondControl)**, you have to separate both values by semicolon, e.g. -1y;1y

Month shortlists: You can use the following default values for a month shortlist:

- Current month: 0m
- Last month: -1m
- Following month: 1m
- 4 months ago: -4m

If you want to preallocate two fields (**ShowSecondControl**), you have to separate both values by semicolon, e.g. -1y;1y.

7.6.7 IsKey

The IsKey column is very important and should be occupied for all key columns of a service, since otherwise data records cannot be clearly identified. Columns including the value 'null' may NOT be defined as keys. The IsKey columns should be identical for all services (insert, update, delete, lock, unlock, copy). These entries are important, so it is best to verify them twice.

This field specifies the positioning of the cursor after an editing operation. If the positioning option OnKeyValue is selected, the client should only request one new data row after editing. You also use values that are marked **IsKey** as selection criteria. **IsKey** must also be set for *delete,* since this data record must be deleted from the view.

IsKey must also be indicated for *list.* If no sorting is given in the *list,* sorting takes place according to **IsKey** fields.

Every parameter which is **IsKey** MUST always be **IsMandatory**. This rule has two exceptions:

- List service
- Wrappers with composed keys.

7.6.8 ShowInGrid

Specifies whether the parameter is to be displayed in tables by default.

7.6.9 ShowInDetail

Specifies whether the parameter is to be displayed in detail views by default.

7.6.10 ShowInSearch

Specifies if the parameter is to be used as selection criterion (i.e. in selection panels) by default.

7.6.11 ColumnCategory

7.6.11.1 General

In the tabular view, the client should provide the option to summarize the columns in the table to categories. You specify a language key that is displayed as title of the summarized columns.

7.6.11.2 Processing in the MOC client

The ColumnCategory is used to assign the parameter to a "strip" in the grid (table view).

7.6.12 Category1, Category2, Category3

7.6.12.1 General

The client processes the columns Category1, Category2, Category3 in order to group fields in applications. The grouping can be performed via tabs or frames for a group of fields. You specify a language key that is displayed as title or label text of the grouped elements.

7.6.12.2 Processing in the MOC client

Category1: Assigns the parameter to a tab in the detail view.

Category2: Grouping options for detail screens.

Category3: Currently not used.

7.6.13 TabOrder

You specify the order of tabs for detail views.

7.6.14 ColumnOrder

You specify the order of columns in tabular views.

7.6.15 ShowSecondControlInSearch

7.6.15.1 General

Specifies whether a second control is to be displayed (from/t0). You can use this setting with selection criteria that include a value range via the operator CanBetween, e.g. "date from/to".

7.6.15.2 Processing in the MOC client

The MOC provides two adjoining fields. The label text of the second field is automatically "to". If it is a field of "date" type, you can predefine a relative date for both fields.

7.6.16 SearchTabOrder

Specifies the tab sequence for the selection panel.

7.6.17 SearchCategory1, SearchCategory2

7.6.17.1 General

The client processes the columns SearchCategory1 and SearchCategory2 in order to group fields in selection panels. The grouping can be performed via tabs or frames for a group of fields. You specify a language key that is displayed as title or label text of the grouped elements.

7.6.17.2 Processing in the MOC client

SearchCategory1: You allocate the parameter to a tab in the selection panel.

SearchCategory2: Grouping options for the selection panel.

7.6.18 ControlType

Use the ControlType to specify which control should be used for the relevant parameter. The client will map the abstract type onto a specific control class. If you do not specify a type, the client uses the data type to decide on the ControlType. Possible values for the ControlType:

CheckEdit: Selects a Boolean value (true/false) or multiple values if a reference to a data source is given.

ColorEdit: Selects a color value.

ComboBoxEdit: Combobox with selection of values from web service or data reference.

DateTimeEdit: Enter a date and/or a time.

MemoEdit: Enter an arbitrary text.

RadioGroup: Selects a Boolean value (true/false) or one of multiple values if a reference to a data source is given.

TextEdit: Standard text input. You can add a button opening a search dialog to this control, if you add a reference to a service in **ControlDataSource**. If you enter the name of a DataLogic in **ControlParameter** and if a mapping is included in **ControlDataSourceResult**, data will be requested upon leaving the control and return values will be mapped appropriately.

7.6.19 ControlTypeMode

7.6.19.1 General

Allows for controlling the input control.

CheckEdit: DualState (default), TriState, J;N;J (checked;unchecked;tristate)

ColorEdit: none

ComboBoxEdit: SingleEdit, Single, Multiple (multiple selection)

DateTimeEdit: Date (date display), Time (time display), DateTime, RelativeDate, RelativeDateTime

MemoEdit: none .

RadioGroup: SingleColumn, SingleRow

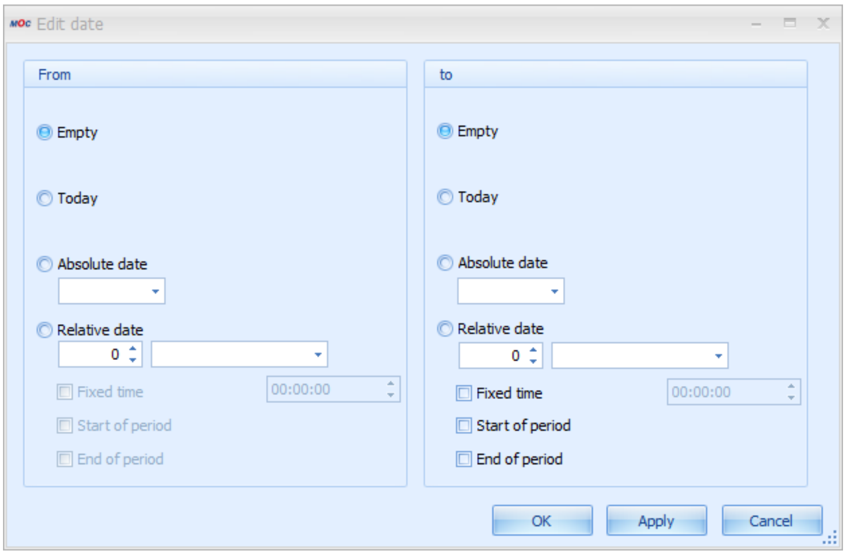
TextEdit:

- Empty: the search button is shown if a **ControlDataSource** is defined.
- "SearchButton": Search button is shown.
- "SearchButtonValidate": Search button is shown. If you enter an invalid value, an error is displayed.
- "OpenFileDialog": opens a file selection dialog.

7.6.19.2 Processing in the MOC client

If you use **DateTimeEdit** including the definition of a relative date **(ControlTypeMode**: RelativeDate or RelativeDateTime), you can enter a relative date.

If **ShowSecondControl** = true, you can predefine the complete relative value range. In this case, a button is displayed behind the second input control. You can use this button to open the following dialog:



Use this dialog to customize the values for **ClientDefaultValue** . The following entries are possible:

- **Empty:** no value is adopted
- **Today:** the current date is adopted
- **Absolute date: you can select** a fixed date value via a calendar control
- **Relative date: you can select and adopt** a date relative to the current date. In this context, "Start of period" means that you additionally go to the start of the selected period. Example: current date is 20-MAY-2010. If you select "- 1 month", 20-APR-2010 is adopted. If you also select "Start of period", the date is changed to 01-APR-2010. The same applies to "End of period". These settings are saved in the mpdvEdit or the selection profiles as **ClientDefaultValue.**

7.6.20 ControlParameter

See **ControlType** → TextEdit

7.6.21 ControlDataSource

Data source for the selection of values. The data source can be:

- **Web service** (configuration: **ControlType** = *ComboBoxEdit*, **ControlDataSourceMode** = *Lookup*, **ControlDataSource** = **Name** of a ControlDataSource. See also section "7.8 ControlDataSource")
- **ReferenceData (**configuration: **ControlType** = *ComboBoxEdit*, **ControlDataSourceMode** = *Reference*, **ControlDataSource** = **Type** of ReferenceData)
- **Search application** (configuration: **ControlType** = *TextEdit*, **ControlDataSourceMode** = *Lookup*, **ControlDataSource** = application name)

- **Script** (configuration: **ControlType** = *ComboBoxEdit*, **ControlDataSourceMode** = *Script*, **ControlDataSource** = Name of script)

7.6.22 ControlDataSourceMode

Data source mode (*Lookup, Reference* or *Script*).

7.6.23 ControlDataSourceParameter

Optional setting of parameters of a **ControlDataSource**. If you make settings here, these settings overwrite the settings in the **ControlDataSource**.

See also the description **ControlDataSource - Parameter**

7.6.24 ControlDataSourceResult

Optional setting of the result of a **ControlDataSource**. If you make settings here, these settings overwrite the settings in the **ControlDataSource**.

The settings in this field provide more options than the **Result** in the **ControlDataSource:**

Result columns are separated by semicolon. Field mapping:

- First entry: Value
- Second entry: Labeling
- Third entry: UnitLabel
- As of the fourth entry, the fields are mapped:
  - Via acronym or semantic type.
  - Field mapping: in **ControlDataSourceResult**, you can enter a mapping in the form "FieldName=ColumnFromResult" as from the fourth entry. For example, you can specify tool.id=resource.id in order to fill the field "tools.id" with the "resource.id" value from the search application. Several mappings are separated by ";" - spaces are not allowed.
  - Asterisk mapping: Instead of mapping, you can also enter * . Subsequently, all return columns of the search application are mapped. The mapping is performed as usual via ID or semantic type.

7.6.25 VisibleCondition

This value decides whether an input field is visible on the client. For customization, see EditableCondition.

7.6.26 EditableCondition

This value decides whether you can edit an input field on the client. There are three possibilities:

- Boolean value: In case of *TRUE* or *FALSE,* the field is always editable / non-editable.
- Binary expression:
  - Field name must be the name of a field that is also located in the ControlPanel.
  - Valid operators: =, <, >, <=, >=, <>, !=
  - The value is written as a string and interpreted depending on the comparative field value.
  - Field, operator and value must be separated by a space!
- Concatenation of binary expressions:
  - You can concatenate an arbitrary number of binary expressions.
  - You can use the operators "&&", "AND", "||", "OR" to link expressions.
  - Here, too, all components of the conditions must be separated by a space.
  - Priority of operators: "AND" or "&&" are evaluated first, then "OR" and "||".
    You cannot use brackets.
  - Example: resource.id = 12345 && resource.costcenter = 20 || resource.id = 60610

The client assigns the default value of the property "ClientDefaultValue" to the field, if the result of an expression in the EditableCondition or the VisibleCondition changes from FALSE to TRUE. The client dynamically evaluates the expressions in the EditableCondition and the VisibleCondition, if the fields of the application change.

7.6.27 ScriptId

7.6.27.1 General

The ID of the script that is allocated to the parameter. If you set the ID, the relevant script is performed upon various events (at present EditValueChanged and Leave).

7.6.27.2 Processing in the MOC client

The method name of the script is ScriptId+EditValueChanged and/or ScriptId+Leave. The script can be included in any DLL that is read by the CodeManager.

7.7 Property

For the acronyms, properties include information on data types, input and output formats, display options, a name (that can be localized) and other settings specifying how ServiceParameters are displayed in the client. Each property has a system-wide unique acronym.

A number of settings exist in the ServiceParameterGui and in the properties. You normally use the properties to define how data is displayed on the client. You only fill the respective field in the ServiceParameterGui if you want to display specific services on the client in a way that is different to the settings in the properties. The ServiceParameterGui fields overwrite the property fields of the same name.

7.7.1 Acronym

Clear identification of the property across all domains.

7.7.2 WebServiceType

Describes the data type used to transfer the property between client and server. The currently supported WebServiceTypes are exclusively

- *binary*
- *boolean*
- *datetime*
- *decimal*
- *integer*
- string

Important: the types **date* and **time* are internal types which are not transferred.

7.7.3 NETType

The data type used by the client. If NETType is empty, the **WebServiceType** is used to automatically identify the data type used by the client. At present, NETType supports the following entries:

- **color:** Use *color* to convert the transferred integer into an RGB code. In this case, the conversion is implemented by the grid.
- **duration:** creates a duration from an integer.
- **image:** either creates an image from a transferred byte array or interprets a transferred string as image name. For example, the *maintenance.active.led* property is transferred as a string including the name of an icon.

- **preview:** Specifies that the contents in the client may be displayed as "preview" (similar to auto-preview outlook) (application e.g. in DevExpress grid).
- **timestamp: Use *timestamp*** to automatically create an additional column for date values in the client in order to process time and date separately.

7.7.4 SemanticType

Use semantic types to inherit semantic properties. The "order.id" is therefore used to identify orders (semantic meaning). The acronym *operation.order.id* includes such an order identification and therefore has the semantic type *order.id.* If an attribute of the property is not set (empty), the respective value from the semantic type is used for the processing in the client.

For example: You must set the semantic type if you want to adopt a value from a lookup screen in the field. For the *workplace* field, enter e.g. resource.id as semantic type in order to adopt the selected workplace from a search screen for workplaces. Refer to the description of the SyntaticType for further information on the priority used to specify the attributes of a Property, the SemanticType and the SyntacticType.

7.7.5 SyntacticType

You mainly use a syntatic type for a uniform presentation of the different properties. The syntactic type does not include any semantic content. For example: The properties booking.begin_ts and booking.shift.start_ts have different semantic meanings, but are presented in a uniform format that can be controlled centrally.

Syntactic types are used to control the characteristics of a *Property:* for example length, input and output screen, tooltip, label, etc. To select the valid value for a characteristic, the client proceeds as follows:

- If the characteristic (e.g. length) is set in Property, the client uses this value.
- Or: If a semantic type is available and the characteristic is set, the client uses this value.
- Or: If a syntactic type is available and the characteristic is set, the client uses this value.

Note:

- You must always enter a **description** for syntactic and semantic types.
- Syntactic types can reference other syntactic types so that "inheritance hierarchies" can be created.
- Create syntactic types as property of the *SyntacticType* domain.
- Semantic types are usually "real" properties of a "normal" domain that are used as semantic type at other places.

7.7.6 Label

7.7.6.1 General

The label includes a language key. Using this language key, the parameter on the user interface is identified (by default), e.g. as label text of a column header. Overwrites the value from the property configuration.

7.7.6.2 Processing in the MOC client

The label is displayed as label text of a field or a column title.

7.7.7 DefaultTooltip

Specifies the default tooltip for the property as language key.

7.7.8 UnitLabel

Text key for unit. The unit is displayed to the right of the input field.

7.7.9 OutputFormat

This field specifies the format that is used to display a value (e.g. for date or quantity values). If you do not enter an **InputFormat** in the repository, the MOC tries to develop an appropriate format from the **OutputFormat.** Enter the value **InputFormat** in the repository only if special masking is required. Find further details in section "7.7.12 Rules for the input/output formatting".

7.7.10 InputFormat

Equivalent to OutputFormat. You can enter a valid regular expression in the field InputFormat. Other entries that are not regular expressions are not permissible. Find further details in section 7.7.12.

7.7.11 Length

The client shows the control for this acronym in the specified width (i.e. the specified number of characters). With Length=0, the control uses the entire width available. If a width is specified but the space available is not sufficient, the control is cut off.

This field also specifies the number of characters that you can enter in an input field with **ControlType = TextEdit,** if no other **InputFormat** is specified.

7.7.12 Rules for the input/output formatting

Overview

In the repository, you define the formatting of the data output and the input dialogs to edit data. The "Properties" of the different acronyms include an **OutputFormat** and an **InputFormat.** The **OutputFormat** defines formatting if you display a value.

Important: If you do not enter an **InputFormat** in the repository, the MOC uses the **OutputFormat** to generate an appropriate formatting. Enter the value InputFormat in the repository only if special masking is required.

In case of strings, you cannot enter the special characters asterisk (*) and pipe (|), if you have not defined any input format. As you use these two special characters as separator and control character, they can cause problems if they are written in the database.

With strings, the maximum number of characters that you can enter is defined by the attribute **Length**, if no other input format is defined.

Syntactic types

The Properties provide so-called "syntactic types" in order to make groups (similar to field types in Delphi). Syntactic types have the same properties as real properties. The real properties have a syntactic type. For example, if the **output format** of the syntactic type includes a value, this value is used wherever this syntactic type is entered.

Example: Industrial minutes

The syntactic type "Durations" has the format {0:mpdv_timespan}. With the different properties showing durations, "Durations" is entered in the column **SyntacticType** and no entries are made in the columns "output format" and "input format". When the property is read - and if no **output format** is available in the property - the format of the syntactic type is used.

If a system displays industrial minutes (no standard function!) and if the syntactic type "Durations" is specified, the **output format** is automatically changed from {0:mpdv_timespan} to {0:mpdv_industrialMinutes}. As a result, all formats including the syntactic type "Durations" are shown in industrial time units.

Times and durations are internally stored in the system as integer seconds. If you convert times or durations during input or output formatting to formats other than hours, minutes and seconds (HH:MM:SS), the conversion may not be possible without losses. For example, this applies to the use of the "mpdv_calc" format and the classic industry minute display:

When converting from seconds to hours (division by 3600), decimal numbers with an infinite number of decimal places can occur, which inevitably have to be rounded when displayed on the client. Example: 20 minutes = 1200 seconds = 0.333333… hours. If the value is rounded to three decimal places, you calculate backward as follows: 0.333 * 3600 = 1198.8 seconds. Depending on how the client rounds, the internal value is then no longer 1200 seconds, but 1999 or 1998 seconds.

If you use less than three decimal places, the conversion error gets even greater:

The system recorded a duration of 123 seconds. The client displays 0.03 hours. If you calculate backwards, the result is 108 seconds.

Output formats

OutputFormat	Automatically created masking (input format)	Examples	Description
Numeric data			
f(number)	None	f3, f1	Numeric value without thousands separator. The number specifies the number of decimal places.
n(number)	None	n0, n2, n5	Numeric value with thousands separator. The number specifies the number of decimal places, even if the data type to be displayed is an integer type. In case of n0, no decimal places will be shown.
#.(##) , #.(0)	None	#.####, #.0000	Arbitrary format
{0:mpdv_timespan}	None	2:33:30	MPDV format provider. Conversion of seconds to hh:mm:ss and vice-versa.
{0:mpdv_timespan_short}	None	2:33	MPDV format provider. Conversion of seconds to hh:mm and vice-versa.
{0:mpdv_timespan_minutes}	None	45	MPDV format provider. Conversion of seconds to minutes and vice-versa.
{0:mpdv_cycletime}	None	1:30:00	MPDV format provider. Hours per 1000 pieces. Conversion into seconds and vice versa.
{0:mpdv_te}	None	2.00	MPDV format provider. Hours per 1000 pieces. Conversion into seconds and vice versa.

<table>
<tr><td colspan="4">Strings</td></tr>
<tr><td>empty</td><td>[^*|]]*</td><td></td><td>Illegal characters begin with ^. In this example * and | * No limitation in length</td></tr>
<tr><td>empty</td><td>[^*|]{0.10}</td><td></td><td>Illegal characters begin with ^. Max. length: 10 characters</td></tr>
<tr><td>empty</td><td>[0-9a-fA-F]</td><td></td><td>Allowed characters 0 through 9, a through f, A through F.</td></tr>
<tr><td colspan="4">Special formats</td></tr>
<tr><td>{0:mpdv_cycletime_sec_cycle}</td><td>None</td><td>29 sec/cycle</td><td>MPDV format provider. Seconds per cycle. Conversion into seconds per 1000.</td></tr>
<tr><td>{0:mpdv_IndustrialMinutes}</td><td>None</td><td>1.50</td><td>MPDV format provider. Conversion of seconds into industrial minutes and vice versa.</td></tr>
<tr><td>{0:mpdv_leadingzeros_order}</td><td>ORDER</td><td></td><td>You must combine this output format with the input format ORDER. The combination is used in the syntactic type "order_id". The basic settings are used to automatically specify the length.</td></tr>
<tr><td>{0:mpdv_leadingzeros_operation}</td><td>ORDER</td><td></td><td>You must combine this output format with the input format ORDER. The combination is used in the syntactic type "operation". The basic settings are used to automatically specify the length.</td></tr>
<tr><td>{0:mpdv_leadingzeros_sequence}</td><td>ORDER</td><td></td><td>You must combine this output format with the input format ORDER. The combination is used in the syntactic type "ordersequence_id". The basic settings are used to automatically specify the length.</td></tr>
</table>

Input formats

The following definitions are available for the input format:

- Leave empty: The input format is implicitly defined using the output format. See table above.
- Use of logical input formats
- Use of regular expressions

Logical input formats

To simplify the definition of input formats and limit the variety of entries in the repository, the logical input formats are provided. These input formats are permanently implemented in the client and can directly be used in the repository. Input formats are customized in the properties. But service parameters specify whether wildcards are allowed. For this reason, the input format actually used can vary depending on the allocated service.

In order to use logical input formats, define the name of the input format in the affected property in the repository. The following formats are currently available:

Name	Input format without wildcard	Input format with wildcard
CHARACTER	[^*][LENGTH]	[^]][LENGTH]
NUMBER_N0	[0-9][LENGTH]	[0-9][LENGTH]
NUMBER_N1	[0-9][LENGTH]\R.?[0-9]{0,1}	[0-9][LENGTH]\R.?[0-9]{0,1}
NUMBER_N2	[0-9][LENGTH]\R.?[0-9]{0,2}	[0-9][LENGTH]\R.?[0-9]{0,2}
NUMBER_N3	[0-9][LENGTH]\R.?[0-9]{0,3}	[0-9][LENGTH]\R.?[0-9]{0,3}
NUMBER_N6	[0-9][LENGTH]\R.?[0-9]{0,6}	[0-9][LENGTH]\R.?[0-9]{0,6}
TIMESPAN_SHORT	[0-9][LENGTH]\R:[0-9]{2,2}	[0-9][LENGTH]\R:[0-9]{2,2}
TIMESPAN	[0-9][LENGTH]\R:[0-9]{2,2}\R:[0-9]{2,2}	[0-9][LENGTH]\R:[0-9]{2,2}\R:[0-9]{2,2}
ORDER	[0-9a-zA-Z.+][LENGTH]	[0-9a-zA-Z.+*][LENGTH]

The placeholder [LENGTH] is replaced with the configured field length at runtime. If the defined length is '0', an '*' is entered. With the logical format "ORDER", the system automatically changes the [LENGTH] according to the basic settings when the output format changes.

Input/Output formats including calculation

If you specify the **output format** *mpdv_calc,* you can include calculations in the formatting. In the format, you can specify a divisor and multiplier and an identifier that specifies if a reciprocal is calculated. You can also specify the number of decimal places. The **OutputFormat** *mpdv_calc* implicitly defines the **InputFormat**. If an input of values is made, the reciprocal value is calculated.

Example: "mpdv_calc;MULT=5;DIV=2;INVERSE=false;FORMAT=n3" (the value is multiplied by 5, divided by 2, then the reciprocal is calculated and the result is displayed with 3 decimal places).

The input/output format including calculation is normally used for the display of cycle times or specifications of single pieces. In the database, these times are always saved in seconds per 1000 pieces. If an input/output format including calculation is used, you can convert the times to hours per 1000 pieces, minutes per piece or with reciprocal also to piece per hour.

Overview of regular expressions

You can find a large amount of information on regular expressions using the search engines on the internet. In the following, the most important aspects are presented.

Meta characters

Represent a range of characters.

Character	Description
.	Matches any character.
[aeiou]	Matches any single character included in the specified set of characters.

[^aeiou]	Matches any single character, which is not included in the specified set of characters.
[0-9a-fA-F]	Use of a hyphen (–) allows specification of contiguous character ranges.
\R.	Matches the decimal separator specified by the System.Globalization.NumberFormatInfo.NumberDecimalSeparator property of the current culture.
\R:	Matches the time separator specified by the DateTimeFormatInfo.TimeSeparator property of the current culture.

Quantifier

Repetition, number of characters

Quantifier	Description	Samples
*****	Specifies zero or more matches.	The "\w*" mask matches a string consisting of zero or more letter characters. It's equivalent to the "\w{0,}" mask.
+	Specifies one or more matches.	The "\w+" mask matches a string consisting of one or more letter characters. It's equivalent to the "\w{1,}" mask.
?	Specifies zero or one match.	The "\w?" mask matches zero or one letter character. It's equivalent to the "\w{0,1}" mask.< /description>
{n}	Specifies exactly *n* matches.	The "\d{4}" mask matches exactly four digits.
{n,}	Specifies at least *n* matches.	The "\d{2,}" mask matches two or more digits.
{n,m}	Specifies at least n, but no more than m, matches.	The "\d{1,3}" mask matches either one, or two, or three digits.

Special characters

Special characters

<table>
<tr><th>Character</th><th>Description</th><th>Samples</th></tr>
<tr><td>|</td><td>Alternation symbol. This can be used to implement a choice between two or more alternatives.</td><td>The "1|2|3" mask matches either "1" or "2" or "3".
The "abc|123" mask matches either "abc" or "123".
The "\d{2}|\p{L}{2}" mask matches either two digits or two letters.</td></tr>
<tr><td>()</td><td>Grouping. You can use parentheses to create sub-expressions, or to limit the scope of the alternation.</td><td>The "(an|ba)t" mask matches either "ant" or "bat".
The "(net)+" mask matches "net", "netnet", "netnetnet", ... strings. Compare with the "net+" mask which matches the "net", "nett", "nettt", ... strings.
The "(0|1)+" mask matches a string of indeterminate length, consisting of "0" and "1".</td></tr>
</table>

Examples

Input 1..9999 => Input format for property : ([1-9]|[1-9][0-9]|[1-9][0-9][0-9]|[1-9][0-9][0-9][0-9])

Input 0..999 => Input format for property : ([0-9]|[1-9][0-9]|100)

Best practice: input of long string fields

The client identifies the width of an input field using the attribute **Length**. In case of long string fields with more than 20 characters, the layout can become confusing because these string fields use the complete width of the layout and are very long compared to other input fields. Very long string fields are cut off on the right-hand side, if the available space is not enough. To avoid this behavior, you can control the displayed field width regardless of the number of characters that you can enter.

- Use the attribute **Length** to specify the **width** of the input field.
- You can use a regular expression in the **InputFormat** to specify the **number of characters that you can enter**.

If you enter strings that are larger than the displayed field, the input field automatically scrolls horizontally.

Examples:

Attribute	Length	InputFormat	Effect
article.designation	50	.{0.250}	The field is displayed with a width of 50 characters. You can enter up to 250 characters.
operation.input_component_list	25	.{0.40}	The field is displayed with a width of 25 characters. You can enter up to 40 characters.

7.7.13 FillChar

Obsolete. This field must be left empty.

7.7.14 Calculation

Obsolete. This field must be left empty.

7.7.15 Further fields see ServiceParameterGui

For a description of the following fields, refer to the data types of the ServiceParameterGui:

ControlType, ControlTypeMode, ControlParameter, ControlDataSource, ControlDataSourceMode, ControlDataSourceParameter, ControlDataSourceResult, VisibleCondition and EditableCondition.

7.8 ControlDataSource

A ControlDataSource defines a data source that you can use to fill selection lists in controls, for example. These can be data logics (service requests) or reference values (see also ReferenceData).

Reference values are usually required to fill selection lists (and/or RadioGroups) with static contents.

You use data logics to request services that identify selection lists (or RadioGroups) dynamically. For example, these lists can include master data that are configured in the database.

The settings made in the columns **Parameter** and **Result** can be overwritten in a **Property** or **ServiceParameterGui**.

7.8.1 Name

Name of the ControlDataSource. The name should be composed of English terms clearly describing the data source. You usually use the camelCase notation.

7.8.2 Source

If the data source is a web service, this field contains the name of the client's data logic. You derive the data logic from the service name. To do so, remove the dot between domain and function and use a capital letter for the first letter of the function:

Service	Data logic
MDUser.list	MDUserList
MDUnits.list	MDUnitsList

In case of reference values, this field includes the **Type** of a **ReferenceData**.

7.8.3 Parameter

A list of parameters. The list does not include spaces, use semicolons to separate parameters. This field is only allowed in combination with web service data sources. A parameter can be allocated dynamically or permanently.

Permanent parameters appear as <acronym>=<value>, e.g.

"dialogconfiguration.type=AIPDEF;dialogconfiguration.type=AIPTNR".

Dynamic parameters are specified as a pair of <acronym1>=**[**<acronym2>**].** e.g.

"resource.id=[resource.id];pdvprocessparameter.evaluation_ts=[pdvsinglevalue.evaluation_ts]"

The acronym in square brackets is replaced with the acronym values from the ControlPanel.

7.8.4 Columns

A list of requested columns. The list does not include spaces. To separate columns, semicolons are used. This is only permissible for web service data sources.

7.8.5 Result

You can enter 1-n acronyms separated by semicolon. The sequence used specifies the importance.

- Position 1 (Value): Name of acronym whose value is entered in the input field.
- Position 2 (ControlValue): Name of acronym whose value is displayed in the selection list. If you do not specify position 2, the acronym of position will be displayed.
- Position 3 (LabelValue): If you specify position 3, the value of the acronym is entered in the label field of the input field and also displayed in the selection list.
- Position 4-n: Use these positions to define additional return values, which are then used to update "dependent" controls in the client ("lookup").
  Only with web service data sources:
  Optional return columns of the data source, separated by semicolons. Without spaces. The return has the format <acronym>=<value> - for acronym pairs, the second acronym is therefore replaced with the result value (e.g. if you enter "operation.resource.id=resource.id", this results in "operation.resource.id=4711").

7.9 ReferenceData

Reference values are usually required to fill selection lists (and/or RadioGroups) with static contents. In contrast to values provided by web services, reference values are fixed and do not change. For this reason, reference values can be entered once in a list and are delivered in this form.

7.9.1 ref_data_key

The **ref_data_key** must be unambiguous for each entry. In special cases, this key is used in the source code (at least in the server).

Usually, the **ref_data_key** is composed of **type** + : + **db_key**; this facilitates its allocation to type and key. An exception occurs if the **db_key** includes a German expression. The **ref_data_key** must then be formed differently. For example, *pwdexclusion:person.firstname* is a super **ref_data_key** for the type *pwdexclusion.pwd* and **db_key** *PNR.PVORNAME*.

7.9.2 Type

Use this field to summarize various ReferenceData entries to a list.

7.9.3 db_key

The **db_key** is the actual value that is selected in the list. This key identifies an entry unambiguously within a **Type.** You cannot freely select the key because the key is often transferred to services and can correspond to the content of a configuration identifier in the database, for example.

7.9.4 is_default

The entry with this key is preallocated as default.

7.9.5 Designation

Text displayed in the selection list. A language key is specified.

7.9.6 sort_key

Specifies the sequence that is used to display the entries in the selection list.

7.10 Authorization

The authorization mechanism
- protects applications and functions against unauthorized use on the client,
- hides fields or field groups on the GUI,
- prevents these fields from being edited.

7.10.1 Authorization type

Controls the type of authorization. Possible values:

- Acronym: enables the authorization of individual fields (properties)
- AcronymGroups: enables the authorization to group fields
- Application: enables the authorization of applications
- Functions: enables the authorization of functions which are e.g. requested from the application toolbar.

7.10.2 Authorization Context

Context where the authorization is intended. If the field is left empty, authorization is always granted, irrespective of the context. You normally use this field to control the authorization of acronyms in the context of special services.

7.10.3 Authorization ID

Identifies the object to be authorized, i.e. the name of the acronym or the ID of an application.

7.10.4 Authorization key

The authorization key that is used to protect the object.

7.10.5 Authorization Designation

(Optional) text description of the authorization.

8 HYDRA Production Data Manager - Preface

The data transfer to HYDRA is performed by “dialog data” in the ASCII format. The string consists of header and field data. Header data are required for HYDRA to identify the data type, whereas field data contain data of the external application.

8.1 Header data of the dialog data

The following data **always** have to be transmitted in a dialog by an external acquisition system to HYDRA. The single fields are separated by the character "|“.

Identification	Content/(type)	Description
DLG	{Dialog ID}	The dialog ID indicates the required functions, e.g.: DLG=P_AN to log staff on D LG=MNR.INSERT to create a machine
USR	{N4}	HYDRA user: This HYDRA user number is the unique ID of a HYDRA client: For terminals the respective terminal number is added to the terminal number 2000. Example: USR=2004 corresponds to terminal 4 **Please note:** Terminal numbers assigned to HYDRA terminals must not be used for the HYDRA-PDM connection. We recommend using a separate number range.
DAT	{mm/dd/yyyy}	Date: current date in the format mm/dd/yyyy“ Example: DAT=03/30/2007
ZEI	{seconds}	Time: point in time in the "seconds" format, i.e. in seconds as of midnight Example: ZEI=36003 for the time: 10:00:03 am

Dialog IDs having the structure **OBJEKT.AKTION** (such as MNR.INSERT = create machine) are BAPI calls, which are described in detail in the sections that follow.

The following data **can** be transferred in a dialog to HYDRA by an external data acquisition system.

ID	Content / {Type}	Description
ID	{Identification number}	When started for the first time the ID=0 is sent. The ID is increased by 1 for each other call. The return string contains the same ID as the dialog data string.
OFF={J|N}	OFF={J|N}	Offline identification. Indicate "*OFF=J*“, if data were captured and buffered offline. In this case the data are processed separately. ("*OFF=N"* doesn't have to be indicated)
DATEI	{C256}	File name: Indication of the file name when lists are requested. The file name is created by means of the Hydra user:

		.\spool\hyu{Hydrauser}.{extension} {Hydrauser} corresponds to the user numbers of the terminal. (The terminal 1 corresponds to the Hydrauser-number 2001 etc.) {extension} can be chosen subject to the data. Standard: “txt“ Example: ...|DLG=.\spool\hyu2004.txt|...

8.2 Common notes

☞ **To avoid errors:**

The SCS-PDM interface requires the dialog data to be in the correct chronological order.

If data are sent and their chronological order is wrong errors in determining quantities and activities will occur. In addition to this, unintentional plausibility errors might arise due to wrong plausibility checks. Consequently, data records might be rejected.

- The header data DLG, USR, DAT and ZEI always have to be indicated completely within dialog data.
- OFF has only to be sent if it deals with data that were collected offline (default is OFF=N).
- It is not mandatory to keep a special order of IDs in dialog data..
- Double data records are filtered out by the process, provided that a valid ID is forwarded within header data.
- The plausibility of data is basically checked prior to posting. Incorrect data records are ignored and recorded in an error log. Processing ignores unknown field identifications.
- The field identifications ANR, AGNR, PNR, KNR and MNR have adjustable lengths and thus may vary in the HYDRA basic settings according to the respective configuration. Checks regarding data field lengths are based on the settings made within the HYDRA setup.

- All fields within a data record are separated by the "|“ character.
  Thus, the length of the data is variable but may not exceed the maximum length specified. If the characters "|“ pipe or "\“ backslash are included the data content they have to be masked by another preceding backslash :
  \ → \\
  | → \|

8.3 Field data

Further IDs (“field data”) have to or can be entered depending on the dialog ID (“DLG“).

8.4 Return values

Each return string includes the following header data:

Identification	Content/(type)	Description
RET	{N8}	Return code: Return value RET = 0: activity executed RET ≠ 0: activity has not been executed; evaluate error code Examples: ...|RET=0|... ...|RET=60|...
KT	{C20}	Short text: description of the error (short) if RET ≠ 0 or optional info if RET = 0 Examples: KT=Already logged on KT=In 2040
LT	{C40}	Long text: description of the error (long) if RET ≠ 0 or optional info if RET = 0 Example: |LT=OP cannot be logged on several times. LT=In Hans Meier

Each return string **may** optionally include the following data

Identification	Content/(type)	Description
ID	{Identification number}	The return string contains the same ID as the dialog data string, provided that it was indicated there.
	{C256}	Dialogdaten über Dialogdatei, nur in Kombination mit RETFILE (nur wenn im BAPI-Call angegeben)
RETFILE	{C256}	Rückgabewerte über Rückgabedatei, nur in Kombination mit DLGFILE (nur wenn im BAPI-Call angegeben)

Depending on the dialog ID ("DLG") additional IDs, which include, e.g., further return values (information, etc.) may also be returned.

Moreover, the following definitions apply for BAPI calls:

→ The BAPI calls ***.NEW, *.SELECT, *.LOCK** return a new or existing data record. The object name "Objekt" (value before the point) is returned as DATA=Objekt, the single values are transferred without the preceding object name.

→ The BAPI call ***.LIST** displays the selected data records in form of a list with the file name transferred in DATEI. The object name does not precede the columns. Colons are displayed as underline "_".

Error messages are described in detail in the section "HYDRA error messages" of the HYDRA-BDE manual.

8.5 BAPI call reference to the data model

The BAPI calls described in the corresponding documentation (MDM - Master Data Maintenance and DDM - Dynamic Data Management) partly refer to the HYDRA data model. For information on the data model, see the CUT-HDB course.

8.6 Lock mechanism for BAPI calls

The lock mechanism of HYDRA locks a virtual dataset on the server (e.g. "lock person 999999" or "lock machine group 145"). The virtual dataset can consist of one data record or of several data records. All BAPIs based on this virtual dataset check whether this dataset has been locked by another HYDRA client (e.g. console) prior to the BAPI calls ***.UPDATE** or ***.DELETE.**

Using the BAPI call ***.Lock** a client may lock a dataset/data record for a HYDRA client and once it has been processed by a BAPI call ***.UPDATE** it can again be released using the BAPI call ***.UNLOCK.**

In case a dataset/data record is locked at another console, the BAPI ***.LOCK** returns the error code 1666 (RET) as well as the HYDRA user (USR), the user (BEARB) and the client function (MODULE, provided that the other client has entered these details for the BAPI call ***.LOCK),** for which the dataset is locked.

To be able to delete a dataset/data record by a BAPI call ***.DELETE,** this data record should not be locked using the BAPI call ***LOCK.** The BAPI call ***.DELETE** checks in any case whether the data record/dataset is locked or not and cancels the process with an error if it is locked.

9 HYDRA Production Data Manager Basis - Data Collection

9.1 Reading time from HYDRA server

Structure of dialog data:

„DLG=SCMD;44|DAT=...|ZEI=...|USR=...|...“

The command does not expect further parameters.

The following values are returned to the terminal:

Field	Description	Example
RET=0	The return value RET is always 0.	RET=0
DAT={mm/dd/yyyy}	Current date in format "mm/dd/yyyy"	DAT=09/06/2001
ZEI={seconds}	Current time in format "seconds since midnight“ (Example: 10:00:03 is *ZEI=36003*)	ZEI=63142
GMTDAT={mm/dd/yyyy}	Current date in format "mm/dd/yyyy" in time zone GMT	GMTDAT=09/06/2001
GMTZEI={seconds}	Current time in format „seconds since midnight“ in time zone GMT	GMTZEI=55942
GMTOFF={±HHMM}	Description of the time zone local time	GMTOFF=+0200
GMTDIFF=[-]{seconds}	Deviation of local time from GMT in seconds	GMTDIFF=7200
WSDAT={mm/dd/yyyy}	Date of time shift from standard time to daylight saving time in the current year.	WSDAT=03/31/2002
WSZEI={seconds}	Time of time shift from standard time to daylight saving time in the current year.	WSZEI=7200
SWDAT={mm/dd/yyyy}	Date of time shift from daylight saving time to standard time in the current year.	WSDAT=10/27/2002
SWZEI={seconds}	Time of time shift from daylight saving time to standard time in the current year.	WSZEI=10800

9.2 Sending terminal status

Use the terminal status command to store information on the terminal status in the database. Requests asking to restart terminal, to download program or to read the authorizations are returned.

Structure of dialog data:

(The dialog ID DLG=SCMD;41 must be transferred as first entry):

„DLG=SCMD;41|DAT=...|ZEI=...|USR=...|TNR=...|...“

The following table contains the data that can be transmitted:

Field	Description	Example
IP={...}	Transfer of the IP-address. A character string of a maximum of 15 characters can be transmitted (could also be the hardware-address). With specific terminals, the hardware-address is configured in the HYDRA GUI; then it is not useful to transfer this field.	IP=192.168.20.234
PROG={...}	Program name of the terminal program or of the function list (max. 20 characters)	PROG=hyterm.exe
VERNR={...}	Version number of the terminal program (max. 10 characters)	VERNR=6.5.1.27
LOKANZ={...}	Number of local data records (after an offline phase)	LOKANZ=0
OFFDAT={...}	Date of the last offline phase	OFFDAT=05/21/1999
OFFZEI={...}	Time of the last offline phase in seconds since beginning of the day	OFFZEI=53214
ZNR={...} bzw. ZGRP={...}	Access number or access group If one of those fields is transferred with the 3 following fields the number of authorized badges, date and time are saved in the access status instead of the terminal status. This is necessary because several accesses can be administered via a terminal. You can make the request for an access or an access group.	ZNR=4 or ZGRP=5

BERANZ={...}	Number of persons authorized to access and to clock (the fields BERANZ, BERDAT and BERZEI are normally always transferred together directly after download of authorizations).	BERANZ=124
BERDAT={...}	Date when the authorizations were last read	BERDAT=05/21/1999
BERZEI={...}	Time when the authorizations were last read	BERZEI=5213
SUBANZ={...}	Number of sub-systems (DS100 or LEGIC-reader)	SUBANZ=0
SUBOK={...}	Number of active sub-systems	SUBOK=0
STA={...}	Terminal status (if possible): O=Online, F=Offline: Terminal answers in the specified cycle "ZYKL" (see below) and is currently "online" or "offline". o=Online, f=Offline: Terminal does not cyclically answer, but answers only when the terminal changes to status "online" or "offline".	STA=O
NEUDAT={...}	Date of last terminal restart	NEUDAT=05/20/1999
NEUZEI={...}	Time of last terminal restart	NEUZEI=27594
PROGDAT={...}	Date of last download of terminal program	PROGDAT=05/13/1999
PROGZEI={...}	Time of last download of terminal program	PROGZEI=43834
VLISTSYNC=N	Must be transferred with terminal type 110 A-SUB (ALS sub-system) if the sequencing list has been updated. This flag is then reset.	VLISTSYNC=N
ALSSYNC=N	Must be transferred with terminal type 111 A-SYN (ALS data synchronization) if the production combinations have been updated. This flag is then reset.	ALSSYNC=N

The following values are returned to the terminal:

Field	Description	Example
RET={...}	The transfer and storage of the terminal status is only successful if RET=0. All further fields are only available if RET=0.	RET=0
ZYKL={...}	Time in seconds after which the next status message has to take place (is not processed with INCA as the time is read every 15 minutes).	ZYKL=900
NEUSTART={J/N}	Do you want to restart terminal (‚J' Yes or ‚N' No)? This flag is reset if a new point in time is passed with the transfer of NEUDAT and NEUZEI. Consequently, the fields NEUDAT and NEUZEI can be passed with each terminal status.	NEUSTART=N
PROGLADEN={J/N}	Do you want to reload the terminal program? This flag is reset if a new point in time is passed with the transfer of PROGDAT and PROGZEI. Consequently, the fields PROGDAT and PROGZEI can be passed with each terminal status.	PROGLADEN=J
BERLESEN={J/N}	Do you want to read the authorizations again? This flag is reset if a new point in time is passed with the transfer of BERDAT and BERZEI. Consequently, the fields BERDAT and BERZEI can be passed with each terminal status.	BERLESEN=J
TNR={...}	Return of the terminal number for which the status was sent.	TNR=101
PROG={...}	Program name transmitted after last loading of program. If a list of the terminal programs is not available, this program name can be used to load the program.	PROG=hydra.exe
VLISTSYNC=J	Must be transferred with terminal type 110 A-SUB (ALS sub-system) if the sequencing list must be updated.	VLISTSYNC=J
ALSSYNC=J	Must be transferred with terminal type 111 A-SYN (ALS data synchronization) if the production combinations must be updated.	ALSSYNC=J

V:{module name}=…	The terminal sends version info of loaded modules (DLLs, scripts, etc.) in terminal status in format ...|V:dllname.dll=modules/function[[;version];date]|... This version information is stored in the software status as type "TERMINAL" and name "USER:{hydrauser}:{modulename}". Version: version number or time stamp in format "mm/dd/yyyy hh:mm:ss" Date: date of module generation in format mm/dd/yyyy (requires hymw.out / exe 7.2.1.353)	V:ftp.dll=FTP;7.2.1.5

9.3 Reloading lists on the terminal

The terminals read their internal lists at regular intervals, e.g. the list of assigned machines, currently running orders and currently logged on persons.

If data, which is included in terminal lists and displayed on the terminal, changes after activities on other terminals or on the HYDRA server, it might be useful to immediately update the data on the terminal and not only after the next cyclic update, which might take some minutes.

Using the string command "DLG=SCMD;53|", you can reload specific lists on the terminal and immediately display the changes.

Structure of dialog data:

„DLG=SCMD;53|TYP=INFO|ACTION=LST_RELOAD|LOAD=…|TNR=…|“

The following table contains the data that can be transmitted:

Field	Description	Example
LOAD=...	Transfer of lists that must be reloaded: • MNR: Machines assigned to the terminal • ANR: Operations logged on • PNR: Staff logged on • MAT: Input materials of assigned machines • RES: Resources of machines assigned to the terminal • PPKT: CAQ inspection points. The parameters RECTYP, BER, PANNR, PAUNR, EINTTYP, EINTNR and CAUSE must additionally be specified.	LOAD=ANR,MNR|
TNR=	Terminal number	TNR=117|

Only use the string command "DLG=SCMD;53|", if necessary. A frequent reload of lists means system strain for terminal and HYDRA server.

To trigger the loading of lists, the HYDRA server sends a message to the terminal using the network. To this end, you must release the port used so it is not blocked by a firewall. Terminals normally use port 9002, the PCC port 9005.

The string command "DLG=SCMD;53|" is not suitable if you want to initialize MDE data like machine status or counter readings on the terminal from the server.

9.4 Generating MLE outbound segments

Purpose

Use this service if you want to generate MLE outbound transactions, which are then transferred to another system (ERP, WMS,...).

BAPI: HSODATA.INSERT

Identifier	Description	Example
HSODATA.SEGNAM	Segment name (CHAR30)	HSODATA.SEGNAM=E2BP_PP_TIMTICKET
HSODATA.HYSYS	Logical target system (CHAR10)	HSODATA.HYSYS=FP
HSODATA.SDATA	Data record (CHAR1000)	HSODATA.SDATA=ABCDE
HSODATA.TYP	Type of data record: "HD" header record "CH" child record	HSODATA.TYP=HD
HSODATA.VERWEIS:HEADER	Reference to master segment when child segments are requested	HSODATA.VERWEIS:HEADER=4711
HSODATA.LCHILD	Identifier "last child segment" "J" only if HSODATA.TYP=CH "N" any other case	HSODATA.LCHILD=J

Example: Generating a master segment

DLG=HSODATA.INSERT|HSODATA.SEGNAM=SAMPLESEGNAMHD|HSODATA.HYSYS=FP|HSODATA.SDATA=THIS IS SAMPLE DATA|HSODATA.TYP=HD|

Example: Generating master segment and child segments

DLG=HSODATA.INSERT|HSODATA.SEGNAM=SAMPLESEGNAMHD|HSODATA.HYSYS=SAP|HSODATA.SDATA=THIS IS SAMPLE DATA|HSODATA.TYP=HD|

The reference to the master segment is transferred in the BAPI result. Use this reference in subsequent requests for the child segments as *HSODATA.VERWEIS:HEADER*.

RET=0|KT=|LT=|DATA=HSODATA|VERWEIS=4711|TYP=HD|ID=|

Child segment - not the last one:

DLG=HSODATA.INSERT|HSODATA.SEGNAM=SAMPLESEGNAMCH|HSODATA.HYSYS=SAP|HSODATA.SDATA=THIS IS SAMPLE DATA CHILD|HSODATA.TYP=CH|HSODATA.VERWEIS:HEADER=**4711**|HSODATA.LCHILD=**N**|

Child segment - the last one:

DLG=HSODATA.INSERT|HSODATA.SEGNAM=ECKTESTCH|HSODATA.HYSYS=SAP|HSODATA.SDATA= THIS IS ANOTHER SAMPLE DATA CHILD |HSODATA.TYP=CH|HSODATA.VERWEIS:HEADER=**4711**|HSODATA.LCHILD=**J**|

9.5 Generating logging entry

Use the dialog described below to generate an entry in the change management (logging). This requires a relevant configuration in the *Logging - configuration*. If you additionally activate the option *Log dialog data* in this configuration, then all data included in the dialog data is transferred to the database. This data can then be displayed in the application *Logging - change management.* If you use the button *Show,* the dialog data is displayed in a separate information window.

Structure of dialog data:

„DLG=SCMD;51|USR=...|BEARB=…|ACTION=…|KEY:1=...|DAT=…|ZEI=…|“

The following table contains the data that can be transmitted:

Identifier	Description
KEY:1=	The value of this identifier must be identical to the content of field *Object* in *Logging - configuration*.
ACTION=	The value of this identifier must be identical to the content of field *Action* in *Logging - configuration*.
BEARB=	The person specified in the *Modified by* field is transferred to the logging entry.
USR=	The specified *User* is transferred to the logging entry

Activate the option *Log dialog data* in the *Logging - configuration* to save all further identifiers except the keys.

Only if the return value is 0 (`RET=0|)`, the entry was properly saved.

Example:

Content of logging configuration

Field	Content
Object	TEST
Action	INSERT
Log dialog data	Activated

Dialog data:

```
DLG=SCMD;51|KEY:1=TEST|ACTION=INSERT|MNR=MyMachine|ANR=MyOperation|CNR=MyBa
tch|USR=2112|BEARB=12345|
```

Result:

For object "TEST", an action "INSERT" is logged. Other dialog data can be viewed as detail data.

9.6 Generating entry for dialog error log

Use the dialog described below to generate an entry in the HYDRA dialog error log.

Structure of dialog data:

„DLG=SCMD;52| . . . "

The following table contains the data that can be transmitted:

Identifier	Type / max. field length	Description
STA=	C1	Status: I=Info, W=Warning, E=Error Note: Default value is E, if ERRCODE does not equal 0. It is I, if ERRCODE is 0.
ERRCODE=	N8	Error code
ERRCLASS=	C10	Error class
EREIG=	C40	Event
BEM=	C80	Comment
USR=	N4	HYDRA user
DAT=	{mm/dd/yyyy}	Date in format mm/dd/yyyy

ZEI=	{seconds}	Time: point in time in format "seconds", i.e. in seconds since midnight Example: ZEI=36003 for the time 10:00:03
BEARB=	C10	Modified by
MNR=	N8/C8	Machine number (numeric/alphanumeric)
MGRP=	C8	Machine group
ANR=	C	HYDRA order number (fully defined key)
CNR=	C20	Batch number
PNR=	C10	Personnel number
KNR=	C10	Badge number

9.7 Triggering escalation

Use the dialog described below to trigger a configured escalation.

Structure of dialog data:

„DLG=ESKMSG.INSERT|ESKMSG.ID=. . .| "

The following table contains the data that can be transmitted:

Identifier	Type / max. field length	Description
ESKMSG.ESKID=	C40	Escalation that must be triggered. The escalations that must be triggered are configured in the database table esk_event_cfg
ESKMSG.RCV:ART=	C1	Recipient type of message P: Person (see ESKMSG.RCV:PNR) F: Function group (see ESKMSG.RCV:FKT)
ESKMSG.RCV:PNR=	N8	Person from HR master data. If you specify the parameter, it overrides the configured recipient.
ESKMSG.RCV:FKT=	C40	Function group. If you specify the parameter, it overrides the configured recipient.
ESKMSG.ATTACH=	C250	If the mail delivery is configured, you can specify a mail attachment here. The HYDRA server must be able to reach the file.
USR=	N4	HYDRA user
DAT=	{mm/dd/yyyy}	Date in format mm/dd/yyyy

ZEI=	{seconds}	Time: point in time in format "seconds", i.e. in seconds since midnight Example: ZEI=36003 for the time 10:00:03
Other identifiers		For each escalation, a fixed amount of variables is defined. The defined variables can be viewed in the application *Escalation configuration.* You can attach these variables here. Note: the total length of the dialog data string is limited to 1000 characters. The available variables of each escalation are stored in the table `esk_event_reg_var`.

Example: Event ANR.REGISTER_REMARK

Available variables

```
select * from esk_event_reg_var where event_id = 'ANR.REGISTER_REMARK' order
by key_nr desc
```

event_id	ref_kenn	f_kennung	use_cond	key_nr
ANR.REGISTER_REMARK	<NULL>	ANR.ANR	J	1
ANR.REGISTER_REMARK	<NULL>	ANR.AUNR	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.AGNR	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.AFOLG	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.UAGNR	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.SPLNR	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.MNR	J	0
ANR.REGISTER_REMARK	<NULL>	BEM	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.PNR	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.KNR	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.DAT	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.ZEI	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.ATK	J	0
ANR.REGISTER_REMARK	<NULL>	PNR.PVORNAME	J	0
ANR.REGISTER_REMARK	<NULL>	PNR.PNAME	J	0

Be careful to always specify the key field variables (`key_nr > 0`) in the dialog string. The key fields uniquely identify an event. As long as an escalation with this key field combination is open, another escalation cannot be generated. Only if the escalation is closed, a new escalation with these key fields can be triggered.

```
DLG=ESKMSG.INSERT|ESKMSG.ESKID=ANR.REGISTER_REMARK|ANR.ANR=222222220100|ANR
.ATK=Testartikel|ESKMSG.RCV:ART=P|ESKMSG.TO:PNR=2|BEM=hello from
PDM|USR=2001|
```

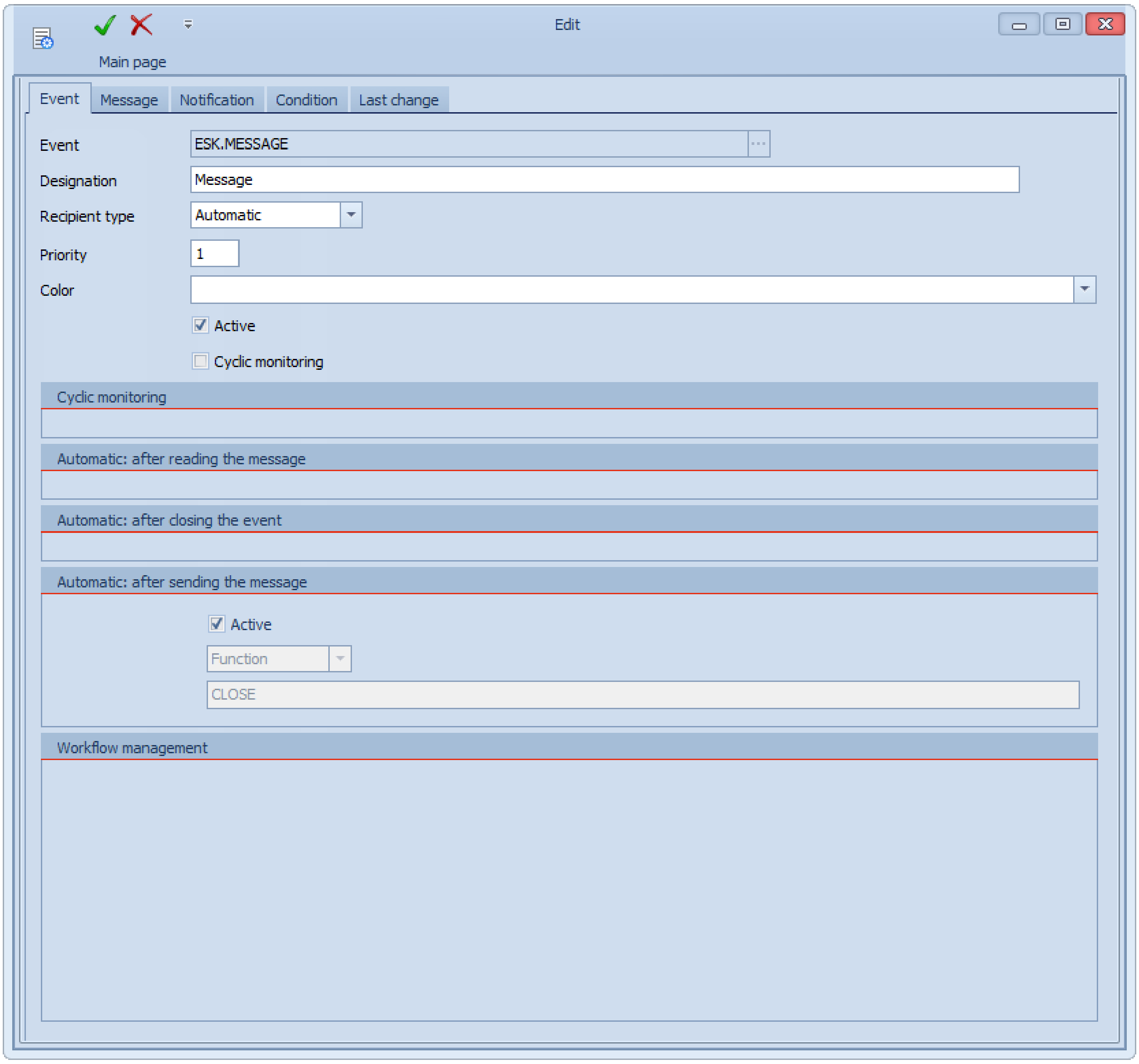
The escalation ANR.REGISTER_REMARK is triggered for operation 222222220100. The comment is "hello from PDM".

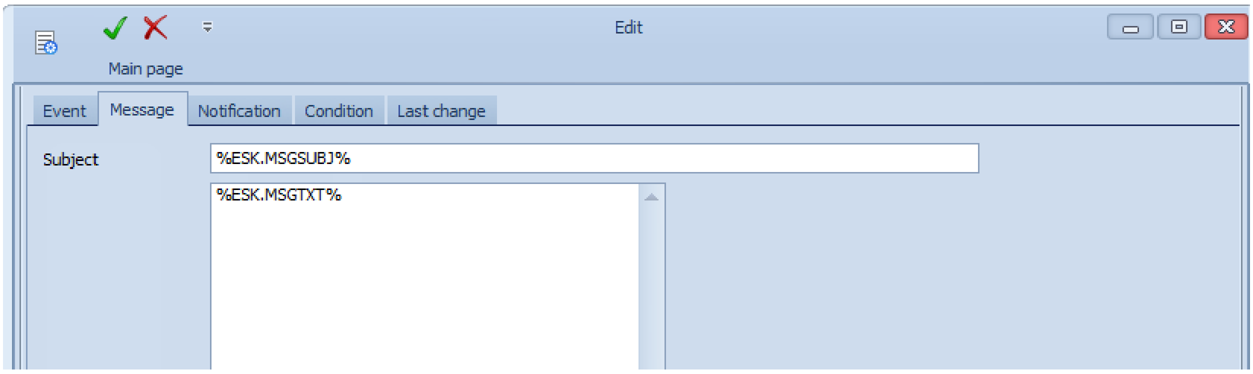
A mail attachment is added in this example.

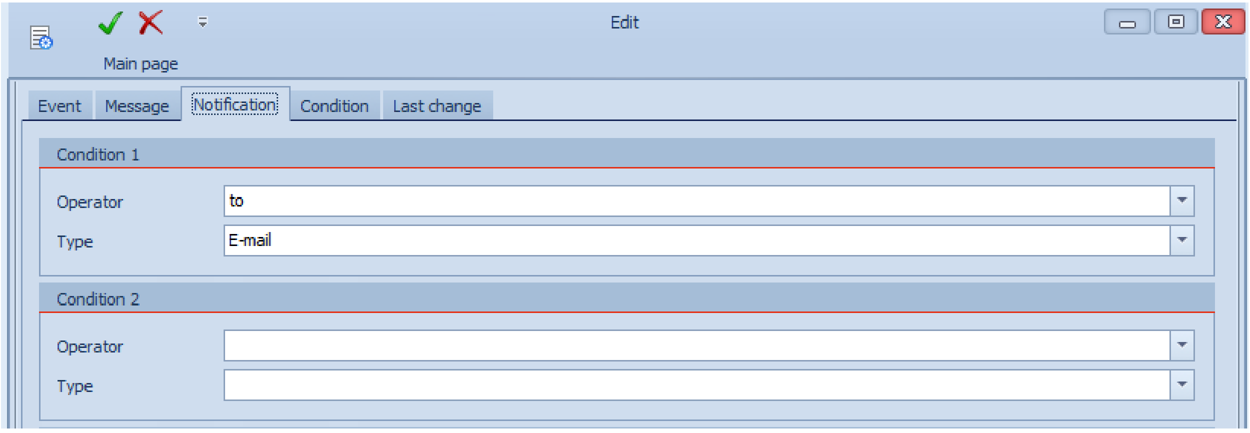
```
DLG=ESKMSG.INSERT|ESKMSG.ESKID=ANR.REGISTER_REMARK|ANR.ANR=222222220100|ANR
.ATK=Testartikel|ESKMSG.RCV:ART=P|ESKMSG.TO:PNR=2|BEM=hello from
PDM|ESKMSG.ATTACH=.\1\grafik\bde\maschine.jpg|USR=2001|
```

Example: Event ESK.MESSAGE

Use the escalation ESK.MESSAGE to send "generic" escalations. You can send mails, for example. To avoid open escalations, activate the option *Automatic: after sending the message* with CLOSE.


Edit
Main page
Event
Message
Notification
Condition
Last change
Event
ESK.MESSAGE
Designation
Message
Recipient type
Automatic
Priority
1
Color
Active
Cyclic monitoring
Cyclic monitoring
Automatic: after reading the message
Automatic: after closing the event
Automatic: after sending the message
Active
Function
CLOSE
Workflow management


Edit
Main page
Event
Message
Notification
Condition
Last change
Subject
%ESK.MSGSUBJ%
%ESK.MSGTXT%



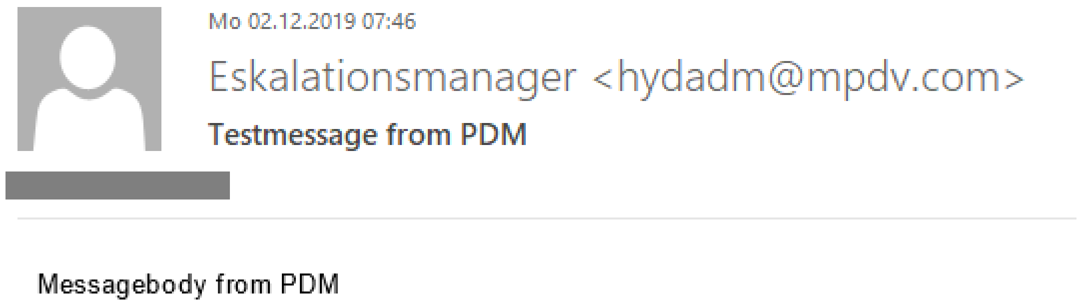
Available variables

```sql
select * from esk_event_reg_var where event_id = 'ESK.MESSAGE' order by key_nr
desc, f_kennung asc
```

EVENT_ID	REF_KENN	F_KENNUNG	USE_COND	KEY_NR
ESK.MESSAGE	<NULL>	ESK.MSGSUBJ	J	0
ESK.MESSAGE	<NULL>	ESKMSG.TO:ESKFKT	N	0
ESK.MESSAGE	<NULL>	ESKMSG.TO:PNR	N	0
ESK.MESSAGE	<NULL>	ESK.MSGTXT	J	0

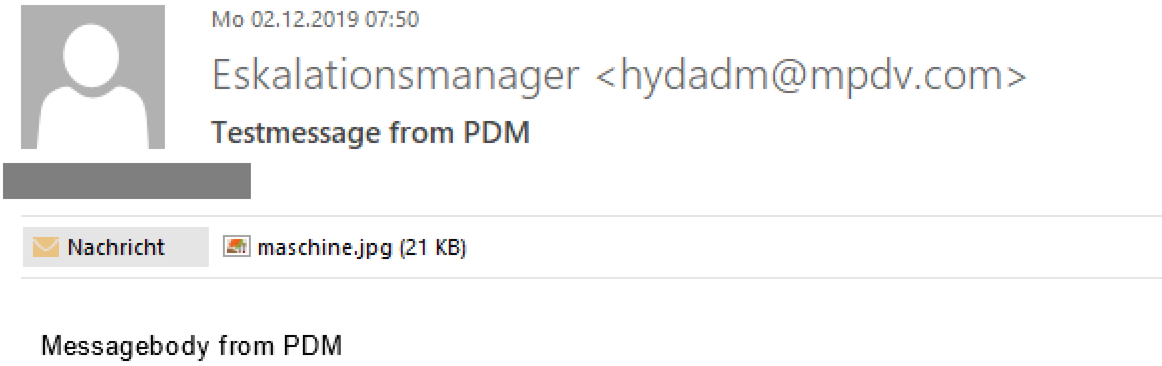
Identifier	Type / max. field length	Description
ESK.MSGSUBJ=	C50	If the mail delivery is configured, you can override the e-mail subject here.
ESK.MSGTXT=	C320	If the mail delivery is configured, you can override the e-mail body here.

```
DLG=ESKMSG.INSERT|ESKMSG.ESKID=ESK.MESSAGE|ESK.MSGSUBJ=Testmessage from
PDM|ESK.MSGTXT=Messagebody from
PDM|ESKMSG.RCV:ART=P|ESKMSG.TO:PNR=2|USR=2001|
```



You can send attachments with this escalation, too.

DLG=ESKMSG.INSERT|ESKMSG.ESKID=ESK.MESSAGE|ESK.MSGSUBJ=Testmessage from PDM|ESK.MSGTXT=Messagebody from PDM|ESKMSG.RCV:ART=P|ESKMSG.TO:PNR=2|**ESKMSG.ATTACH=.\1\grafik\bde\maschine.jpg**|USR=2001|



10 HYDRA Production Data Manager Basis - Master Data

10.1 Note on the descriptions of the basic dialogs

All mandatory fields that must be specified have the addition PK (primary key). All other fields are optional and are processed if they are transferred.

10.2 Terminal configuration

10.2.1 Edit terminal configuration (DLG=TNR.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

You can edit the terminal configuration using the BAPI calls described in this chapter.

Tables

Table	Key field	Description
terminals	terminal no. TNR.TNR	Terminal number (PK) for the terminal label 1
pzt_kenn	user no. TNR.TNR	Terminal number (PK) for the terminal label 2
terminal_status	terminal no. TNR.TNR	Terminal number (PK) for the terminal status

BAPI call

Acronym	Content / {type}	Description
DLG	TNR.INSERT	Create terminal configuration
	TNR.UPDATE	Change terminal configuration
	TNR.DELETE	Delete terminal configuration
	TNR.COPY	Copy terminal configuration
	TNR.LOCK	Lock terminal configuration for processing
	TNR.UNLOCK	Unlock terminal configuration after processing
	TNR.NEW	Read specification for new terminal configuration
	TNR.SELECT	Select terminal configuration

TNR.TNR	{N3}	PK Terminal number
TNR.TNR:Z	{N3}	PK new (target) terminal number to copy
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Returned

Acronym	Content / {type}	Description
TNR.TNR	{N3}	TNR.INSERT, TNR.COPY: Return of the terminal number for the created configuration

Validation checks

Error codes	Description
1680	Parameter with label TNR.TNR must be specified (SELECT, LOCK, UNLOCK, INSERT, UPDATE, DELETE, COPY)
1668	The terminal must be set up in the database (UPDATE).
1683	The terminal number (TNR.TNR) must be between 1 and 999. (INSERT, UPDATE)
1683	The terminal type (TNR.ART) must be specified. (INSERT, UPDATE)
1692	The terminal type 110 and 10 are only available with license HYD-ALS and are used for the ALS sub-system and the ALS data synchronization. (INSERT, UPDATE)
1684	In certain system constellations, several terminals per machine are permitted. No machine may be assigned to this terminal which is assigned to another MDE terminal. This means that the terminal must not be a MDE terminal if an assigned machine is also assigned to another MDE terminal.
1668	The terminal with the terminal number TNR.TNR is not existent. (SELECT, LOCK, UPDATE, DELETE, COPY)
764	The terminal with the terminal number TNR.TNR exists already. (INSERT)
764	The terminal with the terminal number TNR.TNR exists already. (COPY)
1666	The data record is currently locked by another user. (UPDATE, DELETE).
1666	The data record is currently locked by another user. (LOCK).

10.2.2 List for terminal configurations (DLG=TNR.LIST)

This BAPI call list the terminals and their configurations in HYDRA.

Tables

Table	Key field	Description
terminals	terminal no TNR.TNR	Terminal number (PK) for the terminal label 1
pzt_kenn	user no TNR.TNR	Terminal number (PK) for the terminal label 2
terminal_status	terminal no TNR.TNR	Terminal number (PK) for the terminal status

BAPI call

Acronym	Contents	Description
DLG	TNR.LIST	List of terminal configuration
TNR	{N3}	PK terminal number if an action is supposed to be carried out for a terminal.
TNR:FROM	{N3}	PK terminal number from/to if the action is supposed to be carried out for several terminals in the area *from* and *to*. (In this case, no plausibility check on the existence of the entry is made).
TNR:BIS	{N3}	
MOD:AKTIV	J|N	Listing active terminals J/N
MOD:INAKTIV	J|N	Listing inactive terminals J/N
FILE	{C256}	Specification of the file name for the list

Returned

Acronym	Contents	Description
—	—	—

Validation checks

Error codes	Description
1656	The file with the name DATEI cannot be written on.

10.2.3 Leave terminal update (DLG=TNR.PROGLADEN)

You can trigger a new installation of the terminal program with this BAPI call. After a successful installation, the option with label "TNR.PROGLADEN" is automatically reset to "N". **Tables**

Table	Key field	Description
terminal_status	terminal no TNR.TNR	Terminal number (PK) for the terminal status

BAPI call

Acronym	Contents	Description
DLG	TNR.PROGLAD EN	Leave or withdraw terminal update
TNR.TNR	{N3}	PK terminal number if an action is supposed to be carried out for a terminal.
TNR.TNR:VON	{N3}	PK terminal number from/to if the action is supposed to be carried out for several terminals in the area *from* and *to*. (In this case, no plausibility check regarding the existence of the entry is made).
TNR.TNR:BIS	{N3}	
TNR.PROGLAD EN	J|N	J: initiate terminal update N: withdraw terminal update

Returned

Acronym	Contents	Description
—	—	—

Validation checks

Error codes	Description
1661	You must specify the parameter with label TNR.TNR or TNR.TNR:VON and TNR.TNR:BIS.
1668	The terminal (if parameter TNR.TNR is specified) must be set up in the database.

10.2.4 Restart terminal (DLG=TNR.NEUSTART)

You can trigger a restart of the terminal with this BAPI call. This setting is transferred to the terminal when the next status message is sent. After restarting the terminal, the option with label "TNR.NEUSTART" is automatically reset to "N":

Tables

Table	Key field	Description
terminal_status	terminal no	Terminal number (PK) for the terminal status

	TNR.TNR	

BAPI call

Acronym	Contents	Description
DLG	TNR.NEUSTART	Initiate or withdraw the terminal restart
TNR.TNR	{N3}	PK terminal number if an action is supposed to be carried out for a terminal.
TNR.TNR:VON	{N3}	PK terminal number from/to if the action is supposed to be carried out for several terminals in the area *from* and *to*. (In this case, no plausibility check regarding the existence of the entry is made).
TNR.TNR:BIS	{N3}	
TNR.NEUSTART	J|N	J: initiate restart terminal N: withdraw terminal restart

Returned

Acronym	Contents	Description
—	—	—

Validation checks

Error codes	Description
1661	You must specify the parameter with label TNR.TNR or TNR.TNR:VON and TNR.TNR:BIS.
1668	The terminal (if parameter TNR.TNR is specified) must be set up in the database.

10.2.5 Terminal administration (DLG=TNR.ADMIN)

You can use this BAPI call to initiate administrative tasks (terminal update, terminal restart, request a diagnosis upload, ...).

Tables

Table	Key field	Description
terminal_status	terminal no TNR.TNR	Terminal number (PK) for the terminal status

BAPI call

Acronym	Contents	Description
DLG	TNR.NEUSTART	Initiate or withdraw the terminal restart

TNR.TNR	{N3}	PK terminal number if an action is supposed to be carried out for a terminal.
TNR.TNR:VON	{N3}	PK terminal number from/to if the action is supposed to be carried out for several terminals in the area *from* and *to*. (In this case, no plausibility check regarding the existence of the entry is made).
TNR.TNR:BIS	{N3}	
TNR.PROGLADEN	J|N|U	J: initiate terminal update N: withdraw terminal update U: do not execute any changes
TNR.NEUSTART	J|N|U	J: initiate restart terminal N: withdraw terminal restart U: do not execute any changes
TNR.TLVL	{C1}	Set trace level (terminal_status.protokoll)
TNR.BERLESEN	J|N|U	J: initiate the reading of the authorizations J: initiate the reading of the authorizations U: do not execute any changes
TNR.DIAGUPLOAD	J|N|U	J: initiate the creation of a diagnosis upload N: withdraw the creation of a diagnostic upload U: do not execute any changes
TNR.AKTIV	J|N|U	J: set terminal to active N: set terminal to inactive U: do not execute any changes
TNR.ERR:TNRIP	N	N: Reset error "double HYDRA user number"

Returned

Acronym	Contents	Description
—	—	—

Validation checks

Error codes	Description
1661	You must specify the parameter with label TNR.TNR or TNR.TNR:VON and TNR.TNR:BIS.
1668	The terminal (if parameter TNR.TNR is specified) must be set up in the database.

10.3 Function authorizations

10.3.1 Edit function authorizations (DLG=BEARBFKT.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

The function authorizations are edited with these BAPI calls.

Tables

Table	Key field	Description
fkt_tab	usr prg berechtigung art BEARBFKT.BEARB BEARBFKT.FKT BEARBFKT.BERECHT BEARBFKT.ART	The combination must be unique.

BAPI call

Acronym	Content / {type}	Description
DLG	BEARBFKT.INSERT	Create function authorization
	BEARBFKT.UPDATE	Change function authorization
	BEARBFKT.DELETE	Delete function authorization
	BEARBFKT.COPY	Copy function authorization
	BEARBFKT.LOCK	Lock function authorization for processing
	BEARBFKT.UNLOCK	Unlock function authorization after processing
	BEARBFKT.NEW	Read specification for new function authorization
	BEARBFKT.SELECT	Select function authorization
BEARBFKT.BEARB	{C10}	User
BEARBFKT.FKT	{C15}	Function
BEARBFKT.BERECHT	{N4}	Authorization

BEARBFKT.AR T	{C15}	Type F = function authorization P = function profile
MOD	C 1	Copy mode E - copy current selected authorization G - copy all authorizations F - copy missing authorizations
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Returned

Acronym	Content / {type}	Description
BEARB:FKT	{C15}	Current data record
FKT	{C15}	Current data record
BERECHT	{N4}	Current data record
ART	{C15}	Current data record

Validation checks

Error codes	Description
1660	The specified agent is invalid.
1661	A value is missing that is required for processing.
1662	A value relevant for processing is invalid.
1666	The function authorization is currently being maintained by another user.
1669	Data with the same key fields already exist.

10.3.2 List of function authorizations (DLG=BEARBFKT.LIST)

The BAPI call returns all specified function authorizations.

Tables

Table	Key field	Description
fkt_tab	usr prg berechtigung art BEARBFKT.BEARB BEARBFKT.FKT BEARBFKT.BERECHT BEARBFKT.ART	The combination must be unique.

BAPI call

Acronym	Contents	Description
DLG	BEARBFKT.LIST	List of function authorizations
FILE	{C256}	Specification of the file name for the list

10.4 Function profiles

10.4.1 Edit function profile (DLG=FKTPROF.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

The function authorizations are edited with these BAPI calls.

Tables

Table	Key field	Description
fkt_profil	profil prg berechtigung FKTPROF.FKTPROF FKTPROF.FKT FKTPROF.BERECHT	The combination must be unique.

BAPI call

Acronym	Content / {type}	Description
DLG	FKTPROF.INSERT	Create function profile
	FKTPROF.UPDATE	Change function profile
	FKTPROF.DELETE	Delete function profile
	FKTPROF.COPY	Copy function profile
	FKTPROF.LOCK	Lock function profile for processing
	FKTPROF.UNLOCK	Release lock for function profile after processing
	FKTPROF.NEW	Read requirement for a new function profile
	FKTPROF.SELECT	Select function profile
FKTPROF.FKTPROF	{C10}	Profiles
FKTPROF.FKT	{C15}	Function
FKTPROF.BERECHT	{C15}	Authorization
MOD	C 1	Copy mode

		G - copy all profiles F - copy missing profiles Delete mode E - delete single functions only G - delete total function
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Returned

Acronym	Content / {type}	Description
FKTPROF	{C15}	Current data record
FKT	{C15}	Current data record
BERECHT	{N4}	Current data record

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
1662	A value relevant for processing is invalid.
1666	The function profile is just edited by another function profile
1669	Data with the same key fields already exist.

10.4.2 List function profile (DLG=FKTPROF.LIST)

The BAPI call returns all defined function profiles.

Tables

Table	Key field	Description
fkt_profil	profil prg berechtigung FKTPROF.FKTPROF FKTPROF.FKT FKTPROF.BERECHT	The combination must be unique.

BAPI call

Acronym	Contents	Description
DLG	FKTPROF.LIST	Function profile list
FILE	{C256}	Specification of the file name for the list

10.5 Responsibility profiles

10.5.1 Edit responsibility profile (DLG=VABPROF.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

This BAPI call is used to edit responsibility profiles.

Tables

Table	Key field	Description
vab_profil	verantw_profil verantw_bereich VABPROF.VABPROF VABPROF.VAB	The combination must be unique.

BAPI call

Acronym	Content / {type}	Description
DLG	VABPROF.INSERT	Create responsibility profiles
	VABPROF.UPDATE	Change responsibility profile
	VABPROF.DELETE	Delete responsibility profile
	VABPROF.COPY	Copy responsibility profile
	VABPROF.LOCK	Lock processing for a responsibility profile
	VABPROF.UNLOCK	Release lock for responsibility profile after processing
	VABPROF.NEW	Read requirement for new responsibility profile
	VABPROF.SELECT	Select responsibility profile
VABPROF.VAB PROF	{C15}	Profile
VABPROF.VAB	{C15}	Area
MOD	{C1}	Copy mode G - copy all profiles F - copy missing profiles Delete mode E - Delete only responsibility area

		G - delete total profile
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Returned

Acronym	Content / {type}	Description
VABPROF	{C15}	Current data record
VAB	{C15}	Current data record
SELECT	{C1}	Current data record
USE	{C1}	Current data record
INSERT	{C1}	Current data record
UPDATE	{C1}	Current data record
DELETE	{C1}	Current data record

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
1662	A value relevant for processing is invalid.
1666	The responsibility profile is currently edited by another user
1669	Data with the same key fields already exist.

10.5.2 Responsibility Profiles (DLG=VABPROF.LIST)

The BAPI call returns all defined responsibility areas. The list can be restricted via parameter VABPROF.VABPROF (with wildcard support).

Tables

Table	Key field	Description
vab_profil	verantw_profil verantw_bereich VABPROF.VABPROF VABPROF.VAB	The combination must be unique.

BAPI call

Acronym	Contents	Description
DLG	VABPROF.LIST	List responsibility
FILE	{C256}	Specification of the file name for the list

10.6 Assignment responsibility areas

10.6.1 Edit assignment responsibility areas (DLG=BEARBVABPROF.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

You can edit the assignment of the responsibility area with these BAP calls.

Tables

Table	Key field	Description
vab_berechtigung	usr berechtigung_art berechtigung_wert BEARBVABPROF.BEARB:VAB BEARBVABPROF.ART BEARBVABPROF.WERT	The combination must be unique.

BAPI call

Acronym	Content / {type}	Description
DLG	BEARBVABPROF.INSERT	Create assignment responsibility areas
	BEARBVABPROF.UPDATE	Change assignment responsibility area
	BEARBVABPROF.DELETE	Delete assignment responsibility area
	BEARBVABPROF.COPY	Copy assignment responsibility area
	BEARBVABPROF.LOCK	Lock assignment for responsibliy area for processing
	BEARBVABPROF.UNLOCK	Release lock for assignment for responsibility are after processing
	BEARBVABPROF.NEW	Read the requirement for new assigment of responsibility area
	BEARBVABPROF.SELECT	Select assignment for responsibility area
BEARBVABPROF.BEARB:VAB	{C15}	User

BEARBVABPROF.ART	{C1}	Type V – responsibility area P - profile
BEARBVABPROF.WERT	{C15}	Responsibility area or profile
MOD	{C1}	Copy mode G - copy all assignments of a user F - copy all missing assignments Delete mode E - delete only separate assignments G - delete all entries of a user
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
1662	A value relevant for processing is invalid.
1666	The assignment is currently edited by another user
1669	Data with the same key fields already exist.
1660	The specified agent is invalid.
425	The responsibility profile is not existent

10.6.2 Assignment list (DLG=BEARBVABPROF.LIST)

The BAPI call returns all defined assignments. The list can be restricted with the following parameter: BEARBVABPROF.BEARB:VAB und BEARBVABPROF.WERT (with wildcard support).

Tables

Table	Key field	Description
vab_berechtigung	usr berechtigung_art berechtigung_wert BEARBVABPROF.BEARB:VAB BEARBVABPROF.ART BEARBVABPROF.WERT	The combination must be unique.

BAPI call

Acronym	Contents	Description
DLG	BEARBVABPROF. LIST	Assignment list
FILE	{C256}	Specification of the file name for the list

10.7 User administration

10.7.1 Edit user (DLG=BEARB.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

You use these BAPI call to edit users.

Tables

Table	Key field	Description
user_tab	usr BEARB.BEARB	PK user

BAPI call

Acronym	Content / {type}	Description
DLG	BEARB.INSERT	Create users
	BEARB.UPDATE	Change user
	BEARB.DELETE	Delete user
	BEARB.COPY	Copy user
	BEARB.LOCK	Lock user for processing
	BEARB.UNLOCK	Release lock for user after processing
	BEARB.NEW	Read requirement for new user
	BEARB.SELECT	Select user
BEARB.BEARB	{C10}	User
BEARB.BEARB:Z	{C10}	Target user during copying
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Returned

Acronym	Content / {type}	Description
BEARB	{C10}	Current data record

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
1662	A value relevant for processing is invalid.
1666	The user data is currently being edited by another user.
1669	Data with the same key fields already exist.
3020	The user account is locked!

Plausibility checks for password guidelines

Error codes	Description
3274	According to the password guideline, the password must not contain the user name.
3704	The entered password is not valid because the password is in the exclusion list.
3275	According to the password guideline, the password is not made-up of enough letters.
3276	According to the password guideline, the password is not made-up of enough numbers.
3277	According to the password guideline, the password is not made-up of enough special characters.
3278	According to the password guideline, the pasword is too short.
2379	According to the password guideline, the password has invalid characters.
3283	The entered password is equivalent to the current one.
3280	The entered password is already in use.
3281	Internal error during the access to the password history

10.7.2 User list (DLG=BEARB.LIST)

The BAPI call returns all defined users.

Tables

Table	Key field	Description
user_tab	usr BEARB.BEARB	PK user

BAPI call

Acronym	Contents	Description
DLG	BEARB.LIST	User list
FILE	{C256}	Specification of the file name for the list

10.7.3 User login (DLG=BEARB.LOGIN)

The BAPI call perfoms the login process of a user

Tables

Table	Key field	Description
user_tab	usr BEARB.BEARB	PK user

BAPI call

Acronym	Content / {type}	Description
DLG	BEARB.LOGIN	Logon user
BEARB.BEARB	{C10}	User
TYP	{C10	T = Hydra Mobile Login

Returned

Acronym	Content / {type}	Description
SYSPRO	{C20}	System logs If full: There is an alarm file alarm.txt available!
LANG	{N4}	Language key
MSGSYNC	{C1}	Escalation management: J = There are new messages including the automatic display of messages when you log on to the console.
PWDW	{C1}	J = password has run out

Validation checks

Error codes	Description

1661	A value is missing that is required for processing.
1667	Number of licenses were exceeded.
1665	Editor/ and or the password is invalid

10.7.4 User logout (DLG=BEARB.LOGOUT)

The BAPI call performs a logout process of a user.

Tables

Table	Key field	Description
user_tab	usr BEARB.BEARB	PK user

BAPI call

Acronym	Content / {type}	Description
DLG	BEARB.LOGOUT	Log off user
BEARB.BEARB	{C10}	User
TYP	{C10	T = Hydra Mobile Login

10.8 Locked data records

10.8.1 Delete locked data records (DLG=BEARBFKT.DELETE)

You can use these BAPI calls to delete locked data records.

Tables

Table	Key field	Description
hyd_lock	key1 key2 key3 key4 key5 param1 param2 hydrauser bearb LOCK.KEY:1 LOCK.KEY:2 LOCK.KEY:3 LOCK.KEY:4 LOCK.KEY:5 LOCK.PARAM:1 LOCK.PARAM:2 LOCK.USR LOCK.BEARB	The combination must be unique. Note: The fields key1..key5 und param1 and param2 can also be zero.

BAPI call

Acronym	Content / {type}	Description
DLG	LOCK.DELETE	Delete the locked data record
LOCK.BEARB	{C10}	User
LOCK.USR	{N4}	User number
LOCK.KEY:1	{C40}	BAPI – Object name e.g. MNR
LOCK.KEY:2	{C40}	Value of the BAPI object e.g. machin number
LOCK.KEY:3	{C40}	
LOCK.KEY:4	{C40}	
LOCK.KEY:5	{C40}	

LOCK.PARAM:1	{C40}	
LOCK.PARAM:2	{C40}	
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
101	General error message that appears if the selected data (tables or files) is not available.

10.9 Paths

10.9.1 Edit paths (DLG=PATH.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

You use these BAPI call to edit HYDRA paths.

Tables

Table	Key field	Description
hy_path	path PATH.PATH	PK path

BAPI call

Acronym	Content / {type}	Description
DLG	PATH.INSERT	Create path
	PATH.UPDATE	Change path
	PATH.DELETE	Delete path
	PATH.COPY	Copy path
	PATH.LOCK	Lock the path for processing
	PATH.UNLOCK	Release lock for path after processing
	PATH.NEW	Read requirement for new path
	PATH.SELECT	Select path
PATH.PATH	{C8}	Path
PATH.SCHEMA	{C10}	Schema
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
1669	Data with the same key fields already exist.

1666	The path is currently edited by another user.

10.9.2 Path list (DLG=PATH.LIST)

The BAPI call returns all defined paths.

Tables

Table	Key field	Description
hy_path	path PATH.PATH	PK path

BAPI call

Acronym	Contents	Description
DLG	PATH.LIST	List paths
FILE	{C256}	Specification of the file name for the list

10.10 Licensing

10.10.1 Edit licenses (DLG=LIC.INSERT, DELETE)

You use these BAPI calls to edit licenses.

Tables

Table	Key field	Description
hyd_license	productkey category value valid_till licensedate majorrelease minorrelease LIC.PROKEY LIC.CAT LIC.VAL LIC.VALIDTILL LIC.LICDAT LIC.VER:MAJ LIC.VER:MIN	PK combination from all fields

BAPI call

Acronym	Content / {type}	Description
DLG	LIC.INSERT	Create license
	LIC.DELETE	Delete license
	LIC.USE	Assign license
LIC.PROKEY	{C10}	Product name
LIC.CAT	{C10}	Category, fix "HYDRA"
LIC.VAL	{C10}	
LIC.VALIDTILL	{DATE}	
LIC.LICDAT	{DATE}	
LIC.VER:MAJ	{N4}	
LIC.VER:MIN	{N4}	
As an alternative, a license file can be transferred.		

MOD	{C1}	D – file
FILE	{C255}	File name
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
1669	Data with the same key fields already exist.
1671	The license data are incorrect or are not suitable to your system.
2027	The processing mode is for this posting not allowed.

10.10.2 License list (DLG=LIC.LIST)

The BAPI call returns licenses.

BAPI call

Acronym	Contents	Description
DLG	LIC.LIST	Liste Lizenzen
FILE	{C256}	Specification of the file name for the list
MOD	{C1}	List mode V - available licenses U- list of used licenses L - list of all licenses

10.11 Client

10.11.1 Client login and logout (DLG=CLIENT.LOGIN, LOGOUT)

These BAPI calls are used to assign a SessionID to a HYDRA USR number.

Tables

Table	Key field	Description
client_status	sessionid usr CLIENT.SID CLIENT.USR	PK SessionID PK HYDRA user number

BAPI call

Acronym	Content / {type}	Description
DLG	CLIENT.LOGIN	Create assignment between a Session ID and a HYDRA user number.
	CLIENT.LOGOUT	Delete assignment between a SessionID and a HYDRA number.
CLIENT.SID	{C100}	SessionID of the client
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Returned

Acronym	Content / {type}	Description
USR	{N5}	HYDRA user number

10.12 INI configuration

10.12.1 INI - edit configuration (DLG=INI.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT, IMPORT, EXPORT)

You use these BAPI calls to edit the INI configuration

Tables

Table	Key field	Description
hyd_ini	hydrauser ininame INI.USR INI.INI	The combination must be unique.

BAPI call

Acronym	Content / {type}	Description
DLG	INI.INSERT	INI - create configuration
	INI.UPDATE	INI - change configuration
	INI.DELETE	INI - delete configuration
	INI.LOCK	INI – configuration for processing
	INI.UNLOCK	Release lock for INI - configuration after processing
	INI.NEW	Read requirement for new INI configuration
	INI.SELECT	INI - select configuration
	INI.IMPORT	Import a complete INI configuration
	INI.EXPORT	Export of a complete INI configuration in a file.
INI.USR	{N4}	HYDRA user number
INI.INI	{C80}	INI - name
FILE	{C255}	Filename for import/export
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
1666	The configuration is currently edited by another user
1669	Data with the same key fields already exist.
101	General error message that appears if the selected data (tables or files) is not available.
1611	General database fields

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10.12.2 INI list - configurations (DLG=INI.LIST)

The BAPI call returns all defined INI configurations. The list can be restricted via parameter INI.INI (with wildcard support).

Tables

Table	Key field	Description
hyd_ini	hydrauser ininame INI.USR INI.INI	The combination must be unique.

BAPI call

Acronym	Contents	Description
DLG	INI.LIST	List INI configurations
FILE	{C256}	Specification of the file name for the list

10.12.3 Edit INI sections (DLG=INIDATA.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

You use these BAPI calls to edit INI sections.

Tables

Table	Key field	Description
hyd_ini_data	ini_verweis section ident INIDATA.INIVERWEIS INIDATA.SECTION INIDATA.KEY	The combination must be unique.

BAPI call

Acronym	Content / {type}	Description
DLG	INIDATA.INSERT	Create INI section

	INIDATA.UPDATE	Change INI section
	INIDATA.DELETE	Delete INI section
	INIDATA.LOCK	Lock INI section for processing
	INIDATA.UNLOCK	Release lock for INI section after processing
	INIDATA.NEW	Read requirement for new INI section
	INIDATA.SELECT	Select INI section
INIDATA.INIVERWEIS	{N8}	Reference to the corresponding INI configuration in the hyd_ini table.
INIDATA.SECTION	{C80}	Section name
INIDATA.KEY	{C80}	Key name
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
1666	The configuration is currently edited by another user
1669	Data with the same key fields already exist.
101	General error message that appears if the selected data (tables or files) is not available.
1611	General database fields

10.12.4 List of the INI sections (DLG=INIDATA.LIST)

The BAPI call returns all defined sections for the INI entry. The list can be restricted via parameter INIDATA.SECTION (with wildcard support). The INI configuration must be specified via parameters INI.INI and INI.USR in the list request.

Tables

Table	Key field	Description
hyd_ini_data	ini_verweis section ident INIDATA.INIVERWEIS INIDATA.SECTION INIDATA.KEY	The combination must be unique.

BAPI call

Acronym	Contents	Description
DLG	INIDATA.LIST	INI sections list
FILE	{C256}	Specification of the file name for the list

10.13 Number ranges

10.13.1 Edit number ranges (DLG=NRKREIS.INSERT, UPDATE, DELETE, LOCK, UNLOCK)

Number ranges are the basis to assign numbers automatically, e.g. order numbers, posting numbers, etc.

Example: When creating an order and no order number was specified, then the order number can be automatically calculated according to the specified number range.

You can use two different keys to process existing data records.

1. The internal number of the data record (NRKREIS.VERWEIS)
2. The combination of the logical key fields Object (NRKREIS.OBJTYP), Type (NRKREIS.ART), Key (NRKREIS.KEY) and Key Value (NRKREIS.VAL).

BAPI call

Acronym	Content / {type}	Description
DLG	NRKREIS.INSERT	Create number range
	NRKREIS.UPDATE	Change number range
	NRKREIS.DELETE	Delete number range
	NRKREIS.LOCK	Release lock number ranges
	NRKREIS.UNLOCK	Release lock for number ranges after processing
NRKREIS.OBJTYP	C 20	Generation object
NRKREIS.ART	C 6	Generation type (V=automatic number assignment)
[NRKREIS.ART:1]	C 1	Optional access to the first sign of the generation type
[NRKREIS.ART:2]	C 1	Optional access to the second sign of the generation type
NRKREIS.KEY	C 20	Optional key The value of the key is managed with identification NRKREIS.VAL NRKREIS.KEY and NRKREIS.VAL can in principle be assigned as required. The following keys are specified for number ranges processed by default in the system: „AART“: order type „SAG“: merged operation (only for object = "AUNR") „MELDART“: posting type (only for object IHNR for WRM-IH)
NRKREIS.VAL	C 20	Wert zu NRKREIS.KEY e.g. PP01 = order type for NRKREIS.KEY=AART. If "*" , then no differentiation is made. All order types without a specific number range get this entry.
NRKREIS.VERWEIS	N	Internal data record number (not for NRKREIS.INSERT)
NRKREIS.PRAEFIX	C 5	Prefix placed before the number, e.g. "GK (overheads)" could be the prefix for GK orders.
NRKREIS.BER:VON	C 20	Start of the value range
NRKREIS.BER:BIS	C 20	End of the value range
NRKREIS.BER:VAL	C 20	Last assigned value
NRKREIS.VERGCODE	C 6	Assignment code "NUM" = numeric assignment

Example

```
DLG=NRKREIS.INSERT|NRKREIS.OBJTYP=MyObj|NRKREIS.ART=V|NRKREIS.KEY=MyKey|NRKREIS.VAL=MyVal|NRKREIS.PRAEFIX=MYPRF|NRKREIS.BER:V
ON=0000000|NRKREIS.BER:BIS=9999999|NRKREIS.BER:VAL=100|NRKREIS.VERGCODE=NUM|
```

10.13.2 Number range list (DLG=NRKREIS.LIST)

The list gets all data records of the number ranges. The identification can be deduced from the editing functions.

10.13.3 Create new numbers (DLG=NRKREIS.CREATENR)

The function NRKREIS.CREATENR creates a new number in the configured number range.

The new number for allocation code "NUM" results from the value +1 last allocated.

Create a type definition for the user field in order to format values.

Field in the type definition	Contents
General / type	Number range object
General / name	Any
General / description	Any
General / length	Required length for formatting the new number. It might be the case that the length requires a prefix included in the number range.
Terminal / fill character	0: the new value is filled with leading zeros at the beginning -: the new value is filled with zeros starting at the end

BAPI call

Acronym	Contents	Description
DLG	NRKREIS.CREATENR	Generate a new number from the number range
NRKREIS.OBJTYP	C 20	Generation object
NRKREIS.ART	C 6	Generation type (V=automatic number assignment)
NRKREIS.KEY	C 20	Key

		Qualification Qualification within a generation object (for order number, order type, merged operation)in a generation object (for order number, e.g. order type, merged operation) AART = order type SAG = merged operation (only for object = "AUNR") „MELDART“ = posting type (only for object IHNR for WRM-IH)
NRKREIS.VAL	C 20	Value e.g. PP01 = order type; If "*" , then no distinction is made or all order types for which no specific number range is defined are assigned this entry.
[NRKREIS.VERWEIS]	N	Internal data record number Alternative key for the combination of NRKREIS.OBJTYP, NRKREIS.ART, NRKREIS.KEY and NRKREIS.VAL.

Returned

Acronym	Contents	Description
BER:VAL	Cx	New number not formatted
NR	Cx	Prefix + new number not formatted
NR:FILLED	Cx	Only if a type definition for the user field exists for the object of the number range:

Validation checks

Error codes	Description
3300	Incorrect assignment code Currently only the assignment code NUM is supported.
3301	If the value range of the number range is exceeded, an error message pops up.

Example

```
DLG=NRKREIS.CREATENR|NRKREIS.OBJTYP=MyObj|NRKREIS.ART=V|NRKREIS.KEY=MyKey|NRKREIS.VAL=MyVal|
```

Rückgabe:Return:

```
RET=0|KT=|LT=|DATA=NRKREIS|OBJTYP=MyObj|ART=V|KEY=MyKey|VAL=MyVal|VERWEIS=19|NR=MYPR106|PRAEFIX=MYPR|BER:VON=0000000|BER:BIS=
9999999|BER:VAL=106|VERGCODE=NUM|BEARB=12345|BEARBDAT=05/29/2019|BEARBZEI=35096|NR:FILLED=MYPR0000000000000106|
```

11 HYDRA Production Data Manager BDE/MDE - Data Collection

11.1 Note on the descriptions of the input dialogs

All fields that are mandatory and must be specified are highlighted using a gray background color. All other fields are optional and are processed when passed.

The MES order number ANR is the *fully defined key* of the object. This number always includes all specifications (in the order displayed)

- Order number
- Sequence (only relevant in combination with the additional function BDE-APF)
- Operation number
- Split number (only relevant in combination with the additional functions BDE-SGG)

The MES order number is of type C (char). The field lengths specified during the HYDRA setup must be respected. The length of the MES order number ANR is the sum total of the lengths of the different fields.

If you use the split functionality (BDE-SSG), the length of the MES order number ANR depends on whether it is a split operation or not. If it is not a split OP, the MES order number is reduced by the length of the split number.

Examples:

- Order number length: 8, operation number length: 4, no sequence, no splits:
  → ANR length: 12
- Order number length: 8, operation number length: 4, sequence number length: 1, no splits:
  → ANR length: 13

- Order number length: 8, operation number length: 4, sequence number length: 0, split number length: 1:
  → ANR length if no split operation: 12
  → ANR length if split operation: 13

- Order number length: 8, operation number length: 4, sequence number length: 1, split number length: 1:
  → ANR length if no split operation: 13
  → ANR length if split operation: 14

11.2 Order, staff and machine postings

11.2.1 Note on automatically recorded quantities

All input dialogs (e.g. A_UN, P_AB), which include manually recorded values (EGR:GUT, EGR:AUS), support the automatically recorded values AGR.C:[1..n], AGE.C:[1..n], AGG.C:[1..n], AGB.C:[1..n], AGR:HUB. For a better overview, these values are *not* displayed in the different input dialogs (exception: M_AST, A_AUN and A_ASW).

The following table generally defines the valid IDs for automatic quantities from counters in the dialog data:

Identifier	Type / max. field length	Description
AGR.C:[1..n]=	Double	Automatically recorded counter delta of counters 1...n
AGE.C:[1..n]=	C4	Unit of counter (e.g. PCS) for future use
AGG.C:[1..n]=	N4	Reason for counter (e.g. scrap reason)
AGB.C:[1..n]=	C10	Evaluation for counter (yield, scrap, rework, sample: GUT, AUS, NCH, PRB) - see note below.
AGR:HUB=	Double	Automatically recorded cycles

Do not use the following acronyms as of MW 4.0pe:		
AGR:GUT=	*Double*	*Automatic yield*
AGR:AUS=	*Double*	*Automatic scrap*
AGG:AUS=	*N4*	*Scrap reason*
AGE:GUT=	*C4*	*Unit of yield quantity*
AGE:AUS=	*C4*	*Unit of scrap quantity*

Example

AGR.C:1=47|AGB.C:1=GUT|AGR.C:2=1|AGB.C:2=AUS|AGG.C:2=12|....|AGR:HUB=..|

Notes

- Automatic counters are not yet used for calculation. A calculation using the partitioning and the pulse factor and other quantity accounts (e.g. when the total quantity is recorded) is only performed in the HYDRA server.
- The counter evaluation – identifier AGB.C:{..} – is an optional identifier and need not be transferred.
  - If the identifier AGB.C:{..} is not included in the dialog data, the counter is posted by default according to the evaluation specified in the counter configuration.
  - If the identifier AGB.C:{..} is included in the dialog data, then the counter is posted according to this evaluation. Here, the evaluation specified in the counter configuration is overridden.

11.2.2 Notes on manually recorded quantities

11.2.2.1 Recording of manual quantities in alternative quantity units

All input dialogs (e.g. A_UN, P_AB), which include manually recorded values in primary quantity unit (EGR:GUT, EGR:AUS,etc.), also support the recording of quantities in the following alternative quantity units:

- Base quantity unit
- Secondary quantity unit
- Tertiary quantity unit

The table below provides an overview of the available and valid IDs of the manual recording of quantities in the dialog data:

Identifier	Type / max. field length	Description
EGR:GUT= or EGR:GUTP=	Double	Yield recorded in primary quantity unit
EGG:GUT= or EGG:GUTP=	N4	Optional reason for the yield recorded in primary quantity unit

Identifier	Type / max. field length	Description
EGR:GUTB=	Double	Yield recorded in base quantity unit
EGG:GUTB=	N4	Optional reason for the yield recorded in base quantity unit
EGR:GUTS=	Double	Yield recorded in secondary quantity unit
EGG:GUTS=	N4	Optional reason for the yield recorded in secondary quantity unit
EGR:GUTT=	Double	Yield recorded in tertiary quantity unit
EGG:GUTT=	N4	Optional reason for the yield recorded in tertiary quantity unit
EGR:AUS= or EGR:AUSP=	Double	Scrap quantity recorded in primary quantity unit
EGG:AUS= or EGG:AUSP=	N4	Optional reason for the scrap recorded in primary quantity unit
EGR:AUSB=	Double	Scrap recorded in base quantity unit
EGG:AUSB=	N4	Optional reason for the scrap recorded in base quantity unit
EGR:AUSS=	Double	Scrap recorded in secondary quantity unit
EGG:AUSS=	N4	Optional reason for the scrap recorded in secondary quantity unit
EGR:AUST=	Double	Scrap recorded in tertiary quantity unit
EGG:AUST=	N4	Optional reason for the scrap recorded in tertiary quantity unit
EGR:NCH= or EGR:NCHP=	Double	Rework quantity recorded in primary quantity unit
EGG:NCH= or EGG:NCHP=	N4	Optional reason for the rework recorded in primary quantity unit
EGR:NCHB=	Double	Recorded rework quantity in base quantity unit
EGG:NCHB=	N4	Optional reason for the rework recorded in base quantity unit
EGR:NCHS=	Double	Rework quantity recorded in secondary quantity unit
EGG:NCHS=	N4	Optional reason for the rework recorded in secondary quantity unit
EGR:NCHT=	Double	Rework recorded in tertiary quantity unit
EGG:NCHT=	N4	Optional reason for the rework recorded in tertiary quantity unit
EGR:PRB= or EGR:PRBP=	Double	Open quantity / problem quantity recorded in primary quantity unit
EGG:PRB= or EGG:PRBP=	N4	Optional reason for the open quantity / problem quantity recorded in primary quantity unit
EGR:PRBB=	Double	Open quantity / problem quantity recorded in base quantity unit

Identifier	Type / max. field length	Description
EGG:PRBB=	N4	Optional reason for the open quantity / problem quantity recorded in base quantity unit
EGR:PRBS=	Double	Open quantity / problem quantity recorded in secondary quantity unit
EGG:PRBS=	N4	Optional reason for the open quantity / problem quantity recorded in secondary quantity unit
EGR:PRBT=	Double	Open quantity / problem quantity recorded in tertiary quantity unit
EGG:PRBT=	N4	Optional reason for the open quantity / problem quantity recorded in tertiary quantity unit

Examples:

- Upload of a part quantity with yield in primary quantity unit and scrap and scrap reason in secondary quantity unit
  `DLG=A_TR|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100473100200|`
  `EGR:GUTP=1|EGR:AUSS=2|EGG:AUSS=1|`
- Upload of a part quantity with yield in primary quantity unit and secondary quantity unit
  `DLG=A_TR|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100473100200|`

  `EGR:GUTP=1| EGR:GUTS=11|`

If quantities are recorded in alternative quantity units, these quantities take priority over the conversion using the conversion factors stored for the operation or over a formula stored in the system. In this case, no conversion into the quantity unit recorded is performed.

But the conversion into configured, but not recorded quantity units is performed.

Example: The yield can be recorded in primary quantity unit (EGR:GUTP) and secondary quantity unit (EGR:GUTS) in a dialog and the quantity is then converted into the base quantity and tertiary quantity if the conversion factors are stored.

The conversion of quantities into quantity units that are not manually recorded is performed according to the following priority:

1. Primary quantity: conversion into secondary quantity and tertiary quantity

2. Secondary quantity: conversion into primary quantity and tertiary quantity

3. Tertiary quantity: conversion into primary quantity and secondary quantity

The base quantity unit is generally used for conversions between primary, secondary and tertiary quantities.

For further information on the conversion of quantities and examples, refer to the document MBL_Quantity_Conversion.pdf.

11.2.2.2 Recording of quantities with different reasons

The recording of scrap, rework and open quantities with different reasons in one dialog is supported. To do so, you must complement the IDs for the scrap quantity and the reason with a numeric and consecutive index – starting with 1.

The below table for scrap also applies for rework (identifier *NCH* instead of *AUS*) and open quantities (identifier *PRB* instead of *AUS*).

Identifier	Type / max. field length	Description
EGR:AUS#[1..n]= or EGR:AUSP#[1..n]=	Double	Scrap quantity recorded in primary quantity unit with consecutive index
EGG:AUS#[1..n]= or EGG:AUSP#[1..n]=	N4	Optional reason for the scrap recorded in primary quantity unit and the relevant index
EGR:AUSB#[1..n]=	Double	Scrap recorded in base quantity unit with consecutive index
EGG:AUSB#[1..n]=	N4	Optional reason for the scrap recorded in base quantity unit and the relevant index
EGR:AUSS#[1..n]=	Double	Scrap recorded in secondary quantity unit with consecutive index
EGG:AUSS#[1..n]=	N4	Optional reason for the scrap recorded in secondary quantity unit and the relevant index
EGR:AUST#[1..n]=	Double	Scrap recorded in tertiary quantity unit with consecutive index
EGG:AUST#[1..n]=	N4	Optional reason for the scrap recorded in tertiary quantity unit and the relevant index

Example:

- Upload of a part quantity with yield quantity and two scrap quantities with reasons 20 and 30

```
DLG=A_TR|ANR=ANR=AAA2100473100200|EGR:GUTP=123|EGR:AUSP#1=2|EGG:AUSP#1=20|
EGR:AUSP#2=1|EGG:AUSP#2=20|
```

11.2.3 Collection of user fields

User fields can be collected in dialogs with the following events:

- Log on operations

- Posting of part quantities for operations
- Interrupt operations
- Log off operations
- Finish operations
- Quantity uploads for operations
- Log off staff

The collected user fields are integrated into the data of the operation and the order-related postings resulting from the posting event.

Sometimes it is unwanted that the collected user fields overwrite the user fields of the operation. If the field identification ANR:SETUSRFLD is transmitted with the value "N", the system does not integrate the collected user fields in the *Operation*, but only in the *Order-related postings*.

Identifier	Data type	Description
FU:1 to FU:6	Date	User fields for a date
FU:7 to FU:22	N (=night)	Integer with value range from -2147483647 to 2147483647
FU:23 to FU:28	DECIMAL(18,6)	Decimal number with a value range of 12 digits before comma and a precision of 6 digits after comma.
FU:29 to FU:44	C1	User field for 1 character
FU:45 to FU:50	C10	User field for a text with a maximum of 10 characters
FU:51 to FU:64	C20	User field for a text with a maximum of 20 characters
FU:65 to FU:66	C40	User field for a text with a maximum of 40 characters
ANR:SETUSRFLD	C1 = {J/N}	With ANR:SETUSRFLD=N, the user fields of the dialog (FU:1 to FU:66) are not integrated into the operation data. The user fields are only transferred to the BDE log record (if a log record is created).

11.2.4 Log operation on (DLG=A_AN)

<table>
<tr><th>Identifier</th><th>Type / max. field length</th><th colspan="2">Description</th></tr>
<tr><td>ANR=</td><td>C</td><td colspan="2">MES order number (fully defined key)</td></tr>
<tr><td>MNR=</td><td>C8</td><td colspan="2">Workplace/machine number</td></tr>
<tr><td>PNR=</td><td>C10</td><td rowspan="2">or</td><td>Personnel number</td></tr>
<tr><td>KNR=</td><td>C10</td><td>Staff badge number</td></tr>
</table>

Identifier	Type / max. field length	Description
CNR=	C20	Batch number (output batch)
MST=	N4	Set new machine status or automatically change into *Production* status (customer-specific)

Example 1: Log OP on

DLG=A_AN|USR=2106|DAT=02/17/2000|ZEI=47972|ANR=0004990701|MNR=4560|KNR=9999 99|

Example 2: Log OP and batch on

DLG=A_AN|USR=2106|DAT=02/17/2000|ZEI=47972|ANR=0004990701|MNR=4560|KNR=9999 99|CNR=CHV0001112|

11.2.5 Log operation and person on (DLG=A_P_AN)

Identifier	Type / max. field length	Description	
ANR=	C	MES order number (fully defined key)	
MNR=	C8	Workplace/machine number	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number
CNR=	C20	Batch number (output batch)	
MST=	N4	Set new machine status or automatically change into *Production* status (customer-specific)	

Example 1: Log operation and person on

DLG=A_P_AN|USR=2106|DAT=02/17/2000|ZEI=47972|ANR=0004990701|MNR=4560|KNR=99 9999|

Example 2: Log batch and person on

DLG=A_P_AN|USR=2106|DAT=02/17/2000|ZEI=47972|ANR=0004990701|MNR=4560|KNR=12 3456|CNR=CHV0001112|

11.2.6 Posting of part quantity (Partial confirmation) (DLG=A_TR)

Identifier	Type / max. field length	Description	
ANR=	C	MES order number (fully defined key)	
MNR=	C8	Workplace/machine number	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number
EGR:GUT=	N8	Yield recorded	
EGR:AUS=	N8	Scrap recorded	
EGG:GUT=	N4	Reason	
EGG:AUS=	N4	Scrap reason (required if EGR:AUS is passed)	
EGE:GUT=	C4	Unit	
EGE:AUS=	C4	Unit of scrap quantity	

Example 1: Posting of part quantity with scrap and scrap reason

DLG=A_TR|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100473100200|
MNR=60610|EGR:GUT=1|EGR:AUS=2|EGG:AUS=1|

11.2.7 Interrupt operation (DLG=A_UN)

Identifier	Type / max. field length	Description	
ANR=	C	MES order number (fully defined key)	
MNR=	C8	Workplace/machine number	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number
EGR:GUT=	N8	Yield recorded	
EGR:AUS=	N8	Scrap recorded	
EGG:GUT=	N4	Reason	
EGG:AUS=	N4	Scrap reason	

Identifier	Type / max. field length	Description
EGE:GUT=	C4	Unit of yield quantity
EGE:AUS=	C4	Unit of scrap quantity
MST=	N4	New machine status

Example 1: Interrupt OP with input of quantity

DLG=A_UN|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|
MNR=60610|EGR:GUT=1|

Example 2: Interrupt OP with input of quantity and unit

DLG=A_UN|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|
MNR=60610|EGR:GUT=1|EGE:GUT=ST|

Example 3: Interrupt OP with input of quantity, scrap and scrap reason

DLG=A_UN|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|
MNR=60610|EGR:GUT=1|EGR:AUS=2|EGG:AUS=1|

11.2.8 Log operation off (DLG=A_AB)

<table>
<tr><th>Identifier</th><th>Type / max. field length</th><th colspan="2">Description</th></tr>
<tr><td>ANR=</td><td>C</td><td colspan="2">MES order number (fully defined key)</td></tr>
<tr><td>MNR=</td><td>C8</td><td colspan="2">Workplace/machine number</td></tr>
<tr><td>PNR=</td><td>C10</td><td rowspan="2">or</td><td>Personnel number</td></tr>
<tr><td>KNR=</td><td>C10</td><td>Staff badge number</td></tr>
<tr><td>EGR:GUT=</td><td>N8</td><td colspan="2">Yield recorded</td></tr>
<tr><td>EGR:AUS=</td><td>N8</td><td colspan="2">Scrap recorded</td></tr>
<tr><td>EGG:GUT=</td><td>N4</td><td colspan="2">Reason</td></tr>
<tr><td>EGG:AUS=</td><td>N4</td><td colspan="2">Scrap reason</td></tr>
<tr><td>EGE:GUT=</td><td>C4</td><td colspan="2">Unit of yield quantity</td></tr>
<tr><td>EGE:AUS=</td><td>C4</td><td colspan="2">Unit of scrap quantity</td></tr>
<tr><td>MST=</td><td>N4</td><td colspan="2">New machine status</td></tr>
</table>

Example 1: Log OP off with input of quantity

DLG=A_AB|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|MNR=60610|EGR:GUT=1|

Example 2: Log OP off with input of quantity and unit

DLG=A_AB|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|MNR=60610|EGR:GUT=1|EGE:GUT=ST|

Example 3: Log OP off with input of quantity, scrap and scrap reason

DLG=A_AB|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|MNR=60610|EGR:GUT=1|EGR:AUS=2|EGG:AUS=1|

11.2.9 Finish operation (DLG=A_BE)

Finish prepared or interrupted operation.

<table>
<tr><th>Identifier</th><th>Type / max. field length</th><th colspan="2">Description</th></tr>
<tr><td>ANR=</td><td>C</td><td colspan="2">MES order number (fully defined key)</td></tr>
<tr><td>MNR=</td><td>C8</td><td colspan="2">Workplace/machine number</td></tr>
<tr><td>PNR=</td><td>C10</td><td rowspan="2">or</td><td>Personnel number</td></tr>
<tr><td>KNR=</td><td>C10</td><td>Staff badge number</td></tr>
<tr><td>EGR:GUT=</td><td>N8</td><td colspan="2">Yield recorded</td></tr>
<tr><td>EGR:AUS=</td><td>N8</td><td colspan="2">Scrap recorded</td></tr>
<tr><td>EGG:GUT=</td><td>N4</td><td colspan="2">Reason</td></tr>
<tr><td>EGG:AUS=</td><td>N4</td><td colspan="2">Scrap reason</td></tr>
<tr><td>EGE:GUT=</td><td>C4</td><td colspan="2">Unit of yield quantity</td></tr>
<tr><td>EGE:AUS=</td><td>C4</td><td colspan="2">Unit of scrap quantity</td></tr>
</table>

The time of the posting (acronym ZEI) must not be at shift change. Background: CORU-Call #118905.

Example 1: Finish OP with input of quantity

DLG=A_BE|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|MNR=60610|EGR:GUT=1|

Example 2: Finish OP with input of quantity and unit

DLG=A_BE|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|MNR=60610|EGR:GUT=1|EGE:GUT=ST|

Example 3: Finish OP with input of quantity, scrap and scrap reason

DLG=A_BE|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|
MNR=60610|EGR:GUT=1|EGR:AUS=2|EGG:AUS=1|

11.2.10 Quantity upload (DLG=A_MR)

You can use the quantity upload to upload quantities for orders, which are not logged on at the moment. This way, you can correct a quantity of an order without having to log the order on and off.

Note:
This dialog does not change the postings performed for the machine performance.

<table>
<tr><th>Identifier</th><th>Type / max. field length</th><th colspan="2">Description</th></tr>
<tr><td>ANR=</td><td>C</td><td colspan="2">MES order number (fully defined key)</td></tr>
<tr><td>MNR=</td><td>C8</td><td colspan="2">Workplace/machine number</td></tr>
<tr><td>PNR=</td><td>C10</td><td rowspan="2">or</td><td>Personnel number</td></tr>
<tr><td>KNR=</td><td>C10</td><td>Staff badge number</td></tr>
<tr><td>EGR:GUT=</td><td>N8</td><td colspan="2">Yield recorded</td></tr>
<tr><td>EGR:AUS=</td><td>N8</td><td colspan="2">Scrap recorded</td></tr>
<tr><td>EGG:GUT=</td><td>N4</td><td colspan="2">Reason</td></tr>
<tr><td>EGG:AUS=</td><td>N4</td><td colspan="2">Scrap reason</td></tr>
<tr><td>EGE:GUT=</td><td>C4</td><td colspan="2">Unit</td></tr>
<tr><td>EGE:AUS=</td><td>C4</td><td colspan="2">Unit of scrap quantity</td></tr>
</table>

The time of the posting (acronym ZEI) must not be at shift change. Background: CORU-Call #118905. We think that the problem that occurred with A_BE can also occur here with A_MR.

Example 1: Quantity upload with scrap and scrap reason

DLG=A_MR|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100473100200|
MNR=60610|EGR:GUT=1|EGR:AUS=2|EGG:AUS=1|

11.2.11 Log person on (DLG=P_AN)

Identifier	Type / max. field length	Description	
MNR=	C8	Workplace/machine number	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number

Example 1: Log person on

DLG=P_AN|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|MNR=871555|

11.2.12 Log person off (DLG=P_AB)

Identifier	Type / max. field length	Description	
MNR=	C8	Workplace/machine number	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number
EGR:GUT=	N8	Yield recorded	
EGR:AUS=	N8	Scrap recorded	
EGG:GUT=	N4	Reason	
EGG:AUS=	N4	Scrap reason	
EGE:GUT=	C4	Unit	
EGE:AUS=	C4	Unit of scrap quantity	

Example 1: Log person off

DLG=P_AB|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|MNR=871555|

Example 2: Log person off with input of quantity, scrap and scrap reason

DLG=P_AB|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|MNR=123456|EGR:GUT=1|
EGR:AUS=2|EGG:AUS=1|

11.2.13 Log off all persons from machine (DLG=P_AAB)

Identifier	Type / max. field length	Description
MNR=	C8	Workplace/machine number

Identifier	Type / max. field length	Description
EGR:GUT=	N8	Yield recorded
EGR:AUS=	N8	Scrap recorded
EGG:GUT=	N4	Reason
EGG:AUS=	N4	Scrap reason
EGE:GUT=	C4	Unit
EGE:AUS=	C4	Unit of scrap quantity

Example 1: Log off all persons from machine

DLG=P_AAB|USR=2106|DAT=02/17/2000|ZEI=47972|MNR=871555|

Example 2: Log all persons off with input of quantity, scrap and scrap reason

DLG=P_AAB|USR=2106|DAT=02/17/2000|ZEI=47972|MNR=123456|EGR:GUT=1|EGR:AUS=2|EGG:AUS=1|

11.2.14 Change machine status (DLG=M_MST)

<table>
<tr><th>Identifier</th><th>Type / max. field length</th><th colspan="2">Description</th></tr>
<tr><td>MNR=</td><td>C8</td><td colspan="2">Workplace/machine number</td></tr>
<tr><td>MST=</td><td>N4</td><td colspan="2">New machine status:
Malfunction reason and/or machine status.
Example: 1 = Production, 99 = general disturbance/malfunction
The definition depends on the machine configuration.</td></tr>
<tr><td>PNR=</td><td>C10</td><td rowspan="2">or</td><td>Personnel number</td></tr>
<tr><td>KNR=</td><td>C10</td><td>Staff badge number</td></tr>
<tr><td>IZY=</td><td>N8</td><td colspan="2">Actual cycle in seconds per 1000 cycles</td></tr>
<tr><td>BEM=</td><td>C40</td><td colspan="2">You can enter a free comment</td></tr>
<tr><td>PROGNDAUER=</td><td>C4</td><td colspan="2">Expected time that the new status will be available.
For documentation purposes, this information is stored with the status event. If the field has the value 0 or is empty, the new status will be available "for an indefinite period".</td></tr>
<tr><td colspan="4">Additional data for the escalation management</td></tr>
<tr><td>MSGPRIO=</td><td>N1</td><td colspan="2">Priority:
1 = highest, 2 = high, 3 = normal, 4 = low, 5 =lowest</td></tr>
</table>

MSGCLASS=	C1	Information class/importance I = Information, W = Warning, E = Error
MSGRCV=	C40	Recipient/addressee, e.g. group of plant managers

Example 1: Person changes machine status

DLG=M_MST|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|MNR=871555|MST=3|

Note:

If the machine status is changed via PDM dialog and if the workplaces/machines are simultaneously managed by terminals (machines with MDE), the machine status change is not automatically displayed on the terminal that manages the machine.

11.2.15 Automatic status update (DLG=M_AST)

Purpose: Continuation of the machine status and posting of automatically recorded quantities and cycles.

Identifier	Type / max. field length	Description
MNR=	C8	Workplace/machine number
AGR:HUB=	N8	Automatically recorded cycles
TRGEN=	C1 = {J/N}	With TRGEN=J, the system triggers operation-related upload(s) of part quantities for the automatic quantities recorded up to now, which result in order-related log records of record type "T". If several operations are active at the machine, one upload of a part quantity is triggered per operation. If scrap counters with different reasons are recorded for the machine, several uploads of part quantities are triggered per operation.
IZY=	N8	Current actual cycle of the machine in seconds per 1000 cycles
AGR.C:[1..n]=	N8	Automatically recorded counter quantity of counter 1...n
AGE.C:[1..n]=	C4	Unit of counter (e.g. PCS) for future use
AGG.C:[1..n]=	N4	Reason for counter (e.g. scrap reason)

Identifier	Type / max. field length	Description
AGB.C:[1..n]=	C10	Evaluation of counter: - GUT = yield - AUS = scrap - NCH = rework - PRB = open quantity (problem quantity)

Example:

DLG=M_AST|USR=2106|DAT=02/17/2000|ZEI=47972|MNR=871555|AGR:HUB=11|AGR.C:1=47|AGB.C:1=GUT| AGR.C:2=1|AGB.C:2=AUS|AGG.C:2=12|

Note:

All input dialogs (e.g. A_UN, P_AB), which include manually recorded values (EGR:GUT, EGR:AUS), support the automatically recorded values AGR.C:[1..n], AGE.C:[1..n], AGG.C:[1..n], AGB.C:[1..n], AGR:HUB.

Important note on counters:
Automatic counters are not yet used for calculation.
A calculation using the partitioning and the pulse factor and other quantity accounts (e.g. when the total quantity is recorded) is only performed in the server.
Example:
AGR.C:1=47|AGB.C:1=GUT|AGR.C:2=1|AGB.C:2=AUS|AGG.C:2=12|....|AGR:HUB=..|

11.2.16 Change of the target quantity (DLG=A_SMG)

Identifier	Type / max. field length	Description	
ANR=	C	MES order number (fully defined key)	
MNR=	C8	Workplace/machine number	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number
SGR:GUT=	N8	New target quantity	
SGE:GUT=	C4	Unit of the target quantity	

Example 1: Person changes target quantity

DLG=A_SMG|USR=2106|DAT=02/17/2000|ZEI=47972|MNR=100|ANR=12Y19D0001|SGR:GUT=123|SGE:GUT=ST|

11.2.17 Change of target cycle (DLG=M_SZY)

Identifier	Type / max. field length	Description	
MNR=	C8	Workplace/machine number	
ANR=	C	MES order number (fully defined key)	
SZY=	N8	Target cycle in seconds per 1000 cycles	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number

Example: Person changes target quantity

DLG=M_SZY|USR=2106|DAT=02/17/2000|ZEI=58371|MNR=100|SZY=3600|

The new target cycle is set for all OPs currently logged on to the machine.

11.2.18 Change of partitioning (DLG=M_TLG)

Identifier	Type / max. field length	Description	
MNR=	C8	Workplace/machine number	
ANR=	C	MES order number (fully defined key)	
TLG=	N8	Partitioning	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number

Example: Person changes partitioning

DLG=M_TLG|USR=2106|DAT=02/17/2000|ZEI=34392|MNR=100|TLG=2|

The new partitioning is set for all OPs currently logged on to the machine.

11.2.19 Logging of the production lock (DLG= M_PSPERRE)

Identifier	Type / max. field length	Description	
PNR=	C10	or optional	Personnel number
KNR=	C10		Staff badge number
ACTIVE=	C1	Setting or resetting the production lock: J … set N … reset	

Example: Person activates the production lock

DLG=M_PSPERRE|USR=2106|DAT=02/17/2000|ZEI=34392|ACTIVE=J|

Note:

If a person (KNR or PNR) is entered, the system checks if the person is authorized to change the production lock. The configuration is made on the MOC in the application HR master data.

11.2.20 BDE comment (DLG=HY_BEM)

<table>
<tr><th>Identifier</th><th>Type / max. field length</th><th colspan="2">Description</th></tr>
<tr><td>ANR</td><td>C</td><td colspan="2">MES order number (fully defined key)</td></tr>
<tr><td>BEM</td><td>C60</td><td colspan="2">The BDE comment must not exceed 60 characters.</td></tr>
<tr><td>MNR</td><td>C8</td><td colspan="2">If no workplace is specified, the BDE comment is saved for the workplace where the operation is planned.</td></tr>
<tr><td>PNR=</td><td>C10</td><td rowspan="2">or</td><td>Personnel number</td></tr>
<tr><td>KNR=</td><td>C10</td><td>Staff badge number</td></tr>
</table>

Example: Person records a BDE comment for the operation.

DLG=HY_BEM|USR=2106|ANR=210047310200|BEM=problems with material supply|KNR=1111|DAT=02/17/2000|ZEI=34392|

Notes:

- The input and display length on the AIP is limited to 60 characters.
- BDE comments can trigger an escalation. If you have configured in the escalation management that the comment text is displayed as subject of the message, then only the first 50 characters are displayed.

11.3 Postings made with shift change

Shift change records must be made in the correct order along with the other postings. All postings for the respective shift must be transferred <u>before</u> the shift end record/shift change record.

11.3.1 Shift end (A_AUN)

Identifier	Type / max. field length	Description
MNR=	C8	Workplace/machine number
DAT=	mm/dd/yyyy	Date of shift end in format "mm/dd/yyyy"

ZEI=	Seconds	Point in time of the shift end in seconds after midnight (example: 10:00:03 is *ZEI=36003*)
AGR:HUB=	N8	Automatically recorded cycles
AGR.C:[1..n]=	N8	Automatically recorded counter quantity of counter 1...n
AGE.C:[1..n]=	C4	Unit of counter (e.g. PCS) for future use
AGG.C:[1..n]=	N4	Reason for counter (e.g. scrap reason)
AGB.C:[1..n]=	C10	Evaluation of counter: - GUT = yield - AUS = scrap - NCH = rework - PRB = open quantity (problem quantity)

Example: **Shift end on 28-APR-2002 at 6:00**

DLG=A_AUN|DAT=04/28/2002|ZEI=21600|MNR=4560

11.3.2 Beginning of shift (A_AAN)

Identifier	Type / max. field length	Description
MNR=	C8	Workplace/machine number
DAT=	mm/dd/yyyy	Date of shift end in format "mm/dd/yyyy"
ZEI=	Seconds	Point in time of the shift end in seconds after midnight (example: 10:00:03 is *ZEI=36003*)

Example: **Beginning of shift on 28-APR-2002 at 6:00**

DLG=A_AAN|DAT=04/28/2002|ZEI=21600|MNR=4560

11.3.3 Shift change (A_ASW)

The shift change triggers a shift end and a beginning of shift.

Identifier	Type / max. field length	Description
MNR=	C8	Workplace/machine number
DAT=	mm/dd/yyyy	Date of shift end in format "mm/dd/yyyy"
ZEI=	Seconds	Point in time of the shift end in seconds after midnight (example: 10:00:03 is *ZEI=36003*)

AGR:HUB=	N8	Automatically recorded cycles
AGR.C:[1..n]=	N8	Automatically recorded counter quantity of counter 1...n
AGE.C:[1..n]=	C4	Unit of counter (e.g. PCS) for future use
AGG.C:[1..n]=	N4	Reason for counter (e.g. scrap reason)
AGB.C:[1..n]=	C10	Evaluation of counter: - GUT = yield - AUS = scrap - NCH = rework - PRB = open quantity (problem quantity)

Example: **Shift change on 28-APR-2002 at 6:00**

DLG=A_ASW|DAT=04/28/2002|ZEI=21600|MNR=4560

11.4 Reading BDE/MDE data

11.4.1 Machine info

The list of machine info is provided by the command DLG=LIST;10 and filed in the directory HYDRADIR\spool\.

The definition of the file name and the respective path is case-sensitive.

Structure of dialog data:

"DLG=LIST;10|DATEI={file name}|DAT=...|ZEI=...|USR=...|...“

The list includes the current quantity of the shift. If the unit changes when the order changes, the shift-related counters are reset to 0. The list includes the following data:

Identifier	Field designation	Description
MNR	Machines	Machine no.
MGRP	Group	Machine group
MBEZK	Mach. des.	Machine name
MBEZL	Det. machine des.	Detailed machine name
KST	Cost center	Cost center of the machine
MST	Status	No. of current machine status or 20000 = no shift 30000 = status not assigned
MSTTXT	Status text	Text of the current machine status

Identifier	Field designation	Description
PKENN	Pchar	Production control characteristic of the current status P: "Production" (defined once per machine) S: Malfunctions (defined x times per machine) A: "General malfunction" (defined once per machine) N: No shift or Status "Not assigned"
MSDATB	Date	Start date of machine status
MSZEIB	Time	Start time of machine status
MSDAUER	Duration	Time that the machine status lasts up to now
BMKNR	RPA no.	Current RPA text number
BMKTXT	RPA	Current RPA abbreviation
BMKTXTL	RPA text	Current RPA text
AGR:BMK01 to AGR:BMK12	RPA1 to RPA 12	Time posted to RPA01 to Time posted to RPA12
SKNR	S	current shift number
SKDATB	Date beg.	Date of beginning of shift
SKZEIB	Time beg.	Time of beginning of shift
SKDATE	Date end	Date of end of shift
SKZEIE	Time end	Time of end of shift
AGR:GUT	Yield	Yield quantity of current shift
AGR:AUS	Scrap	Scrap quantity of current shift
AGR:HUB	Strokes	Cycles of current shift
TLG	Partitioning	Current partitioning
SZY	Target cycle	Current target cycle
KFGKZ:1 to KFGKZ:5	Cfg_ch1 to Cfg_ch5	Configuration indicator where machine configuration data is filed
ZLO	Target	Default for: - Prepare order - Start order
ABFALL	Waste	Customer-specific
MEHRFKZ	MKz.	Customer-specific
MPUFF	Input buffer	Input buffer
MDE_MASCH	MDE machine	MDE machine

Identifier	Field designation	Description
MPL_MOD	Batch mode	Batch mode
AUTOMENGE	AutoQuantity	Autom. quantity
AGE:GUT	Unit	Unit of yield quantity
AGE:AUS	Unit	Unit of scrap quantity
IZYSM	Piece / minute	Actual cycle piece per minute
MNR.PARAM:1 MNR.PARAM:15	Mach. parameter 1 to Mach. parameter 15	General machine parameters without parameters 5 and 11
IMPFAKT	Mach. parameter 5	is machine parameter 5
STLG	Machine partitioning	is machine parameter 11
LINIE	Line	Line of the aggregate
LINIE.REF	Reference aggregate	Reference aggregate Y/N
MART	Machine type	Machine type E: Single workplace G: Group workplace
EXTTYP	External device	K: No external device J: DS100 N: MT3 E: Engel interfacing A: Arburg interfacing
EXTSNR	External device: serial number	depending on EXTTYP: A: Serial number E: Serial number
EXTID	External device: Device address / class	depending on EXTTYP Y/N: ID of master terminal is number of MT3/DS100 A: Machine class - is ARBURG control system
ADEMSDATB	Date	
ADEMSZEIB	Time	
MULTIAG	Multiple ch.	
MATPUF:OUT	Target	

Identifier	Field designation	Description
MATPUF:IN	Input buffer	
OPT:CNRAUTOGEN	Batch no. autom.	
OPT:CNRAGAN	Log on OP CEAN/UMB	
TYP	Machine type	
OPT:CHV	Batch management	
OPT:CNRPRN	Ticket print	From product version **MPL 7.2.5** no longer supported.
DIGIN:CAWL	Digital Input batch change	From product version **MPL 7.2.5** no longer supported.
MNR.VISVERBRBLZ	Consumption balance	New as of product version **MPL 7.2.5**
DIGOUT:SMENGE	Output target qty. reached	
TNR	Terminal number	
IZYABW	Cycle extension	
OPT:AUS	Scrap processing	
OPT:AUSAUTO	Scrap automatic	
OPT:AUSMANU	Scrap manual	
OPT:AUSPROT		
OPT:AUSRS232		
DIGIN:GUT		
DIGIN:AUS	Scrap counter	
DIGOUT:MSPERRE		
ANZSTAKT		
OPT:GUTMANU	Yield manual	
PARAM:12	Customer-specific	
OPT:GUTRS232		
DIGIO	Free input/output	

Identifier	Field designation	Description
CAT		
OPT:PDV	Activation of process data processing	
BDEJMOD	BDE year model	
UEBART		
DLGSTRG	Dialog control	
ICON	Symbol	
OPT:GUTAUTO		
UEBDAUER		
ANZPALMNR	No. of processing stations/palett machines	
MNR.LISTSUFFIX	Dialog control	
RESVERWEIS	Resource ID	
RESWART.BEZ	Designation	
RESWART.ART	Maintenance type	
RESWART.WARTSTA	Status	
RESWART.WARTKL	Class	
RESWART.AKTWERT	Current value	
RESWART.NAEWERT	Value next maintenance	
TNRMSPERRE	TNRMSPERRE	Internal use for CTAIP
TNRPSPERRE	TNRPSPERRE	Internal use for CTAIP

Example:

DLG=LIST;10|DAT=02/22/2005|ZEI=40000|USR=2101|DATEI= ./spool/masch_list.dat|

11.4.1.1 Dynamic extension of machine list

You can dynamically extend the machine list described and add further acronyms, if required. You can then add further fields to the standard list.

Structure of dialog data:

"DLG=LIST;10|DATEI={file name}|MOD={mode}|AKRO=OPT:VLISTMOD;ANZPALMNR;DLGSTRG;.....|....“

In the dialog data, the list of additional fields is started with the identifier AKRO=. The listed acronyms are then separated by semicolon.

MPDV Mikrolab provides a list a acronyms that can be used.

11.4.1.2 Extended machine list

Structure of dialog data:

"DLG=LIST;10|DATEI={file name}|MOD={mode}|....“

Identifier	Field designation	Description
AGR:GUTP AGR:GUT	Machine-specific yield	Primary quantity
AGR:GUTS		Secondary quantity
AGR:GUTT		Tertiary quantity
AGR:GUTB		Basic quantity
AGR:AUSP AGR:AUS	Machine-specific scrap	Primary quantity
AGR:AUSS		Secondary quantity
AGR:AUST		Tertiary quantity
AGR:AUSB		Basic quantity
AGR:NCHP	Machine-specific rework quantity	Primary quantity
AGR:NCHS		Secondary quantity
AGR:NCHT		Tertiary quantity
AGR:NCHB		Basic quantity
AGR:PRBP	Machine-specific open quantity (problem quantity)	Primary quantity
AGR:PRBS		Secondary quantity
AGR:PRBT		Tertiary quantity

Identifier	Field designation	Description
AGR:PRBB		Basic quantity
CTR:1 CTR:2 CTR:3 CTR:4 CTR:5 CTR:6	Counter 1...6	
DIV	Pulse factor	
OPT:MDETLG	Option "order-specific partitioning"	
EGE:GUTP	Primary quantity unit	
EGE:GUTS	Secondary quantity unit	
EGE:GUTT	Tertiary quantity unit	
EGE:GUTB	Base quantity unit	
UMRFAKTP:Z	Conversion of primary quantity	See note **Conversion factors for base quantity**
UMRFAKTP:N	Conversion of primary quantity	
UMRFAKTS:Z	Conversion of secondary quantity	
UMRFAKTS:N	Conversion of secondary quantity	
UMRFAKTT:Z	Conversion of tertiary quantity	
UMRFAKTT:N	Conversion of tertiary quantity	

Identifier	Field designation	Description
VERB:GUT	Offset of manual yield quantity against - AUS - NCH - PRB - empty = no offset	See note **Offset of manual quantities**
VERB:AUS	Offset of manual scrap against - GUT - NCH - PRB - empty = no offset	
VERB:NCH	Offset of manual rework quantity against - GUT - AUS - PRB - empty = no offset	
VERB:PRB	Offset of manual problem quantity = open quantity against - GUT - AUS - NCH - empty = no offset	
OPT:GUTMANUTAKT	Calculation of the relevant manual quantity additionally as cycles (i.e. additionally as AGR:HUB)	

Identifier	Field designation	Description
OPT:AUSMANUTAKT	Calculation of the relevant manual quantity additionally as cycles (i.e. additionally as AGR:HUB)	
OPT:NCHMANUTAKT	Calculation of the relevant manual quantity additionally as cycles (i.e. additionally as AGR:HUB)	
OPT:PRBMANUTAKT	Calculation of the relevant manual quantity additionally as cycles (i.e. additionally as AGR:HUB)	
VISLIST3	Display 3rd list	The third list on the terminals is dynamically controlled using this indicator.
VISFHMTNRAAN	Show material/PRT list when OP is logged on	

Conversion factors for base quantity

You use the conversion factors to convert the primary, secondary and tertiary quantities against the base quantity. You use these conversion factors, for example, when updating target quantities.

You use a numerator and denominator for the conversion factors. This way, you can also use a decimal value (meaning a figure with decimal places) as conversion factor.

Example:
- Base quantity unit: square meter M2
- Primary quantity unit: piece ST
- 1 piece = 2 square meters.

In this case to be stored as
- numerator (= UMRFAKTP:Z) 2 and
- denominator (= UMRFAKTP:N) 1.

Note: Offset of manual quantities

When a quantity is offset against another quantity account, the quantity recorded is subtracted from the other quantity account.

Example:
Total quantity and scrap is recorded → configuration VERB:AUS=GUT

11.4.2 Reading the shift calendar

The shift calendar of all machines of a terminal is provided via the command DLG=LIST;38. The calender is filed in the directory HYDRADIR\spool\. The shift calendar contains the data starting from the day before for 5 days.

Structure of dialog data:

"DLG=LIST;38|DATEI={file name}|TNR={terminal number}|DAT=…|ZEI=...|USR=...|..“

The definition of the file name and the respective path is case-sensitive.

The list includes the following data:

Identifier	Field designation	Description	
MNR	Machine	alter-native	If the mode MOD={..} is not specified, the column MNR is output with the machine number that is valid for the model.
BDEJMOD	Year model		With mode T or H MOD={..} the column BDEJMOD is output with the year model. The machine is not specified with this mode.

Identifier	Field designation	Description
SKDAT	Shift date	Date of beginning of shift
SKNR	Shift number	Shift number
SKZEIB	Beginning of shift	Time in seconds
SKZEIE	End of shift	Time in seconds
SKPAUB:1	Start of shift break 1	Time in seconds
SKPAUE:1	End of shift break 1	Time in seconds
…		
SKPAUB:6	Start of shift break 6	Time in seconds
SKPAUE:6	End of shift break 6	Time in seconds
SKAMST	Automatic machine status	Internal use in the Hydra terminal
MST999SKB	Automatic machine status	Activate/deactivate weekend status 999
SKART	Shift ID	Shift ID
BEGINNDAT	Start date	Date of beginning of shift

11.4.3 Order list

The list of order info is provided by the command DLG=LIST;11 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

„DLG=LIST;11|DATEI={file name}|DAT=...|ZEI=...|USR=...|MOD=...“

The definition of the file name and the respective path is case-sensitive.

a) Reading the sequencing lists of the terminal:

Parameter: MOD=T(terminal from user)

The list includes the data of all machines assigned to this terminal.

The list contains planned (customer-specific status X), prepared (status V), interrupted (status U) and running (status L) production orders.

Overhead orders are only included in the list when they are running (status L).

For the specific machine, the results of MOD=V and MOD=L are therefore output together.

Note: Because of the different configuration options, the list might only output a limited solution set.

b) Reading the info of an order:

Parameter: MOD=A|ANR=order+OP

c) Reading the sequencing list of a machine

Parameter: MOD=V|MNR=machine

The list includes all OPs currently planned for the machine (the configuration of the sequencing list specifies if the OPs of the machine or the OPs of the machine group are selected).

d) List of running orders on the terminal

Parameter: MOD=L

The list includes the running orders (status L) of all machines assigned to this terminal.

The list includes the following relevant data:

Identifier	Field designation	Description
ROW.IDX	ROWINDEX	
ANR	Complete order number	Complete order number contains: AUNR, AGNR, AFOLG, SPLNR
AUNR	Order number	Order number
AGNR	OP	Operation number
AFOLG	Sequence	
SPLNR	Split number	
AUART	Order type	Order type
MNR	Planned single machine	Machine number
POS	Display position	
MGRP	Group	Machine group
MDE_MASCH	MDE machine	MDE machine
ATK	Article	Material number (for output batch)
ATKBEZ	Final article des.	Article name (from order header)
AST	Status	current order status "X", "V", "L", "U"
ASTTXT	Status text	current order status - text dependent of status: planned, prepared, running, interrupted
EGR:BMK01 to EGR:BMK12	Performance posted to RPA 01 to RPA 12	
ANR:MENGEPROZ_UEBLI	Overdelivery	
ANR:OPT_UEBLI	Reaction to overdelivery	

Identifier	Field designation	Description
ANR:MENGEPROZ_UNTLI	Underdelivery	
ANR:OPT_UNTLI	Reaction to underdelivery	
EGS:GUT	since login	
SGE:B	Base quantity unit	
SGE:P	Primary input quantity unit	
SGE:S	Secondary input quantity unit	
SGE:T	Tert. Input quantity unit	
SGR:AUSB	Target scrap	
SGR:AUSP	Target scrap (prim.)	
SGR:AUSS	Target scrap (sec.)	
SGR:AUST	Target scrap (tert.)	
SGR:GUTB	Target quantity (base quantity unit)	
SGR:GUTP	Target quantity (prim.)	
SGR:GUTS	Target quantity (sec.)	
SGR:GUTT	Target quantity (ter.)	
EGR:GUTB	Yield (base quantity unit)	
EGR:GUTP	Yield (prim.)	
EGR:GUTS	Yield (sec.)	
EGR:GUTT	Yield (tert.)	
EGR:AUSB	Scrap (base quantity unit)	
EGR:AUSP	Scrap quantity (prim.)	
EGR:AUSS	Scrap quantity (sec.)	
EGR:AUST	Scrap quantity (ter.)	
EGR:NCHB	Rework quantity	
EGR:NCHP	Rework quantity (prim.)	
EGR:NCHS	Rework quantity (sec.)	
EGR:NCHT	Rework quantity (ter.)	
EGR:PRBB	Problem quantity	
EGR:PRBP	Problem quantity (prim.)	
EGR:PRBS	Problem quantity (sec.)	
EGR:PRBT	Problem quantity (ter.)	
AGE:B	Quantity unit	
AGE:P	Quantity unit (prim.)	
AGE:S	Quantity unit (sec.)	

Identifier	Field designation	Description
AGE:T	Quantity unit (ter.)	
TLG	Partitioning	Partitioning
SZY	Target cycle	Cycle time in seconds per 1000 cycles
AGBEZ	Name/designation of the OP	Operation designation/name.
ASTV	VStatus	Status of the previous OP
AGMFKZ	Parallel production	
CHPFL	Batch management required	
APRIO	External priority	
FIX	fix	Operation fixed Yes/No
DSBEZ	Data ID for ALS	
OPT:SNR	Serial numbers required	
ANR_BEARBZ	Processing time	
ANR_ANDATB	Date of logon	
ANR_ANZEIB	Time of logon	
AST_OPT_PKENN	Control characteristic "production"	
AKTIV	active	
VERARBCODE_PLAUS_MENGE	100% quantity inspection	
EGR:DAUER	Duration	
EGR:PDAUER	Workforce planning	
EGI:GUT	Yield	
EGI:AUS	Scrap	
BEM1	Comment 1	From user field 53 of operation (ANR.FU:53)
BEM2	Comment 2	From user field 54 of operation (ANR.FU:54)
PANZ	Number of staff	
MST	Machine status	
HZTYP	Semi-finished type	
HZBEZ	Name of semi-finished type	
TECHINFO	Technical information	From user field 56 of operation (ANR.FU:56)
TPE	TPUnit [OP]	
CNR	Batch number	
DLL	Customer batch number	

Identifier	Field designation	Description
DLLKZ	Customer batch identification	
LOSGR	Batch size	
HZTYP:EINH	Unit	
PLAUS:MATOK	Plaus. MatOK	
MANR	CBM: Mother operation	
RF:AGTYP	CBM: Operation type	

Example: List of all running orders

DLG=LIST;11|DAT=10/11/2000|ZEI=40000|USR=2101|MOD=L|DATEI= ./spool/auft_list.101|

11.4.3.1 Dynamic extension of the order list

As of HYDRA MW 2.0, you can dynamically complement the order list and add further acronyms, if required. You can then add further fields to the standard list.

Structure of dialog data:

"DLG=LIST;11|DATEI={file name}|MOD={mode}|AKRO=MNR_MSTDSATZ;AUART_PLAUS_MNR;AUART_VISCODE;MNR_EXTSNR; ...;.....|....“

Same procedure as for the machine list.

MPDV Mikrolab provides a list a acronyms that can be used.

11.4.3.2 Controlling the sequencing list

The sequencing list on the BDE terminal is flexible and can be adjusted to meet the customer's requirements.

The settings described below control how data is provided in the sequencing list.

Dialog	Option	Description	Possible settings
Machine configuration	Sequencing list	Via the configuration of the *sequencing list*, you can define the order pool used to generate the sequencing list. The planning in the HYDRA shop floor scheduling or in the PPS system specifies the pool of orders. The pool of orders is different if the OP is planned directly for a machine or for a machine group.	S Basic setting The value of the option with the same name in the HYDRA basic settings is used. M Pool of workplaces The terminal sequencing list only shows the operations planned for the workplace. G Pool of workplaces and groups The terminal sequencing list shows operations that are: - planned for the current workplace or - for another workplace of the group or - that are still located in the pool of groups. K Pool of workplaces and categories The sequencing list on the terminal only shows the operations that are planned for workplaces of the selected machine category. H Group control The terminal sequencing list shows the operations that are - planned for the current workplace or - for another workplace of the group.

Machine configuration	Number of OPs	You can configure the maximum number of OPs included in the sequencing list. The OPs are selected in ascending order based on how the sequencing list is sorted. By default, HYDRA sorts by the following columns: 1. sort_dat 2. sort_zeit 3. auftrag_nr of the OP	0 = no restriction (display of all existing OPs) 1-999 = maximum number of OPs
Order types	Sequencing list	On the level of orders, you use the option *Sequencing list* in the application *Order types* to configure the list. If you define an "N", all OPs of an order with this order type are not displayed in the sequencing list. Example: You do not want to show waiting period OPs in the sequencing list.	J The OP should be displayed in the sequencing list. F The OP should only be displayed in the sequencing list if it is fixed. wurde. N The OP should not be displayed in the sequencing list.
Processing codes	Sequencing list	Also in the configuration of the processing codes, you can define if an OP with this processing code is displayed in the sequencing list or not.	J The OP should be displayed in the sequencing list. N The OP should not be displayed in the sequencing list.
Status assignment	Sequencing list	Also on the level of statuses, you can make configurations. You can define if OPs with a specific status are displayed in the sequencing. If an "N" is defined here, an OP in this status is not displayed in the sequencing list.	J The OP should be displayed in the sequencing list. N The OP should not be displayed in the sequencing list.

		This is used as a standard feature to ensure that only prepared or interrupted OPs appear in the sequencing list. You can also configure that running OPs are displayed in the sequencing list.	

The following operations are not displayed in the sequencing list:

- Locked operations
- The different operations included in merged operations generated on the MOC.
- The original operation of split operations

11.4.4 Personnel list

The list of personnel is provided by the command DLG=LIST;12 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

„DLG=LIST;12|DATEI={file name}|DAT=...|ZEI=...|USR=...|MOD=...|...“

The definition of the file name and the respective path is case-sensitive.

If MOD ist not set, the list includes all persons logged on to the machines of the terminal.

If MOD=V, the list additionally includes all persons logged on in advance to the machines of the terminal.The list includes the following data:

Identifier	Field designation	Description
MNR	Machine	Machine number
ANR	Order	Order number (with all details)
AGNR	OP	Only operation number
AUNR	Order number	
AFOLG	Sequence	
SPLNR	Split number	
PNR	Personnel number	Personnel number
PNAME	Name	Name of the person
PVORNAME	First name	First name of the person
BPOS	Usr.	Operator position/function
BPBEZL	Operator position/function	Name of the operator's function
LPKZ	Wages/premium characteristics	Wages/premium indicator
KNR	Badge number	Staff badge number
VORAN	Logged on in advance	Persons logged on in advance (only with mode V)
ASTUFE	BDE authorization	

Example:

DLG=LIST;12|DAT=10/11/2000|ZEI=40000|USR=2101|DATEI= ./spool/pers_list.101|

11.4.5 Operator positions/functions of machines

The list of operator positions/functions is provided by the command DLG=LIST;14 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;14|DATEI={File name}|DAT=...|ZEI=...|USR=...|..."

The definition of the file name and the respective path is case-sensitive.

The list includes all operator positions of the machines assigned to this terminal.

The list includes the following data:

Identifier	Field designation	Description
MNR	Machine	Machine number
BPSCHL	Operator pos.	Key
BPBEZL	Operator position/function	Description
BPTXTK	BPOS text	Short text
BPFKT	BPOS fct.	Function
BPOS	Usr.	Operator position/function

Example: Operator position/function

DLG=LIST;14|DAT=10/11/2000|ZEI=40000|USR=2101|DATEI= ./spool/bp_list.101|

11.4.7 Machine status list

The list of machine statuses is provided by the command DLG=LIST;16 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;16|DATEI={File name}|DAT=...|ZEI=...|USR=...|..."

The definition of the file name and the respective path is case-sensitive.

a) List of all statuses of the machines assigned to the terminal.

Parameter: MOD=T|USR=HYDRA-User number|

The list is sorted by machine and list of the machine is sorted by malfunction number.

b) List of all statuses of a machine.

Parameter: MOD=M|MNR=machine number|

The list includes the following data:

Identifier	Field designation	Description
MNR	Machine	Machine number
KSTART	Cost center type	Type of cost center
MST	Status	Number of malfunction ID
ZUNR	ToNo.	Assignment number of input
MSTTNR	StNo.	Number of the malfunction text.
MSTTXT	Status text	Malfunction text
BMKTXT	RPA	RPA abbrev.
STKL	Dist. class	Class of malfunction
PKENN	Pchar	Production ID P: flags "Production"; (per machine defined once) S: flags malfunction (per machine defined x times) A: flags "general malfunction" (per machine defined once)
ZUMAN	Man.	Manual assignment J/N
ZUAUTO	Auto.	Automatic assignment J/N
MSPERR	Lock	Machine lock
PABKZ	Log off persons	Log off persons J/N
MSTUFE	Auth.lev.	Authorization level
KST01	Ccr01	Customer-specific Cost center 01
KST02	Ccr02	Customer-specific Cost center 02
KST03	Ccr03	Customer-specific Cost center 03
KST04	Ccr04	Customer-specific Cost center 04

Identifier	Field designation	Description
KST05	Ccr05	kst05[1]J=Prod. lock active kst05[2-8] time of flashing symbol of the machine status in graphic machinery
HARC:ID	Hierarchy level	Configuration **Hierarchical malfunction reasons**
HARC:TYP	Type of hierarchy level	
HARC:TXTKENN	ID: hierarchical level text	
HARC:FLDKENN	ID status field	
AUSNR:AUTO	Scrap reason	Configuration **Status-related collection of scrap reasons**
AUSNR:PSperre	Scrap reason during production lock	
PROGNDAUER	Estimated downtime	See section **Change of machine status (M_MST)**
COLOR	Color code	Configuration of the status text color RGB color - format: RRGGBB:

Example

DLG=LIST;16|DAT=10/11/2000|ZEI=40000|USR=2101|MOD=M|DATEI= ./spool/mstat_list.101|

11.4.8 Deviation reasons

The list of deviation reasons is provided by the command DLG=LIST;23 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;23|DATEI={File name}|DAT=...|ZEI=...|USR=...|..."

The definition of the file name and the respective path is case-sensitive.

The list includes all deviation reasons created.

The list includes the following data:

Identifier	Field designation	Description
WERK	Plant	
MNR	Machine	Machine number
EGG:GUT	Deviation reason	Number of deviation reason
ABWGRTXT	Text of deviation reason	Name of deviation reason

Example:

DLG=LIST;23|DAT=10/11/2000|ZEI=40000|USR=2101|DATEI= ./spool/abgr_list.101|

11.4.9 Premium indicator

The list of premium indicators is provided by the command DLG=LIST;24 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;24|DATEI={File name}|DAT=...|ZEI=...|USR=...|..."

Parameter: USR=HYDRA user number

The definition of the file name and the respective path is case-sensitive.

The list includes all premium indicators created for all machines of the terminal.

The list includes the following data:

Identifier	Field designation	Description
MNR	Machine number	Machine number
LGRP	Wages/premium characteristics	Wage group
LGRPTXT	Designation	Name of the wage group

Example:

DLG=LIST;24|DAT=10/11/2000|ZEI=40000|USR=2101|DATEI= ./spool/lgrp_list.101|

11.4.10 Terminal list

The list of terminals is provided by the command DLG=LIST;45 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;45|DATEI={File name}|DAT=...|ZEI=...|USR=...|..."

Parameter: none

The definition of the file name and the respective path is case-sensitive.

The list includes all terminals created - active and inactive terminals.

The list includes the following data:

Identifier	Field designation	Description
TNR	Terminal number	
TYP	Terminal type	
CFG:1	Configuration 1	
HWADR	Hardware address	
TZ	Time Zone	
BEZK	Location	
BEZL	Designation	
BART:MDE	MDE active	
BART:ADE	ADE active	
BART:PZE	PZE active	
BART:CAQ	CAQ active	
BART:PDV	PDV active	
LANG	Language	
AKTIV	Active	
NEUSTART	Restart	
LEN_KNR	Length of badge number	
PZEBART	Operation mode PZE	
PZESTA:VORG	Default status	
SYSNR	Company number/system number	
TTXT:KOM	IN key	
TTXT:GEH	OUT key	
TTXT:PAU	Break key	
TTXT:INFO	Info key	
FGR:1	Absence reason 1	
TTXT:FGR1	Key Absence reason 1	
...to	...to	
FGR:4	Absence reason 4	
TTXT:FGR4	Key Absence reason 4	

Identifier	Field designation	Description
FGRCFG	Entry absence reason	
OPT:FGRAUTO	Absence reason collection	
PLAUS:PZEONL	Online check	
PLAUS:PZESTA	Status sequence	
ZYKLLOAD:PZE	Cycl. loading PZE	
ZEI:BERLESEN1	Time 1 load PZE	
ZEI:BERLESEN2	Time 2 load PZE	
RELZUG:DAUER	Duration opener	
PZESTA:DAUER	Duration status	
PZEINFO:DAUER	Duration info	
PZEINFO:CFG	Info display	
PZEINFO:NR	Info number	
INFOZEIB:1	Info from 1	
INFOZEIE:1	Info to 1	
...to	...to	
INFOZEIB:5	Info from 5	
INFOZEIE:5	Info to 5	
OFSLST	Delay with shift change	
NEUZEIN	Time for next restart	
NEUDATN	Date for next restart	
OPT:CNRPRN	Ticket print	
CNR:TNR	Batch number	
TGRP	Terminal group	
SAGART	MOP generat. type	
LEN:AUNR	Order length	
OPT:MDEGEN	Generation MDE postings	
OPT:RMIRUEZ	Upload Setup time	
OPT:VGZ	Process. Standard time	
PRJ	Customer project	
OPT:MNRFTYP	Machine number format	
OPT:SAG	Proces. merged operations	
OPT:PZEPLAUS	Plaus. PZE-IN	
SYSNR:SETUP	System number	
MAXDAUERSTEMP	Time clocking supplement	
OPT:PZEAUTO	PZE controls ADE	

Identifier	Field designation	Description
ADEKAR:AKTIV	Waiting time treatment ADE	
ADEKAR:DAUER	Waiting time	
ADEKAR:BMKNR	Waiting time RPA	
OPT:VLIST	Sequencing list	
OPT:AUSNR	Coll. interruption reason	
OPT:ABBRNR	Coll. scrap reason	
OPT:ERFPMENGE	Coll. person-related quantity	
OPT:ERFAMENGE	Coll. order-related quantity	
OPT:GUTMANU	Coll. yield	
OPT:AUSMANU	Coll. scrap	
OPT:CHV	Batch management	
MDEKAR:AKTIV	Waiting time treatm. MDE	
MDEKAR:UG	Lower waiting time limit MDE	
MDEKAR:BMKNPLAN	RPA for times not yet scheduled	
MDEKAR:OG	Upper waiting time limit MDE	
MDEKAR:BMKPLAN	RPA for scheduled times	
OPT:ADEKNRPLAUS	PNO entry (BDE postings)	
OPT:MDEKNRPLAUS	PNO entry (MDE postings)	
LEN:KNR	Badge number length	
VER	Version	
OPT:KDVLIST	Customer sequencing list	
OPT:WILLESYS	Wille system	
OPT:UCHGEN	Creating unknown batches	
OPT:TRMGEN	Generation partial confirmation (upload part quantity)	
OPT:KORRMNGEN	Hide correction messages	
OPT:EDITBRMSPERR	Lock edit RM-data	
OPT:AUSPAUDAUER	Hide breaks in pers. duration	
BDEOPT:9 to BDEOPT:12	BDE-CH 9 to BDE-CH 12	
OPT:VLISTMOD	Sequencing list	
LEN:PNR	Personnel No. length	
OPT:PNRFUELL	PersonnelNofillsign	
OPT:ZST	Set access statuses	

Identifier	Field designation	Description
OPT:WSTEMPFGR	Change clocking absence reason	
OPT:ADATF	FLS earliest start date	
OPT:AART	FLS display order type	
KERNELVER	Kernel version	
KDNR	Customer number	
OPT:ANTVERBPM	Proportionate posting labor times	
OPT:PZPLAUS	Process check digit	
OPT:WSTEMP	Alternate clocking	
OPT:TGLKTOBEGR	Daily account limit	
OPT:EVENTEDIT	Event update active	
VER	HYD:HYDRA version	
KK1:AKTIV	PZE:PZE as SAP subsystem	
KK2:AKTIV	BDE:BDE as SAP subsystem	
KK_KD1:AKTIV	KK_KD1:AKTIV	
KK_KD2:AKTIV	KK_KD2:AKTIV	
LEVEL	Level	
HYDDI:LEVEL	Hyddi level	
LEN:AFOLG	Length of sequence	
LEN:AGNR	Length of OP number	
LEN:SPLNR	Number of splits	
LEN:BCNR	Length of barcode	
LEN:CNR	CNR length	
OPT:CNRAUTOGEN	Generate CNR autom.	
CNRFIX:WE	Fixed incoming goods batch	
CNRFIX:PR	Fixed production batch	
OPT:CNRAUTOAB	Log input batch autom. off	
OPT:MDETLG	Multiple partit.	
ZYKLLOAD:BDE	(empty)	
BDEJMOD	Shift model with HUPE terminal	
FIR	Comment	
ABT	Department where terminal is located	
CFG:2	Configuration flag 2	

Identifier	Field designation	Description
PLAUS:OFF	Plaus. checks	
PARAM1 to PARAM5	Parameter 1 to Parameter 5	
MELDART	Control of postings	
PLAUS:ADEKNR	Configuration ADE card number	
PLAUS:MDEKNR	Configuration MDE card number	
PLAUS:MNR	Machine configuration	
PLAUS:MST	Configuration of machine status	
PLAUS:AART	Configuration of opt. entry of order type	
OPT:BERFKT	Area function active	
ART	Terminal type	
DLGSTRG	Dialog control	

Example:

DLG=LIST;45|DAT=02/11/2005|ZEI=40000|USR=2101|DATEI= ./spool/trm_list.101|

11.4.11 Comments on operations

The list of comments on operations is provided by the command DLG=LIST;61 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;61|DATEI={File name}|DAT=...|ZEI=...|USR=...|..."

Parameter: none

The definition of the file name and the respective path is case-sensitive.

The list includes the following data:

Identifier	Field designation	Description
MNR	Machine	
ANR	Order	
AUNR	Inspection order number	
AGNR	OP	
AFOLG	Sequence	

Identifier	Field designation	Description
SPLNR	Split number	
PNR	Personnel number	
PNAME	Name	
PVORNAME	First name	
DAT	Date	
ZEI	Time	
BEM	Comment	

Example:

DLG=LIST;61|DAT=02/11/2005|ZEI=40000|USR=2101|DATEI= ./spool/agkom_list.101|

11.4.12 Order components (BOM)

The list of order components ("Bill of materials") is provided by the command DLG=LIST;74 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;74|DATEI={File name}|DAT=...|ZEI=...|USR=...|...“

The definition of the file name and the respective path is case-sensitive.

Parameter: ANR order number (mandatory)

Parameter: ART Filter indicator for components (optional parameters)

B = production resource
M = material
V = equipment
W = tools
Q = measuring equipment
D = documents
C = NC programs
U = unknown resource

The list includes the following data:

Identifier	Field designation	Description
ART	Type	Component type B = production resource M = material V = equipment W = tools Q = measuring equipment D = documents C = NC programs U = unknown resource
ATK	Article	With materials: material number
BEZ	Designation	Material designation/name
BEZ:2	Designation	Material name 2
SGR:GUT	Input quantity	Input quantity to produce 1 article in primary quantity unit in the operation. With production resources and tools, this is the quantity required of a resource for this operation.
MENGE:BED	Demand	Required quantity
SGE:GUT	Unit	Unit of the input quantity
ATKBEZ	Article	Optional component name (of the material, document, etc.)
SLP	BOM item	Item number of component (BOM item)
LAGORT	Storage location	Reserved
LAGPZ	Storage compartm.	Reserved
PATH	Path	With documents: path
FILE	File name	With documents: file name
RESTYP	Type	WNR, DOC, etc. = resource type MAT = material
MOD	Extended type	Extended type: Document :DOC Material: MAT Other: NOMAT

Example:

DLG=LIST;74|DAT=02/11/2005|ZEI=40000|USR=2101|ANR=AAA2100473100200|DATEI=./spool/mat_list.101|

11.4.13 Scrap reason list

The list of scrap reasons is provided by the command DLG=LIST;84 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;84|DATEI={File name}|DAT=...|ZEI=...|USR=...|..."

Parameter: MOD T terminal
M machine

ART in mode M (machine) restricted to the type
(A = scrap, N = rework, P = problem quantity, G = yield)

The definition of the file name and the respective path is case-sensitive.

The list includes the following reasons and provides the following data:

Identifier	Field designation	Description
MNR	Machine	
ART	Type	
GR	Reason	
GRTXTNR	Reason text	
BEZK	Designation	
MGR	Mother reason	
GRTXT	Scrap reason	
ATK	Article	
OPT:ATKVORG	Default article	
TNR	Terminal number	with MOD=T

Example:

DLG=LIST;84|DAT=02/11/2005|ZEI=40000|USR=2101|DATEI= ./spool/gr_list.101|MOD=T|TNR=6

11.4.14 BDE order types

The list of BDE order types is provided by the command DLG=LIST;87 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;87|DATEI={File name}|DAT=...|ZEI=...|USR=...|..."

Parameter: none

The definition of the file name and the respective path is case-sensitive.

The list includes the following order types and provides the following data:

Identifier	Field designation	Description
AUART	Order type	
CAT	Category	The category is used to classify and to combine similar order types. The category is a logical umbrella term. Possible values: "PO" production order "PJ" Project order "PM" maintenance order "KP" capacity order "GK" overhead cost order
BEZL	Designation	
OPT:RM	Option "Upload"	With this order type, the data recorded is uploaded to the PPS system or not.
OPT:PLAN	Option "Plan"	
OPT:PLANTERM	Option "Planned dates"	
ICON	Symbol	
OPT:AUBUCH		
OPT:VLIST		
OPT:ERF		
OPT:ABE		
OPT:APAN		
OPT:AANSKBAUTO		
OPT:PANSKBAUTO		
OPT:SNR		
OPT:SNRVERG		
OPT:EDITRM		
PLAUS:STAV		
PLAUS:MINMENGEV		
PLAUS:WEIGMENGE		
FKT		
PLAUS:MNR		
PLAUS.PNR		
OPT:PRIOSTRG		
OPT:APPRN		
OPT:FERTVAR		
DLGSTRG		
VISCODE		
OPT:SYS		
AKTIV		

Example:

DLG=LIST;87|DAT=02/11/2005|ZEI=40000|USR=2101|DATEI= ./spool/auart_list.101|

11.4.15 List of counters

The counter configuration is provided in a list. You use the command DLG=LIST;131 to enable the list request for the machines according to the mode (MOD). The list is filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;131|DATEI={File name}|DAT=...|ZEI=...|USR=...|..."

Parameters:

- MOD=T … List of all counters of all machines assigned to the terminal. In this mode, the acronym USR=HYDRA user number must be specified.
- MOD=M … List of all counters of a machine. In this mode, the acronym MNR=machine number must be specified.

The definition of the file name and the respective path is case-sensitive.

The list includes the following data:

Identification	Field designation	Description
MNR	Machine	
CTR	counters	Unique counter identification
BEZ	Designation	
EINH	Unit	
GR	Reason	Configuration of a reason, e.g. scrap reason
TYP	Assessment	The counter quantity is booked as yield, scrap, rework or open quantity (problem quantity).
VERB	Offset against quantity account	e.g. scrap is subtracted from the total quantity.
VERB:TLG	Allocation with partitioning	If VERB.TLG=J, you use the partitioning of machine and order to book the counter pulses.
VERB:DIV	Allocation with pulse factor	If VERB.DIV=J, you use the pulse factor of machine and order to book the counter pulses.

Identification	Field designation	Description
OPT:TAKT	Posting as cycles	If OPT:TAKT=J, the pulse factors recorded are also booked as cycles. The quantity recorded is transferred to the HYDRA server via the acronym AGR:HUB.
OPT:UEB	Monitoring	Indicator that specifies if the counter is relevant for cycle monitoring.

Example:

DLG=LIST;131|MOD=T|DAT=02/11/2005|ZEI=40000|USR=2101|DATEI= ./spool/trm_list.101|

Note:

The list is sorted by machine and counter number in ascending order.

11.4.16 List providing the assignment of workplaces to MDE shop floor clients

The list provides all assignments of workplaces of this shop floor client to another shop floor client (MDE terminal or PCC) with MDE operation mode.

The list only provides assignments that are fixed in the Workplace terminal assignment.

Workplaces, which are assigned to the executing shop floor client (parameter USR) as so-called "MDE machines", are not included in the list.

The list is provided by the command DLG=LIST;140 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

DLG=LIST;**140**|DATEI={file name}|DAT=...|ZEI=...|**USR**=...|...

Parameters:

- USR=: Terminal number of the shop floor client requesting the list.
  If the list is requested without USR=, then the system provides an empty list.

The definition of the file name and the respective path is case-sensitive.

The list includes the following data:

Identification	Field designation	Description
ID	Object ID	TYP=M: Work center numbers/machine numbers
TYP	Type	Object type: • "M" for workplace/machine
TNR	MDE terminal ID	TYP=M: Terminal number of the shop floor client where the workplace is assigned as "MDE machine".

The list is available from the following program version onwards: hymwmde72.dll / so 8.1.1.143

11.5 Reading HLS data

11.5.1 Production variants

The list of production variants is provided by the command DLG=LIST;50 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

„DLG=LIST;50|DATEI={file name}|MOD={mode}|DAT=...|ZEI=...|USR=...|...“

The definition of the file name and the respective path is case-sensitive.

The list includes all production variants (mode MOD=A) or only the changed production variants (mode MOD=G). If the mode is written in lower case letters (e.g. MOD=a), then the system only selects released production variants (STA=F).

The list includes the following data:

Identifier	Field designation	Description
VERWEIS	Reference	Unique key
VER	Version	Version in a production variant
STA	Status	F=released, S=locked, L=(logically) deleted
FIR:ATK	Company of article	Company of article/material type
ATK	Article	Article
MATTYP	Material type	Material type

Identifier	Field designation	Description
FIR:MNR	Company of machine	Company of machine
MNR	Machine	Machine number of the machine
MGRP	Group	Machine groupe of the machine
MANZ	Number of machines	in format N8.2
FIR:WNR	Company of the tool	Company of the tool
WNR	Tool	Tool
WFAM	Tool family	Tool family
WANZ	Number of tools	Number of tools in format N8.2
SZY	Target cycle	Target cycle
TLG	Partitioning	Partitioning in format N8.2
RUEZ	Setup time	Setup time
ABRZ	Teardown time	Teardown/retooling time
PRIO	Prio.	Priority
BEM	Comment	Comment
DATB	Start date	Valid from
DATE	End date	Valid until
DSBEZ	Data ID	Data ID
BEARBDAT	Changed on	Date of the most recent modification
BEARBZEI	Processing time	Time of the most recent modification
BEARB	Modified by	Modified by

Example:

DLG=LIST;50|DAT=10/11/2000|ZEI=40000|USR=2101|DATEI= ./spool/fertvar.101|

11.7 Annex

11.7.1 Overview of field data BDE/MDE

The following table provides an overview of the field data with structure and short example.

Identifier	Description	Structure	Example
MNR	Machine number	...|MNR={machine number}|...	...|MNR=100|...
MGRP	Machine group	...|MGRP={machine group}|...	...|MGRP=100|...
ANR	Fully defined key	...|ANR={key}|...	...|ANR=4711001001|...
AUNR	Order number	...|AUNR={order number}	...|AUNR=4711|...

Identifier	Description	Structure		Example
AGNR	Operation number	...|AGNR={OP}|...		...|AGNR=0010|...
AFOLG	Sequence	...|AFOLG={sequence}|...		...|AFOLG=01|...
SPLNR	Split number	...|SPLNR={split number}|...		...|SPLNR=01|...
AUART	Order type	...|AUART={type}|...		...|AUART=0|...
ATK	Article number	...|ATK={article number}|...		
EGR:*	Recorded value Manually recorded value	...|EGR:{type}={value}|... Types of recorded values: RPA01, RPA02,..., RPA12 DAUER, PDAUER HUB, GUT, AUS, LEN, GEW The value recorded is added to the relevant account in the DB.		..|EGR:GUT=10. |. ...|EGR:AUS=1 |...
EGR:*	New quantity types as of MW 2.0	GUTB	Yield (base quantity unit)	
		GUTP	Yield (primary quantity unit)	
		GUTS	Yield (secondary quantity unit)	
		GUTT	Yield (tertiary quantity unit)	
		AUSB	Scrap (base quantity unit)	
		AUSP	Scrap (primary quantity unit)	
		AUSS	Scrap (secondary quantity unit)	
		AUST	Scrap (tertiary quantity unit)	
		NCHB	Rework quantity (base quantity unit)	
		NCHP	Rework quantity (primary quantity unit)	
		NCHS	Rework quantity (secondary quantity unit)	
		NCHT	Rework quantity (tertiary quantity unit)	
		PRBB	Problem quantity (base quantity unit)	

Identifier	Description	Structure		Example
		PRBP	Problem quantity (primary quantity unit)	
		PRBS	Problem quantity (secondary quantity unit)	
		PRBT	Problem quantity (tertiary quantity unit)	
AGE:*	Units of automatically recorded values	B	Base quantity unit	
		P	Primary quantity unit	
		S	Secondary quantity unit	
		T	Tertiary quantity unit	
SGE:*	Units of target values	B	Base quantity unit	
		P	Primary quantity unit	
		S	Secondary quantity unit	
		T	Tertiary quantity unit	
EGE:*	Unit of recorded values	...|EGE:{type}={unit}|... Types see EGR		EGE:GUT=ST
EGG:*	Reasons of recorded values	...|EGG:{type}={reason}|... Types see EGR		EGG:AUS = 1
SGR:*	Target value	...|SGR:{type}={value}|... SGR:GUT =target quantity order/OP Types see EGR **New types 7.2: AUS*, GUT* (see EGR)**		...|SGR:GUT=1126|
TLG	Partitioning	...|TLG={partitioning}|...		...|TLG=1|... ...|TLG=0.25|...
SZY	Target cycle	...|SZY={target cycle};{unit}|... First, only unit "s"! Later conversions of the values into seconds.		...|SZY=36000;s|... ...|SZY=10;h|...
MST	Machine status	...|MST={machine status }|...		..|MST=1 |..
PNR	Personnel number	...|PNR={personnel number}|...		...|PNR=999999|...
KNR	Staff badge no.:	...|KNR={badge number}|...		...|KNR=9999|...

Identifier	Description	Structure	Example
BPOS	Operator position/function	...|BPOS={operator function}|...	...|BPOS=A1|...
PMK	Memorize persons	...|PMK={Y/N}|...	...|PMK=J|...
CNR	Batch number	...|CNR={batch number}	...|CNR=998877|...
KST	Cost center	Cost center	
BZW	Posting required	Optional parameter. If the option is set, several validation checks are disabled. If option is not sent: = N [...|BZW={posting required}|...]	...|BZW=J|...
		Field data for BATCH	
LHW	Info on batch	Info on batch (C20)	|LHW={batch info}|...
ZLO	Target	Target	...|ZLO=121234|...
TPE	Transport unit	Transport unit (C10)	...|TPE=1234|...
SLP	BOM item	BOM item (C6)	...|SLP=AA34|...
ATTR:*	Additional attributes of a batch	...|ATTR:{no}]={attribute recorded}|.. no = 1 ..11 field types; 1 - 4 : Integer 4 – 6 : Double 7 : Char 4 8, 9 : Char 10 10, 11: Char 20	
STN	Station number	...|STN ={ station number }|...	...|STN=1|..
LOSANZ	Number of parallel output batches	...|LOSANZ ={ number of batches }|...	...|LOSANZ=5|...

11.7.2 Optional field data for the premium and incentive wages LLE

The following data fields can optionally be used for the premium and incentive wages LLE in the BDE input dialogs.

You require the licenses to calculate single or group incentives.

The listed fields are optional. In most cases, these fields are not explicitly filled during data collection, but the fields are usually transferred automatically from the operation or the LLE assignment of premium groups to the BDE postings or the fields are required in special cases only.

Identifier	Type / max. field length	Description
LPGRP=	C8	Optional only with order logon: In case of Incentive Wage LLE with group bonus: The premium group can optionally only be recorded with machines having the incentive wage indicator "G"=group piecework. Here, the premium group, which is assigned to the machine in the Incentive Wage LLE, is overwritten in the U/E and B records.
LART=	C4	In case of staff logon with LLE: Optional recording of the wage type for the BDE staff postings (B record)
EGR:TE=	N8	Optional entry of the single piece specification for the person t_e in seconds per 1000 pieces.
EGR:TR=	N8	Optional entry of the setup specification for the person t_r in seconds.
EGR:TEB=	N8	Optional entry of the single piece specification for the production resource t_{eb} in seconds per 1000 pieces.
EGR:TRB=	N8	Optional entry of the setup specification for the production resource t_{rb} in seconds.
BPOS=	C10	Optional entry of an operator position/function with staff logon
LPKZ=	C10	Optional entry of a premium indicator with staff logon

11.7.3 Tips and tricks

11.7.3.1 Transfer of machine data and shift change information

If you use the PDM dialogs to integrate the machine data collection including shift change function, then you must assign the workplace/machine, for which you want to send PDM dialogs, to a terminal.

Without this assignment, the system cannot identify an active shift change via PDM dialog for this machine. Result: The system performs proportionate allocations in the order shift log that might have unexpected results.

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12.1 Note on the Descriptions of the Basic Dialogs

All mandatory fields that must be specified have the addition PK (primary key). All other fields are optional and are processed if they are transferred.

12.2 BDE Log Records

12.2.1 Create log record (DLG=ADEPRO.INSERT, COPY)

You can use the BAPI calls described in this section to create or copy log records.

Tables

Table	Key field	Description
ade_protokoll	Reference ADEPRO.VERWEIS	Reference of the posting record

BAPI call

ID	Content / {type}	Description
DLG	ADEPRO.INSERT	Create posting
	ADEPRO.COPY	Copy posting
ADEPRO.VERWEIS:Q	{N8}	PK only with ADEPRO.COPY Refers to the posting that you want to copy.
ADEPRO.SART	{C1}	PK Log record of the posting U = Order interruption E = End of order B = Staff posting H = Batch record
ADEPRO.ANR	{C40}	PK Combined order/OP number of the posting
Other option		
ADEPRO.AUNR	{C40}	PK Order number of the posting
ADEPRO.AGNR	{C40}	PK Operation number of the posting
ADEPRO.AFOLG	{C40}	PK Order sequence number of the posting (only if configured in HYDRA)

ID	Content / {type}	Description
ADEPRO.UAGNR	{C40}	PK Sub operation number of the posting (only if configured in HYDRA)
ADEPRO.SPLNR	{C40}	PK Split number of the posting (only if configured in HYDRA)
ADEPRO.DATB	{MM/DD/YYYY}	PK Start date of the posting
ADEPRO.DATE	{MM/DD/YYYY}	PK End date of the posting
…	…	For further fields, that are not MANDATORY, refer to section 12.2.4 *List of fields (acronyms)*.

Return

ID	Content / {type}	Description
VERWEIS	{N8}	Returned reference of the created posting. Example: **RET_DATA:** RET=0|KT=|LT=|DATA=ADEPRO|**VERWEIS=15673259**|RET=0|KT=|LT=|

Plausibility checks

Error codes	Description
10	The operation (ADEPRO.ANR) must be available in the backlog of orders (online dataset).
90	The workplace (ADEPRO.MNR) must be available in the workplace configuration.
1803	Check is performed if only "BEARB" is passed and BEARB does not equal "HYDRA". No responsibility area authorization for the workplace is available (ADEPRO.MNR).
1030	If ADEPRO.SART =B The person (ADEPRO.PNR) must be part of the staff.
1641	If ADEPRO.SART =H The operation (ADEPRO.ANR) must be subject to batch management.

Error codes	Description
1900	If the license LLE-GRB is active and the premium group is specified (ADEPRO.LEISTGRP). The premium group (ADEPRO.LEISTGRP) must be defined and valid.
1958	ADEPRO.SART =U or ADEPRO.SART =E Within the selected period of time an overlapping log record is already available for the operation and workplace, i.e. a U or E record, which reaches into or lies completely within the period of time of the new record to be created, is already available when trying to insert a log record of record type U or E.
1952	Start date (ADEPRO.DATB / ADEPRO.ZEIB) must not be greater than the end date (ADEPRO.DATE / ADEPRO.ZEIE).
806	If ADEPRO.SART =T or H The new record to be created must be within an already existing U/E record. You can deactivate this check using the parameter ADEPRO.PLAUS:AGUN=N.
806	If ADEPRO.SART = B and ADEPRO.PLAUS:AGUNB=J The center of time of the record to be created must be within an already existing log record of record type U or E.
814	If ADEPRO.SART = H The log record of record type H may only include values for one of the two quantity types: • either yield (ADEPRO.GUT/GUTP/GUTS/GUTT/GUTB) • or scrap (ADEPRO.AUS/AUSP/AUSS/AUST/AUSB)
815	If ADEPRO.SART = H A log record of record type H exists for the specified period of time. You can deactivate this check using the parameter ADEPRO.PLAUS:MULTICNR=J.
1661	You must specify the parameter with the ID ADEPRO.SART.
1661	You must specify the parameter with the ID ADEPRO.ANR (or the IDs AUNR,AGNR,AFOLG,UAGNR,SPLNR).
1661	You must specify the parameter with the IDs ADEPRO.DATB and ADEPRO.DATE.

12.2.2 Edit log record (DLG=ADEPRO.UPDATE, DELETE, LOCK, UNLOCK, SELECT)

You can use the BAPI calls described in this section to edit log records.

Tables

Table	Key field	Description
Ade_protokoll	Reference ADEPRO.VERWEIS	Reference of the log record

BAPI call

ID	Content / {type}	Description
	ADEPRO.UPDATE	Change log record
	ADEPRO.DELETE	Delete log record
	ADEPRO.LOCK	Lock log record for editing MUST be performed before ADEPRO.UPDATE
	ADEPRO.UNLOCK	Unlock log record after editing MUST be performed after ADEPRO.UPDATE
	ADEPRO.SELECT	Select log record
ADEPRO.VERWEIS	{N8}	PK Reference of the log record
ADEPRO.SART	{C1}	PK Record type U = Order interruption E = End of order B = Staff posting H = Batch record
…	…	ADEPRO.UPDATE: For further fields, that are not MANDATORY, refer to section 12.2.4 *List of fields (acronyms)*

Return

ID	Content / {type}	Description
ADEPRO.VERWEIS	{N8}	Return of reference of the changed log record. Note: This reference might not be equal to the reference of ADEPRO.UPDATE if the log record has already been uploaded and a cancel posting must be created.

Plausibility checks

Error codes	Description
10	The operation (ADEPRO.ANR) must be available in the backlog of orders.
90	The workplace (ADEPRO.MNR) must be available in the workplace configuration.
1803	Check is performed if only "BEARB" is passed and BEARB does not equal "HYDRA". No responsibility area authorization for the workplace is available (ADEPRO.MNR).
1030	Only if ADEPRO.SART=B: The person (ADEPRO.PNR) must be part of the staff.
1641	Only if ADEPRO.SART=H: The operation (ADEPRO.ANR) must be subject to batch management.
1958	Within the selected period of time, an overlapping log record is already available for the operation and workplace, i.e. a U or E record, which reaches into or lies completely within the period of time of the new record to be created, is already available when trying to insert a log record of record type U or E.
1952	Start date (ADEPRO.DATB / ADEPRO.ZEIB) must not be greater than the end date (ADEPRO.DATE / ADEPRO.ZEIE).
1900	If the license LLE-GRB is active and the premium group is specified (ADEPRO.LEISTGRP): The premium group (ADEPRO.LEISTGRP) must be defined and valid.
806	If ADEPRO.SART = T The time of the log record must be within an already existing log record of record type U or E.
806	If ADEPRO.SART=H The time of the log record must be within an already existing log record of record type U or E.
1956	UPDATE,DELETE If ADEPRO.SART=T The upload of a part quantity (partial confirmation) cannot be changed because the respective log record of record type U or E has not been generated.
814	If ADEPRO.SART =H The log record of record type H may only include values for one of the two quantity types: • either yield (ADEPRO.GUT/GUTP/GUTS/GUTT/GUTB) • or scrap (ADEPRO.AUS/AUSP/AUSS/AUST/AUSB).
815	A log record of record type H exists for the specified period of time.

Error codes	Description
1954	UPDATE.LOCK The cancellation log record (resp. ADEPRO.VERWEIS) cannot be changed.
1955	UPDATE.LOCK The original log record (resp. ADEPRO.VERWEIS) cannot be changed.
1957	UPDATE The record type cannot be changed (ADEPRO.SART): • the record type of a log record of record type B cannot be changed into the record type U or E. • the record type of a log record of record type U or E cannot be changed into the record type B. Note: It is possible to change the record type of a log record of record type U into the record type E and vice versa.
101	UPDATE,LOCK,UNLOCK The log record (if the parameter ADEPRO.VERWEIS is specified) must be created in the database.
1666	The log record is currently locked by another user. (UPDATE, DELETE,LOCK).
1661	UPDATE.LOCK,UNLOCK You must specify the parameter with the ID ADEPRO.VERWEIS.
1661	UPDATE.LOCK,UNLOCK You must specify the parameter with the ID ADEPRO.SART.
1661	UPDATE.LOCK,UNLOCK You must specify the parameter with the ID ADEPRO.ANR (or the IDs AUNR,AGNR,AFOLG,UAGNR,SPLNR).

Plausibility checks in the MOC

Error codes	Description
	Order type configuration: If upload cannot be changed. → Event is locked because the posting has already been uploaded.
	Order type configuration: If editing authorization is enabled and the user does not have the specified function authorization. → no authorization to change something with this order type
	Setup setting: If "Maintenance of events" is active: → BDE maintenance based on events is active. Automatically generated postings cannot be edited!

Error codes	Description
	Setup setting: If "Maintenance of events" is active: → BDE maintenance based on events is active. Automatically generated postings cannot be deleted!

12.2.3 Sign log record (DLG=ADEPRO.SIGN)

You can use this BAPI call to sign a posting.

Table	Key field	Description
Ade_protokoll	Reference ADEPRO.VERWEIS	Reference of the posting

BAPI call

ID	Content / {type}	Description
DLG	ADEPRO. SIGN	PK Sign posting
ADEPRO.VERWEIS	{N8}	PK Reference
BEARB	{C10}	PK Signed by

Note: The database fields sign_dat, sign_zeit are populated using the timestamp passed (DAT/ZEI).

Return

ID	Contents	Description
—	—	—

Plausibility checks

Error codes	Description
1661	You must specify the parameter with the ID ADEPRO.VERWEIS.
101	The posting record (if the parameter ADEPRO.VERWEIS is specified) must be available in the database.

Notes on the processing

- You can insert U records even if the OP has already been finished.

- If you insert an E record, the operation status is set to finished.

- If you create or change a B record, the person-related RPAs of the respective U/E record and the order status are revised.

- If you change an H record (batch posting), the quantities of the respective U/E record and the order status are revised.

- If you change a T record (upload of part quantity), the quantities of the respective U/E record and the order status are revised.

12.2.4 List of fields (acronyms) for the ADEPRO dialog

ID	Description	DB type	Length
ADEPRO.ANR	MES order number	char	40
ADEPRO.SART	Record type: U: Order interruption; Upload of part quantity E: Order end B: Staff logoff T: Uploads of part quantities for order H: Batch logoff	char	3
ADEPRO.MST	Status text number Machine status that was active or was set when the OP was interrupted.	integer	7
ADEPRO.PANZ	reserved	smallint	7
ADEPRO.PNR	Person who triggered the log record	char	10
ADEPRO.PGRP	reserved	char	20
ADEPRO.ZEIB	Point in time of logon (time)	integer	7
ADEPRO.DATB	Point in time of logon (date)	sqldate	7
ADEPRO.ZEIE	Point in time of logoff (time)	integer	7
ADEPRO.DATE	Point in time of logoff (date)	sqldate	7
ADEPRO.EGR:DAUER	Total time of the posting, compared to the shift calendar	integer	7
ADEPRO.TNR	Terminal that made the posting	smallint	7
ADEPRO.MNR	HYDRA workplace making the posting	char	20
ADEPRO.TLG	Valid partitioning of the posting	decimal	7

ID	Description	DB type	Length
ADEPRO.SKNR	Shift number; assignment of the shift that logs off	smallint	7
ADEPRO.SKZEIB	Beginning of shift of the assigned shift	integer	7
ADEPRO.SKZEIE	End of shift of the assigned shift	integer	7
ADEPRO.SKPAUSE	Shift break in seconds	smallint	7
ADEPRO.EGR:BMK01	Time recorded, splitted onto RP accounts Activity posted in RPA 01	integer	7
ADEPRO.EGR:BMK02	Activity posted in RPA 02	integer	7
ADEPRO.EGR:BMK03	Activity posted in RPA 03	integer	7
ADEPRO.EGR:BMK04	Activity posted in RPA 04	integer	7
ADEPRO.EGR:BMK05	Activity posted in RPA 05	integer	7
ADEPRO.EGR:BMK06	Activity posted in RPA 06	integer	7
ADEPRO.EGR:BMK07	Activity posted in RPA 07	integer	7
ADEPRO.EGR:BMK08	Activity posted in RPA 08	integer	7
ADEPRO.EGR:BMK09	Activity posted in RPA 09	integer	7
ADEPRO.EGR:BMK10	Activity posted in RPA 10	integer	7
ADEPRO.EGR:BMK11	Activity posted in RPA 11	integer	7
ADEPRO.EGR:BMK12	Activity posted in RPA 12	integer	7
ADEPRO.EGR:PDAUER	Posted staff time	integer	7
ADEPRO.KST	Cost center of the workplace	char	10
ADEPRO.VERWEIS	Unique ID of the log record, important for the editing of log records	sqlserial	7
ADEPRO.RCK	J/N specifies if the log record has already been uploaded to PPS system	char	1
ADEPRO.BEARB	Initials of the user who manually makes changes	char	10
ADEPRO.BEARBDAT	Date of the manual change	sqldate	7

ID	Description	DB type	Length
ADEPRO.SKDAT	Date of beginning of shift. In case of ADE records that extend over several shifts, the date is identified using the end posting.	sqldate	7
ADEPRO.RCK:01	customer-specific	char	1
ADEPRO.RCK:02	customer-specific	char	1
ADEPRO.ERFART	Is assigned if the log record is manually created using the function "Maintenance of postings" on the console: M : created manually on the console P : created automatically by the PZE. You must not change the data record in the maintenance of postings.	char	1
ADEPRO.ABREDAT	Evaluation date PZE	sqldate	7
ADEPRO.EGR:GUTB	Yield (base quantity unit)	decimal	7
ADEPRO.EGR:GUTP	Yield (primary quantity unit)	decimal	7
ADEPRO.EGR:GUTS	Yield (secondary quantity unit)	decimal	7
ADEPRO.EGR:GUTT	Yield (tertiary quantity unit)	decimal	7
ADEPRO.EGR:AUSB	Scrap (base quantity unit)	decimal	7
ADEPRO.EGR:AUSP	Scrap (primary quantity unit)	decimal	7
ADEPRO.EGR:AUSS	Scrap (secondary quantity unit)	decimal	7
ADEPRO.EGR:AUST	Scrap (tertiary quantity unit)	decimal	7
ADEPRO.EGR:NCHB	Rework quantity (base quantity unit)	decimal	7
ADEPRO.EGR:NCHP	Rework quantity (primary quantity unit)	decimal	7
ADEPRO.EGR:NCHS	Rework quantity (secondary quantity unit)	decimal	7
ADEPRO.EGR:NCHT	Rework quantity (tertiary quantity unit)	decimal	7
ADEPRO.EGR:PRBB	Problem quantity (base quantity unit)	decimal	7
ADEPRO.EGR:PRBP	Problem quantity (primary quantity unit)	decimal	7
ADEPRO.EGR:PRBS	Problem quantity (secondary quantity unit)	decimal	7
ADEPRO.EGR:PRBT	Problem quantity (tertiary quantity unit)	decimal	7
ADEPRO.EGE:GUTB	Base quantity unit	char	3
ADEPRO.EGE:GUTP	Primary quantity unit	char	3

ID	Description	DB type	Length
ADEPRO.EGE:GUTS	Secondary quantity unit	char	3
ADEPRO.EGE:GUTT	Tertiary quantity unit	char	3
ADEPRO.GGR:HUB	Total of clocks recorded	decimal	7
ADEPRO.EGR:HUBG	Recorded clocks - yield	decimal	7
ADEPRO.EGG:GUTB	Deviation reason Reference: ADE_GRUND_ZUORD	integer	7
ADEPRO.EGT:GUTB	Reason text number for yield Reference: ADE_GRUND_TEXTE	integer	7
ADEPRO.EGG:AUSB	Scrap reason Reference: ADE_GRUND_ZUORD	integer	7
ADEPRO.EGT:AUSB	Reason text number for scrap Reference: ADE_GRUND_TEXTE	integer	7
ADEPRO.EGG:NCHB	Rework reason Reference: ADE_GRUND_ZUORD	integer	7
ADEPRO.EGT:NCHB	Reason text number for rework quantity Reference: ADE_GRUND_TEXTE	integer	7
ADEPRO.EGG:PRBB	Problem reason Reference: ADE_GRUND_ZUORD	integer	7
ADEPRO.EGT:PRBB	Reason text number for problem quantity Reference: ADE_GRUND_TEXTE	integer	7
ADEPRO.EGR:LST01	Posted free activity 1	decimal	7
ADEPRO.EGR:LST02	Posted free activity 2	decimal	7
ADEPRO.EGR:LST03	Posted free activity 3	decimal	7
ADEPRO.EGR:LST04	Posted free activity 4	decimal	7
ADEPRO.EGR:LST05	Posted free activity 5	decimal	7
ADEPRO.EGR:LST06	Posted free activity 6	decimal	7
ADEPRO.EGR:LST07	Posted free activity 7	decimal	7
ADEPRO.EGR:LST08	Posted free activity 8	decimal	7
ADEPRO.EGR:LST09	Posted free activity 9	decimal	7

ID	Description	DB type	Length
ADEPRO.EGR:LST10	Posted free activity 10	decimal	7
ADEPRO.RGR:LST01	Current remaining activity 1	decimal	7
ADEPRO.RGR:LST02	Current remaining activity 2	decimal	7
ADEPRO.RGR:LST03	Current remaining activity 3	decimal	7
ADEPRO.RGR:LST04	Current remaining activity 4	decimal	7
ADEPRO.RGR:LST05	Current remaining activity 5	decimal	7
ADEPRO.RGR:LST06	Current remaining activity 6	decimal	7
ADEPRO.RGR:LST07	Current remaining activity 7	decimal	7
ADEPRO.RGR:LST08	Current remaining activity 8	decimal	7
ADEPRO.RGR:LST09	Current remaining activity 9	decimal	7
ADEPRO.RGR:LST10	Current remaining activity 10	decimal	7
ADEPRO:EGE:LST01	Unit of recorded activity 1	char	3
ADEPRO:EGE:LST02	Unit of recorded activity 2	char	3
ADEPRO:EGE:LST03	Unit of recorded activity 3	char	3
ADEPRO:EGE:LST04	Unit of recorded activity 4	char	3
ADEPRO:EGE:LST05	Unit of recorded activity 5	char	3
ADEPRO:EGE:LST06	Unit of recorded activity 6	char	3
ADEPRO:EGE:LST07	Unit of recorded activity 7	char	3
ADEPRO:EGE:LST08	Unit of recorded activity 8	char	3
ADEPRO:EGE:LST09	Unit of recorded activity 9	char	3
ADEPRO:EGE:LST10	Unit of recorded activity 10	char	3
ADEPRO.FU:1	User-specific field	sqldate	7
ADEPRO.FU:2	User-specific field	sqldate	7
ADEPRO.FU:3	User-specific field	sqldate	7
ADEPRO.FU:4	User-specific field	sqldate	7
ADEPRO.FU:5	User-specific field	sqldate	7

ID	Description	DB type	Lengt h
ADEPRO.FU:6	User-specific field	sqldate	7
ADEPRO.FU:7	User-specific field	integer	7
ADEPRO.FU:8	User-specific field	integer	7
ADEPRO.FU:9	User-specific field	integer	7
ADEPRO.FU:10	User-specific field	integer	7
ADEPRO.FU:11	User-specific field	integer	7
ADEPRO.FU:12	User-specific field	integer	7
ADEPRO.FU:13	User-specific field	integer	7
ADEPRO.FU:14	User-specific field	integer	7
ADEPRO.FU:15	User-specific field	integer	7
ADEPRO.FU:16	User-specific field	integer	7
ADEPRO.FU:17	User-specific field	integer	7
ADEPRO.FU:18	User-specific field	integer	7
ADEPRO.FU:19	User-specific field	integer	7
ADEPRO.FU:20	User-specific field	integer	7
ADEPRO.FU:21	User-specific field	integer	7
ADEPRO.FU:22	User-specific field	integer	7
ADEPRO.FU:23	User-specific field	decimal	7
ADEPRO.FU:24	User-specific field	decimal	7
ADEPRO.FU:25	User-specific field	decimal	7
ADEPRO.FU:26	User-specific field	decimal	7
ADEPRO.FU:27	User-specific field	decimal	7
ADEPRO.FU:28	User-specific field	decimal	7
ADEPRO.FU:29	User-specific field	char	1
ADEPRO.FU:30	User-specific field	char	1

ID	Description	DB type	Length
ADEPRO.FU:31	User-specific field	char	1
ADEPRO.FU:32	User-specific field	char	1
ADEPRO.FU:33	User-specific field	char	1
ADEPRO.FU:34	User-specific field	char	1
ADEPRO.FU:35	User-specific field	char	1
ADEPRO.FU:36	User-specific field	char	1
ADEPRO.FU:37	User-specific field	char	1
ADEPRO.FU:38	User-specific field	char	1
ADEPRO.FU:39	User-specific field	char	1
ADEPRO.FU:40	User-specific field	char	1
ADEPRO.FU:41	User-specific field	char	1
ADEPRO.FU:42	User-specific field	char	1
ADEPRO.FU:43	User-specific field	char	1
ADEPRO.FU:44	User-specific field	char	1
ADEPRO.FU:45	User-specific field	char	10
ADEPRO.FU:46	User-specific field	char	10
ADEPRO.FU:47	User-specific field	char	10
ADEPRO.FU:48	User-specific field	char	10
ADEPRO.FU:49	User-specific field	char	10
ADEPRO.FU:50	User-specific field	char	10
ADEPRO.FU:51	User-specific field	char	20
ADEPRO.FU:52	User-specific field	char	20
ADEPRO.FU:53	User-specific field	char	20
ADEPRO.FU:54	User-specific field	char	20
ADEPRO.FU:55	User-specific field	char	20
ADEPRO.FU:56	User-specific field	char	20

ID	Description	DB type	Lengt h
ADEPRO.FU:57	User-specific field	char	20
ADEPRO.FU:58	User-specific field	char	20
ADEPRO.FU:59	User-specific field	char	20
ADEPRO.FU:60	User-specific field	char	20
ADEPRO.FU:61	User-specific field	char	20
ADEPRO.FU:62	User-specific field	char	20
ADEPRO.FU:63	User-specific field	char	20
ADEPRO.FU:64	User-specific field	char	20
ADEPRO.FU:65	User-specific field	char	40
ADEPRO.FU:66	User-specific field	char	40
ADEPRO.EGR:PBMK01	Personnel activity posted in RPA 1	integer	7
ADEPRO.EGR:PBMK02	Personnel activity posted in RPA 2	integer	7
ADEPRO.EGR:PBMK03	Personnel activity posted in RPA 3	integer	7
ADEPRO.EGR:PBMK04	Personnel activity posted in RPA 4	integer	7
ADEPRO.EGR:PBMK05	Personnel activity posted in RPA 5	integer	7
ADEPRO.EGR:PBMK06	Personnel activity posted in RPA 6	integer	7
ADEPRO.EGR:PBMK07	Personnel activity posted in RPA 7	integer	7
ADEPRO.EGR:PBMK08	Personnel activity posted in RPA 8	integer	7
ADEPRO.EGR:PBMK09	Personnel activity posted in RPA 9	integer	7
ADEPRO.EGR:PBMK10	Personnel activity posted in RPA 10	integer	7
ADEPRO.EGR:PBMK11	Personnel activity posted in RPA 11	integer	7
ADEPRO.EGR:PBMK12	Personnel activity posted in RPA 12	integer	7
ADEPRO.RGR:PBMK01	Current remaining personnel activity RPA 1	integer	7
ADEPRO.RGR:PBMK02	Current remaining personnel activity RPA 2	integer	7
ADEPRO.RGR:PBMK03	Current remaining personnel activity RPA 3	integer	7

ID	Description	DB type	Lengt h
ADEPRO.RGR:PBMK04	Current remaining personnel activity RPA 4	integer	7
ADEPRO.RGR:PBMK05	Current remaining personnel activity RPA 5	integer	7
ADEPRO.RGR:PBMK06	Current remaining personnel activity RPA 6	integer	7
ADEPRO.RGR:PBMK07	Current remaining personnel activity RPA 7	integer	7
ADEPRO.RGR:PBMK08	Current remaining personnel activity RPA 8	integer	7
ADEPRO.RGR:PBMK09	Current remaining personnel activity RPA 9	integer	7
ADEPRO.RGR:PBMK10	Current remaining personnel activity RPA 10	integer	7
ADEPRO.RGR:PBMK11	Current remaining personnel activity RPA 11	integer	7
ADEPRO.RGR:PBMK12	Current remaining personnel activity RPA 12	integer	7
ADEPRO.RGR:PDAUER	Current remaining labor time	integer	7
ADEPRO.RGR:BMK01	Current remaining activity RPA 1	integer	7
ADEPRO.RGR:BMK02	Current remaining activity RPA 2	integer	7
ADEPRO.RGR:BMK03	Current remaining activity RPA 3	integer	7
ADEPRO.RGR:BMK04	Current remaining activity RPA 4	integer	7
ADEPRO.RGR:BMK05	Current remaining activity RPA 5	integer	7
ADEPRO.RGR:BMK06	Current remaining activity RPA 6	integer	7
ADEPRO.RGR:BMK07	Current remaining activity RPA 7	integer	7
ADEPRO.RGR:BMK08	Current remaining activity RPA 8	integer	7
ADEPRO.RGR:BMK09	Current remaining activity RPA 9	integer	7
ADEPRO.RGR:BMK10	Current remaining activity RPA 10	integer	7
ADEPRO.RGR:BMK11	Current remaining activity RPA 11	integer	7
ADEPRO.RGR:BMK12	Current remaining activity RPA 12	integer	7
ADEPRO.RGR:DAUER	Remaining run time	integer	7
ADEPRO.ERFART	"-": original data record of data collection O: original data record if it is edited E: manually created data record (edited) S: cancellation for PPS (posting is deleted because of upload)	char	3

ID	Description	DB type	Lengt h
	X: posting is locked; no booking is made because of plausibility error.		
ADEPRO.SKART	Shift ID of the assigned shift	char	1
ADEPRO.SKMOD	Shift model used (day type)	smallint	7
ADEPRO.LART	reserved	char	4
ADEPRO.EGR:TE	Target time per unit *te* from backlog of orders	decimal	7
ADEPRO.EGR:TR	Target setup time *tr* from backlog of orders	decimal	7
ADEPRO.EGR:TEB	Target machine time per unit *teb* from backlog of orders	decimal	7
ADEPRO.EGR:TRB	Target specified setup time *trb* from backlog of orders	decimal	7
ADEPRO.BPOS	Operator position	char	10
ADEPRO.LPKZ	Wage/premium indicator	char	10
ADEPRO.LEISTGRP	Premium group •	char	10
ADEPRO.CNR	Batch number	char	20

12.3 Configuration of Data Collection Reason Texts

12.3.1.1 Edit reason texts (DLG=GRTXT.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

You use these BAPI calls to edit reason texts.

Tables

Table	Key field	Description
ade_grund_texte	grundtext_nr GRTXT.GRTXTNR	Reason text number (PK)

BAPI call

ID	Content / {type}	Description
DLG	GRTXT.INSERT	Create reason text

ID	Content / {type}	Description
	GRTXT.UPDATE	Change reason text
	GRTXT.DELETE	Delete reason text
	GRTXT.COPY	Copy reason text
	GRTXT.LOCK	Lock reason text for editing
	GRTXT.UNLOCK	Unlock reason text after editing
	GRTXT.NEW	Read specification for new reason text
	GRTXT.SELECT	Select reason text
GRTXT.GRTXT NR	{N4}	PK reason text number
GRTXT.GRTXT NR:Z	{N4}	Target - reason text number
…	…	For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
GRTXTNR	{N4}	Current data record

Plausibility checks

Error codes	Description
1661	A value is missing that is required for processing.
1666	The reason text is currently edited by another user.

12.3.1.2 List of reason texts (DLG=GRTXT.LIST)

The BAPI call returns all defined reason texts.

Tables

Table	Key field	Description
ade_grund_texte	grundtext_nr GRTXT.GRTXTNR	Reason text number (PK)

BAPI call

ID	Contents	Description
DLG	GRTXT.LIST	List of reason texts
DATEI	{C256}	Specification of the file name for the list

12.3.2 Reasons

12.3.2.1 Edit reasons (DLG=GR.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

You use these BAPI calls to edit reasons.

Tables

Table	Key field	Description
ade_grund_zuord	masch_nr art grund GR.MNR GR.ART GR.GR	The combination must be unique.

BAPI call

ID	Content / {type}	Description
DLG	GR.INSERT	Create reason
	GR.UPDATE	Change reason
	GR.DELETE	Delete reason
	GR.COPY	Copy reason
	GR.LOCK	Lock reason for editing
	GR.UNLOCK	Unlock reason after editing
	GR.NEW	Read specification for new reason
	GR.SELECT	Select reason
GR.MNR	C20	PK machine number

ID	Content / {type}	Description
GR.ART	C1	PK type A = scrap N = rework P = problem quantity G = yield L = batch reason
GR.GR	N8	PK reason number
GR.MNR:Z	C20	Target machine number
GR.ART:Z	C1	Target type
GR.GR:Z	N8	Target reason number
MOD	C1	Copy mode E- Copy currently selected reason G- Copy all reasons F- Copy missing reasons M- Copy a reason for all machines
…	…	For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
MNRMNR	C20	Current data record
ART	C1	Current data record
GR	N8	Current data record

Plausibility checks

Error codes	Description
1661	A value is missing that is required for processing.
1666	The reason text is currently edited by another user.
712	Processing mode is invalid.
1401	Invalid deviation reason

Error codes	Description
1803	No responsibility area authorization. If you create workplace/machine-related reasons, the system checks the responsibility area of the workplace. To override this check, you can set the parameter VABCHECK=N.
3273	Not possible - A reason can either be assigned to SYSTEM or to a random number of machines.
709	Copy: The target machine has not been specified.
710	Copy: The target status has not been specified.

12.3.2.2 List of reasons (DLG=GR.LIST)

The BAPI call returns all defined reasons.

Tables

Table	Key field	Description
ade_grund_zuord	masch_nr art grund GR.MNR GR.ART GR.GR	The combination must be unique.

BAPI call

ID	Contents	Description
DLG	GR.LIST	List of reason texts
DATEI	{C256}	Specification of the file name for the list

12.4 Order Management and Order Sequencing

12.4.1 Lock order/operation for editing (ANR.LOCK)

You lock the operation to prevent the operation from being edited by another user.

Tables

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number if ANR.ATYP = OP

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number if ANR.ATYP = AU (order)

BAPI call

ID	Content / {type}	Description
DLG	ANR.LOCK	Plan operation
ANR.ATYP	{C4}	Order type {AU} Order type {AG}
ANR.ANR	{C40}	Combined order/OP number if ANR.ATYP=OP
ANR.AUNR	{C40}	Order number if ANR.ATYP=AU (order)
BEARB	{C10}	Modified by
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Return

ID	Content / {type}	Description
*		All attributes of the order or the operation

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked BEARB

12.4.2 Unlock order/operation for editing (ANR.UNLOCK)

Unlock an operation locked by a user.

Tables

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number if ANR.ATYP = OP

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number if ANR.ATYP = AU (order)

BAPI call

ID	Content / {type}	Description
DLG	ANR.LOCK	Plan operation
ANR.ATYP	{C4}	Order type {AU} Order type {AG}
ANR.ANR	{C40}	Combined order/OP number if ANR.ATYP=OP
ANR.AUNR	{C40}	Order number if ANR.ATYP=AU (order)
BEARB	{C10}	Modified by
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified

12.4.3 Plan operation (ANR.EINPLANEN)

Plan an operation for a workplace.

Tables

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number

BAPI call

ID	Content / {type}	Description
DLG	ANR.EINPLANEN	Plan operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.MNR	{C8}	Workplace
BEARB	{C10}	Modified by
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
90	Machine is not available ANR.MNR is not specified
1658	No cost center authorization for machine
1666	Object is locked BEARB

12.4.4 Deallocate operation (ANR.AUSPLANEN)

Deallocate an operation from a workplace

Tables

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
…		

BAPI call

ID	Content / {type}	Description
DLG	ANR.AUSPLANEN	Deallocate operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.MGRP	{C8}	Group
BEARB	{C10}	Modified by
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1658	No cost center authorization for group
1666	Object is locked BEARB

12.4.5 Block operation (ANR.SPERREN)

Block the data collection for an operation

Tables

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
…		

BAPI call

ID	Content / {type}	Description
DLG	ANR.SPERREN	Block operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number

ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
101	No data available! ANR.ANR is not available

12.4.6 Unlock operation (ANR.ENTSPERREN)

Unlock an operation that is blocked for data collection.

Tables

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
…		

BAPI call

ID	Content / {type}	Description
DLG	ANR.ENTSPERREN	Unlock operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
101	No data available!

	ANR.ANR is not available

12.4.7 Update operation (ANR.AKTUALISIEREN)

Tables

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number
…		

BAPI call

ID	Content / {type}	Description
DLG	ANR.AKTUALISIEREN	Update order
ANR.ATYP	{C4}	Order type {AU}
ANR.AUNR	{C40}	Order number
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
101	No data available! ANR.AUNR is not available

12.4.8 Release operation (ANR.FREIGEBEN)

Set the operation status to "Prepared".

Tables

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
…		

BAPI call

ID	Content / {type}	Description
DLG	ANR.ENTSPERREN	Unlock operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
101	No data available! ANR.ANR is not available

12.4.9 Change order status (ANR.SETSTATUS)

Change order status

Tables

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number
…		

BAPI call

ID	Content / {type}	Description
DLG	ANR.SETSTATUS	Change order status
ANR.ATYP	{C4}	AU – Type
ANR.AUNR	{C40}	Order number
ANR.TABLE	{C1}	A – Backlog of orders (default value)

<table>
<tr><td>MOD</td><td>{C1}</td><td colspan="2">S → Sest status
E → Finish order
R → Reactivate order</td></tr>
<tr><td>ANR.OPT:PKENN</td><td>{C2}</td><td rowspan="2">alternative</td><td>Production ID of the new status
with MOD=S
only "E" is permitted</td></tr>
<tr><td>ANR.AST</td><td>{C2}</td><td>New status
with MOD=S
only status with production identifier = "E" is permitted</td></tr>
</table>

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
101	No data available! ANR.AUNR is not available
2026	Order is still running
712	Processing mode is invalid if MOD <> 'S'

Processing notes

Reactivate (MOD=R):

If you reactivate the order header, ALL operations of the order are reactivated.

The status of the order header is set as follows:

- Status V if no OP of the order has been started
- otherwise status L

"Finish order" (MOD=E):

If you finish the order header, ALL operations of the order are finished.

12.4.10 Change operation status (ANR.SETSTATUS)

Change operation status

Tables

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
…		

BAPI call

<table>
<tr><th>ID</th><th>Content / {type}</th><th colspan="2">Description</th></tr>
<tr><td>DLG</td><td>ANR.SETSTATUS</td><td colspan="2">Change order status</td></tr>
<tr><td>ANR.ATYP</td><td>{C4}</td><td colspan="2">OP – Type</td></tr>
<tr><td>ANR.ANR</td><td>{C40}</td><td colspan="2">MES order number</td></tr>
<tr><td>ANR.TABLE</td><td>{C1}</td><td colspan="2">A – Backlog of orders (default value)</td></tr>
<tr><td>MOD</td><td>{C1}</td><td colspan="2">S → Set status/secondary status/resource status for the operation
E → Finish operation
R → Reactivate operation</td></tr>
<tr><td>MOD:2</td><td>{C1}</td><td colspan="2">R → Set resource status for the operation
With MOD=S; ANR.STATUSTYP must not be set</td></tr>
<tr><td>ANR.STATUSTYP</td><td>{C1}</td><td colspan="2">S → Set secondary status
With MOD=S; MOD2 must not be set</td></tr>
<tr><td>ANR.OPT:PKENN</td><td>{C2}</td><td rowspan="2">Other option</td><td>Control indicator of the new status
with MOD=S</td></tr>
<tr><td>ANR.AST</td><td>{C2}</td><td>New status with MOD=S
➔ You may only set statuses with control indicator=S.
New secodary status with MOD=S and ANR.STATUSTYP=S
The statuses must be configured in the system.</td></tr>
<tr><td>ANR.RESSTA:1</td><td>{C2}</td><td colspan="2">New resource status "Person OK"
With MOD=S and MOD2:R</td></tr>
<tr><td>ANR.RESSTA:2</td><td>{C2}</td><td colspan="2">New resource status "Tool OK"
With MOD=S and MOD2:R</td></tr>
<tr><td>ANR.RESSTA:3</td><td>{C2}</td><td colspan="2">New resource status "Material OK"
With MOD=S and MOD2:R</td></tr>
</table>

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
101	No data available! ANR.ANR is not available
2026	Order is still running
2807	Status cannot be changed, as sequence is inactive

Examples

Set operation to the status with control indicator V

DLG=ANR.SETSTATUS|ANR.ATYP=AG|ANR.ANR=S30002000700|**MOD=S|ANR.OPT:PKENN=V**|BEARB=12345|DAT=08/31/2018|ZEI=80806|

Finish operation

DLG=ANR.SETSTATUS|ANR.ATYP=AG|ANR.ANR=S30002000700|**MOD=E**|BEARB=12345|DAT=08/31/2018|ZEI=81180|

Set secondary status

DLG=ANR.SETSTATUS|ANR.ATYP=AG|ANR.ANR=S30002000700|**MOD=S|ANR.STATUSTYP=S**|ANR.AST=801|BEARB=12345|DAT=08/31/2018|ZEI=80632|

Set resource status

DLG=ANR.SETSTATUS|ANR.ATYP=AG|ANR.ANR=S30002000700|**MOD=S|MOD:2=R**|ANR.RESSTA:1=OK|ANR.RESSTA:2=OK|ANR.RESSTA:3=OK|BEARB=12345|DAT=08/31/2018|ZEI=80740|

Reactivate operation

DLG=ANR.SETSTATUS|ANR.ATYP=AG|ANR.ANR=S30002000700|**MOD=R**|BEARB=12345|DAT=08/31/2018|ZEI=81180|

12.4.11 Create order (ANR.INSERT)

Order (ANR.TABLE=A)

Work plan (ANR.TABLE=P)

Template (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number

auftrags_leistung	ANR.AUNR	
auftrag_status	ANR.AUNR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.AUNR	PK Order number
arbplan_leistung	ANR.AUNR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.INSERT	Create order
ANR.ATYP	{C4}	Order type {AU}
ANR.AUNR	{C40}	Order number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.AUART	{C5}	Order type
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
AUNR	{C40}	Order number
AST	{C5}	Status
OPT:PKENN	{C2}	Control characteristic "production"

ID	Content / {type}	Description
TABLE	{C1}	
ATYP	{C4}	

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1669	Data is already available → ANR.AUNR is already available
2812	Order is already available in the archive
50	Order status is not available → ANR.AUART is not specified

12.4.12 Edit order (ANR.UPDATE)

Order (ANR.TABLE=A)

Work plan (ANR.TABLE=P)

Template (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number
auftrags_leistung	ANR.AUNR	
auftrag_status	ANR.AUNR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.AUNR	PK Order number
arbplan_leistung	ANR.AUNR	

arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.UPDATE	Edit order
ANR.ATYP	{C4}	Order type {AU}
ANR.AUNR	{C40}	Order number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
*		All attributes of the order

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
101	No data available. ANR.AUNR is not available.
1803	No responsibility area authorization
50	Order status is not available
900	Unit is not available
2809	Order type cannot be changed because order has started

Error codes	Description
3241	User field key is not defined
1989	Maximum number of orders is exceeded for priority The maximum number of orders is exceeded for priority when using the priority management

12.4.13 Copy order (ANR.COPY)

Order (ANR.TABLE=A)

Work plan (ANR.TABLE=P)

Template (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number
auftrags_leistung	ANR.AUNR	
auftrag_status	ANR.AUNR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.AUNR	PK Order number
arbplan_leistung	ANR.AUNR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.COPY	Copy order
ANR.ATYP	{C4}	Order type {AU}
ANR.AUNR	{C40}	Order number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.TABLE:Z	{C1}	Target type: A – Order P – Work plan
ANR.AUNR:Z	{C40}	New order number
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
ANR		Order number of the new order

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
1669	Data is already available → ANR.AUNR is already available
2812	Order is already available in the archive
1855	Target data record is already available
900	Unit is not available
1997	Order type is not equal If you copy an order, the order types of the source and the target order must be identical.
1998	Order template is not available

Notes on the processing

Copying a complete order:

If you copy a complete order, all values specified in the dialog data string are passed for ALL operations!

12.4.14 Delete order (ANR.DELETE)

Order (ANR.TABLE=A)

Work plan (ANR.TABLE=P)

Template (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number
auftrags_leistung	ANR.AUNR	
auftrag_status	ANR.AUNR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.AUNR	PK Order number
arbplan_leistung	ANR.AUNR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.DELETE	Delete order

ANR.ATYP	{C4}	Order type {AU}
ANR.AUNR	{C40}	Order number
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
101	No data available. ANR.AUNR is not available.
1803	No responsibility area authorization
1990	Order header must not be deleted

12.4.15 Select order (ANR.SELECT)

Order (ANR.TABLE=A)
Work plan (ANR.TABLE=P)
Template (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number
auftrags_leistung	ANR.AUNR	
auftrag_status	ANR.AUNR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.AUNR	PK Order number
arbplan_leistung	ANR.AUNR	

arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.SELECT	Select order
ANR.ATYP	{C4}	Order type {AU}
ANR.AUNR	{C40}	Order number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.ARCHIV	{C1}	The archive is used

Return

ID	Content / {type}	Description
*		All attributes of the order

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
101	No data available. ANR.AUNR is not available.

12.4.16 Order – Select all operations (ANR.LIST)

Order (ANR.TABLE=A)

Work plan (ANR.TABLE=P)

Template (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number
auftrags_leistung	ANR.AUNR	
auftrag_status	ANR.AUNR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.AUNR	PK Order number
arbplan_leistung	ANR.AUNR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.LIST	Select all operations of the order
ANR.ATYP	{C4}	Order type {AU}
ANR.AUNR	{C40}	Order number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.ARCHIV	{C1}	The archive is used

Return

ID	Content / {type}	Description
*		List including all operations and all attributes of the operations

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
101	No data available. ANR.AUNR is not available.

12.4.17 Create operation (ANR.INSERT)

Operation (ANR.TABLE=A)

Work plan-operation (ANR.TABLE=P)

Template-operation (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.ANR	PK Combined order/OP number
arbplan_leistung	ANR.ANR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.INSERT	Create operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.MNR ANR.MGRP	{C8} {C8}	Workplace Group One of the two values must be specified
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
ANR	{C40}	Combined order/OP number
AUNR	{C40}	Order number
AGNR	{C40}	OP number
AFOLG	{C40}	Sequence number
UAGNR	{C40}	Suboperation number
SPLNR	{C40}	Split No.
TABLE	{C1}	
ATYP	{C4}	Order type {AG} (OP)
AUART	{C5}	Order type
AST	{C2}	Status
OPT:PKENN	{C2}	Control characteristic "production"

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified

1669	Data is already available → ANR.ANR is already available
2812	Order is already available in the archive
94	OP ➔ Machine group is not available

12.4.18 Edit operation (ANR.UPDATE)

Operation (ANR.TABLE=A)

Work plan-operation (ANR.TABLE=P)

Template-operation (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.ANR	PK Combined order/OP number
arbplan_leistung	ANR.ANR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.UPDATE	Edit operation

ID	Content / {type}	Description
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
*		All attributes of the order

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
101	No data available. ANR.ANR is not available.
1803	No responsibility area authorization
50	Order status is not available
900	Unit is not available
2809	Order type cannot be changed because order has started
3241	User field key is not defined
1989	Maximum number of orders is exceeded for priority The maximum number of orders is exceeded for priority when using the priority management
911	Processing code is not available
94	Machine group ######### is not available
2808	Start date is after end date
2813	Activity code is not defined

12.4.19 Copy operation (ANR.COPY)

Operation (ANR.TABLE=A)

Work plan-operation (ANR.TABLE=P)

Template-operation (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.ANR	PK Combined order/OP number
arbplan_leistung	ANR.ANR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.COPY	Copy order
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.TABLE:Z	{C1}	Target type:

ID	Content / {type}	Description
		A – Order P – Work plan
ANR.ANR:Z	{C40}	New combined order/OP number
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
ANR		New combined order/OP number

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
1669	Data is already available → ANR.ANR is already available
2812	Order is already available in the archive
1855	Target data record is already available
900	Unit is not available
1997	Order type is not equal If you copy an order, the order types of the source and the target order must be identical.
1998	Order template is not available

12.4.20 Delete operation (ANR.DELETE)

Operation (ANR.TABLE=A)

Work plan-operation (ANR.TABLE=P)

Template-operation (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.ANR	PK Combined order/OP number
arbplan_leistung	ANR.ANR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.DELETE	Delete operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked

101	No data available. ANR.ANR is not available.
1803	No responsibility area authorization
1986	Operation cannot be deleted

12.4.21 Select operation (ANR.SELECT)

Operation (ANR.TABLE=A)

Work plan-operation (ANR.TABLE=P)

Template-operation (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.ANR	PK Combined order/OP number
arbplan_leistung	ANR.ANR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.SELECT	Select operation

ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.ARCHIV	{C1}	The archive is used

Return

ID	Content / {type}	Description
*		All attributes of the order

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
101	No data available. ANR.ANR is not available.

12.4.22 Split operation (ANR.SPLITCREATE)

Operation (ANR.TABLE=A)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

BAPI call

ID	Content / {type}	Description
DLG	ANR.SPLITCREATE	Split operation

ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.ANR	{C40}	Combined order/OP number
ANR.ANZSPLIT	{N8}	Number of splits

Return

ID	Content / {type}	Description
ANR.MAXANZSPLIT	{N8}	Maximum split number of operation

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
101	No data available!
1859	OP must not be a split master This operation is a so-called "split master", i.e. it is an operation that has been split up into several operations.
1860	OP must not be OP of a split OP
1862	OP must not be OP of a merged OP
1867	The OP may not be split.
1868	Maximum number of splits of the OP has been exceeded The OP that you want to split exceeds the maximum number of splits specified for the operation.
1869	The maximum number of slits has been exceeded.
30	Order is finished
20	Order is running
85	OP is locked
80	invalid OP status
31	Order is interrupted
1666	Object is locked

Error codes	Description
	BEARB

12.4.23 Cancel operation split (ANR.SPLITDELETE)

Operation (ANR.TABLE=A)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

BAPI call

ID	Content / {type}	Description
DLG	ANR.SPLITDELETE	Cancel operation split
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.ANR	{C40}	Combined order/OP number + split number

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
101	No data available!
20	Order is running

12.4.24 Enhanced operation split (ANR.ADVSPLITCREATE)

Requirements

- If you want to use the split function via PDM interface, you must configure the enhanced split function in HYDRA.

Table

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrag_status	ANR.ANR	

BAPI call

ID	Content / {type}	Description
DLG	ANR.ADVSPLITCREATE	Enhanced split function
ANR.SPL	Operation (ANR)	The operation to be split (complete operation number). • An OP that is configured as OP that may be split • Or an already split OP (complete operation number incl. split)
SPL:MOD	PDM	Mode for the PDM interface
SPL:DIFFQUANTITY	J/N	Default = J: **Relevant only if you create splits using the master OP (OP has not yet been split).** Automatically creates a further split using the difference quantity. I.e. one additional split is created compared to the number specified in the IDs SPL.EGRGUTP:X,… To create a split using the difference quantity, the sum total of the target quantities of all splits to be created must be smaller than the target quantity of the master OP.
ANR.OPT:SPL_EGR_UPD	J/N	Default = J: Relevant if a split is split up a further time (MOC client function "Offset target quantity of the split OP"). The target quantity of an OP (here it is already a split), which is to be split, is reduced respectively.

ID	Content / {type}	Description
ANR.OPT:PLAN	M/G	Splits planned for machine or group. If not specified, the value of the OP to be split (master OP or split) is taken over.
SPL.EGRGUTP:X	Primary target quantity included in the split that is to be created	Target quantity (primary quantity) of the individual split. (X) stands for continuous numbers from 1...99
SPL.RUEZ:X	Duration in seconds	Setup time of the individual split. (X) stands for continuous numbers from 1...99, if not specified (ID is not included) the setup time of the OP is used that has been split.
BEARB	Modified by	
DAT	Date	(Example: 09/24/2014)
ZEIT	Time in seconds since midnight.	

Note

X in the IDs of the above table must be replaced successively with 1...99 for each split to be created. The ANR-BAPI automatically assigns the split numbers. For example, if an OP that has already been split up is split a new time, the split numbers are already assigned. The BAPI then automatically searches for the next free split number. If an ascending number is not assigned, the processing ends (e.g. in the string that follows, only the first two splits are created). ...|SPL.EGRGUTP:1=10|SPL.EGRGUTP:2=20| SPL.EGRGUTP:5=50|…)

Plausibility checks

Error codes	Description
10	Order not available
1859	No split possibility OP must not be a split master
1971	Split number < Minimum below min. number of splits
424	Split function is not active

Examples:

```
Split operation (use setup time of the master, no split with the difference quantity is created):
DLG=ANR.ADVSPLITCREATE|ANR.SPL=160100400020|SPL:MOD=PDM|ANR.OPT:PLAN=G|SPL.EGRGUTP:1=15|SPL.EGRGUTP:2=10|SPL:DIFFQUANTITY=N|BEARB=12345|DAT=09/29/2014|ZEIT=44421|
```

```
Existing split is again split up (specified setup time, quantities are not reduced by the OP to be split. I.e. the total target quantity of the operation is increased by the two new splits.):
DLG=ANR.ADVSPLITCREATE|ANR.SPL=16010040002002|SPL:MOD=PDM|ANR.OPT:PLAN=G|SPL.EGRGUTP:1=5|SPL.RUEZ:1=1200|SPL.EGRGUTP:2=100|SPL.RUEZ:2=1200|ANR.OPT:SPL_EGR_UPD=N|BEARB=12345|DAT=09/29/2014|ZEIT=44621|"
```

12.4.25 Create merged operation (ANR.SAGINSERT)

Operation (ANR.TABLE=A)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

BAPI call

ID	Content / {type}	Description
DLG	ANR.SAGINSERT	Create merged operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.SANR	{C40}	Merged operation Combined order/OP number
ANR.ANR	{C40}	Reference OP Combined order/OP number
ANR.ANR:1	{C40}	Operations that you want to combine into one merged operation.

ID	Content / {type}	Description
ANR.ANR:2 .. ANR.ANR:n		
ANR.SGR:GUTB	{DEC13,3}	Target quantity yield
ANR.SGR:AUSB	{DEC13,3}	Target quantity scrap
ANR.SGE:B	{C1}	Base quantity unit of the merged OP
ANR.RUEZ	{N8}	Setup time
ANR.BEARBZ	{N8}	Processing time

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1669	Data is already available → ANR.ANR is already available
2812	Order is already available in the archive
101	No data available! → This message also pops up if no reference OP is specified.

12.4.26 Delete merged operation (ANR.SAGDELETE)

Operation (ANR.TABLE=A)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

BAPI call

ID	Content / {type}	Description
DLG	ANR.SAGDELETE	Delete merged operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.SANR	{C40}	Merged operation Combined order/OP number
MOD	{C1}	E – Remove single OP from MOP G – Remove all OPs from MOP

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
101	No data available! → This message also pops up if no reference OP is specified.
1866	Specified OP is no OP of the merged OP
1865	Specified MOP is no merged OP

12.4.27 Lock order network for editing (ANETZ.LOCK)

Table	Key field	Description
ade_auftragsnetz	ANETZ.ANRV ANETZ.ANRN	PK Predecessor + successor

BAPI call

ID	Content / {type}	Description
DLG	ANETZ.LOCK	Lock order network
ANETZ.ANRV	{C40}	Predecessor Combined order/OP number
ANETZ.ANRN	{C40}	Successor

		Combined order/OP number

Return

ID	Content / {type}	Description
*		All attributes of the order network

Plausibility checks

Error codes	Description
1661	Missing parameter
1666	Object is locked BEARB

12.4.28 Unlock order network for editing (ANETZ.UNLOCK)

Table	Key field	Description
ade_auftragsnetz	ANETZ.ANRV ANETZ.ANRN	PK Predecessor + successor

BAPI call

ID	Content / {type}	Description
DLG	ANETZ.UNLOCK	Unlock order network
ANETZ.ANRV	{C40}	Predecessor Combined order/OP number
ANETZ.ANRN	{C40}	Successor Combined order/OP number

Plausibility checks

Error codes	Description
1661	Missing parameter

12.4.29 Create order network (ANETZ.INSERT)

Table	Key field	Description
ade_auftragsnetz	ANETZ.ANRV ANETZ.ANRN	PK Predecessor + successor

BAPI call

ID	Content / {type}	Description
DLG	ANETZ.INSERT	Create order network
ANETZ.ANRV	{C40}	Predecessor Combined order/OP number
ANETZ.ANRN	{C40}	Successor Combined order/OP number
ANETZ.AOB	{C2}	Relationship ES = End-Start
ANETZ.AKTIV	{C1}	J - Active
ANETZ.HERK	{C1}	Origin E = Explicit (via interface)
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
ANRV	{C40}	Predecessor
ANRN	{C40}	Successor

Plausibility checks

Error codes	Description
1661	Missing parameter
1669	Data is already available

12.4.30 Edit order network (ANETZ.UPDATE)

Table	Key field	Description
ade_auftragsnetz	ANETZ.ANRV ANETZ.ANRN	PK Predecessor + successor

BAPI call

ID	Content / {type}	Description
DLG	ANETZ.UPDATE	Edit order network
ANETZ.ANRV	{C40}	Predecessor Combined order/OP number
ANETZ.ANRN	{C40}	Successor Combined order/OP number
MOD	{C1}	V – predecessor – update relationships
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Plausibility checks

Error codes	Description
1661	Missing parameter
1666	Object is locked

12.4.31 Delete order network (ANETZ.DELETE)

Table	Key field	Description
ade_auftragsnetz	ANETZ.ANRV ANETZ.ANRN	PK Predecessor + successor

BAPI call

ID	Content / {type}	Description
DLG	ANETZ.UPDATE	Delete order network
ANETZ.ANRV	{C40}	Predecessor Combined order/OP number

ANETZ.ANRN	{C40}	Successor Combined order/OP number
ANETZ.AUNR	{C40}	Order number with MOD=A
MOD	{C1}	A – Delete complete network

Plausibility checks

Error codes	Description
1661	Missing parameter
1666	Object is locked

12.4.32 Update order network (ANETZ.AKTUALISIEREN)

Table	Key field	Description
ade_auftragsnetz	ANETZ.ANRV ANETZ.ANRN	PK Predecessor + successor

BAPI call

ID	Content / {type}	Description
DLG	ANETZ.AKTUALISIEREN	Update order network
ANETZ.ANR	{C40}	Combined order/OP number
ANETZ.AUNR	{C40}	Order number
ANETZ.AGNR	{C40}	OP number
ANETZ.AFOLG	{C40}	Sequence number
MOD	{C1}	**MOD=I :** Recalculation of the ANETZ relationship of the added OP to the preceding and succeeding operation. **MOD=D :** Recalculation of the ANETZ relationship of the deleted OP to the preceding and succeeding operation. **MOD=A :** Recalculation of the activated ANETZ relationship of a sequence **MOD=N :** Complete order network recalculation of an order

Plausibility checks

Error codes	Description
712	Processing mode is invalid
1661	Missing parameter
1669	Data is already available with MOD=I

12.4.33 Lock material list for editing (MATLIST.LOCK)

Operation - material list (MATLIST.TABLE=A)

Workplan-operation - material list (MATLIST.TABLE=P)

Table (MATLIST.TABLE=A)

Table	Key field	Description
mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

Table (MATLIST.TABLE=P)

Table	Key field	Description
arbplan_mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

BAPI call

ID	Content / {type}	Description
DLG	MATLIST.LOCK	Lock material list
MATLIST.ANR	{C40}	Combined order/OP number
MATLIST.ATK	{C40}	Material
MATLIST.RESTYP	{C4}	Resource type
MATLIST.SLP	{C10}	BOM item (default=1)

Return

ID	Content / {type}	Description
MATLIST.*		All attributes of the newly created entry in the material list

Plausibility checks

Error codes	Description
1661	Missing parameter
101	No data available!
1666	Object is locked

12.4.34 Unlock material list for editing (MATLIST.UNLOCK)

Operation - material list (MATLIST.TABLE=A)

Workplan-operation - material list (MATLIST.TABLE=P)

Table (MATLIST.TABLE=A)

Table	Key field	Description
mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

Table (MATLIST.TABLE=P)

Table	Key field	Description
arbplan_mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

BAPI call

ID	Content / {type}	Description
DLG	MATLIST.UNLOCK	Unlock material list
MATLIST.ANR	{C40}	Combined order/OP number

MATLIST.ATK	{C40}	Material
MATLIST.RESTYP	{C4}	Resource type
MATLIST.SLP	{C10}	BOM item (default=1)

Plausibility checks

Error codes	Description
1661	Missing parameter

12.4.35 Create material list (MATLIST.INSERT)

Operation - material list (MATLIST.TABLE=A)

Workplan-operation - material list (MATLIST.TABLE=P)

Table (MATLIST.TABLE=A)

Table	Key field	Description
mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

Table (MATLIST.TABLE=P)

Table	Key field	Description
arbplan_mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

BAPI call

ID	Content / {type}	Description
DLG	MATLIST.INSERT	Create material list
MATLIST.ANR	{C40}	Combined order/OP number
MATLIST.ATK	{C40}	Material
MATLIST.RESTYP	{C4}	Resource type
MATLIST.SLP	{C10}	BOM item (default=1)

…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
MATLIST.*		All attributes of the newly created entry in the material list

Plausibility checks

Error codes	Description
1661	Missing parameter
1669	Data is already available
3284	BOM level is invalid if MATLIST.SLS:M has been specified
1987	Operation cannot be changed
10	Order not available

12.4.36 Edit material list (MATLIST.UPDATE)

Operation - material list (MATLIST.TABLE=A)

Workplan-operation - material list (MATLIST.TABLE=P)

Table (MATLIST.TABLE=A)

Table	Key field	Description
mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

Table (MATLIST.TABLE=P)

Table	Key field	Description
arbplan_mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

BAPI call

ID	Content / {type}	Description
DLG	MATLIST.UPDATE	Edit material list
MATLIST.ANR	{C40}	Combined order/OP number
MATLIST.ATK	{C40}	Material
MATLIST.RESTYP	{C4}	Resource type
MATLIST.SLP	{C10}	BOM item (default=1)
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
MATLIST.*		All attributes of the entry in the material list

Plausibility checks

Error codes	Description
1661	Missing parameter
101	No data available!
1666	Object is locked
3284	BOM level is invalid if MATLIST.SLS:M has been specified
1987	Operation cannot be changed
10	Order not available

12.4.37 Delete material list (MATLIST.DELETE)

Operation - material list (MATLIST.TABLE=A)

Workplan-operation - material list (MATLIST.TABLE=P)

Table (MATLIST.TABLE=A)

Table	Key field	Description
mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

Table (MATLIST.TABLE=P)

Table	Key field	Description
arbplan_mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

BAPI call

ID	Content / {type}	Description
DLG	MATLIST.DELETE	Create order network
MATLIST.ANR	{C40}	Combined order/OP number
MATLIST.ATK	{C40}	Material
MATLIST.RESTYP	{C4}	Resource type
MOD		**A** - Delete all components of an OP **E** - Delete individual components **R** - Delete all resource components of an OP **M** - Delete all material components of an OP

Plausibility checks

Error codes	Description
1661	Missing parameter
2027	Processing mode is invalid
1986	Operation cannot be deleted
1666	Object is locked

13 HYDRA Production Data Manager MDE - Master Data

13.1 Note on the descriptions of the basic dialogs

All mandatory fields that must be specified have the addition PK (primary key). All other fields are optional and are processed if they are transferred.

13.2 Machine configuration

13.2.1 Edit machine configuration (DLG=MNR.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

Use this BAPI call to create a machine or a workplace. The machine statuses 20000 "NO SHIFT" and 30000 "NOT ASSIGNED" are automatically assigned.

Tables

Table	Key field	Description
maschinen	masch_nr MNR.MNR	Machine number (PK) in machine configuration
maschinen_status	masch_nr MNR.MNR	Machine number (PK) in machine status

BAPI call

Identification	Content / {type}	Description
DLG	MNR.INSERT	Create machine/workplace
	MNR.UPDATE	Change machine/workplace
	MNR.DELETE	Delete machine/workplace
	MNR.COPY	Copy machine/workplace
	MNR.LOCK	Lock machine/workplace for editing
	MNR.UNLOCK	Unlock machine/workplace after editing
	MNR.NEW	Read specification for new machine/new workplace
	MNR.SELECT	Select machine/workplace
MNR.MNR	{C8}	PK: workplace/machine number

		The valid characters are listed in the document MOC_ResourceConfiguration.pdf. The number must not exceed 8 characters.
MNR. MNR:Z	{C8}	PK: new (target) workplace/machine number for COPY The valid characters are listed in the document MOC_ResourceConfiguration.pdf. The number must not exceed 8 characters.
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
923	The cost center of the group must match the cost center of the machine or workplace.
1611	When you delete (MNR.DELETE), the machine cannot be removed from the group assignment.
1611	When you create a machine (MNR.INSERT) and the specified machine group (MNR.MGRP) has not yet been created as capacity group, this capacity group is then automatically created in the group assignment. Creating the group MNR.MGRP as capacity group failed.
1611	When you create a machine (MNR.INSERT), this machine is automatically assigned to the group assignment of the capacity group (MNR.MGRP). The automatic assignment failed.
1661	• The value MNR.MNR has not been specified. • The value MNR.MNR:Z has not been specified with MNR.COPY • The value DAT has not been specified with MNR.SKINFO
1803	The authorization for the responsibility area is not available and the specified machine cannot be edited.
1666	The machine is currently edited by another user.

13.2.2 List of machines/workplaces (DLG=MNR.LIST)

This list shows all machines or workplaces defined in the system.

Tables

Table	Key field	Description
maschinen	masch_nr MNR.MNR	Machine number (PK) in machine configuration

BAPI call

Identification	Contents	Description
DLG	MNR.LIST	List of machines
DATEI	{C256}	Specification of the file name for the list

13.2.3 Read shift information of the machine (DLG=MNR.SKINFO)

This BAPI call returns the current shift information of the machine. Depending on the configuration of the shift calender, up to 10 shifts (index 1-10) can be returned.

BAPI call

Identification	Contents	Description
DLG	MNR.SKINFO	Read shift information
MNR.MNR	{C8}	Machine number
MNR.MOD	{C1}	1: Identify shifts following the current machine date. 2: Identify shift times of a specific shift

Return mode 1

Identification	Contents	Description
MNR	{C8}	Machine number
DAT	{mm/dd/yyyy}	Current log time for the machine
ZEI	{seconds}	Current log time for the machine
SKDAT	{mm/dd/yyyy}	Current shift date for the machine
SKNR	{N1}	Current shift number for the machine (1-4)
SKZEIB	{seconds}	Current start of shift
SZEITE	{seconds}	Current end of shift
SKDAUER:RES T	{seconds}	Remaining time of the current shift
SKDAT:1 … SKDAT:10	{mm/dd/yyyy}	Date of the shift selected

SKNR:1 … SKNR:10	{N1}	Number of the shift selected (1-4)
SKZEIB:1 … SKZEIB:10	{seconds}	Start of the shift selected
SKZEIE:1 … SKZEIE:10	{seconds}	End of the shift selected
SKDAUER:1 … SKZEIB:10	{seconds}	Duration of the shift selected

Return mode 2

Identification	Contents	Description
MNR	{C8}	Machine number
DAT	{mm/dd/yyyy}	Current log time for the machine
ZEI	{seconds}	Current log time for the machine
SKDAT	{mm/dd/yyyy}	Current shift date for the machine
SKNR	{N1}	Current shift number for the machine (1-4)
SKZEIB	{seconds}	Current start of shift
SZEITE	{seconds}	Current end of shift

13.3 Status texts

13.3.1 Edit status texts (DLG=MSTTXT.INSERT, UPDATE, DELETE, LOCK, UNLOCK, SELECT)

Use these BAPI calls to edit status texts that are assigned to the machine statuses.

Tables

Table	Key field	Description
stoertexte	stoertxt_nr MSTTXT.STNR	Status text number (PK) in status texts

BAPI call

Identification	Content / {type}	Description
DLG	MSTTXT.INSERT	Create status text
	MSTTXT.UPDATE	Change status text
	MSTTXT.DELETE	Delete status text
	MSTTXT.LOCK	Lock status text for editing
	MSTTXT.UNLOCK	Unlock status text after editing
	MSTTXT.SELECT	Select status text
MSTTXT.STNR	{N4}	PK: status text number
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
734	The parameter MSTTXT.STNR has not been specified.
735	The machine status text does not exist.
733	The machine status text is referenced in the machine status configuration and cannot be deleted.
933	The status texts 20000 and 30000 must not be deleted.
736	A machine status text with this number already exists.
1666	The machine status text is currently edited by another user.

13.3.2 List of status texts (DLG=MSTTXT.LIST)

The BAPI call returns all defined status texts (incl. the special texts 20000 "NO SHIFT" and 30000 "NOT ASSIGNED").

Tables

Table	Key field	Description
stoertexte	stoertxt_nr MSTTXT.STNR	Status text number (PK) in status texts

BAPI call

Identification	Contents	Description
DLG	MSTTXT.LIST	List of status texts
DATEI	{C256}	Specification of the file name for the list

13.4 Status classes

13.4.1 Edit status classes (DLG=STKL.INSERT, UPDATE, DELETE, LOCK, UNLOCK, SELECT)

Use these BAPI calls to edit status classes that are assigned to the machine statuses.

Tables

Table	Key field	Description
mz_stklasse	nummer STKL.STKLNR	Status class number (PK) in status class configuration
mz_stklasse	kuerzel STKL.STKL	Status class (PK) in status class configuration

BAPI call

Identification	Content / {type}	Description
DLG	STKL.INSERT	Create status class
	STKL.UPDATE	Change status class
	STKL.DELETE	Delete status class
	STKL.LOCK	Lock status class for editing
	STKL.UNLOCK	Unlock status class after editing
	STKL.SELECT	Select status class
STKL.STKLNR	{N4}	PK: status class number (unique)
STKL.STKL	{C3}	PK: status class (unique)
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description

1661	• The status class number (STKL.STKLNR) has not been specified • The status class (STKL.STKL) has not been specified
719	The status class is not available
750	The status class cannot be deleted because it is assigned to at least one machine status
749	The status class/status class number already exists
1666	The machine status text is currently edited by another user.

13.4.2 List of status classes (DLG=STKL.LIST)

The BAPI call returns all status classes defined.

Tables

Table	Key field	Description
mz_stklasse	nummer STKL.STKLNR	Status class number (PK) in status class configuration
mz_stklasse	kuerzel STKL.STKL	Status class (PK) in status class configuration

BAPI call

Identification	Contents	Description
DLG	STKL.LIST	List of status classes
DATEI	{C256}	Specification of the file name for the list

13.5 Machine status configuration

13.5.1 Edit valid machine statuses (DLG=MST.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

Use this BAPI call to configure the valid machine statuses for a machine or workplace. The status texts are assigned here.

Tables

Table	Key field	Description
stoer_tabelle	masch_nr MST.MNR	Machine number (PK) in machine status configuration
stoer_tabelle	stoernr MST.MST	Machine status (PK) in machine status configuration
stoer_tabelle	stoertxt_nr MST.STNR	Status text number (FK) in status texts

BAPI call

Identification	Content / {type}	Description
DLG	MST.INSERT	Create machine status configuration
	MST.UPDATE	Change machine status configuration
	MST.DELETE	Delete machine status configuration
	MST.COPY	Copy machine status configuration
	MST.LOCK	Lock machine status configuration for editing
	MST.UNLOCK	Unlock machine status configuration after editing
	MST.NEW	Read specification for new machine status configuration
	MST.SELECT	Select machine status configuration
MST.MNR	{C8}	PK: machine number
MST.MST	{N4}	PK: machine status
MST.MNR:Z	{C8}	PK: new (target) machine number for COPY with MOD=E
MST.MST:Z	{N4}	PK: new (target) machine status for COPY with MOD=E
MOD	{C1}	Copy/delete mode: G = all statuses of machine with DELETE/COPY F = missing statuses of machine with COPY E = single statuses of machine with DELETE/COPY M = single status of all machines with DELETE/COPY **Note:** The statuses 20000 and 30000 are not deleted with MST.DELETE and MOD=G because these statuses are automatically created when the machine is created (MNR.INSERT) and automatically deleted when the machine is deleted (MNR.DELETE).

…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
707	The machine number (MNR.MNR) has not been specified.
708	The machine status (MST.MST) has not been specified.
712	The parameter MOD is invalid or has not been specified.
709	The (target) machine number has not been specified.
710	The (target) machine status has not been specified.
1803	The authorization for the responsibility area is not available and the specified machine cannot be edited.
713	The machine status specified is not available.
706	The machine status specified cannot be deleted because this status is currently assigned to the machine.
714	The machine status already exists for this machine.
715	The production flag (MST.OPT:PKENN) has not been specified.
716	A status with the specified production flag (MST.OPT:PKENN) already exists for this machine.
717	The specified resource performance account (MST.BMKNR) is not available.
718	The specified status text (MST.STNR) is not available. You must create this status text in the configuration of status texts (see MSTTXT.INSERT).
719	The disturbance class (MST.STKL) is not available. You must create this class in the configuration of disturbance classes.
720	You can only set the option "Status transfer of aggregates" (MST.OPT.LINIE=J) with machines of type "Line".
809	Invalid superior machine status (MST.MST:UEB). The flag "Hierarchy level" is not set in the superior status.
810	Invalid superior machine status (MST.MST:UEB). The status is not available for the machine selected!
811	Invalid superior machine status (MST.MST:UEB). The superior and the inferior statuses are identical.
812	Invalid superior machine status (MST.MST:UEB). The higher-level/superior status refers to the lower-level/inferior status at the end of the hierarchy.
802	Status already assigned for "no shift".

1666	The machine status is currently edited by another user.

13.5.2 List machine status (DLG=MST.LIST)

This list shows all machine statuses defined in the system.

Tables

Table	Key field	Description
stoer_tabelle	masch_nr MST.MNR	Machine number (PK) in machine status configuration
stoer_tabelle	stoernr MST.MST	Machine status (PK) in machine status configuration
stoer_tabelle	stoertxt_nr MST.STNR	Status text number (FK) in status texts

BAPI call

Identification	Contents	Description
DLG	MST.LIST	List of machine statuses
DATEI	{C256}	Specification of the file name for the list

13.6 Counter configuration

13.6.1 Edit counter configuration (DLG=MNRCTR.INSERT, UPDATE, MODIFY, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

Use this BAPI call to create and edit the configuration for 1...n counters per machine or workplace.

Tables

Table	Key field	Description
maschinen_zaehler	masch_nr MNRCTR.MNR	Machine number (PK)
maschinen_zaehler	zaehler MNRCTR.CTR	Counter number (PK)

BAPI call

Identification	Content / {type}	Description
DLG	MNRCTR.INSERT	Create counter configuration
	MNRCTR.UPDATE	Change counter configuration
	MNRCTR.DELETE	Delete counter configuration
	MNRCTR.COPY	Copy counter configuration
	MNRCTR.LOCK	Lock counter configuration for editing
	MNRCTR.UNLOCK	Unlock counter configuration after editing
	MNRCTR.NEW	Read specification for new counter configuration.
	MNRCTR.SELECT	Select counter configuration
MNRCTR.MNR	{N3}	PK: machine number
MNRCTR. MNR:Z	{N3}	PK: new (target) machine number for COPY
MNRCTR.CTR	{N3}	PK: counter number
MNRCTR. CTR:Z	{N3}	PK: new (target) counter number for COPY
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
1662	The parameter **Allocation with** (MNRCTR:VERB) must not have the same value as the parameter **Evaluation** (MNRCTR.TYP).
1662	The parameter **Evaluation** (MNRCTR.TYP) is not specified and the value is invalid. Possible values: GUT, AUS, NCH, PRB and empty
1662	The parameter **Allocation with** (MNRCTR:VERB) is specified and the value is invalid. Possible values: GUT, AUS, NCH, PRB and empty
1662	The parameter **Allocation with partitioning** (MNRCTR.VERB:TLG) is specified and the value is not valid. Possible values: J, N and empty
1662	The parameter **Posting as cycles** (MNRCTR.OPT:TAKT) is specified and the value is invalid. Possible values: T, N and empty
1662	The parameter **For monitoring** (MNRCTR.OPT:UEB) is specified and the value is invalid. Possible values: Z, N and empty
931	Auto-allocation is not possible.

935	The current counter configuration is only supported with CT-8xx or C-76x terminals.
1669	Counter configuration is already available (MNRCTR.UPDATE)
90	Machine does not exist (MNRCTR.MNR / MNRCTR.:Z)
930	The maximum number of counters at the machine has been exceeded.
1662	The key fields are not correctly specified: - MNRCTR.MNR - MNRCTR.MNR:Z (with MNRCTR.COPY) - MNRCTR.CTR - MNRCTR.CTR:Z (with MNRCTR.COPY)
101	No counter configuration available (with MNRCTR.UPDATE and MNRCTR.DELETE)
1669	Counter configuration is already available (with MNRCTR.INSERT)
1666	The data record is currently locked by user %s (with MNRCTR.UPDATE and MNRCTR.DELETE)
712	The parameter MOD is not valid. Possible values: E, G and F
101	Source counter configuration is not available (MNRCTR.COPY)
1855	Target counter configuration is already available (MNRCTR.COPY)

13.6.2 List of counter configuration (DLG=MNRCTR.LIST)

This list shows all defined counters of the counter configuration.

Tables

Table	Key field	Description
maschinen_zaehler	masch_nr MNRCTR.MNR	Machine number (PK)
maschinen_zaehler	zaehler MNRCTR.CTR	Counter number (PK)

BAPI call

Identification	Contents	Description
DLG	MNRCTR.LIST	List of counter configuration
DATEI	{C256}	Specification of the file name for the list

13.7 Assignment of machine/workplace to terminal

13.7.1 Edit assignment (DLG=MNRTNR.INSERT, DELETE,SELECT)

Use these BAPI calls to edit the assignments of machines to terminals.

Tables

Table	Key field	Description
masch_term_zuord	masch_nr terminal_nr MNRTNR.MNR MNRTNR.TNR	Machine and terminal must be unique.
masch_term_zuord	terminal_nrterminal_nr anzeige_pos MNRTNR.TNR MNRTNR.POS	Terminal and display position must be unique.

BAPI call

Identification	Content / {type}	Description
DLG	MNRTNR.INSERT	Create assignment
	MNRTNR.DELETE	Delete assignment
	MNRTNR.SELECT	Select assignment
MNRTNR.MNR	C 20	Machine
MNRTNR.TNR	N8	Terminal number
MNRTNR.POS	N8	Display position
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
707	Machine has not been specified.
746	Terminal number has not been specified.
747	Position has not been specified.

510	Person is not authorized.
1672	Assignment does already exist.
1682	Assignment is not available.

13.7.2 List of assignments (DLG=MNRTNR.LIST)

The BAPI call returns all assignments for all terminals.

BAPI call

Identification	Contents	Description
DLG	MNRTNR.LIST	List of assignments
DATEI	{C256}	Specification of the file name for the list

13.8 Group assignment

13.8.1 Edit group assignment (DLG=GRPRES.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

Use these BAPI calls to edit the group assignments.

Tables

Table	Key field	Description
hy_gruppen_zuord	group res_nr res_typ GRPRES.GRP GRPRES.RESNR GRPRES.RESTYP	The combination must be unique.

BAPI call

Identification	Content / {type}	Description
DLG	GRPRES.INSERT	Create assignment
	GRPRES.UPDATE	Edit assignment
	GRPRES.DELETE	Delete assignment
	GRPRES.COPY	Copy assignment

	GRPRES.SELECT	Select assignment
	GRPRES.COPY	Copy assignment
	GRPRES.LOCK	Lock assignment for editing
	GRPRES.UNLOCK	Unlock assignment after editing
	GRPRES.NEW	Read specification for new assignment
GRPRES.GRP	C 20	Group
GRPRES.RESNR	C 40	Resource
GRPRES.RESTYP	C 4	Resource type
MOD	C 1	Copy mode E- Copy currently selected resource G- Copy all resources of the group M- Move all resources of the group

Return

Identification	Content / {type}	Description
GRP	C 20	current data record
RESNR	C 40	current data record
RESTYP	C 4	current data record
POS	N8	current data record

Validation checks

Error codes	Description
1661	• The value GRPRES.GRP has not been specified. • The value GRPRES.RESTYP has not been specified. • The value GRPRES.RESNR has not been specified. • The value GRPRES.GRP:Z has not been specified with GRPRES.COPY.
414	The group specified does not exist.
1669	Assignment is already available (GRPRES.UPDATE)
94	With RESTYP = MGRP Machine group does not exist.
918	With RESTYP = MGRP A resource of type MGRP must be a capacity group.

415	The RESTYP specified does not match the existing group containing only one type.
712	Processing mode is invalid.

13.8.2 List of group assignments (DLG=GRPRES.LIST)

The BAPI call lists the group assignments.

BAPI call

Identification	Contents	Description
DLG	GRPRES.LIST	List of assignments
DATEI	{C256}	Specification of the file name for the list
GRPRES.GRP	C 20	Group {wildcard permitted}
GRPRES.RESNR	C 40	Resource {wildcard permitted}
GRPRES.RESTYP	C 4	Resource type {wildcard permitted}

13.9 Groups

13.9.1 Edit groups (DLG=GRP.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

Use these BAPI calls to edit groups.

Tables

Table	Key field	Description
hy_gruppen	group user GRP.GRP GRP.BEARB	The combination must be unique.

BAPI call

Identification	Content / {type}	Description
DLG	GRP.INSERT	Create group
	GRP.UPDATE	Edit group
	GRP.DELETE	Delete group

	GRP.COPY	Copy group
	GRP.SELECT	Select group
	GRP.COPY	Copy group
	GRP.LOCK	Lock group for editing
	GRP.UNLOCK	Unlock group after editing
	GRP.NEW	Read specification for new group
GRP.GRP	C 20	Group
GRP.BEARB	C10	User
MOD	C 1	Copy mode E- copy currently selected group Z- copy currently selected group with all resources assigned

Return

Identification	Content / {type}	Description
GRP	C 20	current data record
BEARB	C10	current data record

Validation checks

Error codes	Description
1661	• The value GRPRES.GRP has not been specified. • The value GRPRES.RESTYP has not been specified. • The value GRPRES.RESNR has not been specified. • The value GRPRES.GRP:Z has not been specified with GRPRES.COPY.
414	The group specified does not exist.
1669	Assignment is already available (GRPRES.UPDATE)
94	With RESTYP = MGRP Machine group does not exist.
918	With RESTYP = MGRP A resource of type MGRP must be a capacity group.
415	The RESTYP specified does not match the existing group containing only one type.

712	Processing mode is invalid.

13.9.2 List of group assignments (DLG=GRPRES.LIST)

The BAPI call lists the group assignments.

BAPI call

Identification	Contents	Description
DLG	GRPRES.LIST	List of assignments
DATEI	{C256}	Specification of the file name for the list
GRPRES.GRP	C 20	Group {wildcard permitted}
GRPRES.RESNR	C 40	Resource {wildcard permitted}
GRPRES.RESTYP	C 4	Resource type {wildcard permitted}

13.10 MDE postings

13.10.1 Create posting (DLG=MDEPRO.INSERT, COPY)

Use the BAPI calls described in this section to create, copy and delete MDE records for postings.

Tables

Table	Key field	Description
event	Internal ID MDEPRO.VERWEIS	Internal ID of the MDE record of the posting (PK)

BAPI call

Identification	Content / {type}	Description
DLG	MDEPRO.INSERT	Create MDE posting
	MDEPRO.COPY	Copy MDE posting
MDEPRO.MNR	{C20}	HYDRA workplace that makes the posting (PK)
MDEPRO.DATB	{MM/DD/YYYY}	Start date of the posting (PK)
MDEPRO.DATE	{MM/DD/YYYY}	End date of the posting (PK)

Identification	Content / {type}	Description
MDEPRO.SART	{C1}	Record type of the posting (PK) P: log record N: end of shift record
MDEPRO.MST	{N6}	only with MDEPRO.INSERT Machine status of posting (PK)
MDEPRO.VERWEIS	{N8}	only with MDEPRO.COPY Internal ID of the posting (PK) that you want to copy.
…	…	For further fields, that are not MANDATORY, refer to section 13.10.3 List of fields (acronyms) for the MDEPRO dialog

Return

Identification	Content / {type}	Description
MDEPRO.VERWEIS	{N8}	Returns the internal ID of the posting created.

Validation checks

Error codes	Description
90	Workplace/machine (MDEPRO.MNR) must be available in the workplace/machine configuration.
1662	The parameter with the ID MDEPRO.SART must have the value "N" or "P".
1803	Check is only performed if "BEARB" is also passed and BEARB does not equal "HYDRA". Responsibility area authorization for the workplace is not available (MDEPRO.MNR).
1952	Start date (MDEPRO.DATB / MDEPRO.ZEIB) must not be greater than end date (MDEPRO.DATE / MDEPRO.ZEIE).
1661	You must specify the parameter with the ID MDEPRO.SART.
1661	You must specify the parameter with the ID MDEPRO.MNR.
1661	You must specify the parameter with the IDs MDEPRO.DATB and MDEPRO.DATE.

13.10.2 Edit posting (DLG=MDEPRO.UPDATE, DELETE, LOCK, UNLOCK)

Use the BAPI calls described in this section to edit MDE records of postings.

Tables

Table	Key field	Description
event	Internal ID MDEPRO.VERWEIS	Internal ID of the posting

BAPI call

Identification	Content / {type}	Description
	MDEPRO.UPDATE	Change posting
	MDEPRO.DELETE	Delete posting
	MDEPRO.LOCK	Lock posting for editing MUST be performed before MDEPRO.UPDATE
	MDEPRO.UNLOCK	Unlock posting after editing MUST be performed after MDEPRO.UPDATE
MDEPRO.VERWEIS	{N8}	Internal ID of the posting (PK)
MDEPRO.SART	{C1}	Record type of the posting P: log record N: end of shift record
…	…	MDEPRO.UPDATE: For further fields, that are not MANDATORY, refer to section 13.10.3 List of fields (acronyms) for the MDEPRO dialog

Return

Identification	Content / {type}	Description
event	Internal ID MDEPRO.VERWEIS	Internal ID of the MDE record of the posting

Validation checks

Error codes	Description
90	Workplace (MDEPRO.MNR) must be available in the workplace/machine configuration.
101	UPDATE,LOCK,UNLOCK The record of the posting (if the parameter MDEPRO.VERWEIS is specified) must be available in the database.

Error codes	Description
1803	Check is only performed if "BEARB" is also passed and BEARB does not equal "HYDRA". Responsibility area authorization for the workplace is not available (MDEPRO.MNR).
1952	Start date (MDEPRO.DATB / MDEPRO.ZEIB) must not be greater than end date (MDEPRO.DATE / MDEPRO.ZEIE).
1666	The data record is currently locked by another user. (UPDATE, DELETE,LOCK).
1661	UPDATE.LOCK,UNLOCK You must specify the parameter with the ID MDEPRO.VERWEIS.
1661	UPDATE You must specify the parameter with the ID MDEPRO.SART.
1661	UPDATE You must specify the parameter with the ID MDEPRO.MNR.
1661	You must specify the parameter with the IDs MDEPRO.DATB and MDEPRO.DATE.
1662	The parameter with the ID MDEPRO.SART must have the value "N" or "P".

		dialog

13.10.3 List of fields (acronyms) for the MDEPRO dialog

Identification	Description		DB type	Length
MDEPRO.MNR	workplace		char	20
MDEPRO.KST	Cost center of the workplace		char	10
MDEPRO.VERWEIS	Unique ID of the log record, important for the editing of log records		sqlserial	7
MDEPRO.SART	Record type P: log record N: end of shift record Original records of corrections: 1: end of shift record 2: log record		char	1
MDEPRO.ZEIB MDEPRO.MSZEIB	or	start date (time) of status	integer	7
MDEPRO.DATB MDEPRO.MSDATB	or	start date (date) of status	sqldate	7

Identification	Description		DB type	Length
MDEPRO.ZEIE MDEPRO.MSZEIE	or	end time (time) of status	integer	7
MDEPRO.DATE MDEPRO.MSDATE	or	end time (date) of status	integer	7
MDEPRO.MSDAUER	Duration of the status, synchronized with the shift calendar		integer	7
MDEPRO.MST	Machine status (table stoer_tabelle)		smallint	7
MDEPRO.BMKNR	Resource performance account of machine status		smallint	7
MDEPRO.STNR	Refers to status text (table stoer_texte)		smallint	7
MDEPRO.SZY	Target cycle at time of logging		decimal	7
MDEPRO.TLG	Partitioning at time of logging		decimal	7
MDEPRO.SKNR	Shift number (1...4)		smallint	7
MDEPRO.SKZEIB	Beginning of shift		integer	7
MDEPRO.SKZEIE	End of shift		integer	7
MDEPRO.SKPDAUER	Total of breaks in this shift		smallint	7
MDEPRO.SKDATB	Beginning of shift – date		sqldate	7
MDEPRO.EGR:GUTB	Yield in base quantity unit Note: delta quantity (no absolute value)		decimal	7
MDEPRO.EGR:GUTP	Yield in primary quantity unit Note: delta quantity (no absolute value)		decimal	7
MDEPRO.EGR:GUTS	Yield in secondary quantity unit Note: delta quantity (no absolute value)		decimal	7
MDEPRO.EGR:GUTT	Yield in tertiary quantity unit Note: delta quantity (no absolute value)		decimal	7
MDEPRO.EGR:AUSB	Scrap in base quantity unit Note: delta quantity (no absolute value)		decimal	7
MDEPRO.EGR:AUSP	Scrap in primary quantity unit Note: delta quantity (no absolute value)		decimal	7
MDEPRO.EGR:AUSS	Scrap in secondary quantity unit Note: delta quantity (no absolute value)		decimal	7
MDEPRO.EGR:AUST	Scrap in tertiary quantity unit Note: delta quantity (no absolute value)		decimal	7
MDEPRO.EGR:NCHB	Rework quantity in base quantity unit Note: delta quantity (no absolute value)		decimal	7
MDEPRO.EGR:NCHP	Rework quantity in primary quantity unit Note: delta quantity (no absolute value)		decimal	7

Identification	Description	DB type	Length
MDEPRO.EGR:NCHS	Rework quantity in secondary quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:NCHT	Rework quantity in tertiary quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:PRBB	Problem quantity in base quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:PRBP	Problem quantity in primary quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:PRBS	Problem quantity in secondary quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:PRBT	Problem quantity in tertiary quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGE:GUTP	Base quantity unit	char	3
MDEPRO.EGE:GUTP	Primary quantity unit	char	3
MDEPRO.EGE:GUTS	Secondary quantity unit	char	3
MDEPRO.EGE:GUTT	Tertiary quantity unit	char	3
MDEPRO.EGR:HUB	Machine strokes Note: delta quantity (no absolute value)	decimal	7
MDEPRO.CTR:1	Counter 1: yield (parts) Note: absolute value (accumulated for the shift)	integer	7
MDEPRO.CTR:2	Counter 2: machine strokes Note: absolute value (accumulated for the shift)	integer	7
MDEPRO.CTR:3	Counter 3: scrap Note: absolute value (accumulated for the shift)	integer	7
MDEPRO.CTR:4	Counter 4 Note: absolute value (accumulated for the shift)	integer	7
MDEPRO.CTR:5	Counter 5 Note: absolute value (accumulated for the shift)	integer	7
MDEPRO.CTR:6	Counter 6 Note: absolute value (accumulated for the shift)	integer	7
MDEPRO.BEM	Comment The comment is created and edited in a separate table: event_dlg_data The table is identified via the ID MDEPRO.VERWEIS	Char	60

Identification	Description	DB type	Length
	<u>Storage:</u> Comment = event_dlg_data.dlg_data Identified via the ID ereignis. dlg_data_verweis = event_dlg_data.verweis or event_dlg_data.ev_verweis= ereignis.verweis		

14 HYDRA Production Data Manager HR – PZE/PZW/ZKS/PEP

14.1 Optional data transfer from SAP

You use the structures described in the following to transfer data from SAP in HYDRA dialog data format using a customer-specific function module.

Message type:	Z*
IDoc type:	Z*01
Segments:	Z2BAPI000

NOTE

To generate segment names in HYDRA inbound processing as described above, the segments must have been created in SAP according to the pattern Z1<segment name>. Versioning in SAP outbound processing then creates the segment names in the form Z2<Segment name><Version>.

Example: **Z1**BAPI becomes **Z2**BAPI**000**

Field name	Type		Description	Example
Transaction	CHAR	20	Transaction ID (dialog ID in HYDRA)	TGERG.MODIFY
Description	CHAR	40	Plain text designation as comment	Zeitwirtschaftsergebnisse.an.HYDRA
Data	CHAR	940	Dialog data string for HYDRA	DLG=TGERG.MODIFY|TGERG.PNR=12345678|… For more details see the sections that follow

14.2 Sending HR data to HYDRA

14.2.1 Transfer of PZE and ZKS time events

The time events listed below can be transferred to HYDRA online.

Structure of dialog data

The different fields are separated by the "|" symbol:

Header data

Field	Description

DLG={P_KOM, P_GEH, ...}	Dialog ID (see below) (e.g. clocking-in *DLG=P_KOM*) The dialog ID should be transferred as the first field.
DAT={mm/dd/yyyy}	Date of the event in the format "mm/dd/yyyy“ (e.g. clocking date *DAT=03/30/1999*)
ZEI={seconds}	Time of the event in seconds after midnight (e.g. clocking time 10:00:03 is *ZEI=36003*)
USR={hy_user}	"Hydra User“ (see above) With terminals, 2000 is added to the terminal number. (terminal 4 is *USR=2004*)
OFF={J|N}	Offline flag. Enter "*OFF=J*“ if data has been entered offline and buffered. In this case, data is processed separately. ("*OFF=N"* need not be specified)

Field data

Field	Description
TNR={...}	Terminal number
KNR={...}	Badge number with a maximum of 10 characters (0...9, A...F). The badge number has to be uploaded with the same number of digits (length) becaus it has been transferred while downloading authorizations.
PNR={...}	Personnel number with a maximum of ten digits (0..9)
FIR={...}	Company with a maximum of four digits. As an alternative to the staff badge number, the personnel number can be transferred with the company with some dialogs.
KST={...}	Cost center with a maximum of eight digits
ZART={...}	Time type (S = daylight saving time, W = standard time). If this field is not transferred, HYDRA automatically identifies the time type using the change between daylight saving time and standard time. Daylight saving and standard time cannot be distinguished properly on the day the time is reset.
FGR={...}	Absence reason for P_VB, P_NB or P_FGR.
PIN={...}	The person's PIN code or password (e.g. for Internet clocking records). If this field is transferred, the PIN code is always checked (against the PIN code of the HR master) and the access authorization is also checked (with Internet, it is always the terminal number 255).

PARAM02={...}	This field can include up to two references (separated by comma) to a message displayed on the terminal. You use this field to count down the number specifying how often a message is displayed (or if escalation management is used, an escalation can be finished by the message display). Examples: "PARAM02=143" or "PARAM02=143,178"

Note

Enter either the personnel number and the company or the staff badge number.

Dialog IDs

Identification	Description
P_KOM	Clocking-in
P_GEH	Clocking-out
P_PAU	Clocking out for break
P_AST	Automatic status clocking (subject to the status, it is like a clocking-in or clocking-out)
P_VB	Advance clocking: Clocking-out including reason. The number of the absence reason is transferred by FGR=x without leading zeros. For business trip clocking records FGR=1 is set.
P_NB	Subsequent clocking: Clocking-in including reason. The number of the absence reason is transferred by FGR=x without leading zeros.
P_FGR	Absence reason clocking (subject to the status, it is like an advance posting or subsequent posting): the number of the absence reason is transferred by FGR=x without leading zeros.
Z_ZUT	Logging of an access. (Using the fields BDAT (date until) and BZEI (time until), you can additionally transfer the time specifying how long the door was open. DAT and ZEI are interpreted as start date and start time).
Z_ZV	Logging of an access attempt (The field ZVG includes the reason for the access attempt: 2001=Unauthorized badge 2002=No badges loaded (number of badges = 0) 2003=Beyond access time model 2004=Beyond opening hours 2005=Wrong PIN code 2006=Wrong company number 2010=Double posting within the blocking time (anti-passback) 2013=Missing PIN code 2014=Badge beyond validity period 2020=Other access point of the security gate opened 2030=Already present in room zone 2031=Not present in room zone 2032=Room zone completely occupied 3000=Wrong badge number length

Example of a clocking-out on 15 January 2010 at 16:32:53:

DLG=P_GEH|KNR=001365|DAT=01/15/2010|ZEI=59573|TNR=17|

14.2.2 Transfer of access statuses and access logs

The status of an access point can be sent to HYDRA using the dialog "Z_STA":

Field	Description
DLG=Z_STA	Dialog ID for the access point status
ZNR={...}	Access point number
DAT={...}, ZEI={...}	Date and time (in seconds since beginning of the day)
DATB={...}, ZEIB={...} DATE={...}, ZEIE={...}	As an alternative to DAT and ZEI, DATB and ZEIB can be used to transfer the start time of the status and DATE and ZEIE can be used to specify the end time. With cyclic status postings, DATE and ZEIE include the current point in time. With this alternative, the fields DATB, ZEIB, DATE and ZEIE must be populated.
STA={...}	Status of the access point: Z: The access point is closed O: The access point is open U: The access point is open without permission L: The access point has been opened too long S: Sabotage: The terminal has been opened A: Malfunction: The terminal is not connected with the reader F: The alarm is suppressed at the access point. One of the statuses "closed" or "open" (Z, O) specify the end of the alert (U, L, S, A).
KNR={...}	Badge that has opened the access point (with status "opened too long").
BERANZ={...}	Number of authorized badges
BERDAT={...}, BERZEI={...}	Point in time when authorizations have last been read.
PRO={J/N}	This button controls whether or not HYDRA logs are generated from access statuses (PRO=J). As an alternative, log records about access point statuses may also be sent to HYDRA (PRO=N).

Access point statuses are sent to HYDRA in cyclic intervals. The cyclic status time is defined for each access point in the access list (ZYKL:STA).

Access status logs are sent to HYDRA using the dialog "Z_PRO“:

Field	Description
DLG=Z_PRO	Dialog ID for access status logs
ZNR={...}	Access point number
DATB={...}, ZEIB={...} DATE={...}, ZEIE={...}	DATB and ZEIB can be used to transfer the start time of the status and DATE and ZEIE can be used to specify the end time.
DAUER={...}	Duration of the status at the access. If this field is not transferred, the duration is calculated automatically.
STA={...}	Status of the access point (the same values as for the access point status)
KNR={...}	Badge that has opened the access point (with status "opened too long").

14.3 Reading HR data from HYDRA

14.3.1 Requesting PZE access authorizations

Access authorizations are provided by the command DLG=LIST;27 and filed in HYDRADIR\spool\{file name}.

Structure of dialog data:

"DLG=LIST;27|DATEI={file name}|DAT=...|ZEI=...|USR=...|TNR=...|..."

{File name}

Specification of ".\spool\hyu{Hydrauser}.c27“; with {Hydra user} ranging between 2001 and 2999 (user numbers of the terminals). The external interface 1 corresponds to the HYDRA user number 3001.

The list is sorted by the badge number and includes the following data:

KNR=Badge number|...	The first row includes the column caption and the field ID	KNR=Badge number|...
{Badge1}|...	The other rows include the values for the field ID. The individual badges are sorted in ascending order in the file.	0001|...
{Badge1}|...		0004|...

When data is interpreted, the field abbreviations of the 1st line must be interpreted. Future versions can have a different column sequence or new columns are inserted at any place. Columns with unknown field abbreviations must be ignored and must not lead to a program error or termination.

The following table shows further columns that are included in each row. The columns are separated from each other by the "|" pipe symbol:

DGBE= Business trip authorization	Business trip authorization (J/N)	e.g. J
PNR=Person	The person's personnel number	e.g. 142234
FIR=Company	Company the person is assigned to	e.g. DBI
KTO:x=Designation	Up to 8 columns including accounts can be listed. The number depends on how many accounts have been activated in HYDRA. Behind the equals sign, the account name is shown that is displayed on the terminal, if possible.	Example for an account balance: 123:43

14.3.2 Requesting the ZKS access authorizations

Access authorizations are provided by the command DLG=LIST;28 and filed in HYDRADIR\spool\{file name}. The file name should be structured as follows: .\spool\hyu{Hydra user}.c{number of the list};

Example: .\spool\hyu2017.c28

Structure of dialog data:

"DLG=LIST;28|DATEI={file name}|DAT=...|ZEI=...|USR=...|ZNR=...|..."

with ZNR being the corresponding access point number.

The list is sorted by the staff badge number and includes the following data:

KNR=Badge number	Valid badge number	e.g. 0111
PIN=PIN code	PIN code; transferred uncoded. The PIN code is only requested during the access periods when PIN code request is enabled.	e.g. 4711
DAT_VON=Date from	Beginning of the validity period of the badge. Access authorizations, which are valid next week, are transferred. For this reason, the validity period must be checked. If this field is empty, the validity of the badge is not restricted in the future.	e.g. 05/30/1999
DAT_BIS=Date until	End of validity of the badge. If this field is empty, the badge is valid for an unlimited period.	e.g. 12/31/1999
WTAG=Weekday	The access authorization is valid on the specified weekdays. This 8-digit field contains a "J" on the days when the authorization is valid. The first digit stands for Sunday, the 7th digit for Saturday and the 8th digit for a public holiday. **Note:** This field is no longer relevant as of HYDRA 7.1, as it has been replaced by the weekdays (WTAG) in the access time models.	e.g. NJJJJJNN
ZZ=access time model	Number of the access time model to specify a time limit for the access authorization.	e.g. 2

14.3.3 Requesting the list of access time models

The list of access time models is provided using the command DLG=LIST;29 and stored in HYDRADIR\spool\{file name}. The list is identical for all access points and, as a result, does not have to be requested for each access point.

Structure of dialog data:

"DLG=LIST;29|DATEI={File name}|DAT=...|ZEI=...|USR=...|..."

The list includes the following data:

ZZ=access time model	Number of the access time model.	e.g. 2
ZEIVON=beginning of the access period	Start time of an interval of the access time model in seconds since beginning of the day. If values < 0 are transferred for this time, they are interpreted as times of the previous day. Example: -3600 is 23:00 on the previous day (only with the terminal programs AIP and ctwin).	e.g. 21600
ZEIBIS=end of the access period	End time of an interval of the access time model in seconds since beginning of the day. If values > 86400 (= 24 hours) are transferred for this time, this end time is interpreted as a time of the following day. Example: 90000 is 01:00 on the next day (only with the terminal programs AIP and ctwin).	e.g. 86400
GETKNR=Requesting the badge number	Defines whether or not the badge is required to open the door during that time interval (J/N). This field is only interesting for opening hours because the check if the badge number request is activated can only be performed when the badge has been read.	e.g. J
GETPIN= Request PIN code	Defines whether or not a PIN code is required to open the door during that time interval (J/N).	e.g. N

PRO=Logging	Specifies the postings that are logged in this access period: - 'Z': accesses and access attempts - 'V': only access attempts - 'N': no logging	e.g. Z
WTG=Weekdays	The access period is valid on the specified weekdays. This 10-digit field contains a "J" on the days when the authorization is valid. The 1st character stands for Sunday, the 7th character for Saturday and the 8th to 10th characters stand for three types of public holidays.	e.g. NJJJJJNNNN

The list can include several intervals for an access time model each having the same number.

14.3.4 Requesting the public holidays

The list of public holidays is provided by the command DLG=LIST;30 and filed in HYDRADIR\spool\{file name}. It includes the public holidays for the next 30 days as of yesterday. The list is made up for an access group. When you make a request for an access, the system must request the list of the relevant access group.

Structure of dialog data:

„DLG=LIST;30|DATEI={file name}|DAT=...|ZEI=...|USR=...|ZNR=...|...“

The list includes the following data:

DAT=Date	Date of the public holiday	e.g. 06/03/1999
BEZ=Designation	Designation of the public holiday	e.g. Feast of Corpus Christi
ART=Type	Type of the public holiday (1/2/3): three types of public holidays are possible as of HYDRA 6.5. In the access time models, you can assign authorizations for these 3 types of public holidays.	e.g. 1

14.3.5 Requesting opening hours

The list of opening hours is provided by the command DLG=LIST;32 and filed in HYDRADIR\spool\{file name}. The list is made up for an access group. When you make a request for an access, the system must request the list of the relevant access group.

Structure of dialog data:

„DLG=LIST;32|DATEI={file name}|DAT=...|ZEI=...|USR=...|ZNR=...|...“

The list includes the following data:

DAT_VON=Date from	Beginning of the validity period for opening hours. Opening hours that are valid next week are transferred. For this reason, the validity period must be checked.	e.g. 05/03/1999
DAT_BIS=Date until	End of validity of opening hours. If this field is empty, the validity of opening hours is unlimited.	e.g. 31/12/1999
WTAG=Weekday	The opening hours are valid on the specified weekdays. This 8-digit field contains a "J" on the days when the opening hours are valid. The first digit stands for Sunday, the 7th digit for Saturday and the 8th digit for a public holiday. **Note:** This field is no longer relevant as of HYDRA 7.1, as it has been replaced by the weekdays (WTAG) in the access time models.	e.g. NJJJJJNN
ZZ=access time model	Number of the access time model to specify a time limit for the opening hours.	e.g. 2

14.3.6 Requesting the terminal list

The list of terminals is provided by the command DLG=LIST;45 and filed in HYDRADIR\spool\{file name}. You can request this list for a specific or for all terminals.

Structure of dialog data:

„DLG=LIST;45|DATEI={file name}|DAT=...|ZEI=...|USR=...|TNR=...|“

The list includes the following data:

TNR=Terminal number	Number of the terminal	e.g. 13
TYP=Terminal type	Terminal type (series). MBB terminals range between 10, 40-44 or 50. Benzing terminals range between 100 and 109 and between 120 and 129. Terminals by MECS range between 201 and 204 (201 = Staff Port 2.0, 202 = Time Port 2.0, 203 = Access Port 2.0, 204 = Job Port 2.0)	e.g. 331 or 380
CFG:1=Configuration 1	Configuration ID 1: includes the terminal class for MBB, Benzing and MECS terminals	e.g. 1
HWADR=Hardware address	Hardware address of the terminal. The bus address is entered after a semicolon for a bus at the COM interface.	e.g COM1;5 e.g. 192.168.10.123
TZ=Time zone	Time zone of the terminal. The following entries are processed: GMT+2DST Eastern Europe (Bucharest, Sofia, ...) GMT+1DST Central Europe (Berlin, Vienna, ...) GMT0DST Great Britain (London, ...) GMT-6DSTU USA Central Standard Time (Chicago, Dallas, Kansas City, Winnipeg) GMT-5DSTU Mexico (Quintana Roo) GMT-6DSTU Mexico (Mexico City, ...) GMT-7DSTU Mexico (Chihuahua, ...) GMT-8DSTU Mexico (Baja California Tijana) If no entry is made in this field, Central Europe is the default value (with change from standard to daylight saving time).	empty
LANG=Language	Language (empty or 1=German, 2=English, 3=Dutch, 4=French, 5=Danish, ...) Please note: With old installations, a letter might be included in this field --> ignore	e.g. 1
BEZK=Location	Location of the terminal (char 8)	e.g. Entrance
BEZL=Designation	Designation of the terminal (char 20)	e.g. Time at the entrance
AKTIV=Active	Specifies whether or not the terminal is active (J/N). Inactive terminals are not polled.	e.g. J

LEN_KNR=Length of badge number	Authorized badges are transferred to the terminal with the length that is configured here. This length must also be integrated when clocking records are uploaded.	e.g. 4
BART:MDE=MDE active	Operation mode of the terminal. This field shows whether or not the machine data collection module (HYDRA MDE) is active on this terminal (J/N).	N
BART:ADE=ADE active	Operation mode of the terminal. This field shows whether or not the shop floor data collection module (HYDRA ADE) is active on this terminal (J/N).	N
BART:PZE=PZE active	Operation mode of the terminal. This field shows whether or not the time and attendance module (HYDRA PZE) is active on this terminal (J/N). Access control (ZKS) terminals are also presented as time and attendance (PZE) terminals.	J
PZEBART=PZE operation mode	Operation mode for PZE and ZKS ("AST"=automatic status (rhythm posting), empty = enter status using the keyboard, "Z*"=ZKS operation modes)	e.g. AST
PZESTA:VORG= Default status	Default status to which the terminal automatically returns (currently not processed).	e.g. "A" for automatic status
SYSNR=Company number/system number	Company number/system number that is to be included in the badge number.	e.g. empty
TTXT:KOM=Key IN	Labeling of the IN key	e.g. clock-in
TTXT:GEH=Key OUT	Labeling of the OUT key	e.g. clock-out
TTXT:PAU=Break key	Labeling of the break key	e.g. break
TTXT:INFO=Key INFO	Labeling of the INFO key	e.g. info
FGR:1=Absence reason 1	1th absence reason (char 3)	e.g. bus
TTXT:FGR1=Key Absence reason 1	Labeling of the first absence reason key	e.g. business trip
FGR:2=Absence reason 2	Second absence reason (char 3)	e.g. 97
TTXT:FGR2=Key Absence reason 2	Labeling of the second absence reason key	e.g. doctor
FGR:3=Absence reason 3	Third absence reason (char 3)	e.g. empty

TTXT:FGR3=Key Absence reason 3	Labeling of the third absence reason key	e.g. empty
FGR:4=Absence reason 4	Fourth absence reason (char 3)	e.g. FGL
TTXT:FGR4=Key Absence reason 4	Labeling of the fourth absence reason key	e.g. list of absence reasons
FGRCFG= Input of absence reason	Configuration of absence reason input: Entry "3" stands for the absence reason list of the 4th function key	e.g. empty
OPT:FGRAUTO=Recording of absence reasons	Automatic input of absence reasons if beyond planned time (entry "G").	e.g. empty
PLAUS:PZEONL=Online check	Online checking of rejected badges (not yet processed)	e.g. empty
PLAUS:PZESTA=Status sequence	Checking of the status sequence: "S"=checking whether the status sequence is correct "P"=checking whether clocking-in is available at the same terminal With the operation mode "AST" the entry "S" results in the status to be requested and displayed online.	e.g. empty
ZYKLLOAD:PZE= cyclic loading PZE	Specification in seconds how often the terminal cyclically reloads its configuration and authorizations.	e.g. 3600
ZEI:BERLESEN1	1. Time for reading PZE authorizations	e.g. 7200
ZEI:BERLESEN2	2. Time for reading PZE authorizations	e.g. 10800
RELZUG:DAUER= Duration of opener	Relay time for door opener	e.g. 3
PZESTA:DAUER= Duration of status	General return time for display outputs to the default status	e.g. 5
PZEINFO:DAUER= Duration of info	Display duration of the information	e.g. 8
PZEINFO:CFG=Information display	"S" for displaying the last clocking records	e.g. empty
PZEINFO:NR=Information number	Number of the information to be displayed	e.g. 1
INFOZEIB:1=Info from 1	Start of the first period for displaying the information on the terminal	e.g. 0
INFOZEIE:1=Info to 1	End of the first period for displaying information on the terminal	e.g. 86400

INFOZEIB:2=Info from 2	Start of the second period for displaying the information on the terminal	e.g. 0
INFOZEIE:2=Info to 2	End of the second period for displaying information on the terminal	e.g. 0
INFOZEIB:3=Info from 3	Start of the third period for displaying the information on the terminal	e.g. empty
INFOZEIE:3=Info to 3	End of the third period for displaying information on the terminal	e.g. empty
INFOZEIB:4=Info from 4	Start of the 4th period for displaying the information on the terminal	e.g. empty
INFOZEIE:4=Info to 4	End of the 4th period for displaying information on the terminal	e.g. empty
INFOZEIB:5=Info from 5	Start of the 5th period for displaying the information on the terminal	e.g. empty
INFOZEIE:5=Info to 5	End of the 5th period for displaying information on the terminal	e.g. empty

14.3.7 Requesting the access list

The list of access points is provided by the command DLG=LIST;46 and filed in HYDRADIR\spool\{file name}. You can request this list for a specific or for all terminals. The list is sorted by the terminal number and by the access point number if several access points are connected to one terminal.

Structure of dialog data:

„DLG=LIST;46|DATEI={file name}|DAT=...|ZEI=...|USR=...|TNR=...|“

The list includes the following data:

ZNR=access point	Number of the access point	e.g. 17
BEZK=Short name	Short name	e.g. gate S
BEZL=Designation	Designation	e.g. Southern gate factory building
ZGRP=Access group	Number of the access group	e.g. 21
AKTIV=Active	Specifies whether or not the access point is active (J/N). Inactive access points are blocked (no access authorization applicable).	e.g. J
RAUM=Room number	Room number (not yet processed at the moment)	e.g. 0
ART=Access type	Access point type: 'E' : Entrance 'A' : Exit 'D' : Passage 'O' : Without check of badge number (check of system number SYSNR only)	e.g. E
TNR=Terminal number	Number of the terminal to which the access point is connected.	e.g. 10
LESER=Reader	Number of the reader. The access point is uniquely defined by the combination of terminal number and reader. The access point number is a logical number for the "physical" address. Local readers at the MBB terminal get an offset of 100. Subbus readers with MBB terminals are without offset.	e.g. 5
ZAUF:DIGOUT=Channel opener	Output (relay) for door opening contact	e.g. 1

ZZU:DIGOUT=Channel closer	Output (relay) for door closing contact	e.g. 2
ALARM:DIGOUT= Channel alarm	Output (relay) for a connected alarm (e.g. warning lights or alarm system).	e.g. 3
ZV:DIGOUT=channel attempt	currently not processed	e.g. 0
ZST:DIGIN=Channel status	Input for monitoring the door status (open/closed)	e.g. 1
ALRAM:AKTIV= Alarm active	Specifies whether the access point generates alarm messages (J/N).	e.g. N
ALRAM:TIMER= Alarm "open"	Delay time in seconds for triggering the alarm.	e.g. 30
OPT:PRO=Trigger posting	'N' : Protocol messages are not written. '0' : (Zero) Alarm messages are written only. '1' : Access attempts are recorded in addition to alerts. '2' : All accesses are recorded in addition to the alerts and access attempts. As of HYDRA 7.1 this setting is overwritten by the logging configuration specified for the access periods of the opening hours.	e.g. 2
OPT:OFFEN=Message "open"	currently not processed	e.g. 2
MAXZAUF=Duration "open"	Maximum duration of "access point open"	e.g. 10
ZAUF:DAUER=Duration of opener	Duration of opener signal in seconds.	e.g. 4
ZZU:DAUER=Duration of closer	Duration of the closer signal in seconds.	e.g. 4
ZV:DAUER=Duration of the attempt	currently not processed	e.g. 0
WDHSPERR= anti repeat function (blocking)	Minimum time to open the access point with the same badge (seconds).	e.g. 600
ZYKL:STA=cyclic status	Time for sending the access point status to the server in cyclic intervals (seconds).	e.g. 300

ZYKL:CFG=cyclic configuration	Time for reading the access configuration from the server in cyclic intervals.	e.g. 3600
ZYKL:BERLESEN= cyclic authorizations	Time for reading authorizations from the server in cyclic intervals.	e.g. 3600
CFGANZTG=Number of days	Specifies for how many days in advance access configurations are to be loaded onto the terminal.	e.g. 7
CFGANZ=Number of records	currently not processed	e.g. 0
OPT:PZEAUTO=Trigger PZE	Specifies whether an access includes a PZE clocking (entry "J") or not (entry "N"). A clocking-in record is generated if it is an entry and a clocking-out record is generated if it is an exit. An automatic status clocking record (P_AST) is generated if it is a passage.	e.g. N
SAB:DIGIN=Channel sabotage	Channel number of the terminal to which a sabotage signal is connected or 0 if not available. This channel indicates when the terminal or badge reader (subject to the model) is opened without permission.	e.g. 0
DIGIN:1=channel in 1	currently not processed	e.g. 0
DIGIN:2=channel in 2	currently not processed	e.g. 0
DIGIN:3=channel in 3	currently not processed	e.g. 0
DIGIN:4=channel in 4	currently not processed	e.g. 0
DIGOUT:1=channel out 1	currently not processed	e.g. 0
DIGOUT:2=channel out 2	currently not processed	e.g. 0
DIGOUT:3=channel out 3	currently not processed	e.g. 0
DIGOUT:4=channel out 4	currently not processed	e.g. 0
CFG=configuration	currently not processed	e.g. "" (empty)

14.4 Online requests

14.4.1 Online check of PZE authorizations

Using the command DLG=SCMD;46, you can check whether or not a badge is allowed to clock at a specific PZE terminal. The return string "RET=0“ indicates that the authorization is available. The badge is not authorized if "RET=“ includes other values..

Structure of dialog data:

"DLG=SCMD;46|DAT=...|ZEI=...|USR=...|TNR=...|KNR=...“

Instead of the badge number "KNR=", it is also possible to transfer the personnel number "PNR=" along with the company "FIR=".

14.4.1.1 Display of messages

When PZE authorizations are checked, it is also checked whether a message is available for the person. This message is written in the file "spool\hyinf<tnr>.dat" on the server, whereas <tnr> stands for the terminal number. The file can be loaded after the online check and, if a message is included, it can be displayed. The file has the following structure:

KNR	Badge number	e.g. 17
TYP	Information about the generation of the message.	e.g. PE
DATB, DATE	Period during which the message is to be displayed.	06/15/2004, 06/30/2004
WTG	Weekday on which the message is to be displayed: 0 = Sunday ... 6 = Saturday 7 = all days of the week	e.g. 7
OPT:KOM=J/N	Specification whether or not the message is to be displayed for a clock-in record.	e.g. J
OPT:GEH=J/N	Specification whether or not the message is to be displayed for a clock-out record.	e.g. J

OPT:PAU=J/N	Specification whether or not the message is to be displayed for clocking a break.	e.g. J
OPT:FGR=J/N	Specification whether or not the message is to be displayed for an absence reason posting.	e.g. J
OPT:INFO=J/N	Specification whether or not the message is to be displayed when information on the account balance is shown.	e.g. J
ANZ	Number specifying how often the message is to be displayed. If this field is populated, the message display with the next clocking must be posted using the field PARAM02.	e.g. 1
VERWEIS	Unique data record number	e.g. 10537
INFO:1, ..., INFO:10	Up to 10 lines of the message to be displayed	e.g. Please contact the payroll office

14.4.2 Online check of ZKS authorizations

Using the command DLG=SCMD;49, you can check whether or not a badge is authorized for a specific access point in ZKS. The return string "RET=0“ indicates that the authorization is available. The badge is not authorized if "RET=“ includes other values..

Structure of dialog data:

"DLG=SCMD;49|DAT=...|ZEI=...|USR=...|ZNR=...|KNR=...“

14.4.3 Online request of a person's account balances

Using the command DLG=SCMD;31 a person's account balances can be requested. The return string includes the designations and values of up to 8 accounts that are to be displayed.

Structure of dialog data:

"DLG=SCMD;31|DAT=...|ZEI=...|USR=...|TNR=...|KNR=...“

Structure of the return string:

"RET=0|KT=|LT=|BEZL:1=Gleitzeit|WERT:1=0:00|BEZL:2=Urlaubskonto|WERT:2=38.00|“

14.5 Pre-processed data from third-party systems

14.5.1 Import of day-related (clocking) data

This section describes how data is transferred from external systems providing preprocessed data for time and attendance. This data can be processed further by HYDRA-PZE (Time & Attendance) or they are directly available for modules affecting several products, e.g. ADE/PZE comparison or HYDRA incentive wages.

Preprocessed clocking records and calculated workday results may be imported in HYDRA. The workday results, e.g. include the determined net attendance time per person and day. In addition to the workday result, one or more clocking records may be available.

Three dialogs are relevant for this purpose:

Include day results:	TGERG.MODIFY
Delete all clocking records of a person of one day:	STMP.DELETE
Load clocking:	STMP.LOAD

The following dialogs are required in the interface file to include a person's data of an entire day:

1) Include the person's day result (header record)
2) Optional: Delete all clocking records of a person at one day
   This avoids double clocking records if data is transferred several times over time.
3) Optional: Insert all individual clocking records of the person at one day (subsequent records)

Only the work day result (header record) is relevant if the "production time statistics" function is required in the first place. The clocking records may optionally be skipped, as they only provide more detailed information.

Below please find a description of the individual dialogs and a sample file.

Example of a dialog (one line of an interface file):

```
DLG=STMP.DELETE|STMP.FIR=012|STMP.PNR=002449|STMP.ABREDAT=03/01/2001|
```

Explanation:

STMP.DELETE is the dialog identification. Vertical lines ("|", ASCII 124) are used to separate the individual parameters that follow (STMP.FIR, STMP.PNR, ...). Select a field width for the different parameters that is wide enough for the presentation. However a fixed file structure can also be realized by adding leading zeros (for numbers) and trailing spaces. The sequence of the parameters is irrelevant. Some parameters must be specified in a dialog and some parameters are optional. You can specify the optional parameters, if required, or you can leave these parameters out. HYDRA than populates these parameters using default values.

Conventions used to present the different data types

<table>
<tr><th>Data type</th><th>Format</th><th>Examples</th></tr>
<tr><td>N<x></td><td>Digits, maximum of <x> places</td><td>|PNR.PNR=2449| or
|PNR.PNR=002449|</td></tr>
<tr><td>N<x>.<y></td><td>Decimal number with a maximum of <x> predecimal places and <y> decimal places. A dot is the decimal separator.</td><td>|PNR.PSTDSATZ=30.5| or
|PNR.PSTDSATZ=00030.5|</td></tr>
<tr><td>C<x>
Texts (character)</td><td>Character string with a maximum length of <x></td><td>|PNR.NAME=Huber| or
|PNR.NAME=Huber |</td></tr>
<tr><td>Date</td><td>MM/DD/YYYY (American format: month, day, year. Slashes are used as separator.)</td><td>|STMP.DAT=12/31/2001|</td></tr>
<tr><td>Times or durations</td><td>Seconds since midnight or
HH:MM or
HH:MM:SS or
HH,DDD or
HH.DDD
H hours (as many places as required)
M Minutes (in groups of 60)
S Seconds
D Industrial or decimal minutes (in groups of 100)</td><td>|STMP.ZEI=52200| or
|STMP.ZEI=14:30| or
|STMP.ZEI=14:30:00| or
|STMP.ZEI=014,5| or
|STMP.ZEI=14.500|</td></tr>
</table>

14.5.1.1 Importing work day result

The dialog for importing the work day results includes the daily times per person and day (header record). Provided that the work day result is not yet available, it will be inserted. In case this dialog already includes a work-day result, it will be updated to avoid multiple entries. HYDRA automatically deletes old work-day results from the data management after a configurable period of time.

Dialog: TGERG.MODIFY				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
TGERG.FIR	C4		Company	Company of the person
TGERG.PNR	N8	X	Personnel number	The person's personnel number
TGERG.ABREDAT	Date	X	Settlement date	
TGERG.IZ	Duration	X	Actual working time	The person's (net) actual working time on the settlement day (breaks are already deducted)
TGERG.SZ	Duration		Target time	Optional: The person's target time on the settlement day
TGERG.FZ	Duration		Absence	Optional: The person's absence time on the settlement day
TGERG.SCHZART	C1		Shift time type	Optional: The person's shift type on the settlement day (F/S/N/…)
TGERG.DATB	Date	*1)	Start of work date	Date of the person's evaluated start of working time
TGERG.ZEIB	Time	*1)	Start of work time	Time of the person's evaluated start of working time
TGERG.DATE	Date	*1)	End of work date	Date of the person's evaluated end of working time
TGERG.ZEIE	Time	*1)	End of work time	Time of the person's evaluated end of working time

*1) If the PZE/ADE comparison function is used, these fields must be filled. Only then, the application *Labor time calculation* can be used to its full extent.

Example:

DLG=TGERG.MODIFY|TGERG.PNR=999998|TGERG.ABREDAT=12/27/2010|
TGERG.DATB=12/27/2010|TGERG.ZEIB=21600|TGERG.DATE=12/27/2010|TGERG.ZEIE=52200|
TGERG.SZ=28800|TGERG.IZ=30600|TGERG=FZ=0|TGERG.SCHZART=F|

14.5.1.2 Deleting all clocking records of a day

This dialog deletes all clocking records of the person on the settlement day. This dialog has been designed to clean the data set before new clocking records are inserted. Old clocking records are deleted automatically in HYDRA after a configurable period of time (at the moment 200 days by default).

Dialog: STMP.DELETE				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
STMP.FIR	C4	X	Company	Company of the person
STMP.PNR	N8	X	Personnel number	The person's personnel number
STMP.ABREDAT	Date	X	Settlement date	

14.5.1.3 Loading a clocking record

This dialog loads a clocking record in HYDRA. Old clocking records are deleted automatically in HYDRA after a configurable period of time (at the moment 200 days by default).

Dialog: STMP.LOAD				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
STMP.FIR	C4	X	Company	Company of the person
STMP.PNR	N8	X	Personnel number	The person's personnel number
STMP.ABREDAT	Date	*1)	Settlement date	
STMP.ABREDATB	Date	*1)	Start date	Rounded start date May differ from the settlement date with night shifts.
STMP.ABREZEIB	Time	*1)	Start time	Rounded start time
STMP.ABREDATE	Date	*1)	End date	Rounded end date May differ from the settlement date with night shifts.
STMP.ABREZEIE	Time	*1)	End time	Rounded end time
STMP.DATB	Date	*2)	Start date (not rounded)	optional: start date that is not rounded May deviate from the settlement date with night shifts.
STMP.ZEIB	Time	*2)	Start time (not rounded)	optional: start time that is not rounded
STMP.DATE	Date	*2)	End date (not rounded)	optional: end date that is not rounded May deviate from the settlement date with night shifts.
STMP.ZEIE	Time	*2)	End time (not rounded)	optional: end time that is not rounded
STMP.IZ	Duration		Actual time	optional: The person's actual working time in the time interval (clocking duration)
STMP.SCHZART	C1	*1)	Shift type	Optional: The person's shift type on the settlement day (e.g. E, L or N)

*1) Rounded time stamps are only required if the transferred clocking data are not to be evaluated by the HYDRA work-day evaluation or the work-day evaluation function is not supposed to round independently. If you want to import clockings as raw data, only the time stamps that are not rounded and marked via *2) may be filled. The fields marked via *1) must be skipped or left empty.

*2) If the clocking records are imported as raw data to be further processed by the work-day evaluation, use time stamps not rounded to fill the fields. The fields marked via *1) must be skipped or left empty.

14.5.1.4 Examples

In the example below, the data of two persons is transferred for a day:

```
DLG=TGERG.MODIFY|TGERG.FIR=012|TGERG.PNR=002449|TGERG.ABREDAT=01/29/2001|TGERG.IZ=7:4
5|TGERG.SZ=7:30|TGERG.FZ=0:00|TGERG.SCHZART=F|

DLG=STMP.DELETE|STMP.FIR=012|STMP.PNR=002449|STMP.ABREDAT=03/01/2001|

DLG=STMP.LOAD|STMP.FIR=012|STMP.PNR=002449|STMP.ABREDAT=01/29/2001|STMP.ABREDATB=01/2
9/2001|STMP.ABREZEIB=5:53|STMP.ABREDATE=01/29/2001|STMP.ABREZEIE=9:57|STMP.SCHZART=F|

DLG=STMP.LOAD|STMP.FIR=012|STMP.PNR=002449|STMP.ABREDAT=01/29/2001|STMP.ABREDATB=01/2
9/2001|STMP.ABREZEIB=10:13|STMP.ABREDATE=01/29/2001|STMP.ABREZEIE=14:11|STMP.SCHZART=
F|

DLG=TGERG.MODIFY|TGERG.FIR=012|TGERG.PNR=002450|TGERG.ABREDAT=01/29/2001|TGERG.IZ=7:3
0|TGERG.SZ=7:30|TGERG.FZ=0:00|TGERG.SCHZART=F|

DLG=STMP.DELETE|STMP.FIR=012|STMP.PNR=002450|STMP.ABREDAT=03/01/2001|

DLG=STMP.LOAD|STMP.FIR=012|STMP.PNR=002450|STMP.ABREDAT=01/29/2001|STMP.ABREDATB=01/2
9/2001|STMP.ABREZEIB=5:58|STMP.ABREDATE=01/29/2001|STMP.ABREZEIE=10:01|STMP.SCHZART=F
|

DLG=STMP.LOAD|STMP.FIR=012|STMP.PNR=002450|STMP.ABREDAT=01/29/2001|STMP.ABREDATB=01/2
9/2001|STMP.ABREZEIB=10:16|STMP.ABREDATE=01/29/2001|STMP.ABREZEIE=14:01|STMP.SCHZART=
F|
```

14.6 Transferring configurations from third-party systems

14.6.1 Transfer of the planned working time

14.6.1.1 Personal working time

As of HYDRA PZE 7.2 it is possible to transfer the planned working time per person and day without having to define day types and models. These working time plans are saved as "personal working time" in HYDRA.

The following key fields need to be assigned to values for the transfer:

Dialogs: GLZTMOD.INSERT, GLZTMOD.UPDATE, GLZTMOD.MODIFY				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
GLZTMOD.PNR	N8	X	Personnel number	
GLZTMOD.DATB	Date	X	Beginning of validity	Start date for the "personal working time".
GLZTMOD.DATE	Date	X	Validity end	The end date for the "personal working time" mostly matches the start date.

GLZTMOD.ART	C1	X	Constantly 'P'	P=Personal working time

The following data fields can be transferred for each person and day:

Parameter	**Type**	**Mandatory**	**Contents**	**Description**
GLZTMOD.BEZK	C6		Short name	
GLZTMOD.BEZL	C20		Detailed designation	
GLZTMOD.VAB	C 15		Responsibility area	
GLZTMOD.SCHZART	C1		Shift type	Only transferred for shift workers.
GLZTMOD.SZ	Time		Target time	Target working time
GLZTMOD.AZMAX	Time		Max. working time per day	A message is triggered after the maximum working time has been exceeded.
GLZTMOD.RAHMB	Time		Skeleton time from	Start of the skeleton time
GLZTMOD.RAHME	Time		Skeleton time until	End of the skeleton time
GLZTMOD.NORMB	Time		Normal time from	Beginning of the normal working time
GLZTMOD.NORME	Time		Normal time until	End of the normal working time
GLZTMOD.KERNB	Time		Core time from	Start of the core working time
GLZTMOD.KERNE	Time		Core time until	End of the core working time. The core time is interrupted by the break frame.
GLZTMOD. PAURAHMB:<1..3>	Time		Frame "from" for the breaks 1 to 3	Beginning of the break frame for the breaks 1 to 3
GLZTMOD. PAURAHME:<1..3>	Time		Frame "until" for the breaks 1 to 3	End of the break frame for the breaks 1 to 3
GLZTMOD. PAUMIN:<1..3>	Time		Minimum duration for the breaks 1 to 3	
GLZTMOD. PAUNORMB:<1..3>	Time		"Normal break from" for the breaks 1 to 3	Beginning of the normal breaks 1 to 3. The normal break is allocated if no break is clocked.
GLZTMOD. PAUNORME:<1..3>	Time		"Normal break until" for break 1 to 3	End of normal break 1 to 3.

GLZTMOD. ENTLTMOD:SCHZART	N4		Payment day type	Payment day type for the shift specified in the parameter SCHZART
GLZTMOD.PAUFREI	Time		Free break	Free break that is deducted irrespective of the spent working time.
GLZTMOD.OPT:SZB	C1		Compensation of target time starting	A=Beginning of working time R=Beginning of skeleton time K=Beginning of core time N=Beginning of normal time
BEARB	C10		Modified by	

The data type "time" can be assigned to seconds since the beginning of the day or in the format HH:MM (with normal minutes) (example 10:30 a.m. = 37800 = 10:30).

Example for transferring an early shift on 17 December 2007 from 6:00 a.m. until 2:00 p.m.:

```
DLG=GLZTMOD.MODIFY|GLZTMOD.PNR=4711|
        GLZTMOD.DATB=12/17/2007|GLZTMOD.DATE=12/17/2007|
        GLZTMOD.SCHZART=F|GLZTMOD.SZ=28800|
        GLZTMOD.RAHMB=21600|GLZTMOD.RAHME=50400|
        GLZTMOD.NORMB=21600|GLZTMOD.NORME=50400|
        GLZTMOD.KERNB=21600|GLZTMOD.KERNE=50400|
```

14.6.1.2 Day types and year models

In HYDRA the working time is defined in the form of day types and year models:

- There is a separate working time day type for each different daily routine.
- The year models define which working time day type and which shift type is to be assigned at which day of the year.
- The defined year models are saved with the person in the HR master (also see the document entitled HYD-LUG).

14.6.1.3 Transfer of working time day types

The following key fields need to be assigned when working time day types are transferred:

Dialogs: GLZTMOD.INSERT, GLZTMOD.UPDATE, GLZTMOD.MODIFY				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**

GLZTMOD.GLZTMOD	N4	X	Number of the flextime day type	
GLZTMOD.DATB	Date	X	Beginning of validity	A working time day type is valid until the next validity period starts for the next day type with the same number.
GLZTMOD.ART	C1	X	Type of the model	G=Flextime day type S=Fixed shift model F=Flexible shift model

The data fields for transferring a working time day type are the same as for the personal working times as described in section "14.6.1.1 Personal working time".

With shift day types, one data record is transferred per shift.

Example:

DLG=GLZTMOD.MODIFY|GLZTMOD.GLZTMOD=7000|GLZTMOD.BEZK=KBez|GLZTMOD.BEZL=Langbez|GLZTMOD.VAB=|GLZTMOD.DATB=01/01/1900|GLZTMOD.ART=G|GLZTMOD.SCHZART=|GLZTMOD.ENTLTMOD:SCHZART=|GLZTMOD.PAUFREI=0|GLZTMOD.OPT:SZB=A|GLZTMOD.SZ=28800|GLZTMOD.AZMAX=36000|GLZTMOD.RAHMB=25200|GLZTMOD.RAHME=68400|GLZTMOD.PAURAHMB:1=28800|GLZTMOD.PAURAHME:1=36000|GLZTMOD.PAURAHMB:2=39600|GLZTMOD.PAURAHME:2=50400|GLZTMOD.PAURAHMB:3=68400|GLZTMOD.PAURAHME:3=70200|GLZTMOD.KERNB=32400|GLZTMOD.KERNE=57600|GLZTMOD.PAUMIN:1=900|GLZTMOD.PAUMIN:2=1800|GLZTMOD.PAUMIN:3=1800|GLZTMOD.NORMB=28800|GLZTMOD.NORME=59400|GLZTMOD.PAUNORMB:1=32400|GLZTMOD.PAUNORME:1=33300|GLZTMOD.PAUNORMB:2=43200|GLZTMOD.PAUNORME:2=45000|GLZTMOD.PAUNORMB:3=68400|GLZTMOD.PAUNORME:3=70200|BEARB=12345

14.6.1.4 Transfer of working time year models

This dialog has been designed to edit working time year models:

Dialog: GLZJMOD.MODIFY				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
GLZJMOD.GLZJMOD	N4	X	Number of the flextime year model	
GLZJMOD.BEZK	C6		Short name	
GLZJMOD.BEZL	C20		Detailed designation	
GLZJMOD.JAHR	N4	*1)	Year	
GLZJMOD.DATB	Date	*1)	Start date of the assignment	
GLZJMOD.DATE	Date	*1)	End date of the assignment	
GLZJMOD.VAB	C 15		Responsibility area	

GLZJMOD.MODELL	Cx		Day types to be assigned	The day types to be entered are included in this parameter and separated by comma. The day type 0 means that no day type is to be defined on this day. The models are entered in the period specified by the parameters JAHR; DATB and DATE. The parameter MODELL specifies how many day types are assigned. If the parameter MODELL includes more day types than can be entered in the date range, they will be ignored.
BEARB	C10		Modified by	

***1) The following combinations are allowed:**

1) Year only
2) Start and end date
3) Year, start and end date

In any case, the dates of these three parameters must all be in the same year!

Example: DLG=GLZJMOD.MODIFY|GLZJMOD.GLZJMOD=7001|GLZJMOD.BEZL=Langbez|
GLZJMOD.BEZK=KBez|GLZJMOD.DATB=01/14/2002|GLZJMOD.DATE=01/18/2002|
GLZJMOD.MODELL=100,100,100,100,100| GLZJMOD.VAB=|BEARB=12345|

14.6.1.5 Transfer of shift rhythm year models

This dialog has been designed to edit shift rhythm year models:

Dialog: SCHZARTJMOD.MODIFY				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
SCHZARTJMOD. SCHZARTJMOD	N4	X	Number of the shift rhythm year model	
SCHZARTJMOD. BEZK	C6		Short name	
SCHZARTJMOD.BEZL	C20		Detailed designation	
SCHZARTJMOD. JAHR	N4	*1)	Year	

SCHZARTJMOD. DATB	Date	*1)	Start date of the assignment	
SCHZARTJMOD. DATE	Date	*1)	End date of the assignment	
SCHZARTJMOD.VAB	C 15		Responsibility area	
SCHZARTJMOD. MODELL	Cx		Day types to be assigned	The shift types to be entered are included in this parameter and separated by comma. The shift type "" (blank) means that no shift type is to be defined for this day. The shift types are entered in the period specified by the parameters JAHR; DATB and DATE. The parameter MODELL specifies how many shift types are assigned. If the parameter MODELL includes more shift types than can be entered in the date range, they will be ignored.
BEARB	C10		Modified by	

***1) The following combinations are allowed:**

1) Year only
2) Start and end date
3) Year, start and end date

In any case, the dates of these three parameters must all be in the same year!

Example: DLG=SCHZARTJMOD.MODIFY|SCHZARTJMOD.SCHZARTJMOD=7001|
SCHZARTJMOD.BEZL=Langbez|SCHZARTJMOD.VAB=|SCHZARTJMOD.BEZK=KBez|
SCHZARTJMOD.DATB=01/14/2002|SCHZARTJMOD.DATE=01/18/2002|
SCHZARTJMOD.MODELL=F,F,F,S,S|BEARB=12345|

14.6.2 Absence planning

14.6.2.1 Overview

When absence plannings are transferred from third-party systems, all absences are generally updated for each person. The following processing applies:

1) All absences of a person are deleted (FZ.DELETE using MOD=G or FZ.DELETE_ALL)
2) All absences of a person are transferred a new time (FZ.INSERT for each absence)

If you have stored the user who created the absences, you can configure that only this user can delete the absence. It is then possible to manually manage the absences in the system in addition to the absences managed via interface.

14.6.2.2 Creating absence planning

You can use this dialog to insert absences of persons for the Time Management or the Personnel planning:

Dialog: FZ.INSERT				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
FZ.PNR	N8	*1)	Personnel number	The person's personnel number (as an alternative to cost center and area)
FZ.KST	C 8	*1)	Cost center	Cost center (as an alternative to person and area)
FZ.BER	C 8	*1)	Area	Area (as an alternative to person and cost center).
FZ.FIR	C4		Company	Company of the person, cost center or area. If no company is indicated, the absence applies for all companies.
FZ.DATB	Date	X	Start date	Start date of the absence
FZ.DATE	Date	X	End date	End date of the absence
FZ.WTG	N1		Weekday	Weekday for which absence planning applies: 0 ... 6 = Sunday ... Saturday 7 = every day (by default)
FZ.ENTLTMOD	N4		Absence payment	Number of the payment day type with which the absence is to be set off. This key determines, for example, the color indicating absences in the graphic absence planning. This key determines, for example, the color indicating absences in the graphic absence planning.

FZ.BEZK	C 3		Comment	The comment is entered in the absence overview. If this field remains empty, the first three letters of the short name of the absence payment will be used.
FZ.BEZL	C20		Comment	Comment defined for the absence
FZ.VERB	C1		Processing	'F': Set off absence from absence planning. 'D': Set off average working time from the HR master. 'S': Set off planned target working time. 'N': Set off planned normal time.
FZ.DAUER	Duration or N6.x		Duration of the absence	If processing "F" is set, this field has to include the duration of the absence.
FZ.TLWABW	N3		Partially absent	For part-day absences, this field specifies the percentage used to fill the time.
FZ.CERTIFY	C1 (J/N)		Subject to approval	Defines whether or not the generated absence is subject to approval.
FZ.LFZ:DATB	Date		Start date of the first pre-existing condition	If continued pay is to be monitored, this field includes the start date of the first pre-existing condition.
FZ.LFZ:DAUER	N4		Duration of pre-existing conditions	If continued pay is to be monitored, this field states the duration of the pre-existing condition in days.
FZ.LFZ:VERWEIS	N8		Reference to the last pre-existing condition	If continued pay is to be monitored, this field includes the reference to the last pre-existing condition.

BEARB	C10		Optional: Modified by	If you transfer the user who modified the entry, you can later on delete the absences of this user.

*1) Only one of the fields FZ.PNR, FZ.KST and FZ.BER must and may be filled.

Examples:

DLG=FZ.INSERT|FZ.PNR=1234|FZ.DATB=12/27/2019|FZ.DATE=12/30/2019|FZ.ENTLMOD=400|

DLG=FZ.INSERT|FZ.PNR=1456|FZ.DATB=08/22/2019|FZ.DATE=09/02/2019|FZ.ENTLTMOD=400|BEARB=MASTERSYS|

14.6.2.3 Deleting an absence planning

Different dialogs are available to delete absences. The dialog used depends on the system used. Does the system use the personnel time management including labor time calculation or does the system only use the absences for display and planning purposes?

- If you use the personnel time management including labor time calculation, use the dialog FZ.DELETE with MOD=G
- If the absences are only displayed or used for the personnel planning, then use the dialog FZ.DELETALL.

The differences between the two dialogs are described below.

14.6.2.4 Deleting absences if you use the labor time calculation

You can use the dialog FZ.DELETE with MOD=G to delete absences of a person. All absence times of the person are deleted. Optionally, you can filter by the user (modified by) or the date range to delete the absences.

If you delete specific absences via the GUI, the data records are internally marked as deleted, but on database level you can track this action in a history in the system. But if you delete all absences of a person, the data records are actually deleted physically because the deletion and the new transfer of all absences require greater data volumes and a useful history is not possible.

If you use the dialog FZ.DELETE with MOD=G, the following actions are performed in addition to the actual delete absence action.

- If labor time calculations have already been performed for the specified person, the results for the absences included in the period of time specified must be recalculated.
- If the deleted absences include requested absences, the absence requests are rejected.

If you do not require these additional actions, use the dialog FZ.DELETE_ALL. This way, the server is not loaded unnecessarily.

Dialog: FZ.DELETE				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
MOD	C1	X	Mode "G" **obligatory** for "global": MOD=G	The mode G specifies that all absence times of a person are deleted. Other modes are not planned.
FZ.PNR	N8	X	Personnel number	The person's personnel number
FZ.BEARB:DEL	C10		The absences of the user (modified by) are deleted.	Optional filter. If the filter is set, then only the absences are deleted that were created or changed by this user.
FZ.DATB	Date		Start date	Optional filter for a date range. To use the filter, enter start and end date. If you use this filter, only the absences are deleted that are fully included in the specified period of time.
FZ.DATE	Date		End date	See FZ.DATB.

Examples:

DLG=FZ.DELETE|MOD=G|FZ.PNR=1456|

DLG=FZ.DELETE|MOD=G|FZ.PNR=1456|FZ.DATB=12/04/2019|FZ.DATE=12/20/2019|

DLG=FZ.DELETE|MOD=G|FZ.PNR=1456|FZ.BEARB:DEL=MASTERSYS|

14.6.2.5 Deleting absences if you do not use the labor time calculation

You can use the dialog FZ.DELETE_ALL to delete absences of a person. All absence times of the person are deleted. Optionally, you can filter by the user (modified by) or the date range to delete the absences.

If you delete specific absences via the GUI, the data records are internally marked as deleted, but on database level you can track this action in a history in the system. But if you delete all absences of a person, the data records are actually deleted physically because the deletion and the new transfer of all absences require greater data volumes and a useful history is not possible.

Contrary to the dialog FZ.DELETE with MOD=G, the dialog FZ.DELETE_ALL does not perform further actions that are integrated in the personnel time management. For this reason, FZ.DELETE_ALL is quicker and unnecessary data changes are avoided.

Dialog: FZ.DELETE_ALL				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
FZ.PNR	N8	X	Personnel number	The person's personnel number
FZ.BEARB:DEL	C10		The absences of the user (modified by) are deleted.	Optional filter. If the filter is set, then only the absences are deleted that were created or changed by this user.
FZ.DATB	Date		Start date	Optional filter for a date range. To use the filter, enter start and end date. If you use this filter, only the absences are deleted that are fully included in the specified period of time.
FZ.DATE	Date		End date	See FZ.DATB.

Examples:

DLG=FZ.DELETE_ALL|FZ.PNR=1456|

DLG=FZ.DELETE_ALL|FZ.PNR=1456|FZ.BEARB:DEL=MASTERSYS|

DLG=FZ.DELETE_ALL|FZ.PNR=1456|FZ.DATB=12/04/2019|FZ.DATE=12/20/2019|

14.6.3 Creating account limits in HYDRA

You can use this dialog to create account limits in HYDRA PZE:

Dialog: PZEKTOG.INSERT				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
PZEKTOG.FIR	C4		Company	Company of the person, cost center or area. If no company is indicated, the account limit applies for all companies.

PZEKTOG.PNR	N8	*1)	Personnel number	Personnel number of the person (as an alternative to reference and value)
PZEKTOG.BZG	C20	*1)	Reference	The account limit applies to the specified organization unit (as an alternative to person): - BER: area - KST: cost center - ABT: department - PKREIS: employee subgroup - TAETIGKEIT: activity - BESCHVERH: employment relationship - NSTMP: person does not clock
PZEKTOG.WERT	C20	*1)	Value	The value refers to the organization unit specified in the previous field (as an alternative to person).
PZEKTOG.KTO	N1	X	Number of the account	Account number of the configuration of accounts
PZEKTOG.JAHR	N4		Year	Year of validity of the account limit
PZEKTOG.PER	N2		Period/month	Validity period for the account limit. It refers to the evaluation periods configured for the month evaluation.
PZEKTOG.KTOG: MAXAKTIV	C1 (J/N)	X	Maximum limit active	Specifies whether or not the account limit includes a maximum limit value.
PZEKTOG.KTOG: MAX	Duration or N6.x		Maximum limit	For time accounts the maximum limit is entered as duration and as decimal number for daily accounts. The maximum number of decimal places results from the configuration of accounts.
PZEKTOG.LART: MAX	C4		Maximum wage type	Wage type for the maximum limit

PZEKTOG.KTOG: MINAKTIV	C1 (J/N)	X	Minimum limit active	Specifies whether or not the account limit includes a minimum limit value.
PZEKTOG.KTOG: MIN	Duration or N6.x		Minimum limit	For time accounts the minimum limit is entered as duration and as decimal number for daily accounts. The maximum number of decimal places results from the configuration of accounts.
PZEKTOG.LART: MIN	C4		Minimum wage type	Wage type for the minimum limit
PZEKTOG.VERB: MOC	C1 (K/B/S)		Processing	"B" specifies that the record is processed as account limit. The entry 'S' is used to set the account balance. If "K" is set for processing, fixed amounts can be posted to (or deducted from) the account, irrespective of the account balance. The account balance that must be set and the fixed amount are stored in the field of the maximum limit value.

*1) Of the fields PZEKTOG.PNR, PZEKTOG.BZG and PZEKTOG.WERT, either the field PZEKTOG.PNR can be populated or the two other fields. All 3 fields can also be empty.

14.6.4 Assigning authorizations for PZE and ZKS terminals

You can use this dialog to assign clocking authorizations at PZE terminals and allow access at ZKS access points:

Dialog: ZBERECHT.INSERT				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
ZBERECHT.FIR	C4		Company	The person's company
ZBERECHT.PNR	N8	X	Personnel number	The person's personnel number
ZBERECHT.ART	C1	X	Type of authorization	PZE: 'T': Terminal 'G': Terminal group ZKS: 'Z': Access profile
ZBERECHT.NR	N4	X	Authorization number	Number of the terminal, terminal group or access profile
ZBERECHT.DATB	Date		Start date of the authorization	If this field remains empty, the validity start date for the authorization will not be restricted.
ZBERECHT.DATE	Date		End date for the authorization	If this field remains empty, the authorization has no end date and is unlimited.

All authorizations of a person are deleted using the dialog ZBERECHT.DELETE:

Dialog: ZBERECHT.DELETE				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
ZBERECHT.FIR	C4		Company	The person's company
ZBERECHT.PNR	N8	X	Personnel number	The person's personnel number
MOD	C1	X	Constant "MOD=G"	Delete all authorizations of the person.

The authorizations are transferred via initial download: all authorizations of a person are deleted and then completely inserted. To make sure that these modifications become active at the same time, use the line "DLG=WORK.BEGIN" to open a transaction before deleting the authorizations. After the last authorization, finish the transaction using the line "DLG=WORK.END". Create a transaction for each person so that is does not get too large.

14.7 Data collection for the incentive wage

14.7.1 List of bonus reasons (ZUSCHLGR.LIST)

You can use the dialog ZUSCHLGR.LIST to request a list of the bonus reasons configured in the system. The list includes all bonus reasons and does not have any selection criteria.

Example:

DLG=ZUSCHLGR.LIST|DATEI=./spool/zuschlgr.dat|DAT=10/23/2017|ZEI=40000|USR=2209|

Dialog: ZUSCHLGR.LIST				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
ZUSCHLGR	N4	X	Reason	
BEZ	C40		Designation	
ASTUFE	C1		BDE authorization	
CERTIFY	C1		Authorization required	J/empty: bonus requires approval N: bonus does not require approval
VERBCERTIF	C1		Allocate if still requires approval	N: bonus is not allocated if still requires approval Other: bonus is used in the incentive wage calculation until it is rejected.
VAB	C 15		Responsibility area	
BEARB	C10		Last modified by	
BEARBDAT	Date		Last modified on	
BEARBZEI	N5		Last modified on	
VERB	C1		Effect on target time/actual time	I: bonus has an effect on the actual time S/Other: bonus has an effect on the target time.
VERBKRIT	C10		Posting indicator	Reserved for customer-specific functions

Example of a file that has been created:

```
ZUSCHLGR|VAB|BEZ|BEARB|BEARBDAT|BEARBZEI|VERB|ASTUFE|CERTIFY|VERBCERTIF|VERBKRIT|
1||Maschinenstrung|@|10/23/2017|44650|S|3|J|N||
2||Materialmangel|@|10/23/2017|44650|S|3|J|N||
3||Transporteur fehlt|@|10/23/2017|44650|S|3|J|N||
10||Nacharbeit/allg.|@|10/23/2017|44650|I|3|J|N||
11||Nacharbeit/Entgraten|@|10/23/2017|44650|I|3|J|J||
12||Nacharbeit/Montage|@|10/23/2017|44650|I|3|J|N||
20||Langsamere Maschine|@|10/23/2017|44650|S|3|J|N||
30||Einrichter fehlt|@|10/23/2017|44650|S|3|J|N||
50||Einarbeitung|@|10/23/2017|44650|S|7|N|N||
55||Einarbeitung Mitarb.|@|10/23/2017|44650|S|7|N|N||
200||Werkzeugschaden|@|10/23/2017|44650|I|7|J|N||
500||Schnellere Maschine|@|10/23/2017|44650|S|3|J|N||
510||Einrichter Mitarbeit|@|10/23/2017|44650|S|3|J|N||
```

14.7.2 Recording of bonuses on the terminal (P_ZUSCHL)

You can use the dialog P_ZUSCHL to record bonuses.

Parameter

DLG=ZUSCHLGR.LIST|DATEI=./spool/zuschlgr.dat|DAT=10/23/2017|ZEI=40000|USR=2209|

Dialog: ZUSCHLGR.LIST				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
DAT	Date	X	Date	
ZEI	N5	X	Time	
ZUSCHLGR	N4	X	Bonus reason	
DAUER:STD or DAUER:MIN or DAUER:SEK	Decimal	*1)	Absolute duration of the bonus	Decimal with 2 decimal places. Optionally in hours with industrial minutes, minutes (60 per hour) or seconds.
DAUER.ANZ:STD or DAUER.ANZ:MIN or DAUER.ANZ:SEK	Decimal	*1)	Relative duration of the bonus	Relative duration of the bonus in combination with a quantity in DAUER.ANZ:ANZ in decimal format with 2 decimal places. Optionally in hours with industrial minutes, minutes (60 per hour) or seconds.
DAUER.ANZ:ANZ	Decimal	*1)	Quantity for the relative duration of the bonus.	Decimal with 2 decimal places in minutes (only in combination with DAUER.ANZ:*).
ANR	C40	X *2)	HYDRA order number (fully defined key)	

KNR	C10	X *2)	Badge number	
LPGRP	C 8	*2)	Premium group •	
KNR:SIGN	C10		Badge number of person that approves	
VORMONAT	C1		Bonus must be booked to the previous month.	J: The bonus date is set to the last day of the previous month in parameter DAT.
BEM:1	C 50		Comment 1	Requirements: *3)
BEM:2 … BEM:5	C 50		Comments 2 to 5	Requirements: *4)
FU:01 … FU:02	Date		User fields date	Requirements: *3)
FU:03 … FU:06	N		User fields integer	Requirements: *3)
FU:07 … FU:08	Decimal		User fields decimal	Requirements: *3)
FU:09	C20		User field text	Requirements: *3)
FU:10	C40		User field text	Requirements: *3)

If the badge number is only entered via bar code, the field must be configured in the field configuration with "field attribute 1" = MANUELL and "field attribute 2" = READONLY.

***1) Recording of the bonus value**

The bonus value is saved in HYDRA in date format "hours" with industrial minutes. In some cases, values of other units are recorded for bonuses, e.g. "piece". Internally, the date format "hours" is always used.

Two options are available:

1. Collection of an absolute value with a parameter DAUER:*
   The value can be collected directly. With parameter DAUER:MIN, the value is divided by 60 to get hours before being saved. With parameter DAUER:SEK, the value is divided by 3600 to get hours before being saved.

2. You use the parameters DAUER.ANZ:ANZ and DAUER.ANZ:STD/MIN/SEK to collect a relative value as time per piece.
   Using the combination of the above parameters, you can record the bonus as "x seconds per piece", for example. You then use the number of pieces and the relative bonus to automatically calculate and save an absolute bonus.

You can record the value either as absolute or as relative bonus.

***2) Assigning the bonus**

You can record a bonus for a premium group (group incentives) or a single person (individual piecework or individual bonus). The following rules apply:

1. You must enter the person (KNR) and the order (ANR). Only then, the system can perform the validation check and assign the bonus of a concrete order processing to a person.
2. If a premium group was collected via LPGRP, the bonus is collected as group incentives. If the bonus is collected as group incentives, the staff badge number and the order are entered to document the context. By default, person and order are not relevant in the processing of the group incentives.

***3) Requirements of comment 1 and user fields**

The parameter BEM:1 and the user fields are only available as of program version hymwlle71.dll 8.1.1.19 (November/2017, SP12). A customer-specific extension must be installed.

***3) Requirements of comment 2 to 5**

For the parameters BEM:2 to BEM:5, you require the program version hymwlle71.dll 8.1.1.19 (November/2017, SP12). A customer-specific extension with database patch must be installed and authorized.

Example

DLG=P_ZUSCHL|DAT=10/24/2017|ZEI=44280|KNR=4125|ANR=4000LL100090|ZUSCHLGR=55|DAUER:STD=3.75|USR=2209|

Validation checks

- The order must exist.

- The person must exist. The person must have at least the BDE authorization level that is stored for the bonus. If no authorization level is stored for the bonus, the person must have at least level 3. This guarantees that only persons with special authorizations can enter bonuses. If the parameter KNR:SIGN is used to record the staff badge number of the person authorizing the bonus, their BDE authorization level is checked.

- If the premium group is specified in the dialog, the group must exist.

- You can record the value either as absolute or as relative bonus. If the three parameters for absolute bonus, quantity and relative bonus include a value that is not 0, the terminal issues an error message with the posting.

- The bonus reason must exist.

Processing

The editing date of the bonus always includes the time of recording. If the bonus is recorded on a terminal, "T<*terminal number*>" is entered for *Modified by* with the bonus.

The bonus requires approval if configured accordingly in the bonus configuration.

If a person approving the bonus is recorded with parameter KNR:SIGN, their personnel number is saved in field "Authorization" of the bonus.

15 HYDRA Production Data Manager MPL - Data Collection

15.1 Please note for the posting dialogs described

The HYDRA-MPL dialogs are only useful in connection with the corresponding HYDRA-BDE dialogs (please also see the associated BDE documentation).

All mandatory fields are highlighted in gray. All other fields are optional and are only processed if they are filled out.

15.2 Batch Postings

15.2.1 Batch change (DLG=CA_WL)

<table>
<tr><th>ID</th><th>Type/max. field length</th><th colspan="2">Description</th></tr>
<tr><td>ANR=</td><td>C16</td><td colspan="2">Order number with operation number</td></tr>
<tr><td>MNR=</td><td>N8/C8</td><td colspan="2">Machine number (numeric/alphanumeric)</td></tr>
<tr><td>PNR=</td><td>C10</td><td rowspan="2">alter-
native</td><td>Personnel number</td></tr>
<tr><td>KNR=</td><td>C10</td><td>Badge number</td></tr>
<tr><td>CNR=</td><td>C20</td><td colspan="2">Batch number</td></tr>
<tr><td>EGR:GUT=</td><td>DEC</td><td colspan="2">Recorded yield</td></tr>
<tr><td>EGE:GUT=</td><td>C4</td><td colspan="2">Unit of yield</td></tr>
<tr><td>KLASSE=</td><td>C1</td><td colspan="2">“G” yield
“A” scrap
“O” on hold
“N” rework</td></tr>
<tr><td>STA=</td><td>C1</td><td colspan="2">Status of output batch
F= with residual quantity (default with KLASSE=G)
S= blocked (default with KLASSE=A)</td></tr>
<tr><td>ZLO=</td><td>C12</td><td colspan="2">Target location, output batch</td></tr>
<tr><td>EGG:AUS=</td><td>NUM</td><td colspan="2">Scrap reason KLASSE=A
KLASSE=O EGG:PRB
KLASSE=N EGG:NCH</td></tr>
</table>

ID	Type/max. field length	Description
TPE=	C10	Transport unit
LHW=	C20	Note, output batch
CALT1= to CALT20=	-	Alternative batch number 1-20 max. field length: CALT1-4 - C20 CALT5-14 - C40 CALT15-18 - C100 CALT19-20 - C512
ATTR:1 = to ATTR:10 =	--	The definition of which values are filed in additional attributes is configured for each material type in HYDRA
ATTR:101= to ATTR:140=	C40	Alphanumeric batch attributes
ATTR:201= to ATTR:220=	NUM	Numeric batch attributes
ATTR:301= to ATTR:320=	DEC	Decimal batch attributes
CNR.USRFLD	C8	User field key
CNR.FU:1 – CNR.FU:66	-	User fields

Please note:

There are other, similar dialogs regarding output batches (CA_AN & CA_AB) in HYDRA. They are only required to keep data within the event maintenance function at the console.

To change output batches, the CA_WL dialog always has to be sent! Except for the batch number (CNR) all information refers to the currently active HYDRA batch, which is to be logged off with this posting. The batch number defines which ID the HYDRA batch gets that is to be started, once the current HYDRA batch has been completed.

The same values are supported for log off operation (DLG=A_AB) and interrupt OP (DLG=A_UN).

1) Example: Change batch and indicate the quantity

```
DLG=CA_WL|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|CNR=X912536177|EGR:
GUT=1|
```

2) Example: Change batch and indicate the quantity and unit

```
DLG=CA_WL|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|CNR=X912536177|EGR:
GUT=1|EGE:GUT=ST|
```

15.2.1.1 Log output batch on (DLG=CA_AN)

ID	Type/max. field length	Description	
ANR=	C16	Order number with operation number	
MNR=	N8/C8	Machine number (numeric/alphanumeric)	
PNR=	C10	alter-native	Personnel number
KNR=	C10		Badge number
CNR=	C20	Batch number	
CALT1= to CALT20=	-	Alternative batch number 1-20 max. field length: CALT1-4 - C20 CALT5-14 - C40 CALT15-18 - C100 CALT19-20 - C512	

Please note: Only possible if the operation runs on the machine.

15.2.1.2 Log output batch off (DLG=CA_AB)

ID	Type/max. field length	Description	
ANR=	C16	Order number with operation number	
MNR=	N8/C8	Machine number (numeric/alphanumeric)	
PNR=	C10	alter-native	Personnel number
KNR=	C10		Badge number
CNR=	C20	Batch number	
EGR:GUT=	DEC	Recorded yield	
EGE:GUT=	C4	Unit of yield	
KLASSE=	C1	“G” yield “A” scrap “O” on hold “N” rework	

ID	Type/max. field length	Description
STA=	C1	Status of output batch F= with residual quantity (default with KLASSE=G) S= blocked (default with KLASSE=A)
EGG:AUS=	NUM	Scrap reason KLASSE=A KLASSE=O EGG:PRB KLASSE=N EGG:NCH
ZLO=	C12	Destination, output batch
TPE=	C10	Transport unit
LHW=	C20	Note, output batch
CALT1= to CALT20=	-	Alternative batch number 1-20 max. field length: CALT1-4 - C20 CALT5-14 - C40 CALT15-18 - C100 CALT19-20 - C512
ATTR:1 = to ATTR:10 =	--	The definition of which values are filed in additional attributes is configured for each material type in HYDRA
ATTR:101= to ATTR:140=	C40	Alphanumeric batch attributes
ATTR:201= to ATTR:220=	NUM	Numeric batch attributes
ATTR:301= to ATTR:320=	DEC	Decimal batch attributes
CNR.USRFLD	C8	User field key
CNR.FU:1 – CNR.FU:66	-	User fields

Please note: Only possible if the operation runs on the machine.

15.2.2 Log input batch on (DLG=CE_AN)

ID	Type/max. field length	Description
ANR=	C16	Order number with operation number

ID	Type/max. field length	Description	
MNR=	N8/C8	Machine number (numeric/alphanumeric)	
PNR=	C10	alter-native	Personnel number
KNR=	C10		Badge number
CNR=	C20	Batch number of the input batch	
SLP	C10	Bill of material item of component	
ATK	C40	Article	

15.2.3 Log input batch off (DLG=CE_AB)

ID	Type/max. field length	Description	
ANR=	C16	Order number with operation number	
MNR=	N8/C8	Machine number (numeric/alphanumeric)	
PNR=	C10	alter-native	Personnel number
KNR=	C10		Badge number
CNR=	C20	Batch number of input batch	
LHW=	C20	Notes on the input batch to be logged off	
STA=	C1	Status of the input batch to be logged off F = with residual quantity S = blocked A = processed	
EGR:REST=	DEC	Residual quantity of the input batch	
EGE:REST=	C4	Unit	
SLP	C10	Bill of material item of the component	
EGR:VERB=	DEC	Consumption of input batch As an alternative to EGR:REST/EGE:REST	
EGE:VERB=	C4	Unit As alternative to EGR:REST/EGE:REST	

If a batch is registered several times in parallel as an input batch and the consumption is to be recorded, the fields EGR:VERB and EGE:VERB must be used for customer-specific implementations.

15.2.4 Repost batch (DLG=C_UMB)

ID	Type/max. field length	Description
ZLO=	C12	Destination, output batch
MNR=	N8/C8	Machine number (numeric/alphanumeric) Only if BEHÄLTERVERWALTUNG (container management) is active
CNR=	C20	Batch
DLL=		Batch number via entry
MOD=	C1	K mode: customer batch mode (objective: assignment of another customer batch) L mode: Delete mode V mode: “forward” mode R mode: “backward” mode
TECHINFO=	C20	Technical info: entry/barcode (mode K only) Will no longer be supported from MPL 7.2.1 on
ATTR:1=	--	G mode: consecutive number, incremented by the terminal
ATTR:2=	--	G mode: Priority R mode: Index from continuous counter
HSDAT=	mm/dd/yyyy	All batches of the same date/time (mode R only)
HSZEIT=	seconds	All batches of the same date/time (mode R only)
STA=	C1	Status (F)free or (S)blocked, default = F

15.2.5 Goods receipt batch (DLG=C_GEN)

<table>
<tr><th>ID</th><th>Type/max. field length</th><th colspan="2">Description</th></tr>
<tr><td>ANR=</td><td>C16</td><td colspan="2">Order number with operation number</td></tr>
<tr><td>MNR=</td><td>N8/C8</td><td colspan="2">Machine number (numeric/alphanumeric)</td></tr>
<tr><td>PNR=</td><td>C10</td><td rowspan="2">alter-
native</td><td>Personnel number</td></tr>
<tr><td>KNR=</td><td>C10</td><td>Badge number</td></tr>
<tr><td>CNR=</td><td>C20</td><td colspan="2">Batch number (*)</td></tr>
<tr><td>EGR:GUT=</td><td>N8</td><td colspan="2">Batch quantity</td></tr>
<tr><td>EGE:GUT=</td><td>C3</td><td colspan="2">Unit of batch quantity</td></tr>
<tr><td>ATK=</td><td>C40</td><td colspan="2">Material/article</td></tr>
<tr><td>HZTYP=</td><td>C10</td><td colspan="2">Material type according to the configuration in HYDRA</td></tr>
<tr><td>KLASSE=</td><td>C1</td><td colspan="2">"G" yield or "A" scrap</td></tr>
<tr><td>EGG:AUS=</td><td>N3</td><td colspan="2">Scrap reason</td></tr>
<tr><td>ZLO=</td><td>C12</td><td colspan="2">Destination</td></tr>
<tr><td>TPE=</td><td>C10</td><td colspan="2">Transport unit</td></tr>
<tr><td>LHW=</td><td>C20</td><td colspan="2">Note, goods receipt batch</td></tr>
<tr><td>STA=</td><td>C1</td><td colspan="2">Batch status (F = free, S = blocked)</td></tr>
<tr><td>QST:2=</td><td>C1</td><td colspan="2">Quality status (G = blocked, F = free)</td></tr>
<tr><td>QST=</td><td>C1</td><td colspan="2">Manual quality status (G = blocked, F = free)</td></tr>
<tr><td>MATST=</td><td>C1</td><td colspan="2">Material status (e.g. V = packed)</td></tr>
<tr><td>CNR:ALT1 – 5 =</td><td>C20 -40</td><td colspan="2">Alternative batch number 1-5
maximum lenght:
CNR:ALT1-4 - C20
CNR:ALT5 - C40</td></tr>
<tr><td>MCNR=</td><td>C20</td><td colspan="2">Mother batch number (e.g. reference from collective batch)</td></tr>
<tr><td>ATTR:1 = to ATTR:11 =</td><td>--</td><td colspan="2">The definition of which values are filed in additional attributes is configured for each material type in HYDRA</td></tr>
</table>

ID	Type/max. field length	Description
ATTR:101= to ATTR:140=	C40	Alphanumeric batch attributes
ATTR:201= to ATTR:220=	NUM	Numeric batch attributes
ATTR:301= to ATTR:320=	DEC	Decimal batch attributes
RESART=	C4	Reservation type (OP = operation, AK = order)
RESVAL=	C40	Reservation value (e.g. operation number)
RESBEM=	C100	Reservation comment
CNR:BREITE=	DEC	Material width (conversion factor for MPLRF-BP)
CNR:RFAGVFA=	DEC	Mass per unit area (conversion factor for MPLRF-BP)
CNR:RFSTKF=	DEC	Surface per piece (conversion factor for MPLRF-BP)
CNR:SAPCNR=	C10	PPS batch number
CNR:TECHINFO =	C20	Technical info
CNR:VVDAT=	DATUM	Availability date
CNR:VVZEI=	ZEIT	Availability time
CNR:VFDAT=	DATUM	Expiry date
CNR:VFZEI=	ZEIT	Expiry time
CNR:WDAT=	DATUM	Warning date
CNR:WZEI=	ZEIT	Warning time
CNR:LAGORT=	C12	PPS storage location
CNR:EXTCNR=	C10	External batch number

(*) = The batch number does not have to be indicated if automatic batch number generation is enabled.

15.2.6 Consumption posting (DLG=A_VERB)

ID	Type/max. field length	Description	
MNR=	N8/C8	Machine number (numeric/alphanumeric) where the OP is running	
ANR=	C16	Order number with operation number (only running OP possible)	
SLP	C10	Bill of material item of the component	
ATK=	C40	Material/article of the component	
PNR=	C10	alter-native	PNR=
KNR=	C10		KNR=
EGR:VERB=	N8	Consumption quantity	
EGE:VERB=	C3	Unit of consumption quantity	
CNR=	C20	Batch number (*) for batch-related input material	

* The batch number is only optionally taken over to the consumption movement.

15.2.7 Create/change batches (DLG=CNR.MODIFY)

By way of the “CNR.MODIFY“ Bapi HYDRA batches can be created or existing batches may be changed.

ID	Type/max. field length	Description
CNR.CNR=	C20	Internal batch number
CNR.DLL=	C20	Run-through batch number
CNR.CNR:Z=	C20	Target batch number with CNR.COPY
CNR.ATK=	C40	Material number
CNR.ATKBEZ=	C40	Material designation
CNR.SGR:GUT=	DEC	Batch quantity
CNR.SGR:REST=	DEC	Residual quantity of the batch
CNR.SGE:GUT=	C3	Unit of the batch quantity
CNR.MNR=	N8/C8	Machine number

ID	Type/max. field length	Description
CNR:ATTR:11=	C40	Order number with operation number
CNR.STA=	C1	Batch status (F = free, S = blocked)
CNR.CKL=	C1	Batch class G = yield A = scrap O = on hold N = rework
CNR.TST=	C1	Transport status (L = delivered)
CNR.QST=	C1	Quality status (G = blocked, F = free)
CNR.QSTMANU=	C1	Manual quality status (G = blocked, F = free)
CNR.MATST=	C1	Material status (e.g. V = packed)
CNR.MATTYPART=	C10	Material type
CNR.OPT:REST=	C1	J – "batch has still got residual quantity" indicator N – batch has no residual quantity
CNR.FIR=	C4	Company
CNR.HSDAT=	DATUM	Manufacturing date
CNR.HSZEI=	ZEIT	Manufacturing time
CNR.VVDAT=	DATUM	Availability date
CNR.VVZEI=	ZEIT	Availability time
CNR.VFDAT=	DATUM	Expiry date
CNR.VFZEI=	ZEIT	Expiry time
CNR.WDAT=	DATUM	Warning date
CNR.WZEI=	ZEIT	Warning time
CNR.CSTWDAT=	DATUM	Date of the last status change
CNR.CSTWZEI=	ZEIT	Time of the last status change
CNR.MATPUF=	C12	Material buffer
CNR.MATTYP=	C10	Material type
CNR.TPE=	C10	Transport unit

ID	Type/max. field length	Description
CNR.BEM=	C20	Batch note
CNR.PNR=	C10	Personnel number
CNR.TECHINFO=	C20	Technical info
CNR.EGG:AUS=	N8	Scrap reason
CNR.GR=	N8	Scrap reason (synonymous with CNR.EGG:AUS)
CNR.GRTXT=	N8	Reference to scrap reason text
CNR.USR=	N8	User number
CNR.LAGORT=	C12	PPS storage location
CNR.LAGPZ=	C12	PPS storage bin
CNR.SAPCNR=	C10	PPS batch number
CNR.RESART=	C4	Reservation type (OP = operation, AK = order)
CNR.RESVAL=	C40	Reservation value (e.g. operation number)
CNR.RESBEM=	C100	Reservation comment
CNR.BREITE=	DEC	Material width (conversion factor for MPLRF-BP)
CNR.RFAGVFA=	DEC	Mass per unit area (conversion factor for MPLRF-BP)
CNR.RFSTKF=	DEC	Surface per piece (conversion factor for MPLRF-BP)
CNR.EGR:1 – 6=	DEC	Quantity, activity 1-6
CNR.RGR:1 – 6 =	DEC	Residual quantity, activity 1-6
CNR.EGE:1 – 6 =	C3	Unit, activity 1-6
CNR.CNR:ALT1 – 20 =	C20	Alternative batch number 1-20
CNR.EXTCNR=	C10	External batch number
CNR.MCNR=	C20	Mother batch number (e.g. reference from collective batch)
CNR.AUART=	C5	Order type
CNR.UMRFAKTP:Z=	DEC	Denominator – conversion factor relating to OP quantity units
CNR.UMRFAKTP:N=	DEC	Numerator – conversion factor relating to OP quantity units
CNR.UMRFAKTS:Z=	DEC	Denominator – conversion factor relating to OP quantity units
CNR.UMRFAKTS:N=	DEC	Numerator – conversion factor relating to OP quantity units

ID	Type/max. field length	Description
CNR.UMRFAKTT:Z=	DEC	Denominator – conversion factor relating to OP quantity units
CNR.UMRFAKTT:N=	DEC	Numerator – conversion factor relating to OP quantity units
CNR.ZUORD:1 – 6 =	C1	Indicator to assign “MPL activity account → ADE quantity unit (e.g. P = primary quantity)
ATTR:101 - ATTR:140	C40	Alphanumeric batch attributes
ATTR:201 - ATTR:220	NUM	Numeric batch attributes
ATTR:301 - ATTR:320	DEC	Decimal batch attributes
CNR.USRFLD	C8	User field key
CNR.FU:1 – CNR.FU:66	-	User fields

15.2.8 Goods movement (DLG=C_MBEW)

Using the command DLG=C_MBEW batches may be created or reposted. Moreover, the quantity and status of an existing batch may be changed as well.

ID	Type/max. field length	Description	
ANR=	C16	Order number with operation number	
MNR=	N8/C8	Machine number (numeric/alphanumeric)	
PNR=	C10	alternative	Personnel number
KNR=	C10		Badge number
CNR=	**C20**	**Batch number (*)**	
SGR:GUT= / EGR:GUT= / RGR:GUT=	DEC	Batch quantity RGR:GUT – change residual quantity only	
SGE:GUT= / EGE:GUT= / RGE:GUT=	C3	Unit of batch quantity RGE:GUT – change residual quantity only	
ATK=	C40	Material/article	

ID	Type/max. field length	Description
ATKBEZ=	C40	Material designation
HZTYP=	C10	Material type of the batch
KLASSE=	C1	"G" yield or “A” scrap
EGG:AUS=	N3	Scrap reason with KLASSE=A
STA=	C1	Status (F) free or (S) blocked, by default = F
QST=	C1	Manual quality status (G – blocked, F – free)
MATST=	C1	Material status (e.g. V = packed)
ZLO=	C12	Destination
CNR:LAGORT=	C12	PPS storage location
TPE=	C10	Transport unit
LHW= / BEM=	C20	Note/comment
ATTR:1 = to ATTR:11 =	--	The definition of which values are filed in additional attributes is defined for each material type in HYDRA
CNR:ALT1 – 5 =	C20	Alternative batch number 1-5
EXTCNR=	C10	External batch number
MCNR=	C10	Mother batch number (e.g. reference from collective batch)
CNR:HSDAT=	DATUM	Manufacturing date
CNR:HSZEI=	ZEIT	Manufacturing time
CNR:VVDAT=	DATUM	Availability date
CNR:VVZEI=	ZEIT	Availability time
CNR:VFDAT=	DATUM	Expiry date
CNR:VFZEI=	ZEIT	Expiry time
CNR:WDAT=	DATUM	Warning date
CNR:WZEI=	ZEIT	Warning time
SAPCNR=	C10	PPS batch number
RESART=	C4	Reservation type (OP = operation, AU = order)
RESVAL=	C40	Reservation value (e.g. operation number)

ID	Type/max. field length	Description
RESBEM=	C100	Reservation comment
CNR:BREITE=	DEC	Material width (conversion factor with MPLRF-BP)
CNR:RFAGVFA=	DEC	Mass per unit area (conversion factor with MPLRF-BP)
CNR:RFSTKF=	DEC	Surface per piece (conversion factor with MPLRF-BP)
UMRFAKTP:Z=	DEC	Denominator – conversion factor relating to OP quantity units
UMRFAKTP:N=	DEC	Numerator – conversion factor relating to OP quantity units
UMRFAKTS:Z=	DEC	Denominator – conversion factor relating to OP quantity units
UMRFAKTS:N=	DEC	Numerator – conversion factor relating to OP quantity units
UMRFAKTT:Z=	DEC	Denominator – conversion factor relating to OP quantity units
UMRFAKTT:N=	DEC	Numerator – conversion factor relating to OP quantity units
ZUORD:1 – 6 =	C1	Indicator to assign “MPL activity account → ADE quantity unit (e.g. P = primary quantity)
MENGE1 – MENGE6 =	DEC	Activity 1-6
EINH1 – EINH6=	C3	Unit of activity 1-6
ATTR:101= to ATTR:140=	C40	Alphanumeric batch attributes
ATTR:201= to ATTR:220=	NUM	Numeric batch attributes
ATTR:301= to ATTR:320=	DEC	Decimal batch attributes

15.2.9 Change batch status (DLG=C_STA)

The status of an existing batch can be changed using the DLG=C_STA command.

ID	Type/max. field length	Description
ANR=	C16	Order number with operation number
MNR=	N8/C8	Machine number (numeric/alphanumeric)
PNR=	C10	Personnel number

<table>
<tr><th>ID</th><th>Type/max. field length</th><th colspan="2">Description</th></tr>
<tr><td>KNR=</td><td>C10</td><td>alternative</td><td>Badge number</td></tr>
<tr><td>CNR=</td><td>C20</td><td colspan="2">Batch number</td></tr>
<tr><td>KLASSE=</td><td>C1</td><td colspan="2">"G" yield or “A” scrap</td></tr>
<tr><td>STA=</td><td>C1</td><td colspan="2">Status (F) free or (S) blocked, by default = F</td></tr>
<tr><td>QST=</td><td>C1</td><td colspan="2">Manual quality status (G – blocked, F – free)</td></tr>
<tr><td>MATST=</td><td>C1</td><td colspan="2">Material status (e.g. V = packed)</td></tr>
<tr><td>EGG:AUS= / GR=</td><td>N3</td><td colspan="2">Scrap reason with KLASSE = A</td></tr>
</table>

15.2.10 Input batch change (DLG=CE_WL)

Input batches can be changed using the command DLG=CE_WL. Input batches are always changed with respect to the bill of material.

<table>
<tr><th>ID</th><th>Type/max. field length</th><th colspan="2">Description</th></tr>
<tr><td>ANR</td><td>C16</td><td colspan="2">Order number with operation number</td></tr>
<tr><td>MNR</td><td>N8/C8</td><td colspan="2">Machine number (numeric/alphanumeric)</td></tr>
<tr><td>PNR</td><td>C10</td><td rowspan="2">alternative</td><td>Personnel number</td></tr>
<tr><td>KNR</td><td>C10</td><td>Badge number</td></tr>
<tr><td>CNRAB</td><td>C20</td><td colspan="2">Batch number of the input batch to be logged OFF</td></tr>
<tr><td>CNR</td><td>C20</td><td colspan="2">Batch number of the input batch to be logged ON</td></tr>
<tr><td>SLP</td><td>C10</td><td colspan="2">Bill of material item of the component (if not transferred the BOM item (SLP) is set to 0)</td></tr>
<tr><td>ATK</td><td>C40</td><td colspan="2">Article</td></tr>
</table>

ID	Type/max. field length	Description
LHW	C20	Note on the input batch to be logged off
STA	C1	Status of the input batch to be logged off F= with residual quantity S= blocked A= processed
EGR:REST	N8	Remaining quantity of the input batch to be logged off
EGE:REST	C4	Unit of remaining quantity
EGR:VERB	N8	Consumption of input batch Alternative to EGR:REST/EGE:REST
EGE:VERB	C4	Unit Alternative to EGR:REST/EGE:REST

15.3 Reading of MPL Data

15.3.1 Material list / batch information

The material/batch list is provided by the command DLG=LIST;13 and filed in the HYDRADIR\spool\ directory.

Structure of dialog data:

“DLG=LIST;13|DATEI={file name }|DAT=...|ZEI=...|USR=...|MOD=...“

a.) Reading of the material list for machine and order:

Parameter: MOD=M|MNR=Machine|ANR=Order number

→ Machine is required for still running batches
→ Order is required for planned materials (if necessary with batch assignment) of the order

b.) Reading of the batch info:

Parameter: MOD=L|CNR=batch number

c.) Reading of preceding batches (output batches) of an order **(including the current output batch):**

Parameter: MOD=A|ANR=order number

d.) Running batches of all machines of the terminal

Parameter: MOD=T

The list includes the following data:

ID	Field designation	Description
MNR	Machine	Machine number
ANR	Order	Order and OP [and type]
ATK	Article	Input material number (**)
ATKBEZ	Des. of final article	Designation of input material (**)
HZTYP	Semi-finished article type	Semi-finished article type (**)
SLP	BOM item	Bill of material item (*)
EMENGE	Required quantity	Quantity of required material for each unit of the output material (*)
EINH	Unit	Unit of required quantity (**)
TECHINFO	Techn.Info	Technical info relating to mat. (**)
DLL	Cs. ba. No.	Run-through batch number
DLLKZ	Cs.-batch ID	Run-through batch ID
CNR	Batch number	Batch number (***)
LHW	Batch note	Batch note (***)
TPE	TPUnit [spec.]	Transport unit (***)
HSDAT	Manuf. date	Manufacturing: date (***)
HSZEI	Manuf. time	Manufacturing: time (***)
VVDAT	Date availability	Availability: date (***)
VVZEI	Time availability	Availability: time (***)
VFDAT	Date expiry	Date of expiry: date (***)
VFZEI	Time expiry	Date of expiry: time (***)
WARNDAT	Warning date	Warning date (***)
WARNZEI	Warning time	Warning time (***)
SGR:GUT	Target	Batch quantity (***) = Target quantity for input batches = Recorded quantity for output batchespos. booking as output batch

SGR:REST	Residue	Remaining quantity of the batch (***) = Actual quantity of the batch: - when logging output batch off or transferring it SS == quantity of batch - neg. booking as input batch
RMENGEKZ	RQ	Remaining quantity indicator (***)
ALTMENGE	Alt.qty	Alternative quantity (***)
ALTEINH	Alt.unit	Alternative quantity unit (***)
ZLO	Destination	Destination (***)
CST	Batch status	Batch status (***)
CSTWDAT	Batch ch.date	Date of status change (***)
CSTWZEI	Batch ch.time	Time of status change (***)
PNR	PersNo.	Personnel number (***)
ABKZ	LogoffID	Logoff ID (****) “X“ for forced logoff
AGCST	OPBST	Batch status of operation: 0 Batch is currently not logged on to order (preceding batches) 1 Batch is currently logged on to order
ATTR:1 to ATTR:11		Customer-specific Batch attribute 1 to Batch attribute 11
HZBEZ	Material type des.	
MINLGZ	Min. storage time	
WARNGRENZ	Warning limit	
VFALLGRENZ	Expiry limit	
HZART	SF type	
LOSGR	Lot size	
LOSEINH	Unit	
LAY	Layout	
TICKETANZ	Number of tickets	
OPT:VFCNRA	Expiry date from input batch	
OPT:VFCNRE	Expiry date for output batch	
OPT:MULTICNR	Can be logged on several times	
OPT:INBESTVERB	Invent. char.	
PROZAGAB	Tolerance limit AGAB	
UTGAGAB	abs. tolerance limit AGAB	

OPT:VERE	Hand batch no. down	
OPT:AGZGVERB	REB-OP-Ref.	OP reference when batches are reposted
OPT:AGZGGEN	GR-OP-Ref.	OP reference when goods receipt batches are created
PLAUS:MATOK	Plaus. MatOK	
PLAUS:EMATOK	Plaus. EMatOK	
OPT:CNREA	Valid for 1 outp. batch	
LOCAL:1 to LOCAL:2	LOCAL	
CKL	Class	
EGG:AUS	Scrap reason	
TST	Transport status	
ART	Indic.	
SCHNEIDNR	Cut number	
SLOS	Collective batch	
LOSANZ	No. of batches	
PBANDNR	Prod. line no.	
CNR:RESVAL	Reservation	
TR:BREITE	Width of DR	
TR:ANZ_GES	Total no. outp. batches	
ISTATK	Material	
ISTATKBEZ	Material des.	
ATKDIFF	Art. diff.	Planned article and actual article are different (J/N/F) J: different N: not different F: free component
ISTANR	Order	
ATK:2	2nd material	
BEDARF	Requirement quantity	
VERB	Consumption	
TREST	Theo. remain. requirements	
REST:VZ	Sign	
STA_KOMBI	Combined status	
SAPCNR	SAP batch	
OPT:VGRCNR	Required qty. batch-related	
OPT:ERSBAR	Replaceable	
OPT:WZW	Bat.change req.	If the input batch of this component is changed the output batch needs to be changed.

STA:GEW	Entry of weight	
AGBEZ.|	Des. of final article	
LS1	Activity 1	
LS1:REST	Rem. quantity 1	
LS1:EINH	Unit 1	
.....	....	
LS6	Activity 6	
LS6:REST	Rem. quantity 6	
LS6:EINH	Unit 6	
EGR:LEN	Rework	
SLS	Level	
SLS:M	Mother level	
BEM	Comment	
ALT1 to ALT5	Altern. batch no. 1 to altern. batch no. 5	
SNR	Serial number	(*****)
HULEVEL	HU level	(*****)

(*) = Fields are not filled out for information relating to batches!!!

(**) = When it comes to information based on batches, the field is determined from the pool of batches, otherwise from material data.

(***) = Fields are not filled out if no batch is currently logged on for input material!!!

(****) = The batch number is determined subject to the command:

Command (a): Available batches and the materials planned for the order are read for the machine (without batch that is currently logged on).

Command (b): is transferred!

Command (c): output batch number

Example:

DLG=LIST;13|MOD=A|ANR=001122330020|DAT=10/11/2000|ZEI=40000|USR=101|DATEI= ./spool/ml_list.101|

(*****) = Only available as of HYDRA-MPL product version 7.2.5.

15.3.2 Material buffer

The list of material buffers is provided using the command DLG=LIST;49 and filed in the HYDRADIR\spool\ directory.

Structure of dialog data:

“DLG=LIST;49|DATEI={file name }|DAT=...|ZEI=...|USR=...|...“

Parameter: none

The list includes the following data:

ID	Field designation	Description
MATPUF	Material buffer	
BEZ	Designation	
OPT:PKORB	Wastebasket	
OPT:INBESTVERB	Calculate as inventory	
ART	Buffer model	
OPT:LAGVERB	Stock posting buffer	
OPT:VIRTLAG	Virtual stock buffer	
ABT	Department	
BER	Area	
KST	Cost center	
FIR	Company	
LAGORT	Storage location	
BEM	Comment	
DAUER	Duration	
TYP	Buffer type	
ZLO	Receiving storage location	
MPUFF	Material buffer	
HARCMATPUF	Hierarchical buffer	

Example:

DLG=LIST;49|DAT=02/11/2005|ZEI=40000|USR=101|DATEI= ./spool/mpuff_list.101|

15.3.3 Material types

Material types are provided using the command DLG=LIST;21 and filed in the HYDRADIR\spool\ directory.

Structure of dialog data:

“DLG= LIST;21|DATEI={file name}|DAT=...|ZEI=...|USR=...“

Parameter: AKRO=<dyn. user field columns e.g. AKRO=FU:23>

The list includes the following data:

ID	Field designation	Description
HZTYP	Material type	Material type
HZBEZ	Material type des.	Designation of the material type
MINLGZ	Min. storage time	Minimum storage time
WARNGRENZ	Warning limit	Warning limit
VFALLGRENZ	Expiry limit	Expiry limit
HZART	SF type	Material type
LOSGR	Lot size	Standard lot size that is suggested as “default value” for a new generation
EINH	Unit	Unit
LAY	Layout	Layout that is used for ticket printing
TICKETANZ	Number of tickets	Number of tickets
OPT:VFCNRA	Expiry date from input batch	Flag whether the individual expiry date can be used to determine the expiry date of the output batch.
OPT:VFCNRE	Expiry date for output batch	Flag whether the least expiry date of appropriate batches is to be used to determine the individual expiry date.
OPT:MULTICNR	Can be logged on several times	Flag whether identical material types can be logged on several times.
OPT:INBESTVERB	Invent. char.	Flag for inventory management
PROZAGAB	Tolerance limit AGAB	Tolerance limit for OP off (%)
UTGAGAB	abs. tolerance limit AGAB	Absolute value for tolerance limit
OPT:VERE	Hand batch no. down	Flag for inheriting the batch number
OPT:AGZGVERB	REB-OP-Ref.	Flag whether an OP reference is required when reposting batches
OPT:AGZGGEN	GR-OP-Ref.	Flag whether an OP reference is required when generating batches
PLAUS:MATOK	Plaus. MatOK	Flag whether all input batches need to be logged on for booking a batch
PLAUS:EMATOK	Plaus. EMatOK	
OPT:CNREA	Valid for 1 outp. batch	Flag whether input batch exactly applies for 1 output batch
TPE	Transp. unit	Transport unit
OPT:VGRCNR	Required qty. batch-related	MPLRF: Material usage relating to output batch or operation

ID	Field designation	Description
OPT:SNR	Serial numbers required	default=N (*)
OPT:GENHU	Generation of HU	default=N (*)

Example:

DLG=LIST;21|DATEI=./spool/hztyp_list.101|DAT=08/16/2007|ZEI=46918|USR=2101

(*) = Only available as of HYDRA-MPL product version 7.2.5

15.3.4 Transport unit

Transport units are provided using the command DLG=LIST;52 and filed in the HYDRADIR\spool\ directory.

Structure of dialog data:

“DLG=LIST;52|DATEI={file name}|DAT=...|ZEI=...|USR=...“

Parameter: none

The list includes the following data:

ID	Field designation	Description
HZTYP	Material type	Material type
TPE	Transp. unit	Transport unit
BEZ	Transp. unit des.	Designation of transport unit
VORG	Default transp. Unit	“Default” transport unit
ANZ	Lot size	Lot size
EINH	Unit	Unit
INBESTVERB	Invent. char.	Inventory flag
TPE.ANZ	No. of transport units	Number of transport units
BREITE	Width	Width dimension
BREITE:EINH	Width unit	Width dimension unit
HOEHE	Height	Height dimension
HOEHE:EINH	Height unit	Height dimension unit
LEN	Length	Length dimension
LEN:EINH	Length unit	Length dimension unit
GEW	Weight	Weight dimension
GEW:EINH	Weight unit	Weight dimension unit
COLOR:F	Foreground color	Foreground text color
COLOR:B	Background color	Background color

Example:

DLG=LIST;52|DATEI=./spool/tpe_list.101|DAT=08/16/2007|ZEI=46918|USR=2101

15.3.5 Component list

Components are provided by the command DLG=LIST;74 and filed in the HYDRADIR\spool\ directory.

Structure of dialog data:

“DLG=LIST;74|DATEI={file name}|DAT=...|ZEI=...|USR=...“

Parameter: ANR=order number

→ Order is required for planned materials (if necessary incl. batch assignment) of the order.

The list includes the following data:

ID	Field designation	Description
ART	Type	Component type
ATK	Article	Material number
BEZ	Designation	Customer-specific
BEZ:2	Designation	Customer-specific
SGR:GUT	Required quantity	Required quantity relating to the production of 1 article in primary quantity unit at the operation
MENGE:BED	Demand	Requirements quantity
SGE:GUT	Unit	Unit of required quantity
ATKBEZ	Article	Designation of the component (material, document, etc.)
SLP	BOM item	Position of components (bill of material item)
LAGORT	Storage location	Storage location SAP or storage type
LAGPZ	Rack compartment	Storage bin SAP
PATH	Path	For documents: reference to HY_PATH
DATEI	File name	For documents: file name
RESTYP	Type	**MAT = material**

Example:

DLG=LIST;74|ANR=TEST1|DATEI=./spool/komp_list.101|DAT=08/16/2007|ZEI=46918|USR=210
1

15.3.6 Batch attributes of a material type

Batch attributes are provided using the command DLG=LIST;67 and filed in the HYDRADIR\spool\ directory.

Table: hz_atgen

Structure of dialog data:

“DLG=LIST;67|DATEI={file name}|DAT=...|ZEI=...|USR=...|MOD=...| MOD2=...|ANR=…“

a.) Start via console

Parameter: MOD2=K

IDs are displayed without ":" → e.g. OPT_PRN

b.) Read batch attributes

Parameter: MOD=L|CNR=12345678

c.) Read operation attributes

Parameter: MOD=A|ANR=1234567890

The list includes the following data:

ID	Field designation	Description
KENNUNG	ID	ID of attribute
ATTR_VAL	Attribute value	Value of the attribute

ID	Field designation	Description
MATTYP	Material type	Material type
IDX	Index	Completes the unique key (with HZ_TYP)
OPT:SIB	Display	Visible on console: J/N flag whether the attribute is to be displayed or not (applies for all console views)
OPT:PRN	Print	Identifier whether the attribute is to be printed on the batch ticket or not (J/N)
POS:SIB	Display position	Position within display order
POS:PRN	Print position	Position within the ticket print order
TXT	Header	Description of the attribute
EINH	Trailer	Unit of the attribute
OPT:ERF	Enter	Identifier how the attribute is recorded (J/N)
POS:ERF	Entry position	Position within the order of entry
TYP:ERF	Type	Type of entry field - N → Numeric - C → Text field - F → Decimal value (This type might cause that a numeric value is saved in a text database field, if intended)
LEN:ERF	Entry length	Field length. The maximum length might be determined by the attribute database field in which the value is to be saved.
IDX:ERF	Entry index	Index of the material attribute to be recorded. Corresponds to the number of the batch attribute (relevant for entry = “J”)
NKS:ERF	Decimal places	Number of decimal places to be recorded for decimal attributes

Example:

DLG=LIST;67|DATEI=spool/hyu2111.tmp|DAT=08/17/2007|ZEI=34897|USR=2111|MOD=A|ANR =005001720290

15.3.7 Batch logs (MPL-PRO)

Batch logs are provided via the DLG=CNRPROT.LIST command and filed in the HYDRADIR\spool\ directory.

Table: mpl_los_prot

License: MPL-PRO

Structure of dialog data:

“DLG=LIST;CNRPROT.LIST|DATEI={file name}|DAT=...|ZEI=...|USR=...“

Parameter: none

The list includes the following data:

ID	Field designation	Description
CNR	-	Batch number
VERWEIS	-	Reference to the data record
DLL	-	Run-through batch no.
GR	-	Reason
GRTEXT	-	Reason text
BEM	-	Comment/commentary
PNR	-	Personnel number
KNR	-	Badge number
ATTR_1	-	Attribute 1
ATTR_2	-	Attribute 2
ATTR_3	-	Attribute 3
ATTR_4	-	Attribute 4
ATTR_5	-	Attribute 5
ATTR_6	-	Attribute 6
ATTR_7	-	Attribute 7
ATTR_8	-	Attribute 8
ATTR_9	-	Attribute 9
ATTR_10	-	Attribute 10
ATTR_11	-	Attribute 11
ATTR_12	-	Attribute 12
ATTR_13	-	Attribute 13
ATTR_14	-	Attribute 14
ATTR_15	-	Attribute 15
MNR	-	Machine
ANR	-	Order + operation
AUNR	-	Order
AGNR	-	Operation
BEARB	-	Editor

ID	Field designation	Description
BEARBDAT	-	Date of last change
BEARBZEI	-	Time of last change

15.4 Process of Changing Output Batches

15.4.1 Input batch data

To find out which input materials are logged on, the material list has to be polled cyclically:

DLG=LISTE;13|MOD=M|MNR=Machine|ANR=order number....

If no input batch is logged on for planned materials the batch number (CNR) fields, etc. are not filled out. These fields are only filled out if input batches are currently logged on.

15.4.2 Output batch data

To read data of the currently running output batch, the batch list has to be read as follows:

DLG=LISTE;13|MOD=A|ANR=order number

15.4.3 Output batch change

The following dialog has to be sent to HYDRA to execute a batch change:

DLG=CA_WL|ANR=order number |CNR=next batch number|....

The batch list has to be requested anew to be able to print tickets, once output batches have been changed ("DLG=LISTE13;MOD=L;CNR=.. ")

15.4.4 Job end

The following dialogs have to be sent to HYDRA to perform a job end:

DLG=CA_WL|....	- to log the last batch off
DLG=A_AB|…	- to log the order off
DLG=A_UN|…	- to interrupt the order

15.5 Packing and Palletizing (MPL-PAL)

Please note: All functions described in this section are only available as of MPL product version 7.2.5.

15.5.1 Assign batches (DLG=CE_AN_PA)

ID	Type/max. field length	Description	
ANR=	C16	Order number with operation number	
MNR=	N8/C8	Machine number (numeric/alphanumeric)	
PNR=	C10	alter-native	Personnel number
KNR=	C10		Badge number
CNR=	**C20**	**Batch number of the input batch**	

15.5.2 Delete batch assignment (DLG=CE_DEL_PA)

ID	Type/max. field length	Description	
ANR=	C16	Order number with operation number	
MNR=	N8/C8	Machine number (numeric/alphanumeric)	
PNR=	C10	alter-native	Personnel number
KNR=	C10		Badge number
CNR:PA=	C20	Batch number TPU batch	
CNR=	C20	Batch number of the assigned input batch	

15.5.3 Complete TPU (DLG=CA_WL_PA)

ID	Type/max. field length	Description	
ANR=	**C16**	**Order number with operation number**	
MNR=	**N8/C8**	**Machine number (numeric/alphanumeric)**	
PNR=	C10	alter-native	PNR=
KNR=	C10		KNR=
CNR=	**C20**	**New batch number for the next TPU**	

ID	Type/max. field length	Description
EGR:GUT=	DEC	Net weight
EGE:GUT=	C4	Unit of net weight
EGR:AUS=	DEC	Net weight for status scrap
EGE:AUS=	C4	Unit of net weight
KLASSE=	C1	"G" yield or “A” scrap
STA=	C1	Status of output batch F= with remaining quantity (by default for KLASSE=G) S= blocked (by default for KLASSE=A)
ZLO=	C12	Destination of output batch
TPE=	C10	Transport unit
LHW=	C20	Optional: output batch note
CALT1= to CALT5=	C20	Alternative batch number 1 - 5
ATTR:1 = to ATTR:11 =	--	The definition of which values are filed in additional attributes is configured for each material type in HYDRA
ATTR:101= to ATTR:140=	C40	Alphanumeric batch attributes
ATTR:201= to ATTR:220=	NUM	Numeric batch attributes
ATTR:301= to ATTR:320=	DEC	Decimal batch attributes

15.5.4 List with assigned batches for active TPU

Data are provided using the command ***DLG=LIST;u_l_mpl_hu_elose*** and filed in the HYDRADIR\spool\ directory.

License: MPL-PAL

Structure of dialog data:

„DLG=LIST;u_l_mpl_hu_elose|DATEI={file name}|DAT=...|ZEI=...|USR=...“

Parameter: MNR=<machine>|ANR=<operation>

The list includes the following data:

ID	Field designation	Description
MNR	Machine	Machine number
ANR	Order	Order and OP
ATK	Article	Material number
ATKBEZ	Des. of final article	Material designation
CNR	-	Batch number
ACNR	Package	Batch number TPU
HSDAT	Date manuf.	Manufacturing: date
HSZEI	Time manuf.	Manufacturing: time
MENGE	Quantity	Batch quantity of individual batch
EINH	Unit	Unit of individual batch
LS1 – LS6	Activity 1 – 6	Activity 1 - 6
LS1:EINH – LS6:EINH	Unit activity 1 – 6	Unit activity 1 - 6

15.6 Annex

15.6.1 Summary of MPL field data

The following table provides an overview of field data including structure and brief example. Some points overlap with descriptions already explained in other PDM documents.

ID	Description	Structure	Example
MNR	Machine number	...|MNR={machine number }|...	...|MNR=100|...
MGRP	Machine group	...|MGRP={machine group }|...	...|MGRP=100|...
ANR	Order number	...|ANR={order}|...	...|ANR=47110010|...
AUNR	Order number without OP	...|AUNR={order number without OP }	...|AUNR=4711|...
AGNR	Operation number	...|AGNR={OP}|...	...|AGNR=0010|...
ATK	Article number	...|ATK={article number }|...	
EGR:*	Entry size Manually recorded size	...|EGR:{type}={value}|... Types of entry sizes: RPA01, RPA02, .., RPA12 DAUER, PDAUER	..|EGR:GUT=10. |. ...|EGR:AUS=1 |...

		HUB, GUT, AUS, LEN, GEW The recorded value is accumulated onto the respective account within the database	
EGE:*	Unit for entry sizes	...|EGE:{type}={unit}|... For types see EGR	EGE:GUT=ST
EGG:*	Reasons for entry sizes	...|EGG:{type}={reason }|... For types see EGR	EGG:AUS=1
SGR:*	Target size	...|SGR:{type}={value}|... SGR:GUT = target quantity order/OP For types see EGR	...|SGR:GUT=1126|
MST	Machine status	...|MST={machine status}|...	..|MST=1 |..
PNR	Personnel number	...|PNR={personnel number }|...	...|PNR=999999|...
KNR	Badge number	...|KNR={badge number}|...	...|KNR=9999|...
CNR	Batch number	...|CNR={batch number}	...|CNR=998877|...
BZW	Booking obligation	Optional parameter if set several plausibility checks are deactivated. If not sent: = N [...|BZW={ booking obligation}|...]	...|BZW=J|...
LHW	Info on batch	Info on batch (C20)	|LHW={info on batch}|...
ZLO	Destination	Destination	...|ZLO=121234|...
TPE	Transport unit	Transport unit (C10)	...|TPE=1234|...
SLP	Bill of material item	Bill of material item (C6)	...|SLP=AA34|...
ATTR:*	Additional batch attributes	...|ATTR:{nr}]={recorded attribute}|.. nr = 1 ..11 Field types; 1 - 4 : Integer 4 – 6 : Double 7 : Char 4	

		8, 9 : Char 10 10, 11: Char 20	
STN	Station number	...|STN ={station number }|...	...|STN=1|..
LOSANZ	Number of parallel output batches	...|LOSANZ ={number of batches}|...	...|LOSANZ=5|...
ATTR:101 - ATTR:140	Alphanumeric batch attributes	C40	...|ATTR:101=TXT| …
ATTR:201 - ATTR:220	Numeric batch attributes	NUM	...|ATTR:201=1| …
ATTR:301 - ATTR:320	Decimal batch attributes	DEC	...|ATTR:301=1.123| …

16 HYDRA Production Data Manager MPL - Master Data

16.1 Please note for the basic dialogs described

All mandatory fields are provided with the addition “PK” (primary key = key field). All other fields are optional and are processed if they are filled out.

16.2 Quantity Changes

16.2.1 Quantity change affecting several products

16.2.1.1 DLG=ACCMNT.CHANGE

Using the BAPI calls described in this section, it is possible to record and post a quantity change for different dependent products. The Bapi is implemented as script Bapi (b_accmnt.hsc).

Dependent products

Product	Bapi	Description
MDE	MDEPRO.CHANGEQTY	Edit MDE log records
ADE	ADEPRO.UPDATE	Edit ADE log records
MPL	CNR.UPDATE	Generate material movements/change batch status

BAPI call

ID	Content / {type}	Description
DLG	ACCMNT.CHANGE	Realize quantity change
ACCMNT.DLG	Dialog	Executing initial activity (e.g. CNR.UPDATE)
ACCMNT.MNR	Machine	Machine number
ACCMNT.ANR	Operation	OP number
ACCMNT.CNR	Batch	Batch number
ADEPRO.EGR:GUTP ADEPRO.EGR:AUSP CNR.SGR:GUT	Quantity	Depends on initial command

Return

ID	Content / {type}	Description

Validation checks

Error codes	Description
2027	Unknown mode
105	Parameters are missing

16.3 MPL – Master Data

16.3.1 MPL Setup

See “setup“

16.3.2 Material types

16.3.2.1 DLG=MATTYP.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT

The configuration of material types can be edited using the BAPI calls described in this section.

Tables

Table	Key field	Description
hz_typen	hz_typ MATTYP.MATTYP	Material type
hz_atgen		Delete corresponding attributes with DELETE

BAPI call

ID	Content / {type}	Description
DLG	MATTYP.INSERT	Create material type
	MATTYP.UPDATE	Change material type
	MATTYP.DELETE	Delete material type
	MATTYP.COPY	Copy material type
	MATTYP.LOCK	Block processing of the material type
	MATTYP.UNLOCK	Unblock material type after processing
	MATTYP.NEW	Read specification for new material type
	MATTYP.SELECT	Select material type
MATTYP.MATTYP	{C10}	Material type
MATTYP.MATTYP:Z	{C10}	New (target) material type for COPY
…	…	For further fields, please refer to the HYD-HDB documentation about above-mentioned tables.

Return

ID	Content / {type}	Description
MATTYP.MATTYP	{C10}	MATTYP.INSERT, MATTYP.COPY: Return of material type of the configuration created

Validation checks

Error codes	Description
2707	MATTYP.DELETE:Reference to the mat_mattyp table still exists

16.3.3 Material buffer

16.3.3.1 DLG=MATPUF.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT

The configuration of material buffers can be edited using the BAPI calls described in this section.

Tables

Table	Key field	Description
mat_puffer	mat_puf MATPUF.MATPUF	Material buffer

BAPI call

ID	Content / {type}	Description
DLG	MATPUF.INSERT	Create material buffer
	MATPUF.UPDATE	Change material buffer
	MATPUF.DELETE	Delete material buffer
	MATPUF.COPY	Copy material buffer
	MATPUF.LOCK	Block processing of the material buffer
	MATPUF.UNLOCK	Unblock material buffer after processing
	MATPUF.NEW	Read specification for new material buffers
	MATPUF.SELECT	Select material buffer
MATPUF.MATPUF	{C10}	Material buffer
MATPUF.MATPUF:Z	{C10}	New (target) material buffer for COPY

…	…	For further fields, please refer to the HYD-HDB documentation about the above-mentioned tables

Return

ID	Content / {type}	Description
MATPUF.MATPUF	{C10}	MATPUF.INSERT, MATPUF.COPY: Return of the material type of the configuration created

Validation checks

Error codes	Description
2700	DELETE Material buffer is still assigned to a machine
2701	DELETE Material buffer still refers to another hierarchy
2702	DELETE Material buffer still refers to a transport unit
2703	MATPUF.INSERT/UPDATE: Reference to hierarchy is wrong
2704	MATPUF.UPDATE: Reference to hierarchy wrong (too small)
2705	MATPUF.INSERT/UPDATE:Target location does not exist in the los_zielorte table
2730	DELETE Material buffer is still used within batch stock

16.3.4 Transport units

16.3.4.1 DLG=TPE.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT

The configuration of transport units can be updated using the BAPI calls described in this section.

Tables

Table	Key field	Description
hz_typen	tpe TPE.TPE	Transport unit

BAPI call

ID	Content / {type}	Description
DLG	TPE.INSERT	Create transport unit
	TPE.UPDATE	Change transport unit
	TPE.DELETE	Delete transport unit
	TPE.COPY	Copy transport unit
	TPE.LOCK	Block processing of the transport unit
	TPE.UNLOCK	Unblock transport unit after processing
	TPE.NEW	Read specification for new transport unit
	TPE.SELECT	Select transport unit
TPE.TPE	{C10}	Transport unit
TPE.TPE:Z	{C10}	New (target) transport unit for COPY
…	…	For further fields, please see the HYD-HDB documentation about above-mentioned tables

Return

ID	Content / {type}	Description
TPE.TPE	{C10}	TPE.INSERT, TPE.COPY: Return of transport unit of the configuration created

Validation checks

Error codes	Description

16.4 HYDRA-MPL – Movement Data

16.4.1 Batch stock

16.4.1.1 DLG=CNR.INSERT, UPDATE, DELETE, MODIFY, COPY, LOCK, UNLOCK, NEW, SELECT

The batch stock can be edited using the BAPI calls described in this section.

Tables

Table	Key field	Description
los_bestand a_los_bestand	losnr CNR.CNR	Internal batch number a_los_bestand → Archive table
los_attribute		Batch attributes for a batch

BAPI call

ID	Content / {type}	Description
DLG	CNR.INSERT	Create batch
	CNR.UPDATE	Change batch
	CNR.DELETE	Delete batch
	CNR.COPY	Copy batch
	CNR.LOCK	Block processing of the batch
	CNR.UNLOCK	Unblock batch after processing
	CNR.NEW	Read specification for new batches
	CNR.SELECT	Select batch
	CNR.MODIFY	Create or change batch
	CNR.SPLITCREATE	Split batch
	CNR. SUMMARIZE	Batch merge
CNR.CNR	C20	Internal batch number ,'/empty – A new batch number (prefix P) is assigned at the server via User=0, if configured like that in the setup
CNR.DLL	C20	Throughput batch number
CNR.CNR:Z	C20	New (target) batch for CNR.COPY

CNR.ATK	C40	Material number
CNR.ATKBEZ	C40	Material designation
CNR.SGR:GUT	DEC	Batch quantity
CNR.SGR:REST	DEC	Remaining quantity of the batch
CNR.SGE:GUT	C3	Unit of batch quantity
CNR.MNR	N8/C8	Machine number
CNR:ATTR:11	C40	Order number with operation number
CNR.STA	C1	Batch status (F = free, S = blocked)
CNR.CKL	C1	Batch class G = yield A = scrap O = on hold N = rework
CNR.TST	C1	Transport status (L = delivered)
CNR.QST	C1	Quality status (G – blocked, F – free)
CNR.QSTMANU	C1	Manual quality status (G – blocked, F – free)
CNR.MATST	C1	Material status (e.g. V = packed)
CNR.MATTYPART	C10	Material type
CNR.OPT:REST	C1	J – “Batch has still got residual quantity“ flag N – “Batch has not got residual quantity”
CNR.FIR	C4	Company
CNR.HSDAT	DATUM	Manufacturing date
CNR.HSZEI	ZEIT	Manufacturing time
CNR.VVDAT	DATUM	Availability date
CNR.VVZEI	ZEIT	Availability time
CNR.VFDAT	DATUM	Expiry date
CNR.VFZEI	ZEIT	Expiry time
CNR.WDAT	DATUM	Warning date
CNR.WZEI	ZEIT	Warning time
CNR.CSTWDAT	DATUM	Date of last status change
CNR.CSTWZEI	ZEIT	Time of last status change

CNR.MATPUF	C12	Material buffer
CNR.MATTYP	C10	Material type
CNR.TPE	C10	Transport unit
CNR.BEM	C20	Info on batch
CNR.PNR	C10	Personnel number
CNR.TECHINFO	C20	Technical info
CNR.EGG:AUS	N8	Scrap reason
CNR.GR	N8	Scrap reason (synonymous to CNR.EGG:AUS)
CNR.GRTXT	N8	Reference to scrap reason text
CNR.USR	N8	User number
CNR.LAGORT	C12	PPS storage location
CNR.LAGPZ	C12	PPS storage bin
CNR.SAPCNR	C10	PPS batch number
CNR.RESART	C4	Reservation type (AG = operation, AK = order)
CNR.RESVAL	C40	Value of reservation (e.g. operation number)
CNR.RESBEM	C100	Reservation comment
CNR.BREITE	DEC	Material width (conversion factor for MPLRF-BP)
CNR.RFAGVFA	DEC	Mass per unit area (conversion factor for MPLRF-BP)
CNR.RFSTKF	DEC	Surface per piece (conversion factor for MPLRF-BP)
CNR.OPT:SLOS	C1	J – batch is a merged batch
CNR.ANZ:SLOS	N8	For a merged batch, the number of the assigned individual batches
CNR.OPT:SLOSTYP	C1	J – Batch is a merged batch
CNR.OPT:SLOSCTRL	C1	J – Batch is a merged batch
CNR.OPT:MBEW	C1	J – Generate goods movement
CNR.OPT:MDC	C1	J – Generate additional goods movement to change quantities
CNR.MOD	C1	For CNR.SPITCREATE only: L – Batch quantity (otherwise split as default residual quantity)
CNR.MENGE	DEC	For CNR.SPITCREATE only:

		Quantity
CNR.EGR:1 – 6	DEC	Quantity activity 1 – 6
CNR.RGR:1 – 6	DEC	Residual quantity activity 1 – 6
CNR.EGE:1 – 6	C3	Unit activity 1 – 6
CNR.CNR:ALT1 – 20	-	Alternative batch number 1 – 20 max. field length: CNR.CNR:ALT1-4 - C20 CNR.CNR:ALT5-14 - C40 CNR.CNR:ALT15-18 - C100 CNR.CNR:ALT19-20 - C512
CNR.EXTCNR	C10	External batch number
CNR.MCNR	C10	Mother batch number (e.g. reference from merged batch)
CNR.AUART	C5	Order type
CNR.UMRFAKTP:Z	DEC	Denominator – conversion factor relating to OP quantity units
CNR.UMRFAKTP:N	DEC	Numerator – conversion factor relating to OP quantity units
CNR.UMRFAKTS:Z	DEC	Denominator – conversion factor relating to OP quantity units
CNR.UMRFAKTS:N	DEC	Numerator – conversion factor relating to OP quantity units
CNR.UMRFAKTT:Z	DEC	Denominator – conversion factor relating OP quantity units
CNR.UMRFAKTT:N	DEC	Numerator – conversion factor relating to OP quantity units
CNR.ZUORD:1 – 6	C1	Flag for assignment “MPL activity account → ADE quantity unit (e.g. P = primary quantity)
ATTR=101 - ATTR=140	C40	Alphanumeric batch attributes (in los_attribute table)
ATTR=201 - ATTR=220	NUM	Numeric batch attributes (in los_attribute table)
ATTR=301 - ATTR=320	DEC	Decimal batch attributes (in los_attribute table)
CNR.USRFLD	C8	User field key
CNR.FU:1 – CNR.FU:66	-	User fields
CNR.PROTEVENT	C10	Event used in the batch history. Using this key, the console can show a text defined for the event (from the reference list ELOS.EREIGNIS1).

CNR.SETEVDATA	C1	J – Transferred data are written in *event_dlg_data*
CNR.EVDATA:1 – CNR.EVDATA:n	C∞ (Restricted to a maximum of 1000 characters by *event_dlg_data.dlg_data*)	These data are written in the file *event_dlg_data*. CNR.SETEVDATA=J must be configured. These data are shown in the detail view of the batch history at the console.
CNR.EVDATABEZ:1 – CNR.EVDATABEZ:n	C∞ (Restricted to a maximum of 1000 characters by *event_dlg_data.dlg_data*)	Designations for CNR.EVDATA:1 – CNR.EVDATA:n. These are console IDs of the reference list for ELOS.EVDATABEZ. These data are **not** shown in the detail view of the batch history at the console.

Return

ID	Content / {type}	Description
CNR.CNR	{C20}	CNR.INSERT, CNR.COPY: Return of batches of the configuration created

Validation checks

Error codes	Description

16.4.1.2 DLG=CNR.SPLITCREATE

BAPI call

ID	Content / {type}	Description
DLG	CNR.SPLITCREATE	Split batch
CNR.CNR	C20	Batch number to be split
CNR.MENGE:**x**	DEC	Batch quantity of future splits e.g. "|CNR.MENGE:1=1.0| CNR.MENGE:2=2.0|"
CNR.GR:**x**	NUM	Scrap reason of future splits
CNR.CKL:**x**	C1	Batch class (G = yield, A = scrap) of future splits
CNR.MOD	C1	L – Batch transfer (a new batch is generated from the original batch using the "same" data) R – Post residual quantity onto a new batch. The original batch is "processed" and has the quantity 0. B – Reduce existing batch by the total of split quantities (remaining quantity and activities)
OPT.CNRGEN_VIA_BAPI	C1	J – New batches are generated by Bapi CNRGEN.CREATECNR

		N – New batches are generated by the internal function “erzeuge_losnr_sicher“.

BAPI call return values

ID	Content / {type}	Description
CNR:**x**	C20	All generated splits

16.4.1.3 DLG=CNR.SUMMARIZE

ID	Content / {type}	Description
DLG	CNR.SUMMARIZE	Merge batch
CNR.CNR	C20	Only with **CNR.MOD=B** Involved batch that is increased by the total quantity.
CNR.CNR:**x**		All batches that are merged. Please note: In **CNR.MOD=B** the batch to be increased is transferred as CNR.CNR.
CNR.MOD	C1	B – Post quantities to existing batch (CNR.CNR) N – Generate new batch and post quantities to this batch

BAPI call return values

ID	Content / {type}	Description
CNR.CNR:NEW	C20	Only with **CNR.MOD=N:** New batch number

16.4.1.4 DLG=CNR.NEWCNR

A new batch number can be requested using the BAPI calls described in this section (number range User=0).

BAPI call

ID	Content / {type}	Description
DLG	CNR.NEWCNR	Request new batch number (prefix P)

16.4.1.5 DLG=CNR.RESTORE

A batch from the batch archive may be saved back into the online table using the BAPI call described in this section.

BAPI call

ID	Content / {type}	Description
DLG	CNR.RESTORE	Retrieve batch from archive

16.4.1.6 DLG=CNRGEN.CREATENR

BAPI call

ID	Content / {type}	Description
DLG	CNRGEN.CREATENR	Generate new batch
CNRGEN.MATTYP	C10	Material type (fix “SYSTEM“) → FOR FUTURE USE
CNRGEN.TYP	C1	Type of the batch to be generated H → HU batch P → Production batch W → Goods receipt batch A prefix that might be transferred will be ignored if a type is transferred. The type takes priority.
CNRGEN.PRAEFIX	C10	Batch number prefix Overwrites setup configuration
CNRGEN.TNR	N8	Terminal user for number range (Default = 0) Overwrites setup configuration

CNRGEN.LEN	N8	Length of batch number Overwrites setup configuration
CNRGEN.TYP	C1	Batch type H → Packing station batch (prefix fix “HU“) P → Produced batch (prefix from setup) W → Goods receipt batch (prefix from setup) A prefix that might be transferred will be ignored if a type is transferred.

Return

ID	Content / {type}	Description
CNR	C20	Generated batch number

16.4.1.7 DLG=CNRBAUM.INSERT

Batch assignments can be created by the BAPI calls described in this section.

Batch assignments describe the connection between input and output batches. Batch assignments have been designed to trace back material relating to batches.

Tables

Table	Key field	Description
los_zuordnung	CNRBAUM.CNR:E CNRBAUM.CNR:A	Batch assignments

BAPI call

ID	Content / {Type}	Description
DLG	CNRBAUM.INSERT	Create batch assignments
CNRBAUM.CNR:E	Batch number {C20}	Internal batch number: input batch
CNRBAUM.CNR:A	Batch number {C20}	Internal batch number: output batch
CNRBAUM.DLL:E	Batch number {C20}	Batch number: input batch
CNRBAUM.DLL:A	Batch number {C20}	Batch number: output batch
CNRBAUM.ATK:E	Article {C40}	Material number: input batch

CNRBAUM.ATK:A	Article {C40}	Material number: output batch
CNRBAUM.HZTYP:E	Material type {C10}	Material type: input batch
CNRBAUM.HZTYP:A	Material type {C10}	Material type: output batch
CNRBAUM.SAPCNR:E	ERP batch {C40}	ERP batch: input batch
CNRBAUM.SAPCNR:A	ERP batch {C40}	ERP batch: output batch
CNRBAUM.MNR	Machine / {C10}	Machine number
CNRBAUM.ANR	Operation / {C40}	Operation number
CNRBAUM.SLP	BOM item / {C10}	BOM item of input batch from component list
CNRBAUM.ART	Type of BOM item / {C1}	Type of BOM item of input batch from component list (e.g. M for material)
CNRBAUM.HSDAT	Date	Date of output batch logoff
CNRBAUM.HSZEI	Time	Time of output batch logoff
CNRBAUM.HSANDAT	Date	Date of output batch logon
CNRBAUM.HSANZEI	Time	Time of output batch logon
CNRBAUM.EANDAT	Date	Date of input batch logon
CNRBAUM.EANZEI	Time	Time of input batch logon
CNRBAUM.EABDAT	Date	Date of input batch logoff
CNRBAUM.EABZEI	Time	Time of input batch logoff

Return

ID	Content / {Type}	Description

Validation checks

Error codes	Description

16.4.2 Material movements

16.4.2.1 DLG=MBEW.INSERT

Material movements may be created by means of the BAPI calls described in this section. The Bapi has been implemented as script Bapi (b_mbew.hsc).

Material movements may only be created by INSERT, as they can be regarded as delta quantity for a stock or status (batch stock).

A negative algebraic sign must precede the quantity for goods issues (movement 261).

Uploads to PPS:

Goods receipts are uploaded to PPS using the ZWEI segment structure.

Goods issues and return transfers are uploaded to PPS using the ZWAU segment structure.

Tables

Table	Key field	Description
event_mlb	--	Material movements

BAP call

ID	Content / {type}	Description
DLG	MBEW.INSERT	Create material movement
MBEW.SAPBWART	Movement type / {C3}	**Goods receipt:** 101 – Goods receipt 102 – Cancellation for goods receipt 262 – Return transfer 525 – Goods receipt, blocking material 531 – Scrap material **Goods issue:** 261 – Goods issue 262 – Cancellation for goods issue
MBEW.EVENT	Event / {C10}	CMM_E - Goods receipt CMM_A – Goods issue CMM_R – Return transfer
MBEW.KLASSE	{C1}	fix “C“ for batch-related movement
MBEW.PRIO	{N4}	fix “9999“
MBEW.DAT	Date	May alternatively also be transferred with DAT=<> / ZEI=<>.

MBEW.ZEI	Time	
MBEW.DLG	Executing command / {C10}	e.g. MBEW
MBEW.MNR	Machine / {C10}	Machine number
MBEW.ANR	Operation / {C40}	Operation number → Determination of SAP order from backlog of orders
MBEW.ATK	Article / {C40}	Article number
MBEW.ATKBEZ	Designation / {C40}	Article designation
MBEW.SAPANR	{C12}	SAP order from backlog of orders ‚'/empty – definition via OP
MBEW.CNR	Batch / {C20}	Batch number for batch relation (internal batch number) Determination of PPS batch results from batch stock
MBEW.DLL	Run-through batch / {C20}	Run-through batch number for batch reference
MBEW.LAGORT	{C10}	Storage location (e.g. from material buffer of machine)
MBEW.ZLO	{C10}	Material buffer (e.g. material buffer of machine)
MBEW.SAPCNR	{C10}	PPS batch ‚'/empty – Determination via batch
MBEW.ATTR:6	Material type / {C10}	Material type to control uploads to PPS Reference check of hz_typen table
MBEW.OPT:RCK	Upload / {C1}	J – Upload movement to PPS N – Do not upload movement to PPS ‚'/empty – Determine upload flag via material type
MBEW.STA:RCK	Uploaded / {C1}	J – identify movement as being uploaded N – Movement has not yet been uploaded
MBEW.CST	{C1}	Batch status (e.g. "F" for free)
MBEW.CKL	{C1}	Batch class (e.g. "G" for yield)
MBEW.PNR	{C10}	Personnel number
MBEW.TNR	{N8}	Terminal user
MBEW.BEARB	{C10}	Editor
MBEW.EGR:1	{DEC}	Quantity in batch quantity unit
MBEW.TYP:1	{C10}	Quantity type e.g. "GUT" for goods receipt, yield (101) "VGR" for consumption quantity (261)

MBEW.EINH:1	{C3}	Quantity unit
MBEW.GR:1	{C10}	Quantity reason (e.g. blocking reason)
MBEW.EGR:2	{DEC}	Movement 531 only: Scrap in batch quantity unit
MBEW.TYP:2	{C10}	Movement 531 only: Quantity type = “AUS“
MBEW.EINH:2	{C3}	Movement 531 only: Quantity unit
MBEW.GR:2	{C10}	Movement 531 only: Scrap reason
MBEW.EGR:5	{DEC}	Quantity of activity field 3 (for batch relation)
MBEW.EGR:6	{DEC}	Quantity of activity field 4 (for batch relation)
MBEW.EGR:7	{DEC}	Quantity of activity field 5 (for batch relation)
MBEW.EGR:8	{DEC}	Quantity of activity field 8 (for batch relation)
MBEW.TYP:5 - 8	{C10}	Quantity type e.g. “GUT“ for goods receipt, yield (101) “VGR“ for consumption quantity (261)
MBEW.EINH:5 – 8	{C3}	Quantity unit (with batch relation from activity field)
MBEW.GR:5 - 8	{C10}	Quantity reason
MBEW.ERWSTA:1 – MBEW.ERWSTA:6	{C1}	Used customer-specifically
MBEW.VERARB:1 – MBEW.VERARB:4	{C1}	Used customer-specifically
MBEW.OPT:EAUS	{C1}	J – Flag for final issue effected When a component is withdrawn the last time in this order
MBEW.OPT:ENDLIEF	{C1}	J – Flag for final delivery for the last goods receipt of the order
…	…	For further fields, please refer to the HYD-HDB documentation about above-mentioned tables

Return

ID	Content / {type}	Description

Validation checks

Error codes	Description
2709	Material type does not exist

16.4.3 Cutting plan

16.4.3.1 DLG=BAHNVERT.CREATE

Using the BAPI calls described in this section, allows for cutting plans to be created for an order. Cutting plans are only used if the “cutting reels” machine type is used along with coil based manufacturing. They describe the web-shaped structure of the output material of an OP. If a cutting plan already exists for an order and machine it will be deleted beforehand.

Tables

Table	Key field	Description
mpl_bahnverteilung		Cutting plan (header record)
mpl_bahnlayout		Cut layout (detailed record)

BAPI call

ID	Content / {type}	Description
DLG	**BAHNVERT.CREATE**	Create cutting plan
BAHNVERT.MANR	Operation / {C40}	Operation number of mother OP
BAHNVERT.MNR	Machine / {C10}	Machine number
BAHNVERT.BAHNVERT	Cutting plan/ {C10}	Unique key of cutting plan
BAHNVERT.ART	Cut type / {C1}	‘V’ – prepared cutting plan ‘M’ – creation of measurement (without OP reference)
BAHNVERT.BEZ	Designation / {C20}	Designation
BAHNVERT.ANZBAHN:GES	Total of cuts / NUM	Total of cuts within the production order (parent and child OP)
ANR#1= to **ANR#n**	OP / {C40}	Operation number (overall key)
TYP#1= to **TYP#n**	Model / {C1}	H – Parent OP L – Child OP
BAHNART#1= to **BAHNART#n**	Type / {C1}	L – Left cut R – Right cut B – Cut /web
BAHNNR#1= to **BAHNNR#n**	Number / NUM	Cut number 1 – n
BREITE:S#1= to	Target width / DEC	Target width of a cut in mm

BREITE:S#n		
BREITE:I#1= to **BREITE:I#n**	Actual width / DEC	Actual width of a cut in mm

Return

ID	Content / {type}	Description

Validation checks

Error codes	Description

16.5 Transport management (MPL-TRA)

16.5.1 Create transport order

16.5.1.1 DLG=TRANR.CREATE

Using the BAPI calls described in this section, it is possible to create a transportation order for batches or resources.

Dependent products

Product	Bapi	Description
ADE	ANR.COPY	Create order
MPL	CNR.UPDATE	Batch status change

BAPI-Call

ID	Content / {type}	Description
DLG	TRANR.CREATE	Create transport order
TRANR.ATK	{C40}	Material number
TRANR.AUART	{C5}	Order type
TRANR.SMP	{C12}	Source material buffer
TRANR.TMP	{C12}	Target material buffer

TRANR.MNR	{C10}	Machine number
TRANR.SANR	{C40}	Trigger operation
TRANR.CNR	{C20}	Batch number for transport
TRANR.RES	{C20}	Resource number for transport
TRANR.RESTYP	{C10}	Resource type for transport
TRANR.SGR:P	{DEC}	Target quantity
TRANR.SGE:P	{C3}	Unit
TRANR.DATFB	{DAT}	Planned start date
TRANR.DATSE	{DAT}	Planned finish date
TRANR.CALLDLG	{C20}	Internal dialog action (e.g. CA_WL)
TRANR.ASYNC	{C1}	J = transport orders are created asynchronously

Return

ID	Content / {type}	Description
RET.AUNR	Order	Order number of created transport order

Validation checks

Error codes	Description
7043	Work plan not found
1690	Material buffer not found
2817	Oder type not found
3201	Assigned resource not found
7039	Resource is not in source material buffer
7038	Batch is not in source material buffer
1612	Batch not found
1594	batch status not valid
3636	Batch is locked
3632	material number of batch is not valid
7041	Batch is still in transport

16.5.2 Reserve transport order

16.5.2.1 DLG=TRANR.RESERVE

Using the BAPI calls described in this section, it is possible to reserve transportation orders for machines.

Dependent products

Product	Bapi	Description
ADE	ANR.SETSTATUS	Change order status
MPL	CNR.UPDATE	Change batch status

BAPI call

ID	Content / {type}	Description
DLG	TRANR.RESERVE	Reserve transport order
TRANR.MNR	Machine {C10}	Machine number
TRANR.ANR	Operation {C40}	Operation of the transport order

Return

ID	Content / {type}	Description

Validation checks

Error codes	Description
3702	Operation not found
7037	Operation is not in status initial
2613	Wrong operation status
107	Order type not valid

16.5.3 Start transportation order

16.5.3.1 DLG=TRANR.START

Using the BAPI calls described in this section, it is possible to start transportation order at machine.

Dependent products

Product	Bapi	Description
ADE	A_AN	Start operation
MPL	CNR.UPDATE	batch status change
WRM	RES_STATUS	resource status change

BAPI-Call

ID	Content / {type}	Description
DLG	TRANR.START	Start transportation order
TRANR.ATK	{C40}	Material number
TRANR.SMP	{C12}	Source material buffer
TRANR.TMP	{C12}	Target material buffer
TRANR.MNR	{C10}	Machine number
TRANR.ANR	{C40}	Operation
TRANR.CNR	{C20}	batch number for transport
TRANR.RES	{C20}	Resource number for transport
TRANR.RESTYP	{C10}	Resource type for transport

Return

ID	Content / {type}	Description

Validation checks

Error codes	Description
7039	Resource is not in source material buffer
7038	Batch is not in source material buffer

16.5.4 Finish transportation order

16.5.4.1 DLG=TRANR.END

Using the BAPI calls described in this section, it is possible to finish a running transportation order at machine.

Dependent products

Product	Bapi	Description
ADE	A_AB	Finish transportation order
MPL	CNR.UPDATE C_UMB	batch status change batch replace
WRM	RES_STATUS RES_UMB	Resource status change Move resource

BAPI-Call

ID	Content / {type}	Description
DLG	TRANR.END	Finish transportation order
TRANR.TMP	{C12}	Target material buffer
TRANR.MNR	{C10}	Machine number
TRANR.ANR	{C40}	Operation of transportation order
TRANR.EGR:GUT	{DEC}	Quantity

Return

ID	Content / {type}	Description

Validation checks

Error codes	Description

17 HYDRA Production Data Manager PDV - Master Data

17.1 Please note for the basic dialogs described

All mandatory fields are provided with the addition “PK” (primary key = key field). All other fields are optional and are processed if they are filled out.

17.2 Events

17.2.1 Edit event (DLG=PDVEVENTCFG.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, SELECT)

PDV events can be edited using the BAPI calls described in this section. HYDRA-PDV events may be recorded as general events at a machine and are saved in a corresponding log as soon as they occur.

Tables

Table	Key field	Description
pdv_event_cfg	event PDVEVENTCFG.EVENT	Event ID (PK) in the configuration table for PDV events

BAPI call

ID	Content/{type}	Description
DLG	PDVEVENTCFG.INSERT	Create PDV event
	PDVEVENTCFG.UPDATE	Change PDV event
	PDVEVENTCFG.DELETE	Delete PDV event
	PDVEVENTCFG.COPY	Copy PDV event
	PDVEVENTCFG.LOCK	Block PDV event to be processed
	PDVEVENTCFG.UNLOCK	Unblock PDV event after processing
	PDVEVENTCFG.SELECT	Select PDV event

PDVEVENTCFG.EVENT	{C50}	PK PDV event ID
PDVEVENTCFG.EVENT:Z	{C50}	PK new (target) PDV event ID for COPY
PDVEVENTCFG.CAPTION	{C100}	Description of the PDV event
PDVEVENTCFG.ALERT	{C1}	Flag whether the event is an alarm or not. Possible values: “Y” (yes, event is an alarm) or “N” (no, event is no alarm)
PDVEVENTCFG.ALERT:DURATION	{N}	Duration for the alarm signal when the PDV event occurs in seconds

Return

ID	Content/{type}	Description
PDVEVENTCFG.EVENT	{C50}	PDVEVENTCFG.INSERT, PDVEVENTCFG.COPY: Return of PDV event ID of the configuration created

Plausibility checks

Error codes	Description

17.2.2 List of events (DLG=PDVEVENTCFG.LIST)

This BAPI call lists the PDV events configured in HYDRA including their configuration.

Tables

Table	Key field	Description
pdv_event_cfg	event PDVEVENTCFG.EVENT	Event ID (PK) in the configuration table for PDV events

BAPI call

ID	Content	Description
DLG	PDVEVENTCFG.LIST	List of PDV events
PDVEVENTCFG.EVENT	{C50}	Optional restriction to a certain PDV event
DATEI	{C256}	Specification of the file name for the list

Return

ID	Content	Description
—	—	—

Plausibility checks

Error codes	Description
1656	The file assigned to the name DATEI cannot be written

17.3 Logical Channels

17.3.1 Edit logical channels (DLG=LOGCHAN.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, SELECT)

Logical channels required for HYDRA-PDV collection may be edited using the BAPI calls described in this section. Logical channels represent the central configuration for assigning physical interfaces of a machine (measuring channels) to logistic HYDRA values (characteristics, PDV events …).

By assigning them to a machine and a terminal, assigned characteristics or PDV events can be accessed at the corresponding place of entry. Master data of logical channels are put to the second under version control.

Tables

Table	Key field	Description
pdv_logic_chan	masch_nr LOGCHAN.MNR	Machine number (PK) of logical channels
pdv_logic_chan	capture_id LOGCHAN.TNR	Terminal number (PK) of logical channels
pdv_logic_chan	chan_nr LOGCHAN.LOGK	Number of the physical channel (PK) of logical channels
pdv_logic_chan	Fkey LOGCHAN.ID	ID of the logistic value (process parameter ID or PDV event) (PK) of logical channels
pdv_logic_chan	type LOGCHAN.TYP	Channel type, which logistic type it is about (process parameter or PDV event) (PK) of logical channels
pdv_logic_chan	valid_from LOGCHAN.VALID FROM	Valid from for master data put under version control (PK)

BAPI call

ID	Content/{type}	Description
DLG	LOGCHAN.INSERT	Create logical channel
	LOGCHAN.UPDATE	Change logical channel
	LOGCHAN.DELETE	Delete logical channel
	LOGCHAN.COPY	Copy logical channel
	LOGCHAN.LOCK	Block logical channel to be processed
	LOGCHAN.UNLOCK	Unblock logical channel after processing
	LOGCHAN.SELECT	Select logical channel
LOGCHAN.MNR	{C20}	PK machine number
LOGCHAN.TNR	{SHORT}	PK terminal number
LOGCHAN.LOGK	{NUM}	PK channel number
LOGCHAN.ID	{C50}	PK reference ID of recorded channel data
LOGCHAN.TYP	{C2}	PK type of reference ID of the recorded channel (process parameter or PDV event)
LOGCHAN.VALIDFROM	{DATETIME}	PK valid from point in time for master data put under version control
LOGCHAN.MNR:Z	{C20}	PK new (target) machine number for COPY
LOGCHAN.TNR:Z	{SHORT}	PK new (target) terminal number for COPY
LOGCHAN.LOGK:Z	{NUM}	PK new (target) channel number for COPY
LOGCHAN.ID:Z	{C50}	PK new (target) reference ID of recorded channel data for COPY
LOGCHAN.VALIDFROM:Z	{DATETIME}	PK new (target) valid from point in time for master data put under version control for COPY

…	…	For further fields, please refer to the HYD-HDB documentation about above-mentioned tables

Return

ID	Content/{type}	Description
LOGCHAN.MNR	{C20}	LOGCHAN.INSERT, LOGCHAN.COPY: Return of the keys of the configured PK channel number
LOGCHAN.TNR	{SHORT}	LOGCHAN.INSERT, LOGCHAN.COPY: Return of the keys of the configured PK channel number
LOGCHAN.LOGK	{NUM}	LOGCHAN.INSERT, LOGCHAN.COPY: Return of the keys of the configured PK channel number
LOGCHAN.ID	{C50}	LOGCHAN.INSERT, LOGCHAN.COPY: Return of the keys of the configured PK channel number
LOGCHAN.TYP	{C2}	LOGCHAN.INSERT, LOGCHAN.COPY: Return of the keys of the configured PK channel number
LOGCHAN.VALIDFROM	{DATETIME}	LOGCHAN.INSERT, LOGCHAN.COPY: Return of the keys of the configured PK channel number

Plausibility checks

Error codes	Description
550	Unknown channel reference type: Only process parameters (PP) and PDV events are supported as references.
551	The PDV event transferred is not available
552	Anonymous characteristics from TNT configuration only support measured values (MW) as channel type
553	Channel numbers between 0 and 9999 are valid only
554	Cycle times greater than 0 are allowed only
555	Wrong channel direction. Input and output channels are supported only.
556	Alarm channels are not supported for the channel data type indicated. This is only possible for OTG, UTG, OPEG or UPEG (UTL, LTL, UPAL, LPAL).
2039	Machine is not available
1668	Terminal is not available

17.3.2 List of logical channels (DLG=LOGCHAN.LIST)

This BAPI call lists the logical channels configured in HYDRA including their configuration.

Tables

Table	Key field	Description
pdv_logic_chan	masch_nr LOGCHAN.MNR	Machine number (PK) of logical channels
pdv_logic_chan	capture_id LOGCHAN.TNR	Terminal number (PK) of logical channels
pdv_logic_chan	chan_nr LOGCHAN.LOGK	Number of the physical channel (PK) of the logical channels
pdv_logic_chan	fkey LOGCHAN.ID	ID of the logistic value (process parameter ID or PDV event) (PK) of logic channels
pdv_logic_chan	type LOGCHAN.TYP	Channel type, which logistic type it is about (process parameter or PDV event) (PK) of logic channels
pdv_logic_chan	valid_from LOGCHAN.VALID FROM	Valid from for master data put under version control (PK)

BAPI call

ID	Content	Description
DLG	LOGCHAN.LIST	List of logical channels
DATEI	{C256}	Specification of the file name for the list

Return

ID	Content	Description
—	—	—

Plausibility checks

Error codes	Description
1656	The file assigned to the name DATEI cannot be written

17.4 Characteristic Attribute

17.4.1 Edit characteristic attributes (DLG=PAUMMAUSP.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

Characteristic attributes can be edited by way of the BAPI calls described in this section. They are required to record the necessary configurations of an inspection order characteristic for the generation of samples. In this context, the different configuration versions of an inspection order characteristic are put under version control for a certain period of time. Thus, it is possible to trace back, which configuration applied for a characteristic at a certain point in time.

Tables

Table	Key field	Description
caq_paumm_ausp	rec_type RECTYP	Rectype in characteristic attribute (PK)
caq_paumm_ausp	bereich BER	Area in characteristic attribute (PK)
caq_paumm_ausp	pruefanf_nr PANNR	Inspection requirement number in characteristic attribute (PK)
caq_paumm_ausp	pruefauft_nr PAUNR	Inspection order number in characteristic attribute (PK)
caq_paumm_ausp	afo AFO	Order sequence in characteristic attribute (PK)
caq_paumm_ausp	maschine_nr MNR	Machine number in characteristic attribute (PK)
caq_paumm_ausp	valid_from_ts VALIDFROM	Valid from point in time in characteristic attribute (PK)

BAPI call

ID	Content/{type}	Description
DLG	PAUMMAUSP.INSERT	Create characteristic attribute
	PAUMMAUSP.UPDATE	Change characteristic attribute

	PAUMMAUSP.DELETE	Delete characteristic attribute
	PAUMMAUSP.COPY	Copy characteristic attribute
	PAUMMAUSP.LOCK	Block characteristic attribute to be processed
	PAUMMAUSP.UNLOCK	Unblock characteristic attribute after processing
	PAUMMAUSP.SELECT	Select characteristic attribute
PAUMMAUSP.RECTYP	{C10}	Rectyp (PK)
PAUMMAUSP.BER	{C10}	CAQ area (PK)
PAUMMAUSP.PANNR	{NUM}	Inspection request number in characteristic attribute (PK)
PAUMMAUSP.PAUNR	{NUM}	Inspection order number in characteristic attribute (PK)
PAUMMAUSP.AFO	{NUM}	Order sequence in characteristic attribute (PK)
PAUMMAUSP.MNR	{C20}	Machine number in characteristic attribute (PK)
PAUMMAUSP.VALIDFROM	{DATETIME}	Valid from point in time in characteristic attribute (PK)
PAUMMAUSP.RECTYP	{C10}	PK new (target) Rectyp for COPY
PAUMMAUSP.BER	{C10}	PK new (target) CAQ area for COPY
PAUMMAUSP.PANNR	{NUM}	PK new (target) inspection request number for COPY
PAUMMAUSP.PAUNR	{NUM}	PK new (target) inspection order number for COPY
PAUMMAUSP.AFO	{NUM}	PK new (target) order sequence for COPY
PAUMMAUSP.MNR	{C20}	PK new (target) machine number for COPY
PAUMMAUSP.VALIDFROM	{DATETIME}	PK new (target) “colname designation” for COPY (column name)

…	…	For further fields, please refer to the HYD-HDB documentation about above-mentioned tables

Return

ID	Content/{type}	Description
caq_paumm_ausp	{C10}	PAUMMAUSP.INSERT, PAUMMAUSP.COPY: Rectype in characteristic attribute (PK)
caq_paumm_ausp	{C10}	PAUMMAUSP.INSERT, PAUMMAUSP.COPY: Area in characteristic attribute (PK)
caq_paumm_ausp	{NUM}	PAUMMAUSP.INSERT, PAUMMAUSP.COPY: Inspection requirement number in characteristic attribute (PK)
caq_paumm_ausp	{NUM}	PAUMMAUSP.INSERT, PAUMMAUSP.COPY: Inspection order number in characteristic attribute (PK)
caq_paumm_ausp	{NUM}	PAUMMAUSP.INSERT, PAUMMAUSP.COPY: Order sequence in characteristic attribute (PK)
caq_paumm_ausp	{C20}	PAUMMAUSP.INSERT, PAUMMAUSP.COPY: Machine number in characteristic attribute (PK)
caq_paumm_ausp	{DATETIME}	PAUMMAUSP.INSERT, PAUMMAUSP.COPY: Valid from point in time in characteristic attribute (PK)

Plausibility checks

Error codes	Description
—	—

17.4.2 List of characteristic attributes (DLG=PAUMMAUSP.LIST)

This BAPI call lists the characteristic attributes configured in HYDRA including their configuration.

Tables

Table	Key field	Description
caq_paumm_ausp	rec_type RECTYP	Rectype in characteristic attribute (PK)
caq_paumm_ausp	bereich BER	Area in characteristic attribute (PK)
caq_paumm_ausp	pruefanf_nr PANNR	Inspection requirement number in characteristic attribute (PK)
caq_paumm_ausp	pruefauft_nr PAUNR	Inspection order number in characteristic attribute (PK)

caq_paumm_au sp	afo AFO	Order sequence in characteristic attribute (PK)
caq_paumm_au sp	maschine_nr MNR	Machine in characteristic attribute (PK)
caq_paumm_au sp	valid_from_ts VALIDFROM	Valid from period of the characteristic attribute (PK)

BAPI call

ID	Content	Description
DLG	PAUMMAUSP.LI ST	List of characteristic attributes
DATEI	{C256}	Specification of the file name for the list

Return

ID	Content	Description
—	—	—

Plausibility checks

Error codes	Description
1656	The file assigned to the name DATEI cannot be written

18 HYDRA Production Data Manager WRM - Data Collection

18.1 Please note for the posting dialogs described

All mandatory fields are displayed in gray; all other fields are optional and are processed, provided that they are filled out.

18.2 WRM Editing Dialogs

This section describes the editing dialogs to map the terminal function with respect to the resource status.

18.2.1 Log resource on (DLG=RES_AN)

Using this dialog a resource can be logged on to an order. Depending on the configuration of the corresponding resource type, quantities and times are then also posted onto the resource that is logged on.

<table>
<tr><th>ID</th><th>Type/max. field length</th><th colspan="2">Description</th></tr>
<tr><td>DLG</td><td>RES_AN</td><td colspan="2">Log resource on</td></tr>
<tr><td>DAT</td><td>{mm/dd/yyyy}</td><td colspan="2">The date needs to be indicated to log a resource on.</td></tr>
<tr><td>ZEI</td><td>{seconds}</td><td colspan="2">Time, please see DAT.</td></tr>
<tr><td>RESID</td><td>N10</td><td colspan="2">Resource ID, unique number to identify the resource in database tables .
Alternative specification for RESTYP and RES</td></tr>
<tr><td>RESTYP</td><td>C4</td><td colspan="2">Key 1 of the double key to uniquely identify a resource. Resource type. Alternative specification for RESVERWEIS</td></tr>
<tr><td>RES</td><td>C40</td><td colspan="2">Key 2 of the double key to uniquely identify a resource. Name of the resource. Alternative specification for RESVERWEIS</td></tr>
<tr><td>MNR</td><td>C20</td><td colspan="2">Machine number</td></tr>
<tr><td>ANR</td><td>C40</td><td colspan="2">Order number</td></tr>
<tr><td>PNR</td><td>C10</td><td rowspan="2">alter-native</td><td>Personnel number</td></tr>
<tr><td>KNR</td><td>C10</td><td>Badge number</td></tr>
<tr><td>RESVER</td><td>C20</td><td colspan="2">Version ID</td></tr>
<tr><td>KOMMENTAR</td><td>C500</td><td colspan="2">Comment on the logon process, is saved for the event (event_res) in the event_dlg_data table .</td></tr>
</table>

18.2.2 Log resource off (DLG=RES_AB)

By way of this dialog a resource can be logged off from an order. Depending on the configuration of the corresponding resource type, quantities and times are also posted onto the resource that is logged off.

ID	Type/max. field length	Description	
DLG	RES_AB	Log resource off	
DAT	{mm/dd/yyyy}	The date needs to be entered to log a resource off.	
ZEI	{seconds}	Time, please see DAT.	
RESID	N10	Resource ID, unique number to identify the resource in database tables. Alternative specification for RESTYP and RES	
RESTYP	C4	Key 1 of the double key to uniquely identify a resource. Resource type. Alternative specification for RESVERWEIS	
RES	C40	Key 2 of the double key to uniquely identify a resource. Resource name. Alternative specification for RESVERWEIS	
MNR	C20	Machine number	
ANR	C40	Order number	
PNR	C10	alter-native	Personnel number
KNR	C10		Badge number
RESVER	C20	Version ID	
KOMMENTAR	C500	Comment on the logoff process is saved for the event (event_res) in the event_dlg_data table.	

18.2.3 Set resource status (DLG=RES_STATUS)

The status of a resource can be reset by way of this dialog. The old and the new status are compared with each other and corresponding plausibility checks are performed. Once the plausibility check has been realized with success, affected tables are updated/specified.

Resource allocation in res_ress_belegung is updated respectively. If a resource is blocked, i.e. it gets a status with the ID "verarb_planung" != "K" an entry is made in res_ress_belegung. In this context, the date (DATB/ZEIB and DATE/ZEIE) is taken into account. If the current point in time is within the specified period of time an entry is made. In case the end date is prior to this period of time, it is ignored. Provided that the start date lies in the future, it is entered with the date lines specified.

In case the “bill of material processing” license (WRM-STL or DNC-STL) is available, superordinate resources are blocked collectively according to the bill of material by increasing the collective block counter by 1 for superordinate resources. When the resource is released, the collective block counter of the superior resource is reduced by 1.

Block: All superordinate resources. Collective block + 1.
Unblock: All superior resources. Collective block - 1..

The PROD=x ID has been configured to switch to a defined status. If this value is entered a transferred status is ignored. The system then searches for the status, which is assigned to exactly this ID in res_status_zuord.prod and takes this status as new specification. The prod ID is always unique for each “type – family” combination. If this is not the case, it is a configuration error. In this case the last status found is used. This technology actually renders the RES_FREI and RES_ABSTA dialogs redundant. They may be replaced by DLG=RES_STATUS|PROD=F (=RES_FREI) or DLG=RES_STATUS|PROD=B (=RES_ABSTA).

ID	Type/max. field length	Description	
DLG	RES_STATUS	Status change	
DAT	{mm/dd/yyyy}	The date specification is very important for setting statuses. As due to this date and the other date specifications it can automatically be recognized whether the status is set immediately or in future.	
ZEI	{seconds}	Time; please see DAT.	
RESID	N10	Resource ID, unique number to identify the resource in database tables. Alternative specification for RESTYP and RES	
RESTYP	C4	Key 1 of the double key to uniquely identify a resource. Resource type. Alternative specification for RESVERWEIS	
RES	C40	Key 2 of the double key to uniquely identify a resource. Resource name. Alternative specification for RESVERWEIS	
RESSTA	N10	Target status of the resource	
PNR	C10	alter-native	Personnel number
KNR	C10		Badge number
RESVER	C20	Version ID	
PROD	C1 {F|B|U|…}	ID whether the status is to be determined in the database or not. If nothing is entered the transferred status applies. If entered the appropriate status is searched in . res_status_zuord.prod. At the moment the following is configured: F = Release status (status when a resource is released) B = Logoff status (status when a resource is logged off - A_AB) U = Upload status (status when a resource is uploaded - RES_UPLOAD)	
DATB	{mm/dd/yyyy}	Start date for realizing a status in the future. Format mm/dd/yyyy in local time zone If nothing is entered or if the ID is left out or the date lies in the past the status is reset immediately.	
ZEIB	{seconds}	Start time for resetting a status in the future. Normally 0.	
DATE	{mm/dd/yyyy}	End date for resetting a status with limited validity. If this date is finally exceeded a monitoring process recognizes that the status has expired and sets it to the only possible status that has the status “free”. Format mm/dd/yyyy in local time zone	

ID	Type/max. field length	Description
		If nothing is entered or the ID is left out an unlimited validity period is set (max. date).
ZEIE	{seconds}	End time for a status change with limited validity. If empty or = 0. Then automatic change over to 86400.
ZLO	C12	Receiving storage location. If empty the default value from status configuration is used as storage location .
KOMMENTAR	C500	Comment on the status change is saved for the event (event_res) in the event_dlg_data table.

18.2.4 Release resource (DLG=RES_FREI)

With this dialog a resource is changed over to a released status. There is only one resource status with the ID “released” for each type or family .

The process exactly corresponds to starting a RES_STATUS dialog with preselected release status. Consequently, the start process and behavior is the same as for RES_STATUS. By using the ID "PROD=**F**" for the dialog RES_STATUS, it is even to be preferred to the RES_FREI dialog. Example: DLG=RES_STATUS|PROD=**F**|…

ID	Type/max. field length	Description
DLG	RES_FREI	Status change
RESSTA	N10	NO ENTRY!!!
Rest like RES_STATUS		

18.2.5 Change resource status to “status after logging OP off” (DLG=RES_ABSTA)

This dialog changes a resource status to a “status after logging OP off”, i.e. a resource switches to this status if it is logged off/interrupted with an order. There is only one resource status with the "Abmelde_Status" ID for each type or family .

The procedure exactly corresponds to starting a RES_STATUS dialog with preselected logoff status. Thus, the start procedure and behavior is just the same as for RES_STATUS. By using the "PROD=**B**" ID for the RES_STATUS dialog it is even to be preferred to the RES_ABSTA dialog. Example: DLG=RES_STATUS|PROD=**B**|…

ID	Type/max. field length	Description
DLG	RES_ABSTA	Status change

ID	Type/max. field length	Description
RESSTA	N10	No Entry!!!
Remainder like RES_STATUS		

18.2.6 Repost resource (DLG=RES_UMB)

This dialog reposts a resource onto another storage location.

ID	Type/max. field length	Description
DLG	RES_UMB	Change of storage location
ZLO	C12	New receiving storage location (target). If nothing is entered the default value from status configuration is used as storage location.
KOMMENTAR	C500	Comment on the changed storage location; is saved for the event (event_res) in the table event_dlg_data.

18.2.7 Mount resource (DLG=RES_EIN)

A resource may be added to another resource using this dialog. From a technical point of view, a BOM relationship is established between mother and daughter resource.
Quantities or times are not posted onto the resource by this dialog. But the event is recorded and shown in many different evaluations/reports.

ID	Type/max. field length	Description
DLG	RES_EIN	Log resource off
DAT	{mm/dd/yyyy}	The date needs to be specified for logging a resource off.
ZEI	{seconds}	Time; see DAT.
RESTYP:M	C4	Key 1 of the double key to uniquely identify a resource. Type of the mother resource.
RES:M	C40	Key 2 of the double key to uniquely identify a resource. Name of the mother resource.
RESTYP:T	C4	Key 1 of the double key to uniquely identify a resource. Type of the daughter resource.
RES:T	C40	Key 2 of the double key to uniqley identify a resource. Name of the daughter resource.
PNR	C10	Personnel number

ID	Type/max. field length	Description	
KNR	C10	alter-native	Badge number
KOMMENTAR	C500	Logoff comment is saved for the event (event_res) in the table event_dlg_data.	

18.2.8 Demount resource (DLG=RES_AUS)

A resource can be demounted from another resource using this dialog. From a technical point of view, the BOM relationship between the transferred mother and daughter resource is removed/deleted.

Quantities or times are not posted onto the resource by this dialog. But the actual event is recorded and shown in different reports/evaluations.

ID	Type/max. field length	Description	
DLG	RES_AUS	Log resource off	
DAT	{mm/dd/yyyy}	The date needs to be specified for logging a resource off.	
ZEI	{seconds}	Time; see DAT.	
RESTYP:M	C4	Key 1 of the double key to uniquely identify a resource. Type of the mother resource.	
RES:M	C40	Key 2 of the double key to uniquely identify a resource. Name of the mother resource.	
RESTYP:T	C4	Key 1 of the double key to uniquely identify a resource. Type of the daughter resource.	
RES:T	C40	Key 2 of the double key to uniquely identify a resource. Name of the mother resource.	
PNR	C10	alter-native	Personnel number
KNR	C10		Badge number
KOMMENTAR	C500	Logoff comment is saved for the event (event_res) in the table event_dlg_data.	

18.3 DNC Dialogs

This section describes special dialogs to manage resources of the DNC type. These are functions to upload and download resources. As DNC resources also include program files in addition to resource data records, these program files are also transferred along with these operations. This establishes the connection between the local data room (local hard disk directory) and the HYDRA server data room. Depending on its configuration, the HYDRA server data room can also be connected with an external programming system.

18.3.1 Load DNC resource to the machine (DLG=RES_DOWNL)

Using this dialog a DNC resource (res_typ.dnc_verarbeitung != "K") is logically “downloaded” to a terminal, i.e. the dialog performs different plausibility checks and the logging function in event_res.

Files are not copied with RES_DOWNL. An external copy process is required for this. RES_DOWNL exclusively provides for database processing.

Consequently, a logical, complete download process looks as follows:

1. Validity check (through RES_DOWNL|VERB=N| …)
2. Files are copied to the terminal, which is performed by external functions (not HYDDI).
3. Posting of the download (through RES_DOWNL, possibly with BZW=J)

Using different “definitions” of the IDs leads to different behaviors of the dialog. The “VERB“ ID differentiates between “J“ = complete download incl. posting (by default) and “N” = only plausibility check without posting. Using the "BZW=J" (booking obligation) ID avoids all plausibility checks. This is reasonable if started several times, after starting with VERB=N.

Several resources can be transferred in one start process. That is why a number variable is planned for RESID:n or RES:n and RESTYP:n. If only one resource is transferred, the number variable can be omitted. If several resources are transferred they have to be indicated completely including their respective RESID or RESTYP-RES combination. At the moment a maximum of 30 resources can be transferred. Further resources are ignored. All plausibility checks are made for the resources specified and all resources are recorded separately in event_res. However, VERB and BZW have to be configured respectively.

The DNC-BP license is required to use this dialog.

ID	Type/max. field length	Description
DLG	RES_DOWNL	Download
VERB	C1	N = only validity check, without posting J = validity check and posting (by default)
DAT	{mm/dd/yyyy}	Date specification for the download.

ID	Type/max. field length	Description
ZEI	{seconds}	Time for the download
RESVERWEIS:n or RESID:n	N10	Resource ID, unique number to identify the resource in database tables. Alternative specification for RESTYP and RES. n = 1..30 (complete entry required). RESID takes priority over RESTYP+RES.
RESTYP:n	C4	Key 1 of the double key to uniquely identify a resource. Resource type. Has always to be indicated together with RES. Alternative specification for RESVERWEIS. n = 1..30 (complete entry required).
RES:n	C40	Key 2 of the double key to uniquely identify a resource. Resource name. Has always to be indicated together with RESTYP. Alternative specification for RESVERWEIS. n = 1..30 (complete entry required).
BEARB	C10	Editor
PNR	C10	alter-native: PNR
KNR	C10	alter-native: KNR
RESVER:n	C20	Version ID (not used as key) n = 1..30 (complete entry required).
BZW	C1	J = Booking obligation, i.e. no plausibility checks N = No booking obligation (by default)
OPT_DNCVERARB	C1	ID whether DNC resources are concerned ("L", "E", "O") or not ("K"). "K"= no DNC resource "L"=local "E"=external "O"=optimized Currently not in use. If not indicated the dialog determines this value on the basis of the resource type of resource 1.
MNR	N8/C8	Machine
KOMMENTAR	C255	Download comment. Is saved for the event (event_res) in the event_dlg_data table. A reference is saved in event_res. If the resource is indicated several times the comment is saved only once. All new event_res entries (for each resource specified) then refer to it.

18.3.2 Upload DNC resource from the machine (DLG=RES_UPLOAD)

This dialog logically uploads a DNC resource (res_typ.dnc_verarbeitung != "K") from a terminal, i.e. the dialog takes over different plausibility checks, creates a new resource incl. attributes, if required, and records it in event_res.

Files are not copied with RES_UPLOAD. This has to be realized by way of an external copy process. RES_UPLOAD only provides for database processing.

Consequently, a logical, complete upload process looks as follows:

1) Validity checking (through RES_UPLOAD|VERB=N| …)

2. Files are copied from the terminal into the HYDRA namespace, which is performed by functions provided externally (not HYDDI).

3. Posting of the upload (through RES_UPLOAD, possibly with BZW=J). If required, a new resource incl. user fields is created in this context. In case of optimization (res_typen.dnc_verarbeitung=”O”), the resource is additionally set to “optimized” and the resource status is switched to the PROD=”U” status (upload).

Using different “definitions” of the IDs leads to different behaviors of the dialog. The “VERB“ ID differentiates between “J“ = complete upload incl. posting and creation of resources (by default) and “N” = only validity checking without posting. Using the "BZW=J" (posting required) ID avoids all validity checks. This is useful if started several times, after starting with VERB=N.

Two different behaviors can be defined by the MOD mode. MOD=N (= new) and MOD=U (= update). When “N” is selected, it is checked, among other things, whether the resource already exists and can be created anew. In case it has not yet been created, the new resource is saved incl. the user fields transferred (RES.INSERT and several RESATTR.INSERT). A new resource does not have to be created for MOD=U, the optimization flag is set in the resource instead (RES.UPDATE) and if defined the status is switched to the status assigned to the PROD=U ID (DLG=RES_STATUS|PROD=U|…).

The upload is recorded in event_res. However, VERB and BZW have to be configured respectively.

This dialog requires the additional function "DNC-VGN".

ID	Type/max. field length	Description
DLG	RES_UPLOAD	Upload
VERB	C1	N = plausibility check only, without posting J = plausibility check and posting (by default)
DAT	{mm/dd/yyyy}	Date specification for the upload.
ZEI	{seconds}	Time for the upload
MOD	C1	ID to differentiate whether a DNC resource is to be “U” = changed or “N” = newly created
RESVERWEIS or RESID	N10	Resource ID, unique number to identify the resource in database tables. Alternative specification for RESTYP and RES. Valid for MOD="U" only (=Update), if "N" this ID is only determined when the resource is created.

ID	Type/max. field length	Description
RESTYP	C4	Key 1 of the double key to uniquely identify a resource. Resource type. Always to be indicated together with RES. Alternative specification for RESVERWEIS if MOD="U".
RES	C40	Key 2 of the double key to uniquely identify a resource. Resource name. Always to be indicated along with RESTYP. Alternative specification for RESVERWEIS if MOD="U".
BEARB	C10	Editor
PNR	C10	alter-native: PNR
KNR	C10	alter-native: KNR
RESVER	C20	Version ID (not used as key)
RESFAMID	N10	Resource family identifier (serial) for creating a new resource. If both values (RESFAMID and RESFAM) remain empty no family is defined for the resource.
RESFAM	C20	Resource family ID, provided that the resource family identifier is not entered.
BZW	C1	J = Booking obligation, i.e. no plausibility checks N = No booking obligation (by default)
OPT_DNCVERARB	C1	Identifier whether it is a DNC resource ("L", "E", "O") or not ("K"). "K"=no DNC resource "L"=local "E"=external "O"=optimized If not indicated the dialog determines this value on the basis of the resource type of the resource. "O" triggers additional activities when it comes to posting.
MNR	N8/C8	Machine
SPEICHORT:DATA	C128	Name of the DNC file without path and file extension
IDX:1..4	N10	A maximum of 4 user fied data. User field index. This field has to be configured correctly by the assigned user field key of the resource. The value entered here corresponds to the field index of the database.
FIL:1..4	C20	A maximum of 4 user field data. Data content of the user field. Please also see info on “IDX1…4”. The value entered here corresponds to the data field that matches the specified field index of the database.
KOMMENTAR	C500	Upload comment. Is saved for the event (event_res) in the event_dlg_data table. A reference is saved in event_res. If the resource is indicated several times the comment is saved only once. All new event_res entries (for each resource specified) then refer to it.

18.4 Lists for DNC-Data

To operate DNC resources there are special lists to read DNC information at the workplace (terminal) of the machine. These lists have also been configured to ensure conformity and to perform required inspections.

18.4.1 Enhancement of BDE lists

18.4.2 Machines – DNC family (DLG=LIST;82)

The machine defines which DNC family is assigned to the machine. Corresponding attribute configurations and, as a result, validity checks and functions are connected by these admissible families.

Moreover, the list provides a complete description of attributes! Consequently, user attributes do not have to be read separately.

Structure of dialog data:

“DLG=LIST;82|DATEI={file name}|DAT=...|ZEI=...|USR=...|MOD=...“

Parameter:

MOD=M|MNR=... Finding the family of a machine

MOD=T|TNR=... Finding the family of a terminal

Result list:

ID	Type/max. field length	Description
MNR	C20	Machine
RESFAM	C20	Resource family
RESFAM:BEZL	C60	Designation
RESTYPID	C4	Resource type identification
VORG	C1	J/N specification (= default family)
IDX:x	Integer	ID of the user field (free user attributes) within this user field key Numeric 1…n → Value corresponds to value FELD IDX of resource attributes
BEZK:x	C20	Designation, e.g. “DIAM”
BEZL:x	C80	Long designation, e.g. “diameter”
POS:x	Short	Display position/sequence
FKT_PLAUS:x	C10	ID for algo. plausibility check „=“ = IST_GLEICH check [level 1, other checks follow]
CFG_1:x … CFG_10:x	C10	1-4 for "DNC" 1: Filter (for filter dialog at the terminal and for the entry when uploading a new program) 2: Mandatory field for the DNC upload 3: Specification for the DNC upload

ID	Type/max. field length	Description
		4: Read-only for the DNC upload
KENN:x	C20	Acronym by which the field is addressed. This acronym is used in formulas to reference it as parameter, for example. Customer IDs are assigned within the individual namespace, i.e. the system automatically prefixes a “U:”. e.g. MNR
EINH:x	C3	Unit
TYP:x	C1	NUM, TEXT; DEC
FMT:x	C20	Format for console display (optional)
NKS:x	Short	If DEC, decimal places
LEN:x	Short	Field length
VON:x	DEC 18,6	For NUM/DEC data type: min. value range
BIS:x	DEC 18,6	For NUM/DEC data type: min. value range
ZEICH:x	C100	Which characters are allowed for the TEXT data type?
VORG_ EXPR:x	C50	Default value that is, for example, used for calculating formulas if the initial field is not available or empty.
VORG_EDIT:x	C80	Default value for new input

18.4.3 Loadable DNC programs (DLG=LIST;83)

By defining attributes of the DNC family and assigning them to the machine, it is determined which DNC programs currently saved in the system may be transferred to the terminal. The list provides all programs that are admissible or currently planned for the terminal. This restriction is improved through the resource attributes defined and the corresponding plausibility checks. In this context, restrictions are also predetermined by the operation that is logged on. The bills of material defined for DNC packages are broken down to their elements and posted in the list.

Structure of dialog data:

„DLG=LIST;83|DATEI={file name}|DAT=...|ZEI=...|USR=...|MOD=...“

Parameter:

MOD=A|... DNC programs filtered by order relation

MOD=P|... DNC programs filtered via the program name

MOD=D|... DNC programs without order relation. Direct reading on the resource stock. Selection via DNC family

The selection parameters are:

- TNR

- MNR
- ANR
- RES

Valid combinations:

	TNR	MNR	ANR	RES
MOD:A	No	No	Yes	No
MOD:P	No	Yes	No	Yes
MOD:D	Yes	Yes	No	No

ID	**Type/max. field length**	**Description**
MNR	C20	**MOD:A:** if started with parameter MNR or TNR then machine on which the OP is planned. **MOD:P:** Transferred machine **MOD:D:** Transferred machine
ANR	C40	**MOD:A:** OP which this resource is required for. **MOD:P:** empty **MOD:D:** empty
RESTYPID	C4	Type ID
RESTYP	C20	Resource type
RESTYP_BEZL	C60	Resource type designation
RES	C40	Resource
VERWEIS	N10	Resource ID
RESFAM	C20	DNC family
RESFAM_BEZL	C60	DNC family
OPT_DATEIVORH	C1	Optimized program available
BEARBDAT	{mm/dd/yyyy}	Date
BEARBZEI	{seconds}	Time

ID	Type/max. field length	Description
OPT_DNCVERARB	C3	DNC processing with optimization procedure
OPT_DATEIBASIERT	C1	Optimization, file-based
OPT_EXTGUELT	C5	Valid file extension
OPT_EXTOPTIM	C5	Optimized file extension
OPT_PATH	C128	Path of opt. file
DATEI	C128	File name
DATEI_SIZE	N	File size
DATEI_LOKAL	C1	Locally available
AKTIV	C1	Active
OPT_VERSION	C1	Optimized
RESSTA	N10	Status
RESSTABEZ	C60	Status designation
OPT_ERF	C1	Blocked
SSPERR	C1	Collective block
SSTA	N10	Collective status
SSTABEZ	C60	Designation of collective status
COLOR		Color
META_RES	C1	Package J/N
RESTYP:M	C4	ResTyp parent
VERWEIS:M	N10	ResID parent
RES:M	C20	ResNr parent

Comments:

- The configuration “anzeige_teminal J/N“ of the resource must be set to “J“. Otherwise, the element or package is not displayed. However, if the element is part of a package, it is displayed anyway.

- For package elements data are taken over 1:1 from resource types, i.e. the package values are not “handed down” to the elements

➔ With respect to the configuration, it has to be ensured that resource types of the main programs and subprograms are configured correctly. The “package” setting at the resource type is not relevant in this context.

18.5 WRM Maintenance Dialog

This section deals with special dialogs for managing resource maintenances. These are functions to set the maintenance status, activate and deactivate maintenances as well as to restart interval maintenances.

18.5.1 Maintenance status and activation (DLG=RES_WART)

By way of this dialog

- the status of a resource maintenance is set (MOD=Z)
- a maintenance is activated or deactivated (MOD=A)
- a resource maintenance is reset and prepared for the next maintenance (MOD=R)

In all 3 modes current maintenance data are first saved as event. These events are used for logging and reports.

Depending on the mode, different functions are performed:

- Set the resource maintenance status (MOD=Z). The WARTSTA parameter transfers the new status that is to be set. The status is set for the maintenance stated.
- Activate or deactivate (MOD=A) the maintenance. The AKTIV (J/N) parameter indicates whether the maintenance is to be activated or deactivated.
- Reset resource maintenance (MOD=R). All actual values are set to 0 and target values are calculated for the next maintenance. The AUFSATZ (S/I) parameter indicates whether the new calculation is to be based on the old target value “S” or on the current actual value “I”. The LWART:DAT, LWART:ZEI and LWART:USR parameters save the date, time and person of the maintenance made.

Finally, a comment is saved for all modes.

ID	Type/max. field length	Description
DLG	RES_WART	Set maintenance data
MOD	C1	Z = set status A = activate/deactivate R = reset
DAT	{mm/dd/yyyy}	Date specification for the download.
ZEI	{seconds}	Time for the download
RESVERWEIS or RESID	N10	Resource ID, unique number to identify the resource in database tables. Alternative specification for RESTYP and RES. RESID takes priority over the RESTYP+RES specification.

ID	Type/max. field length	Description	
RESTYP	C4	Key 1 of the double key to uniquely identify a resource. Resource type. Always to be stated along with RES. Alternative specification for RESVERWEIS.	
RES	C40	Key 2 of the double key to uniquely identify a resource. Resource name. Always to be stated along with RESTYP. Alternative specification for RESVERWEIS.	
WARTVERWEIS	N10	Reference to the defined maintenance → event_res.info_03	
WARTVERWEIS	N10	Reference to the defined maintenance → event_res.info_03	
PNR	C10	alter-native	Personnel number
KNR	C10		Badge number
RESVER	C20	Version ID (not used as key)	
ZUSTAND	N10	New status [value range 0 to 3]	
AKTIV	C1	New active status: J=enable, N=disable	
KOMMENTAR	C255	Comment	
LWART:DAT	DATE	Date of last maintenance	
LWART:ZEI	TIME	Time of last maintenance	
LWART:USR	C10	Person of last maintenance	

18.6 WRM Measures Dialog

This section describes special dialogs that have been configured to manage resource measures. At the moment this is a function to define a new measure for a resource. At the same time this activity is recorded for reports in HYDRA.

18.6.1 Activate measure (DLG=RES_MASS)

Using this dialog, a measure is activated, i.e. the entered measure is defined within the resource status.

The provided comment is saved.

<table>
<tr><th>ID</th><th>Type/max. field length</th><th colspan="2">Description</th></tr>
<tr><td>DLG</td><td>RES_MASS</td><td colspan="2">Set measure</td></tr>
<tr><td>DAT</td><td>{mm/dd/yyyy}</td><td colspan="2">Date specification for the download.</td></tr>
<tr><td>ZEI</td><td>{seconds}</td><td colspan="2">Time for the download</td></tr>
<tr><td>RESVERWEIS
or
RESID</td><td>N10</td><td colspan="2">Resource ID, unique number to identify the resource in database tables. Alternative specification for RESTYP and RES.
RESID takes priority over the RESTYP+RES specification.</td></tr>
<tr><td>RESTYP</td><td>C4</td><td colspan="2">Key 1 of the double key to uniquely identify a resource. Resource type. Always to be entered along with RES. Alternative specification for RESVERWEIS.</td></tr>
<tr><td>RES</td><td>C40</td><td colspan="2">Key 2 of the double key to uniquely identify a resource. Resource name. Always to be entered along with RESTYP. Alternative specification for RESVERWEIS.</td></tr>
<tr><td>PNR</td><td>C10</td><td rowspan="2">alter-native</td><td>Personnel number</td></tr>
<tr><td>KNR</td><td>C10</td><td>Badge number</td></tr>
<tr><td>RESVER</td><td>C20</td><td colspan="2">Version ID (not used as key)</td></tr>
<tr><td>MASSNR</td><td>N10</td><td colspan="2">Measure number</td></tr>
</table>

18.7 Lists for Resource Data

18.7.1 Resource list (DLG=LIST;115)

The list is used to select a resource.

Structure of dialog data:

"DLG=LIST;115|DATEI={file_name}|DAT=...|ZEI=...|USR=...|RESTYP=...|RESFAM=...|RES=…|ANR=…|AKRO=…|MOD=…|"

Parameter:

Parameters are optional and support wildcard characters (%).

RESTYP =... restrict the list by a resource type

RESFAM =... restrict the list by a resource family. The designation is to be indicated here, not the ID.

RES =... Restrict the list by a resource.

ANR =... Show only resources that are planned for the order.

AKRO=… Fast user fields are available as dynamic fields. When starting the list, they can be requested, separated by semicolon using the AKRO=FU:1 to FU:66 ID.

MOD=M : As set union the list includes all resources that are either logged on/active to/on the transferred order (ANR=) or that are planned for this order. In this case, only resources are selected, which may be logged on explicitly (configuration of the resource stock) or that are to be displayed in the resource list (configuration of the resource type).

OPT:BEDRES=N: The list does **not** include required resources. This option allows for the resource list to be requested without required resources. All other values, except for "N" mean that the list also includes required resources. This function cannot be combined with MOD=M.

Example:

hymw -u2020

-c"DLG=LIST;115|DATEI=l.lst|DAT=today|ZEI=now|USR=2020|RESTYP=WNR| RESFAM= Soft%|RES=066% |AKRO=FU:12;FU:20|"

Result list:

ID	Type/max. field length	Description
RESTYP	C4	Resource type
RESFAM	C20	Resource family
RES	C40	Resource number
BEZ	C40	Designation
RESSTA	Integer	Current status
RESVERWEIS	Integer	Internal ID
RESVER	C20	Resource version
RESSTATXT	C20	Resource status text
RESTYPBEZ	C20	Resource type designation
RESFAMBEZ	C20	Resource family designation
COLOR	Integer	Color
MATPUF:I	12	Current storage location
MNR	C20	Active machine
ANR	C40	Active operation
BZG:RESTYP	C4	Reference resource type
BZG:RES	C40	Reference resource
SLP	Integer	Bill of material item

18.7.2 Resource Status List (DLG=LIST;116)

List of all possible statuses for a resource.

Structure of dialog data:

"DLG=LIST;116|DATEI={file_name}|DAT=...|ZEI=...|USR=...|RESTYP=...|RESFAM=...|RES=…"

Parameter:

Parameters are optional.

RESTYP =... Restrict the list by a resource type

RESFAM =... Restrict the list by a resource family. The designation is to be indicated here, not the ID.

RES =... Restrict the list by a resource.

Example:

hymw -u2020

-c"DLG=LIST;116|DATEI=l.lst|DAT=today|ZEI=now|USR=2020|RES=0668258-WKS|RESTYP=WNR|RESFAM= Soft|"

Result list:

ID	Type/max. field length	Description
OPT:SIBTNR	C1	Display of the resource at the terminal
STUFE	Integer	Authorization level
COLOR	Integer	Status color
MATPUF	C12	Storage location when setting the status
PRIO	Integer	Priority
PKENN	C1	Posting indicator
RESFAMID	Integer	Resource family ID
RESTYP	C4	Resource type
RESSTA	Integer	Status
BEZ	C20	Status designation
OPT:ERF	C1	Collection/processing
OPT:PLAN	C1	HLS assignment

18.7.3 List of measures (DLG=LIST;117)

List of all possible measures for a resource.

Structure of dialog data:

“DLG=LIST;117|DATEI={file_name}|DAT=...|ZEI=...|USR=...|RESFAM=...|RESFAMID=...|RES=…“

Parameter:

Parameters are optional.

RESFAM =... Restrict the list by a resource family (designation).

RESFAMID =... Restrict the list by a resource family (ID).

RES =... Restrict the list by a resource.

Example:

hymw -u2020

-c"DLG=LIST;117|DATEI=l.lst|DAT=today|ZEI=now|USR=2020|RES=0668257-WKS|RESFAMID=10001|RESFAM=Handih.|"

Result list:

ID	Type/max. field length	Description
BEZK	C20	Short designation
BEZL	C60	Long designation
DATE	Datum	Date valid till
DATB	Datum	Date valid from
ZEIE	Integer	Time valid till
ZEIB	Integer	Time valid from
BEM	C200	Comment
MASSNR	Integer	ID of measure
RESFAM	C20	Resource family
RESFAMID	Integer	Reference to resource family
RESTYP	C4	Resource type
TYP	C1	Type of measure
VAB	C15	Responsibility area

18.7.4 Resource types (DLG=LIST;118)

List of all possible resource types.

Structure of dialog data:

“DLG=LIST;118|DATEI={file name}|DAT=...|ZEI=...|USR=...|“

Parameter:

none

Example:

hymw -u2020 -c "DLG=LIST;118|DATEI=l.lst|DAT=today|ZEI=now|USR=2020|"

Result list:

ID	Type/max. field length	Description
OPT:AZSWSIB	C10	Show in reports
OPT:VERB	C1	Posting of the resource
PLAUS:BEL	C1	Assignment within HYDRA-HLS
RESTYPEXT	C10	Designation
BEZL	C60	Long designation
VERB:BMK1..11	Integer	Posting for operating hours
OPT:DATEI	C1	File-based resource
OPT:DNC	C1	Indicator for DNC processing
EXT:GUELITG	C5	File extension
EXT:OPTIM	C5	File extension, optimized files

ID	Type/max. field length	Description
OPT:AUTOANMELD	C1	Log resource on/off with order
CFG:1..10	C1	Configuration settings
PATH	C8	Reference to the hy_path table
AUTOANLAG	C1	Create resource automatically
RESTYP	C4	Resource type
OPT:VBR	C1	Control
USRFLD	C20	Assigned user field ID

18.7.5 Resource family (DLG=LIST;119)

List of all possible resource families.

Structure of dialog data:

“DLG=LIST;119|DATEI={file name}|DAT=...|ZEI=...|USR=...|RESTYP=...|“

Parameter:

Parameters are optional.

RESTYP =... Restrict the list by a resource type

Example:

hymw -u2020

-c"DLG=LIST;119|DATEI=cb.lst|DAT=today|ZEI=now|USR=2020|RESTYP=WNR|"

Result list:

ID	Type/max. field length	Description
RESFAM	C20	Resource family
BEZL	C60	Long designation
VORG:RES	C50	Default specification of resource name
OPT:KOPIE	C1	Performance when copying the resource
OPT:EINST	C1	Parameters for Viewer
FKT:SPEICHORT	Integer	Autom. generation of names
CFG:1..10	C1	Configuration parameters
RESFAMID	Integer	ID of the resource family
RESTYP	C4	Resource type
SPEICHART	C1	Storage type of the family
USRFLD	C20	Assigned user field ID
VAB	C15	Responsibility area
OPT:RESVER	C10	Version procedure
VIEWER:DATA	C10	Viewer to display the resource file
VIEWER:EINST	C10	Parameter fields for the viewer

18.7.6 Resource maintenance (DLG=LIST;120)

List of all possible resource maintenances.

Structure of dialog data:

“DLG=LIST;120|DATEI={file name}|DAT=...|ZEI=...|USR=...|ANR=…|AKTIV=…|BEZ=...|“

Parameter:

Parameters are optional and support wildcard characters (%).

ANR =... Restrict the list by an order

AKTIV =... Restrict the list by active or inactive

BEZ =... Restrict the list using the maintenance designation

Example:

hymw -u2020

-c"DLG=LIST;119|DATEI=cb.lst|DAT=today|ZEI=now|USR=2020|RESTYP=WNR|"

Result list:

ID	Type/max. field length	Description
AKTIV	C1	Active maintenance
APUNR	C40	Work plan
ANR	C40	Order number
BEZ	C60	Maintenance designation
OPT:BZG	C1	Reference of values
DATE	Datum	For maintenance monitoring
DATB	Datum	For maintenance monitoring
INFO:1..7	C80	Info value
ANZ:I	Integer	Actual number
BSTD:I	Integer	Actual operating hours
TAKT:I	Integer	Actual cycle
BDEJMOD	Integer	Year model
WARTKL	C10	Classification of maintenance
LWART:DAT	Datum	Date of last maintenance
LWART:ZEI	Integer	Time of last maintenance
LWART:USR	C10	Person of last maintenance
ANZ:N	Integer	Quantity till next maintenance
BSTD:N	Integer	Operating hours, next maintenance
TAKT:N	Integer	Cycles, next maintenance
TG:N	Datum	Date of next maintenance
RESVERWEIS	Integer	Reference to resource
WARTG:1	Integer	Threshold value for level 1
WARTG:2	Integer	Threshold value for level 2
WARTG:3	Integer	Threshold value for level 3
ANZ:S	Integer	Target number

ID	Type/max. field length	Description
BSTD:S	Integer	Target operating hours
TAKT:S	Integer	Target cycles
TG:S	Integer	Target days
ART	C2	Maintenance type
VERWEIS	Integer	Consecutive no. of maintenance activity
WARTSTA	Integer	Maintenance status

18.7.7 List of maintenance activities (DLG=LIST;91)

List of all possible maintenance activities.

Structure of dialog data:

"DLG=LIST;91|DATEI={file name}|DAT=...|ZEI=...|USR=...|MNR=…|RES=…|RESTYP=...|MOD=…|"

Parameter:

In the machine mode (MOD=M) :

MNR =... Generate list for the transferred machine

In the resource mode (MOD=R) :

RES =... Generate list for the transferred resource + resource type

RESTYP =... Generate list for the transferred resource + resource type

Example:

hymw -u2020

-c"DLG=LIST;91|DATEI=wtk.lst|DAT=today|ZEI=now|USR=2020|MOD=M| MNR=MASCH100|"

Result list:

ID	Type/max. field length	Description
MNR	C40	Machine
RESVERWEIS	Integer	Resource ID
VERWEIS	Integer	Maintenance number
BEZ	C20	Designation of the maintenance
ART	C2	Type of maintenance
WARTSTA	Integer	Status
WARTKL	C10	Class
AKTWERT	Integer	Current value
NAEWERT	Integer	Next value
WARTG:1 … 3	Integer	Threshold value 1 – 3
INFO:1 … 7	C80	Info text 1 – 7
LWART:DAT	Datum	Date of last maintenance
LWART:ZEI	Integer	Time of last maintenance
LWART:USR		User of last maintenance
RES	C40	Resource no.
RESTYP	C4	Resource type

ID	Type/max. field length	Description
KST	10	Cost center
OPT:ONCE	C1	Non-recurring maintenance (J/N)

18.7.8 List of resource comments(DLG=LIST;133)

List of all comments for a resource.

Structure of dialog data:

“DLG=LIST;133|DATEI={FileName}|DAT=...|ZEI=...|USR=...|RESTYP =...|RES=…|DATB=…|ANZ=…|“

Parameter:

RES =... Generate list for resource.

RESTYP =... Restrict list by resource type.

DATB =... Restrict list by date range (only show comments that are “younger” than DATB).

ANZ =... Restrict the number of columns of the list (by default 100).

Example:

hymw -u2020

-c"DLG=LIST;133|DATEI=kommentare.lst|DAT=today|ZEI=now|USR=2020| RES=0668257-WKS|RESTYP=WNR|DATB=12/12/2007|ANZ=10"

Result list:

ID	Type/max. field length	Description
RES	C40	Resource number
RESTYP	C4	Resource type
BEZ	C20	Resource name
DAT	Datum	Date of the comment
ZEI	Integer	Time of the comment
PNR	C10	The editor’s personnel number
PNAME	C40	The editor’s name
BEM	C500	Comment

18.7.9 List of registered resources (DLG=LIST;129)

The list can be used to determine the resources that are logged on.

Structure of dialog data:

“DLG=LIST;129|DATEI={FileName}|DAT=...|ZEI=...|USR=...|MOD=…|“

Parameter:

MOD=M : In the M mode only the topmost machine designations of resources are selected for the machine.

Without MOD: If the list is requested without Mod=M all resources are displayed that are logged on to the machine. Duplicates are removed (e.g. if the resource refers to a machine and is logged on to an order).

Example:

hymw -u2020

-c"DLG=LIST;129|DATEI=l.lst|DAT=today|ZEI=now|USR=2020|MOD=M|MNR=100|"

Result list:

ID	Type/max. field length	Description
MNR	C20	Machine
ANR	C40	Order
RES	C40	Resource number
RESBEZ	C40	Resource name
RESTYP	C4	Resource type
RESTYPBEZ	C20	Resource type name
RESFAMBEZ	C20	Resource family name
RESSTA	Integer	Current status

18.7.10 Combined list of production resources and tools: batch and resources (DLG=LIST;132)

Materials/batches and resources can also be requested in a combined list:

- Resources from the resource list (DLG=LIST;115)
- Materials/batches from the list “material list/batch information“ (DLG=LIST;13)

For further details on these two lists, please refer to the section entitled “resource list” in this documentation or the section “material list/batch information” in the documentation entitled MPL-PDM-DC.

Structure of dialog data:

“DLG=LIST;132|DATEI={FileName}|MNR=…|ANR=…|FHM.ID=…|DAT=...|ZEI=...|USR=...|MOD=…|“

Parameter:

MOD=
K(OMBI)
List of all production resources and tools – material/batches and resources – planned for the OP and logged on to the OP/machine

R(ES)
The list only includes the production resources and tools resources planned for the OP and logged on to the OP/machine

M(AT)
The list only includes the production resources and tools material/batches planned for the OP and logged on to the OP/machine

I(NFO)
List including a single-row information (if the key is not unique, several rows, if necessary) for a specific production resource and tool – batches and/or resources

FHM.ID=
With single-row information it is the key for the selection (resource or batch number)

MNR=
Requesting machine

ANR=
Requesting order/OP

Example:

hymw -u2020

-c"DLG=LIST;132|DATEI=l.lst|DAT=today|ZEI=now|USR=2020|MOD=K|MNR=100|ANR=47110100"

Provides all materials/batches and resources logged on to machine 100 and OP 47110100 or that are planned for OP 47110100.

The result list includes the fields of both lists:

Prefix	Description
MAT.*	The prefix MAT provides all data fields of the batch list DLG=LIST;13. For further information on the individual fields refer to the section “material list/batch information” in the documentation entitled MPL-PDM-DC. These fields are initially assigned (0 or empty) for resource entries.
RES.*	The prefix RES provides all data fields of the resource list DLG=LIST;115 For further information on the individual fields refer to the section entitled “resource list “ in this documentation. These fields are initially assigned (0 or empty) for material/batch entries.
FHM.IDTYP	Key field resource type MAT for materials/batches != MAT for resource (can be WNR, DOC, etc.)
FHM.ID	Key field ID For batches → batch number CNR from list 12 For resources → resource number RES from list 115
FHM.SLP	Key field BOM item BOM item from list 13 or 115
FHM.ATK	Key field For batches → Material ATK from 13 For resources → reference resource BZG:RES from list 115
FHM.BEZ	Key field designation For batches → ATKBEZ from list 13 For resources → BEZ from list 115
RES.AKTIV	J → Resource has already be logged on N → Resource not yet logged on
FHM.VIS	Processing flag for the HYDRA terminal

19 HYDRA Production Data Manager WRM - Master Data

19.1 Note on the descriptions of the input dialogs

All fields that are mandatory and must be specified are highlighted using a gray background color or marked using (PK). All other fields are optional and are processed when passed.

19.2 Resources

19.2.1 Edit resources (DLG=RES.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

You use these dialogs to edit the master data of resources and free resource attributes. You can create and change resources.

Tables

Table	Key field	Description
res_bestand	res_typ RES.RESTYP	Resource type (PK) in resource stock
res_bestand	res_nr RES.RES	Resource number (PK) in resource stock
res_bestand	verweis RES.VERWEIS	Internal serial number
res_status	res_typ RES.RESTYP	Resource type (PK) in resource status
res_status	res_nr RES.RES	Resource number (PK) in resource status
res_status	verweis RES.VERWEIS	Internal serial number, reference to res_bestand.verweis

BAPI call

Identification	Content / {type}	Description
DLG	RES.INSERT	Create resource
	RES.UPDATE	Change resource
	RES.DELETE	Delete resource
	RES.COPY	Copy resource

	RES.LOCK	Lock resource for editing
	RES.UNLOCK	Unlock resource after editing
	RES.NEW	Read specification for new resource
	RES.SELECT	Select resource
RES.RESTYP	{C4}	PK resource type
RES.RES	{C40}	PK resource number
RES.VERWEIS	{N10}	Instead of using RES.RESTYP and RES.RES, you can also call the resource using the serial key RES.VERWEIS.
RES.RESTYP:Z	{C4}	PK new (target) resource type for COPY
RES.RES:Z	{C40}	PK new (target) resource number for COPY
RES.BEZ:Z	{C40}	New (target) resource name for COPY
…	…	For information on further fields, refer to the documentation of the database schema of the above-listed tables. For further information, see the section above.

Validation checks

Error codes	Description
3204	The resource specified in the dialog is already stored in the resource stock.
3220	Resource is logged on or locked
3253	The resource cannot be deleted because it is still used in a resource list relation.
4110	For this resource, a requirement is still available.
3254	The resource cannot be deleted because it is still used as production resource/tool.
7000	Resource records of type "resource logon" or "resource logoff" are available for the resource. For this reason, the resource cannot be deleted.
3261	The resource cannot be deleted because a maintenance calendar is stored for the resource.
3214	The resource type specified is not available in the system.
3200	No resource or resource ID has been transferred to the dialog.
1661	A value relevant for processing is missing.
101	General error message that is displayed when the selected data (tables or files) is not available.

Error codes	Description
1669	Data with the same key fields already exist. It is possible that you cannot see the data because you are not authorized.
3231	The check confirming that the resource is not active fails; i.e. the resource is currently used.
1803	You are not authorized for this responsibility area.
3263	The resource family status is now invalid because the resource family has been changed.
4101	The specified resource is included in a resource list.
3228	The check confirming that the resource is logged on fails; i.e. the resource is currently logged off.
3260	You cannot create or change the resource because the DNC file name is already assigned.

19.2.2 Resource list (DLG=RES.LIST)

This BAPI call creates a resource list. You can narrow down the selection and call the list for one resource, a family, a resource type or combinations of these fields. The free attributes must be read afterwards.

Tables

Table	Key field	Description
res_bestand	res_typ RES.RESTYP	Resource type (PK) in resource stock
res_bestand	res_nr RES.RES	Resource number (PK) in resource stock
res_bestand	verweis RES.VERWEIS	Internal serial number
res_status	res_typ RES.RESTYP	Resource type (PK) in resource status
res_status	res_nr RES.RES	Resource number (PK) in resource status
res_status	verweis RES.VERWEIS	Internal serial number, reference to res_bestand.verweis

BAPI call

Identification	Contents	Description
DLG	RES.LIST	Resource list

RES.RESTYP	{C4}	Search criterion resource type
RES.RES	{C40}	Search criterion resource number
RES.RESFAMID	{N10}	Search criterion resource family
RES.VERWEIS	{N10}	Search criterion serial key
…	…	…
DATEI	{C256}	Specification of the file name for the list

19.3 Free attributes

19.3.1 Edit free attributes (DLG=RESATTR.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

Use the resource type or the resource family to define the resource attributes that you want to edit. The field contents of these fields are entered and managed separately.

When a resource is created (RES.INSERT), a field definition is assigned to this resource, which depends of the relevant resource family or resource type. The key to this field definition is stored in the data field RES.USRFLD,

The field definition of the resource attributes are read via list USRFLDELEM.LIST.

The values entered for the resource are listed in RESATTR.LIST. Both lists have USRFLD and IDX as key. The field definition and the entered attributes are parallel entries; i.e. in one list the column names are listed, in the other the entered values.

Use the dialogs RESATTR.INSERT and RESATTR.UPDATE to enter attribute values, or RESATTR.DELETE to delete values.

If free attributes are used, these attributes must be created after the resource or deleted before the resource.

Tables

Table	Key field	Description
res_attribute	res_id RESATTR.RESVERWEIS	Serial ID (PK) of resource in the resource attributes FK (foreign key) of resource stock (res_bestand.verweis)

Table	Key field	Description
res_attribute	feld_id RESATTR.IDX	Field ID (PK) in the resource attributes Foreign key (FK) of user fields (hyd_userfieldelem.feldid)

BAPI call

Identification	Content / {type}	Description
DLG	RESATTR.INSERT	Create resource attribute
	RESATTR.UPDATE	Change resource attribute
	RESATTR.DELETE	Delete resource attribute
	RESATTR.COPY	Copy resource attribute
	RESATTR.LOCK	Lock resource attribute for editing
	RESATTR.UNLOCK	Unlock resource attribute after editing
	RESATTR.NEW	Read specification for new resource attribute
	RESATTR.SELECT	Select resource attribute
RESATTR.RESVERWEIS	{N10}	Serial ID (PK) of the resource of the resource attribute
RESATTR.IDX	{N2}	Field ID (PK) in the resource attributes
RESATTR.RESTYP	{C4}	Instead of using RESATTR.RESVERWEIS (serial key), you can also call the resource using the resource type RESATTR.RESTYP and the resource number RESATTR.RESNR.
RESATTR.RESNR	{C40}	See above
RESATTR.RESVERWEIS:Z	{N10}	New serial ID (PK) of the resource of the resource attribute for COPY
RESATTR.IDX:Z	{N2}	New field ID (PK) of the resource attribute
RESATTR.ATTR	{C80}	Attribute value

Identification	Content / {type}	Description
…	…	For information on further fields, refer to the documentation of the database schema of the above-listed tables. For further information, see the section above.

Validation checks

Error codes	Description
3201	You must transfer an existing resource number with resource type or a valid resource ID to the dialog.
1661	A value relevant for processing is missing: • You must either specify the serial ID (RESATTR.RESVERWEIS) or the resource type (RES.RESTYP) and the resource number (RES.RES). • You must specify the field ID (RESATTR.IDX) • You must specify the serial ID (RESATTR.RESVERWEIS:Z) with COPY
101	An attribute for the specified key does not exist.
1669	An attribute for the specified key does already exist.
1666	The data record is currently locked by another user.

Note

Make sure that the below configurations are made before you create attributes (RESATTR.INSERT):

1. Create a user field key for the resource type (e.g. DNC) of the resource.
2. To this user field key, assign the user fields for which you want to create attributes. Make sure that the field ID of the user field matches the field ID of the attribute (RESATTR.IDX).
3. You can assign the user field key via the resource type (e.g. DNC) or the resource family(ies).

19.3.2 List of resource attributes (DLG=RESATTR.LIST)

This BAPI call generates a list with the user-defined attributes of the resource. The defined attributes are numbered. The description of the attributes can be read using the field definitions.

Tables

Table	Key field	Description
res_attribute	res_id RESATTR.RESVERWEIS	Serial ID (PK) of resource in the resource attributes Foreign key (FK) of resource stock (res_bestand.verweis)
res_attribute	feld_id RESATTR.IDX	Field ID (PK) in the resource attributes

Table	Key field	Description
		Foreign key (FK) of user fields (hyd_userfieldelem.feldid)

BAPI call

Identification	Contents	Description
DLG	RESATTR.LIST	List of resource attributes
RESATTR.RESVERWEIS	{N10}	Serial ID (PK) of the resource of the resource attribute
RES.RESTYP	{C4}	Instead of using RESATTR.RESVERWEIS (serial key), you can also call the resource using the resource type RESATTR.RESTYP and the resource number RESATTR.RES.
RES.RES	{C40}	See above
DATEI	{C256}	Specification of the file name for the list

19.3.3 List of field definitions (DLG= USRFLDELEM.LIST)

This BAPI call creates a list of field definitions. You can optionally narrow down the list to USRFLD.

Tables

Table	Key field	Description
hyd_userfieldelem	objekt_typ USRFLDELEM.OBJTYP	WRM: resource type of the resource
hyd_userfieldelem	usrfld_key USRFLDELEM.USRFLD	WRM: user field key

BAPI call

Identification	Contents	Description
DLG	USRFELDELEM.LIST	User field list
OBJTYP	{C20}	WRM: The object type is identical to the resource type.
USRFLD	{C6}	WRM: user field key (optional)
DATEI	{C256}	Specification of the file name for the list

19.4 Resource list

19.4.1 Edit resource list (DLG=RESLIST.INSERT, DELETE)

You can create a multi-level structure for the resources in HYDRA. To this end, you can specify a resource list for a resource. This list, also called "package", specifies the components included in the resource. The resource list also integrates the existing relations between the components. A complete tree structure with all components is integrated. A faulty use of this function can cause structural errors in the resource function, which can lead to a hang in HYDRA. We highly recommend to consult MPDV.

To read the resource list entries, a list command is available. This list command can be used at one level of the list or recursively for the complete structure of the resource list.

Note: The above is only true for the master data of the resource list. Once you have created an order and you are interested in the bill of materials, you must call the relevant lists of the BDE.

BAPI call

Identification	Content / {type}	Description
DLG	RESLIST.INSERT	Create resource list entry
	RESLIST.DELETE	Delete resource list entry
RESLIST.RESTYP:M	{C4}	PK: resource type of higher-level resource (M is for "mother")
RESLIST.RESNR:M	{C40}	PK: resource number of higher-level resource (M is for "mother") - alternative entry instead of RESVERWEIS:M
RESLIST.RESVERWEIS:M	{N10}	PK: ID of higher-level resource - alternative entry instead of RESNR:M
RESLIST.RESTYP:T	{C4}	PK: resource type of lower-level resource

Identification	Content / {type}	Description
RESLIST.RESNR:T	{C40}	PK: resource number of lower-level resource (T is for "*Tochter* = daughter") - alternative entry instead of RESVERWEIS:T
RESLIST.RESVERWEIS:T	{N10}	PK: ID of lower-level resource - alternative entry instead of RESNR:T
RESLIST.POS	{N10}	Display position / sequence (with RESLIST.INSERT)
RESLIST.MENGE	{N18.6}	Quantity ratio to mother (with RESLIST.INSERT) May only be set with anonymous resources to a value > 1. Otherwise the value must be 1.
MOD	{C1}	Delete mode (with RESLIST.DELETE) E: Single resource list relation is deleted. G: The complete resource list below a specified mother resource is deleted.
DELFIRSTLEVEL	{C1}	Can only be used with MOD=G. The lower-level resource lists are not deleted. J: Only the first level of the resource list is deleted.

Validation checks

Error codes	Description
3202	The resource specified in the dialog (entry without resource type) is available multiple times in the resource stock with different resource types each.
1666	The data record is currently locked by another user.
1661	A value relevant for processing is missing. • Invalid mode (MOD) is specified. • The lower-level resource has not been specified (RESLIST.RESVERWEIS:T or RESLIST.RESTYP:T and RESLIST.RESNR:T) • The higher-level resource has not been specified (RESLIST.RESVERWEIS:M or RESLIST.RESTYP:M and RESLIST.RESNR:M) • The resource type has not been specified (RESLIST.RESTYP:T or RESLIST.RESTYP:M)
	RESSOURCE_NICHT_VORHANDEN
4101	The specified resource is included in a resource list.
101	General error message that is displayed when the selected data (tables or files) is not available.
1669	Data with the same key fields already exist. It is possible that you cannot see the data because you are not authorized.

19.4.2 Resource list (DLG=RESLIST.LIST)

This BAPI call creates a resource list.

BAPI call

Identification	Contents	Description
DLG	RESLIST.LIST	Resource list
RESLIST.RESTYP:M	{C4}	PK: resource type of higher-level resource (M is for "mother")
RESLIST.RESNR:M	{C40}	PK: resource number of higher-level resource (M is for "mother") - alternative entry instead of RESVERWEIS:M
RESLIST.RESVERWEIS:M	{N10}	PK: ID of higher-level resource - alternative entry instead of RESNR:M
MOD	{C1}	MOD=E: Resource list with one level only from the mother on. Here you can find the resources of the direct level below. MOD=B: Complete structure of the resource list starting from mother. All relations below the mother are recursively read. Here, you can find all related resources in the resource list.
…	…	…
DATEI	{C256}	Specification of the file name for the list

Result list

The result list provides the data fields listed in section 19.4.1 Edit resource list (DLG=RESLIST UPDATE,).

19.5 Assignment to required resources

19.5.1 Edit assignments to required resources (DLG=RESBEDRES.INSERT, DELETE, COPY,)

You can manage required resources in HYDRA that are used as replacements. When the resources are logged on, concrete resources are logged on for these required resources, which are assigned to the required resource. You can assign one or several (replacement) resources to one required resource.

BAPI call

Identification	Content / {type}	Description
DLG	RESBEDRES.INSERT	Create an assignment to a required resource
	RESBEDRES.DELETE	Delete an assignment to a required resource
	RESBEDREST.COPY	Copy an assignment to a required resource
RESBEDRES.RESTYP:M	{C4}	PK: resource type of the required resource
RESBEDRES.RES:M	{C40}	PK: resource number of required resource
RESBEDRES.RESTYP:T	{C4}	PK: resource type of the assigned resource
RESBEDRES.RES:T	{C40}	PK: resource number of the assigned resource
RESBEDRES.BEM	{C100}	Comment
RESBEDRES.RESTYP:MZ	{C4}	With COPY: resource type of the target required resource
RESBEDRES.RES:MZ	{C40}	With COPY: resource number of the target required resource
RESBEDRES.RESTYP:TZ	{C4}	With COPY: resource type of the assigned target resource
RESBEDRES.RES:TZ	{C40}	With COPY: resource number of the assigned target resource
MOD	{C1}	Delete mode (with RESBEDRES.DELETE) E: Single assignment to a required resource is deleted. G: For a specified required resource, all assigned (replacement) resources are deleted

		Copy mode (with RESBEDRES.COPY) E: Single assignment to a required resource is copied. F: All missing assignments of a specified required resource are copied to the target required resource. A: All assignments of a specified required resource are copied to the target required resource.

Validation checks

Error codes	Description
4103	The required resource specified in the dialog is not configured as required resource in the resource stock.
3201	The required resource specified in the dialog or the assigned resource is not available in the resource stock.
4101	The required resource specified in the dialog is already included in a resource list.
4102	The assigned resource specified in the dialog has the type anonymous resource or required resource.
1666	The data record is currently locked by another user.
1661	A value relevant for processing is missing. • Invalid mode (MOD) is specified. • The lower-level resource has not been specified (RESBEDRES.RESTYP:T and RESBEDRES.RESNR:T) • The higher-level resource has not been specified (RESBEDRES.RESTYP:M and RESBEDRES.RESNR:M). • The resource type has not been specified (RESBEDRES.RESTYP:T or RESBEDRES.RESTYP:M)
101	General error message that is displayed when the selected data (tables or files) is not available.
1669	Data with the same key fields already exist. It is possible that you cannot see the data because you are not authorized.

19.6 Resource families

19.6.1 Edit resource families (DLG=RESFAM.INSERT, UPDATE, DELETE, LOCK, UNLOCK, NEW, SELECT)

You use these dialogs to edit the master data of resource families. You can create and change resource families. You can only delete a resource family, if no resource is assigned to this family.

Tables

Table	Key field	Description
res_familien	bezeichnung RESFAM.RESFAM	Resource family (PK). The ID must be unique in all resource types.
res_familien	res_typ RESFAM.RESTYP	If you create a resource family, you must define the resource type.

BAPI call

Identification	Content / {type}	Description
DLG	RESFAM.INSERT	Create resource family
	RESFAM.UPDATE	Change resource family
	RESFAM.DELETE	Delete resource family
	RESFAM.LOCK	Lock resource family and prevent editing
	RESFAM.UNLOCK	Remove lock of resource family after editing
	RESFAM.NEW	Read specification for new resource family
	RESFAM.SELECT	Select resource family
RESFAM.RESFAM	{C8}	PK "resource family"
RESFAM.RESFAMID	{N8}	Internal ID of resource family.
RESFAM.RESTYP	{C4}	Resource type of the resource family. If you create a new resource family, you must define the resource type.
RESFAM.BEZL	{C30}	Name of the resource family
RESFAM.VAB	{C15}	Responsibility area
RESFAMB.USRFLD	{C8}	Reference to a user field key

Validation checks

Error codes	Description
1661	A value is missing that is required for processing. Missing parameter if no short description has been transferred (RESFAM.BEZK).
1661	A value is missing that is required for processing. Missing parameter if no resource type (RESFAM.RESTYP) has been passed on creating a resource family (RESFAM.INSERT).
101	General error message that is displayed when the selected data (tables or files) is not available.
1669	Another resource family with the same short description already exists (RESFAM.BEZK). Data with the same key fields already exist. It is possible that you cannot see the data because you are not authorized.
3243	The resource family is assigned to at least one resource. To delete a resource family, you must first remove all assignments in the resource configuration.
3241	The user field key transferred (RESFAM.USRFLD) does not exist.
1803	You are not authorized for this responsibility area (RESFAM.VAB).

19.7 Resource maintenances

19.7.1 Edit resource maintenances (DLG=RESWART.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK)

You use these dialogs to edit the master data of activities in the *Activity calendar.* You can create, change, copy and delete activities.

Note

To reset a maintenance, use the dialog RES_WART.

Tables

Table	Key field	Description
res_wartungen		The table contains the master data of the activities.

BAPI call

Identification	Content / {type}	Description
DLG	RESWART.INSERT	Create activity
	RESWART.UPDATE	Change activity
	RESWART.DELETE	Delete activity
	RESWART.COPY	Copy activity
	RESWART.LOCK	Lock activity for editing
	RESWART.UNLOCK	Unlock activity after editing
RESWART.AKTTYP	{C1}	Activity type Possible values: empty = maintenance, K = calibration
RESWART.WARTKL	{C10}	Maintenance category
RESWART.KTR	{C40}	Specification of the cost object
RESWART.TAKT:I	{N18.6}	Cycles recorded so far (for the specified resource). Must be set if RESWART-ART = T
RESWART.TAKT:S	{N18.6}	Interval target cycles. Must be set if RESWART-ART = T
RESWART.TAKT:N	{N18.6}	Target cycles (absolute); the next maintenance activity is due after the specified number of target cycles. Must be set if RESWART-ART = T
RESWART.BSTD:S	{N10}	Hours of operation recorded so far (for the specified resource). Must be set if RESWART-ART = B
RESWART.BSTD:N	{N10}	Interval hours of operation. Must be set if RESWART-ART = B
RESWART.BSTD:I	{N10}	Hours of operation (absolute); the next maintenance activity is due after the specified number of operating hours. Must be set if RESWART-ART = B
RESWART.TG:S		Date of next activity. Must be set if RESWART-ART = Z
RESWART.TG:N	{N4}	Interval days. Must be set if RESWART-ART = Z
RESWART.BEZ	{C60}	Name of activity

RESWART.PLANAUNR	{C40}	Specification of planned order to which the activity is assigned
RESWART.PRJNR	{C25}	Specification of project number to which the activity is assigned.
RESWART.OPT:BZG	{C1}	Specification of the reference. Possible values: G = total or A = refers to order
RESWART.RESVERWEIS	{N10}	Internal resource identification number the activity refers to
RESWART.WARTG:1	{N10}	Threshold value for level 1
RESWART.WARTG:2	{N10}	Threshold value for level 2
RESWART.WARTG:3	{N10}	Threshold value for level 3
RESWART.ART	{C2}	Maintenance type T = based on cycles B = based on hours of operation Z = based on time (days)
RESWART.DATB	mm/dd/yyyy	Specification of date for "Valid from"
RESWART.DATE	mm/dd/yyyy	Specification of date for "Valid to"
RESWART.INFO:1 bis 5	{C80}	Information fields 1 to 5
…	…	For information on further fields, refer to the documentation of the database schema of the above-listed tables. For further information, see the section above.

Validation checks

Error codes	Description
3200	No resource or resource ID has been transferred to the dialog.
1661	A value relevant for processing is missing.
101	General error message that is displayed when the selected data (tables or files) is not available.

Error codes	Description
1669	Data with the same key fields already exist. It is possible that you cannot see the data because you are not authorized.
3231	The check confirming that the resource is not active fails; i.e. the resource is currently used.
1803	You are not authorized for this responsibility area.
3263	The resource family status is now invalid because the resource family has been changed.
4101	The specified resource is included in a resource list.